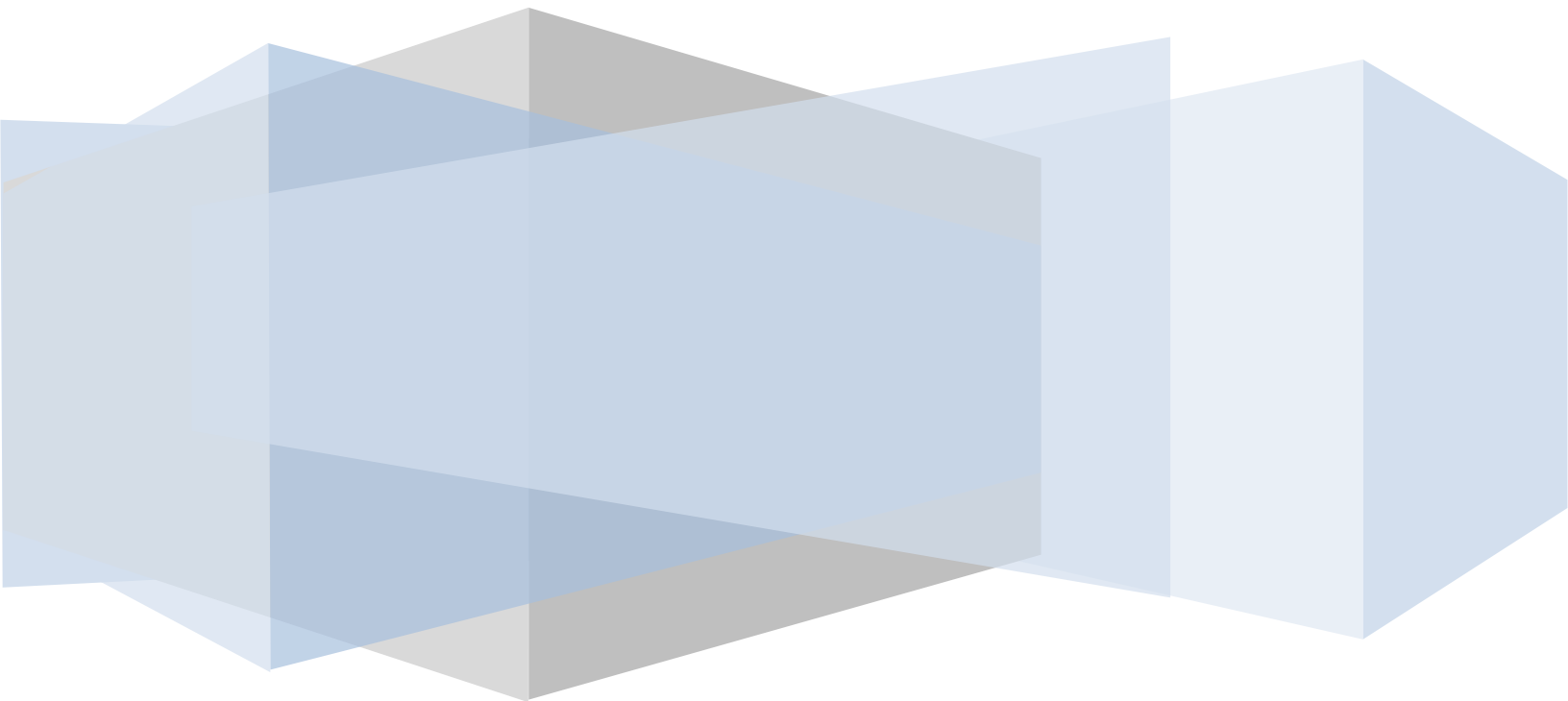


Competency-Based
Federal Democratic Republic of Ethiopia
Revised in 2021 G.C
Ministry of Health in Collaboration with
Ministry of Science and Higher Education
Integrated, Modular
Medical Curriculum



Preface

Curriculum development is defined as planned, a purposeful, progressive, and systematic process to create positive improvements in the educational system. Every time there are changes or developments happening around the world, the school curricula are affected. There is a need to update them to address the society's need.

Curriculum development has a broad scope because it is not only about the school, the learners, and the teachers. It is also about the development of society in general. In today's knowledge economy, curriculum development plays a vital role in improving the economy of a country. It also provides answers or solutions to the world's pressing conditions and problems, such as environment, politics, socio-economics, and other issues of poverty, climate change, and sustainable development.

The development process was led by the Ethiopian Federal Ministry of Health (FMoH). CIRHT and ICAP Colombia University has fully supported the different activities through technical and financial support from inception to finalization.

In 2015, a technical working group (TWG) having members from MoH, MoSHE, Higher Education Relevance and Quality assurance Agency (HERQA), ICAP, CIRHT and different Universities was entrusted to develop the initial competency based integrated curriculum. In 2020/21, it was revised to give space for freshman courses.

The curriculum is intended to be used by all Ethiopian Medical schools to promote the highest standards of medical practice through the delivery of first class medical training.

Acknowledgement

The Federal Ministry of Health would like to recognize the medical education team, Human Resources directorate for taking the initiative to lead the curriculum development process from inception to the end.

The gratitude goes to the Council of Medical Schools in Ethiopia (COME) for the continuous feedback throughout the development process.

The Ministry is pleased to acknowledge CIRHT and ICAP Colombia University for supporting the different workshops of the curriculum development process through technical and financial support.

The ministry of health is very greatfult to experts from Addis Ababa University, Ambo University, University of Gondar, Hawassa University, Haramaya University, Debre-Tabor University, St. Paul Hospital Millenium Medical College, and Yekatit-12 Hospital Medical College for revising the curriculum in 2021.

Finally, it would not have been possible to have achieved this without the extremely valuable and constructive suggestions made during its planning. The research, expertise, dedication and time given so willingly by so many have resulted in the production of this curriculum.

FMoH wishes to express it's deep gratitude to all those who have made it possible

Competency Based Integrated Modular Medical curriculum

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Acronyms

ACGME-	Accreditation Council for Graduate Medical Education
CIRHT -	Center for International Reproductive Health Training
CBE	Community Based Education
CPD -	Continuing Professional Development
DOCS-	Direct Observation of Clinical Skills
DOPS-	Direct observation of Practical skills
EMA-	Ethiopian Medical Association
ENA-	Ethiopian Nurses Association
ENT -	Ear, Nose, Throat
EPA-	Ethiopian Pharmacists Association

EPH- Ethiopian Public Health Association

GPs- General Practitioners

HERQA Higher Education Relevance and Education Agency

HRD- Human Resource Development

HSDP- Health Sector Development Programs

IIME -International Institute of Medical Education

MD – Medical Doctor

MoE- Ministry of Education

NMEI- New Medical Education Initiative

OSCE-Objective Structured Clinical Examination

OSPE- Objective Structured Practical Examination

PBL-Problem-based Learning

PCD -Professional Competency Development

PRRE- Personal Research and Reflection Exercise

SDL- Self-Directed Learning

SGS- Small Group Sessions

SPH- Social and Population Health

SPICES – Student Centered, Problem Based, Integrated, Community Based, Elective and Systematic

UK – United Kingdom

WGS- Whole Group Sessions

WHO- World Health Organization

Background

It is of paramount importance that the value of education is a base for socioeconomic development of a country. The Government of Ethiopia has opened several higher learning institutions over the last decade.

These institutions were established with two main purposes:

- 1) To create opportunities for students to learn by practicing and having real experiences on the teaching-learning process
- 2) To integrate education, research and service aimed at improving the lives of the community at large.

In Ethiopia, the shortage of Health Care Work Force is critical for medical doctors, midwives and anesthesia professionals. In order to address the severe shortage, the Government of Ethiopia has scaled up medical education by expanding number of medical schools and annual student enrollment to 27 and 3000,

respectively as of 2017. Despite the significant stride made to date, the doctor to population ratio continues to lag far behind the WHO benchmark of 1 in 10,000 warranting further scale-up of medical training.

This curriculum is designed and produced as an integrated, competency based, modular approach in which the basic sciences are taught simultaneously with clinical and community medicine. It is described in terms of learning outcomes, which define the knowledge, skills and attitudes (behaviors) so as to produce competent, compassionate, respectful caring and community oriented doctors for Ethiopia with internationally accepted standard of excellence.

The curriculum review was undertaken with the intention of creating common framework for all Ethiopian public universities which are supposed to implement a modular approach to curriculum implementation as of 2017. This necessitates the assessment of the prevailing global, regional and national educational concerns towards producing competent and capable graduates to address the current social and economic dynamics.

The concept of modularization evolved in higher education institutions in the United States in the second half of the nineteenth century (Theodossin, 1986). The underlying conception is that mind itself is modular (Brewer, Nakamura, 1984). Since the early 1980s, in particular, a variety of national and regional reforms as well as many local variations, have involved the introduction and development of modules (Bruijn, 1995).

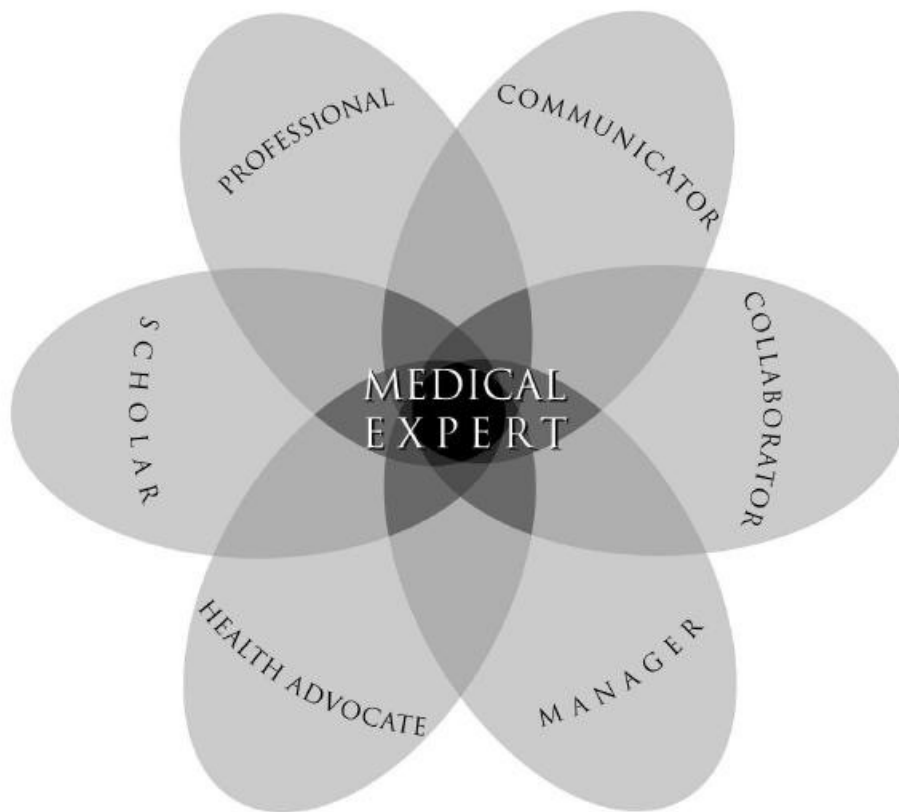
In the reengineering of the Learning-teaching Core Process, modularization was proposed as a best way for the implementation of curricula and the production of competent global graduates.

There are a number of reasons why Federal Ministry of Health opted for modularization.

- Existing curriculum is discipline-based and the courses are fragmented.
- They are not organized around competencies.
- Existing curriculum does not say anything about students' work load which is very important for students' success in their academic life. Hence, student workload is one of the central points in the modularization.
- There is a loose connection between the world of education and the world of work
- Traditional curriculum focused on the teacher rather than the learner. In recent years, there has been a paradigm shift taking place, moving the emphasis from teaching to learning and a more student-centered curriculum. The focus is on how learners learn and the design of effective learning strategies. To this end, it requires the reorganization of the higher education curriculum towards competence-based education.

The curriculum was produced following a Can-Med competency model where the graduate profile (Educational outcomes) is categorized into 7 major physicians' competencies.

These are:



Curriculum designing process

The initiative to have harmonized competency based integrated curriculum was started in 2014 during the medical school's consortium meeting facilitated by the Ministries of Health and Education. The task force was nominated by the consortium in the meeting that was held on August 2015 at Ghion hotel. The task force was represented by Medical Schools, Ministries of Health and Education and Non-Governmental Organizations who are involved in improving quality of medical education in the country including Jhpiego, ICAP and CIRHT. This taskforce was mandated to develop an updated curriculum that has innovative features while maintaining the strength from the conventional one.

The curriculum design process sought to assure that it meets the needs of the society and be responsive to the expectation of the population. To achieve this, local morbidity and mortality data were reviewed; National Health Policies and strategies were considered and National Task Analysis survey findings were used.

Overall the curriculum development process took about 8 months with workshop-based intense engagement and individual assignments completed by members before, in-between and after workshops. Essentially, eight steps were followed:

- Preparation and desk review of documents and resource materials

- First curriculum development workshop
- Second curriculum development workshop
- Individual assignments
- Third Curriculum Development Workshop
- Compilation of draft curricula for review
- Curriculum review workshop
- Finalization of the curriculum

The first meeting was held in November 2015. Faculty from biomedical, clinical and public health disciplines of various senior and junior medical schools have taken part.

The first step in developing the curriculum was identifying core competencies for medical doctors in Ethiopia. The decision was to follow the Can-Med competency framework as it was deemed to have a well-structured none overlapping set of competencies based on 7 roles of physician applicable to our country.

The discussion helped to create clarity, deepen understanding and reach consensus. The task force members agreed to build on strengths of the conventional curriculum and develop a competency-based curriculum with emphasis on constructive alignment of learning outcomes, teaching/learning strategies and assessment methods, learning domains (knowledge, skill and attitude required to be competent) and hierarchy of learning (learning outcomes of appropriate complexity expected of tertiary education). The task force also agreed to use SPICES² Model for the curriculum. This was accomplished through review and critical adaptation of local and international documents. The taskforce reviewed the main existing medical curricula from Ethiopia (the 2011 innovative medical curriculum (NMEI), Hybrid Curriculum of Debre-Tabor and Addis Ababa University integrated curriculum, St. Paul Millennium medical school curriculum and harmonized conventional curriculum (2008) and syllabus of Community based education of Jimma University). Members of the task force discussed and debated the contemporary views vis-à-vis the conventional curriculum and also used the experience in the implementation of the new curricula. Regarding global resources, learning outcomes defined by the International Institute of Medical Education (IIME), Accreditation Council by Graduate Medical Education (ACGME), the royal college physician and surgeon of Canada (Can Meds), tuning Africa harmonization higher education, General Medical Council of UK (Tomorrow's Doctors), Competency based curriculum development in medical education of World Health Organization (WHO) guideline.

Possible topics for PRRE

- Medical interns will select a topic for their PRRE based on their experience in the field

² SPICES- Student Centered, Problem Based, Integration, Community Based, Electives and Systematic

In the second workshop, after having a presentation on how to align the module objectives to content, method and assessment, the course syllabi of the integrated pre-clerkship modules, clerkship and internship modules were drafted. A team composed of all the biomedical sciences has taken the responsibility in identifying the module objectives and agree on list of the topics that should go in the modules allocating time. They have also worked to rectify the confusion in deciding where to put some concepts with overlapping features across disciplines. The teams have to take assignment to work on their time to their respective institutions.

In the third workshop, fine tuning has been done to each of the modules though having more people to review the documents. Course syllabi have been prepared for some courses which course developer was not found in the first meetings like forensic medicine, anesthesia and critical care, dermatology, ENT and radiology. Finally, those who lead the curriculum development have put the material together for medical schools to have one last look before calling it final.

Finally, the curriculum was inaugurated in August 2017 in the 5th COME meeting at Bahir-Dar in the presence of the then State Minister of Health – Dr. Amir Aman.

In 2020, as per the recommendation on Education Roadmap, MoSHE introduced 17 additional courses to all curricula. Accordingly, several workshops (Addis Ababa, Adama, Bishoftu, and Gondar) were conducted engaging experts from MoSHE, MoH, Universities, Professional Associations and partners with the aim of integrating the newly added courses in the Competency-based integrated curriculum.

Rationale

With the aim of increasing the availability of physicians in the country, the government has embarked on massive expansion of medical schools both private and public. However, the current competency of graduates in the country to provide quality health care service is doubted by many. Evidences from the recent findings of national task analysis survey that assessed health professionals' competency and graduate competency assessment studies revealed that physician are not getting the competencies they require before entry in to practice. This can be attributed to a lot of factors – the curriculum design, the execution, lack of training facilities, shortage and inexperienced faculty etc.

Parallel to increasing the number, quality of the graduates that comes as a result of quality training is equally important to improve the health care. One way of assuring quality of training is through designing a curriculum that has effective educational strategies. A well-crafted curriculum can have a lot of impact in assuring quality of graduates.

As each modules are designed by integrating all Biomedical sciences subjects in Pre-clerkship, it is beneficiary to bring an effective, focused and skilled individuals by providing ample time for experimental and PBL related learning activities and to deliver various subjects using the resource and time that has been used in the conventional curriculum.

For example to look over the time management of traditional curriculum system and the new modularization system within a week time, it is true that the new modularized curriculum with delivering variety of teaching learning activities including practical sessions helps students to become competent, self-focused and skill oriented individuals throughout each module. One of the evidences that concretize the point is to look over the weekly time comparison between the new proposed modularized system and the current method of teaching learning process for medical students. For example, in pre-clerkship the teaching methodology of old curriculum needs around 20 lecture hours in a week from the total of 168 Hrs. But the new modularized system-based learning will reduce the lecture hours to 12hours with the same contents that has been delivered in the old system. The practical session proposed in the old curriculum consumes around 12 hrs in a week and even some of the contents are not up-to-date and relevant for medical students. But the new modularized system will bring up a proper lab practical sessions/experiments relevant to medical students and increases the time from 12 to 16hrs to allow students to be skilled than purely theory oriented. This is also in the presence of other teaching and learning opportunities through PBL, and SPH.

This curriculum is, therefore, designed based on the strengths of the traditional curriculum and incorporating innovative features highlighted in the SPICES model (student-centered, problem-based, integrated, community-based, elective and systematic) and other global recommendations on instructional design. It integrates the three arms of medicine (biomedical science, clinical sciences and public health) both horizontally and vertically starting year one to final year of the academic program.

Degree Nomenclature

Upon successful completion of the program, a graduate will be awarded with the degree of **DOCTOR OF MEDICINE (MD)** (in Amharic, “የሕክምና ዶክተራት ዲግሪ”)

Mission and Objectives of the Program

Mission

- To produce competent, compassionate and committed doctors that would meet the present and future priority health needs of the Ethiopian population and fulfill international standards for medical education

Objectives

- To prepare medical doctors equipped with solid knowledge of basic, clinical, behavioral, social and public health sciences
- To prepare medical doctors as clinicians who provide patient-centered care in the spirit of inter-professional collaboration and teamwork
- To prepare medical doctors who address health priority needs of Ethiopian population and its health system challenges
- To prepare medical doctors who promote and protect the health of individuals and populations
- To prepare medical doctors who act according to ethical and legal principles
- To prepare medical doctors who are leaders, problem-solvers, communicators and educators
- To prepare medical doctors who embrace critical thinking, evidence-based practice and research
- To inculcate in medical doctors the ability for lifelong learning
- To produce medical doctors meeting global standards

Core competencies

MEDICALEXPERT

Definition

As Medical Experts, physicians integrate all of the Roles, applying medical knowledge, clinical skills and professional attitudes in their provision of high-quality and safe patient-centered care.

Description

As Medical Experts who provide high-quality, safe, patient-centered care, physicians draw upon an evolving body of knowledge, their clinical skills, and their professional attitudes. They collect and interpret information, make clinical decisions, and carryout diagnostic and therapeutic interventions. They do so within their scope of practice and with an understanding of the limits of their expertise. Their decision-making is informed by best practices and research evidence, and considers the patient's circumstances and preferences as well as the availability of resources. Their clinical practice is up-to-date, ethical, and resource-efficient, and is conducted in collaboration with patients and their families, other health care professionals, and the community. The Medical Expert Role is central to the function of physicians and draws on the competencies included in the Intrinsic Roles (Communicator, Collaborator, Leader, Health Advocate, Scholar, Manager and Professional).

Physicians are able to:

1. Practice medicine within their defined scope of practice and expertise

Enabling competencies

- 1.1. Demonstrate a commitment to high-quality care of their patients

- 1.2. Integrate other intrinsic roles into their practice of medicine
- 1.3. Apply knowledge of the clinical and biomedical sciences relevant to different disciplines they are exposed to
- 1.4. Apply concepts and principles of public health sciences to protect and promote individual and population health
- 1.5 Perform appropriately timed clinical assessments with recommendations that are well organized and properly documented in written and/or oral form
- 1.6 Carry out professional duties in the face of multiple, competing demands
- 1.7 Recognize and respond to the complexity, uncertainty, and ambiguity inherent in medical practice

2. Perform a patient-centered clinical assessment and establish appropriate management plans

- 2.1. Identify and prioritize issues to be addressed in a patient encounter
- 2.2. Elicit history, perform a complete physical exam, make a differential diagnosis, select appropriate investigations, perform basic investigations and interpret their results for the purpose of diagnosis and management, disease prevention, and health promotion
- 2.3. **Diagnose and** establish a patient-centered management plan **based on available evidences**
- 2.4. Provide clinical care within the practical context of patient physician relationship taking into account patient's culture and adapting to economic limitations, and encounter time constraints
- 2.5. Establish goals of care in collaboration with patients and their families, which may include slowing disease progression, achieving cure, improving function, treating symptoms, and palliation

3. Plan and perform procedures and interventions for the purpose of assessment and/or management

- 3.1 Determine the most appropriate procedure(s) for the purpose of assessment and/or management
- 3.2 Obtain and document informed consent, explaining the risks and benefits of, and the rationale for the proposed options
- 3.3 Prioritize procedures, taking into account clinical urgency, potential for deterioration, and available resources
- 3.4 Perform procedures in a skillful and safe manner, adapting to unanticipated findings or changing clinical circumstances

4. Establish plans for ongoing care and, when appropriate, timely consultation

- 4.1 Establish the roles of physicians, other healthcare professionals, and the patient in the provision of a patient-centered care plan that supports ongoing care, including follow-up on investigations, response to treatment, and further consultation.

5. Actively participate as an individual and as a member of a team providing care, in the continuous improvement of health care quality and patient safety

- 5.1 Recognize and respond to adverse events and near-misses
- 5.2 Contribute to a culture that promotes patient safety
- 5.3 Adopt strategies that promote patient safety and mitigate negative human and system factors

Communicator

Definition

Communication Skill: refers to interpersonal and communication skills that result in effective information exchange and teaming with patients, patients' families, professional associates and community at large.

Description

Physicians enable patient centered therapeutic communication by exploring the patient's symptoms, which may be suggestive of disease and by actively listening to the patient's experience of his or her illness. Physicians explore the patient's perspective, including his or her fears, ideas about the illness, feelings about the impact of the illness, and expectations of healthcare and healthcare professionals. The physician integrates this knowledge with an understanding of the patient's context, including socio-economic status, medical history, family history, stage of life, living situation, work or school setting, and other relevant psychological and social issues. Central to a patient- centered approach is shared decision-making: finding common ground with the patient in developing a plan to address his or her medical problems and health goals in a manner that reflects the patient's needs, values, and preferences. This plan should be informed by evidence and guidelines.

Because illness affects not only patients but also their families, physicians must be able to communicate effectively with every one involved in the patient's care.

Key competencies

1. Establish professional therapeutic relationships with patients and their families

Enabling competencies

Physicians are able to:

- 1.1. Communicate using a patient-centered approach that encourages patient trust and autonomy and is characterized by empathy, respect, and compassion
- 1.2. Optimize the physical environment for patient comfort, dignity, privacy,

engagement, and safety

1.3. Recognize when the values, biases, or perspectives of patients, physicians, or other health care professionals may have an impact on the quality of care, and modify the approach to the patient accordingly

1.4. Respond to patients' non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients

1.5. Manage disagreements and emotionally charged conversations

1.6. Adapt to the unique needs and preferences of each patient and to his or her clinical condition and circumstances

2. Elicit and synthesize accurate and relevant information, incorporating the perspectives of patients, their families and communities

2.1. Use patient-centered interviewing skills to effectively gather relevant biomedical and psychosocial information

2.2. Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information

2.3. Provide a clear structure for and manage the flow of an entire patient encounter

2.4. Seek and synthesize relevant information from other sources, including the patient's family, with the patient's consent.

3. Share health care information and plans with patients and their families

3.1 Skillfully share information and explanations that are clear, accurate, timely, and adapted to patient's and his or her family's level of understanding and need

3.2 Communicate appropriately in difficult circumstances, such as when breaking bad news, and when discussing sensitive issues, such as alcohol consumption, smoking or obesity

3.3 Communicate appropriately with difficult/ violent patients and people with mental illness

4. Engage patients and their families in developing plans that reflect the patient's healthcare needs and goals

4.1 Facilitate discussions with patients and their families in a way that is respectful, non-judgmental, and culturally safe

4.2 Assist patients and their families to identify, access and make use of information and communication technologies to support their care and manage their health

4.3 Use appropriate communication skills and strategies to help patients and their families make informed decisions regarding their health

5. Document and share written and electronic information about the medical encounter to optimize clinical decision-making, patient safety, confidentiality, and privacy

5.1 Document clinical encounters in an accurate, complete, timely, and accessible manner, in compliance with legal and regulatory requirements

5.2 Communicate effectively using a written health record, electronic medical record, or other digital technology

Collaborators

As Collaborators, physicians work effectively with other health care professionals and support staff to provide safe, high-quality and compassionate patient-centered care.

Description

Collaboration is essential for safe, high-quality, patient-centered care, and involves patients and their families, Physicians and other colleagues in the health care professions, community partners, and health system stakeholders. Collaboration requires relationships based in trust, respect, and shared decision-making among a variety of individuals with complementary skills in multiple settings across the continuum of care. It involves sharing knowledge, perspectives, and responsibilities, and a willingness to learn together. This requires understanding the roles of others, pursuing common goals and outcomes, and managing differences. Collaboration skills are broadly applicable to activities beyond clinical care, such as administration, education, advocacy, and scholarship.

1. Work effectively with physicians and other colleagues in the health care professions

- 1.1 Establish and maintain healthy relationships with physicians and other colleagues in the health care professions to support relationship-centered collaborative care
- 1.2 Negotiate overlapping and shared responsibilities with physicians and other colleagues in the health care professions
- 1.3 Engage in respectful shared decision-making with other colleagues, patients and their Families
- 1.4 Recognize one's own professional roles and limits
- 1.5 Share knowledge and information to other colleagues

2. Work with physicians and other colleagues in the health care professions to prevent misunderstandings, manage differences, and resolve conflicts

- 2.1 Show respect toward other physicians and members of the health care team
- 2.2 Respect and value diversity among other colleagues
- 2.3 Implement strategies to resolve conflicts in a manner that supports a collaborative culture

3. Effectively and safely transfer care to another health care professional

- 3.1 Assess when care should be transferred to another physician or health care professional
- 3.2 Demonstrate safe transfer of care, using both verbal and written communication, during a patient transition to a different health care professional, setting, or stage of care

Leader

Definition

As Leaders, physicians develop a vision of a high-quality health care system and, in collaboration with other health care leaders, take responsibility for effecting change to move the system toward the achievement of that vision.

Description

Leadership and management ability as core requirements for the practice of medicine is vital. Physicians and others exercise collaborative leadership within the complex health care systems that form their specific work environments. At a system level, physicians contribute to the development and delivery of continuously improving health care and engage others to work with them toward this vision. Physicians must balance their personal lives with their responsibilities as leaders and managers in their everyday clinical, administrative, research, and teaching activities. They function as individual care providers, as members of teams or groups, and as participants and leaders in the health care system locally, regionally, nationally, and globally. The Leader Role describes the active engagement of all physicians as leaders and managers in decision making in the operation and ongoing evolution of the health care system.

Core competencies

1. Contribute to the improvement of health care delivery in teams, organizations, and systems

- 1.1 Apply the science of quality improvement to contribute to improving systems of patient care
- 1.2 Analyze adverse events and near misses to enhance systems of care
- 1.3 Use health informatics to improve the quality of patient care and optimize patient safety

2. Engage in the stewardship of health care resources

- 2.1 Allocate health care resources considering justice, efficiency, and effectiveness for optimal patient care
- 2.2 Apply evidence and management processes to achieve cost-appropriate care

3. Demonstrate leadership in professional practice

- 3.1 Develop their leadership skills

3.2 Design and organize elements of health care delivery

3.3 Facilitate change in health care to enhance services and outcomes

4. Manage their practice and career of themselves, other health care providers and support staff

4.1 Set priorities and manage time to balance practice and personal life

4.2 Manage career planning, finances, and health human resources in a practice

4.3 Implement processes to ensure personal practice improvement

4.4 Promote occupational health for sustainable practice

Health Advocate

Definition

As Health Advocates, physicians contribute their expertise and influence as they work and be **role model** for communities or patient populations to improve health. They work with those they serve to determine and understand needs, speak on behalf of others when needed, and support the mobilization of resources to effect change. **Physicians take the cultural context of the community whenever they engage in health advocacy and promotion.**

Description

Physicians recognize their duty to participate in efforts to improve the health and well-being of their patients, their communities, and the broader populations they serve.* Physicians possess medical knowledge and abilities that provide unique perspectives on health. Physicians also have privileged access to patients' accounts of their experience with illness and the health care system. Improving health is not limited to mitigating illness or trauma, but also involves disease prevention, screening, health promotion, surveillance, and health protection. Improving health also includes promoting health equity, whereby individuals and populations reach their full health potential without being disadvantaged by, for example, race, ethnicity, religion, gender, age, social class, economic status, or level of education.

Physicians leverage their position to support patients in navigating the health care system and to advocate with them to access appropriate resources in a timely manner. Physicians seek to improve the quality of both their clinical practice and associated organizations by addressing the health needs of the patients, communities, or populations they serve. Physicians promote healthy communities and populations by influencing the system (or by supporting others who influence the system), both within and outside of their

work environments. Advocacy requires action. Physicians contribute their knowledge of the determinants of health to positively influence the health of the patients, communities, or populations they serve. **Take steps to maintain their own physical and mental health, using coping strategies and seeking appropriate assistance as needed, by this they serve as role models by advocating for and promoting healthy life style.** Physicians gather information and perceptions about issues, working with patients and their families to develop an understanding of needs and potential mechanisms to address these needs. Physicians support patients, communities, or populations to call for change, and they speak on behalf of others when needed. Physicians increase awareness about important health issues at the patient, community, or population level. They support or lead the mobilization of resources (e.g., financial, material, or human resources) on small or large scales.

- **Respond to the individual patient's health needs by advocating with the patient within and beyond the clinical environment**
 - 1.1 Work with patients to address determinants of health that affect them and their access to needed health services or resources
 - 1.2 Work with patients and their families to increase their opportunities to adopt healthy behaviors
 - 1.3 Incorporate disease prevention, health promotion, and health surveillance into interactions with individual patients
 - 1.4. Lead a healthy life style to be role models to promote population health
- **Respond to the needs of the communities or patient populations they serve by advocating with them for system-level change**
 - 2.1 Work with a community or population to identify the determinants of health that affect them
 - 2.2 Improve clinical practice by applying a process of continuous quality improvement to disease prevention, health promotion, and health surveillance activities
 - 2.3 Participate in a process to improve health in the community or population they serve. Physician advocacy occurs within complex systems and thus requires the development of partnerships with patients, their families and support networks, or community agencies and organizations to influence health determinants. Advocacy often requires engaging other health care professionals, community agencies, administrators, and policy-makers.

SCHOLAR

Definition

As Scholars, physicians demonstrate a lifelong commitment to excellence in practice through continuous learning, the teaching of others, the evaluation of evidence, and contributions to scholarship.

Description

Physicians acquire scholarly abilities to enhance practice and advance health care. Physicians pursue excellence by continually evaluating the processes and outcomes of their daily work, sharing and comparing their work with that of others, and actively seeking feedback in the interest of quality and patient safety. Using multiple ways of learning, they strive to meet the needs of individual patients and their families* and of the health care system. Physicians strive to master their domains of expertise and to share their knowledge. As lifelong learners, they implement a planned approach to learning in order to improve in each role. They recognize the need to continually learn and to model the practice of lifelong learning for others. As teachers they facilitate, individually and through teams, the education of students and residents, colleagues, co-workers, the public, and others. Physicians are able to identify pertinent evidence, evaluate it using specific criteria, and apply it in their scholarly activities and practice. Through their engagement in evidence-informed and shared decision-making, they recognize uncertainty in practice and formulate questions to address knowledge gaps. Using skills in navigating information resources, they identify evidence syntheses that are relevant to these questions and arrive at clinical decisions that are informed by evidence while taking patient values and preferences into account. Finally, physicians' scholarly abilities allow them to contribute to the application, dissemination, translation, and creation of knowledge and practices applicable to health and health care.

Key competencies	Enabling competencies
Physicians are able to:	Physicians are able to:
1. Engage in the continuous enhancement of their professional activities through ongoing learning	1.1 Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
	1.2 Identify opportunities for learning and improvement by regularly assessing their performance using various internal and external data sources
	1.3 Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
2. Facilitate the learning of students, the public, and other health care professionals	2.1 Recognize the power of role-modeling and the impact of the hidden curriculum on learners
	2.2 Promote a safe learning environment
	2.3 Ensure patient safety is maintained when learners are involved

	2.4 Collaboratively identify the learning needs of others and prioritize learning outcomes
	2.5 Demonstrate effective teaching to facilitate learning
	2.6 Provide effective feedback to enhance learning and performance
	2.7 Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
3. Integrate best available evidence, contextualized to specific situations, into real-time decision-making	3.1 Recognize practice uncertainty and knowledge gaps in clinical and other professional encounters and generate focused questions that can address them
	3.2 Demonstrate proficiency in identifying, selecting, and navigating pre-appraised resources
	3.3 Integrate evidence into decision-making in their practice
4. Critically evaluate the integrity, reliability, and applicability of health-related research and literature	4.1 For a given practice scenario, formulate well-structured scholarly questions
	4.2 Identify scholarly sources that shed light on a given professional question
	4.6 Evaluate the applicability (external validity or generalizability) of evidence from a wide range of information resources
	4.7 Discuss the barriers to and facilitators of applying study findings to professional practice
	4.8 Use quality-appraised evidence-alerting services that highlight new evidence appropriate to their scope of professional practice
	4.3 Interpret study findings, including a discussion and critique of their relevance to professional practice
	4.4 Determine the validity and risk of bias in a wide range of scholarly sources

	4.5 Describe study results in both quantitative and qualitative terms
5. Contribute to the dissemination and/or creation of knowledge and practices applicable to health	5.1 Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
	5.2 Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
	5.3 Contribute to the work of a research program
	5.4 Pose questions amenable to scholarly investigation and select appropriate methods to address them
	5.5 Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

PROFESSIONAL

Definition

As Professionals, physicians are committed to the health and well-being of individual patients and society through ethical practice, high personal standards of behavior, dedication to the profession, profession-led regulation, and maintenance of personal health.

Description

Physicians serve an essential role as professionals dedicated to the health and care of others. Their work requires mastery of the art, science, and practice of medicine. The professional identity of physicians has developed over millennia, and the Professional Role reflects contemporary society's expectations of physicians, which include clinical competence, a commitment to ongoing professional development, promotion of the public good, and adherence to ethical standards and values, such as integrity, honesty, altruism, respect for diversity, and transparency with respect to potential conflicts of interest. It is also recognized that, to provide optimal patient care, physicians must take responsibility for their own health and well-being and that of their colleagues. The professionalism of physicians is the basis of the implicit contract between society and the medical profession, by which physicians are granted the privilege of profession-led regulation with the understanding that they are accountable to those served, to society, to their profession, and to themselves.

Key competencies: Physicians are able to:	Enabling competencies: Physicians are able to:
1. Demonstrate a commitment to patients by applying best practices and adhering to high ethical	1.1 Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
	1.2 Demonstrate a commitment to excellence in all aspects of practice

standards	1.3 Recognize and respond to ethical issues encountered in practice
	1.4 Recognize and manage conflicts of interest
	1.5 Exhibit professional behaviors in the use of technology enabled communication
2. Demonstrate a commitment to society by recognizing and responding to the social contract in health care	2.1 Demonstrate accountability to patients, society, and the Profession
3. Demonstrate a commitment to the profession by adhering to standards and participating in physician-led regulation	3.1 Fulfill and adhere to the professional and ethical codes, standards of practice, and laws governing practice
	3.2 Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
	3.3 Participate in peer assessment and standard-setting
4. Demonstrate a commitment to physician health and well-being to foster optimal patient care	4.1 Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
	4.2 Manage personal and professional demands for a sustainable practice throughout the physician life cycle
	4.3 Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Curriculum Model and Educational Strategies

The curriculum model is competency (outcomes)-based with three cardinal features. Firstly, clear, measurable and observable learning outcomes are stated at program, module and rotation levels considering learning domains (knowledge, skill and attitude) and learning hierarchy suitable for higher order learning requiring application and problem-solving. Secondly, appropriate and

experiential teaching and learning strategies that are aligned with defined learning outcomes are selected to maximize attainment of the desired competencies by all students. Thirdly, assessment methods that are authentic and require students to perform the tasks defined by the learning outcomes are suggested. To the extent possible, there is constructive alignment among learning outcomes, teaching and learning strategies and assessment methods.

Within the framework of the SPICES Model, the following innovative educational strategies are used in the development of the curriculum.

Student-centered

In this curriculum, students are given greater responsibility for their own learning by integrating a more learner—centered teaching and learning and assessment methods such as problem-based learning, personal research and reflection exercise and portfolio-based learning and assessment. This has the advantage of putting the emphasis on the student, increasing students' motivation and preparing them to become lifelong learners.

Problem-based learning

Recognizing the deficiencies of the information gathering approach in the traditional curriculum, problem-based learning (PBL) is introduced as a complementary learning strategy in the first two years of the curriculum. PBL has the advantages of developing problem-solving skills, developing an integrated body of knowledge, facilitating student motivation in learning and retention of what is learned and reducing curriculum overcrowding.

Integration

Recognizing the shortfalls of the subject-based organization of the traditional/conventional curriculum, related subjects and learning experiences are integrated throughout this curriculum. There is both horizontal and vertical integration. Integration is done using body systems and thematic areas as an organizing framework. Regarding basic and clinical sciences, the emphasis is on basic sciences during pre-clerkship while clinical sciences take the upper hand in later years. Clinical exposure starts in year one with progressive increase in intensity, complexity and student responsibility over the years. During clerkship, there is vertical integration of clinical and basic sciences but there is no horizontal integration of clinical disciplines. Introduction of clinical medicine early will provide context and relevance to the learning of basic sciences. Going back to basic sciences during clerkship years will enable students to relate clinical data to biological

principles and mechanisms. Social and population health subjects are organized around competency functions and run as a longitudinal thread throughout the medical curriculum. The advantages of integration are reducing fragmentation of medical courses and demonstrating unity of medicine, motivating students and shaping their attitude towards the medical profession, improving educational effectiveness of teaching and learning, encouraging development of higher level objectives, longitudinal coverage of all eight domains of competencies, promoting staff communication and collaboration and bringing about rationalization of teaching resources.

Team teaching

Team teaching will be utilized in this curriculum that is teachers from different disciplines will be teaching in the integrated modules as a team. This will avoid unnecessary repetition, disjointed teaching because of isolation from other 'subjects', and confusion because of departmental differences of opinion. See below a rough depiction of the longitudinal design of the curriculum.

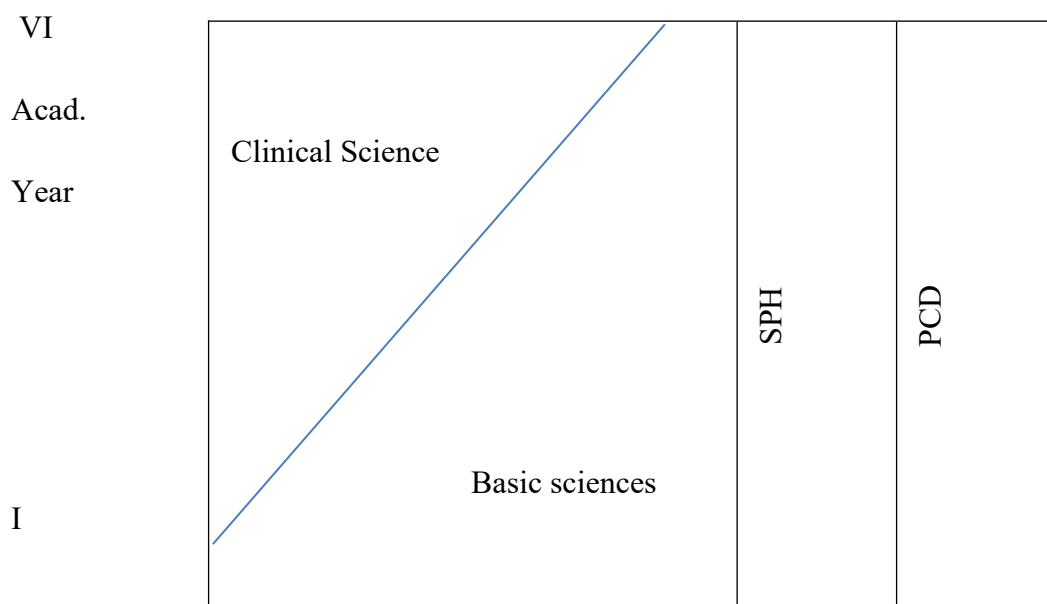


Figure 1: Integration and longitudinal coverage of major components of the undergraduate medical curriculum

Community Based Education

To prepare doctors who are able and willing to address priority health needs of their communities, community-based learning is added to complement the hospital-based learning. To this end, students have scheduled visits to primary health care facilities and communities, community-based training programs and a team training program during internship. Advantages of community-based learning include providing community-orientation to the curriculum, providing useful learning experiences for competencies that may not be adequately developed in a hospital setting, making use of untapped resources, encouraging active learning, exposing students to patients that have not been seen by healthcare providers and introduction to the healthcare system.

Systematic and planned learning

In this curriculum, simulation, clinical and community teaching and learning activities are planned and so that all students gain the necessary experiences for developing essential competencies. To this end, model schedules of essential learning activities in the skills learning labs, clinical settings and communities are developed, systematic rotations are proposed and lists of essential skills and procedures to be performed by each student are specified.

Program Requirements

Admission Requirements

Each medical school selects meritorious and passionate students based upon academic achievement and informed self-selection. Given the importance of having valid, defensible, comprehensive and fair student selection criteria, it is recommended that medical schools develop a defensible guideline on the same. The criteria may include

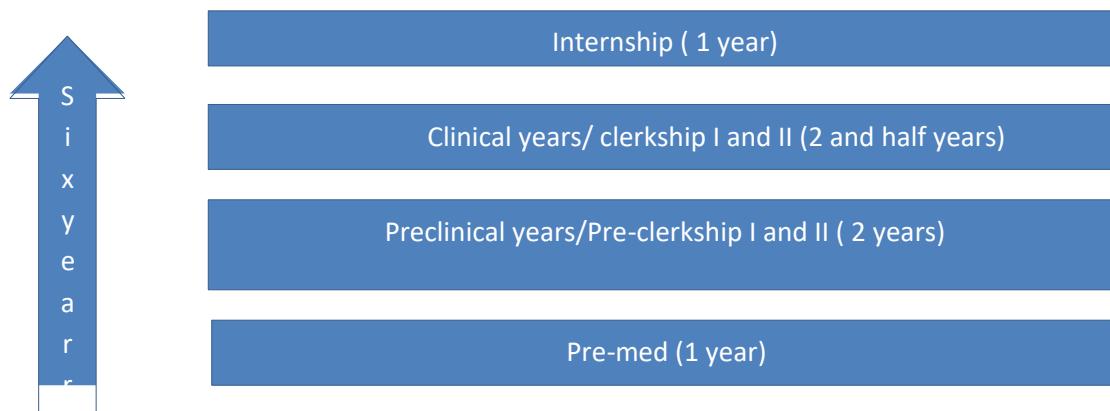
- Academic achievement
- Aptitude test result
- Personality test
- Interview result
- diversity (sex, disability and geography) and
- Absence of **major** mental and physical disabilities that could affect studying and practicing medicine. This has to be substantiated by expert's opinion

Program Structure, Composition and Duration

The duration of the medical education is six years with the following components: Pre-medicine, pre-clerkship I, pre-clerkship II, clerkship I, clerkship II and internship. Pre-clerkship I and II cover system-based modules (integrating biomedical, clinical, professional ethics and generic skills), social and population health modules and common courses. System-based modules instill solid knowledge of scientific basis of medicine and develop early clinical skills. The social and population health modules enable students to measure, analyze and promote population health. Progression from pre-clerkship to clerkship is determined by successfully passing a comprehensive internal and external exam. During clerkship, students develop clinical knowledge, skills and attitude through intensive patient-based teaching and guided practice in different clinical departments with opportunities to review pertinent basic biomedical sciences. They also acquire knowledge and skills of health systems, management and medical research. Then, students must pass a final qualification exam including comprehensive internal and external exam to progress to internship. Internship provides students with the opportunity to work under supervision in different clinical departments including a primary health care setting preparing them for autonomous practice. Before the end of each academic year in years 2 to 4, students have a month long community attachment to offer opportunities for application and deepening of knowledge and skills acquired in the respective years in the community setting.

Based on the task analysis and national health policies and strategies, Emergency medicine was newly designed.

Psychiatry, Emergency Medicine, Ophthalmology, Dermatology, ENT and Dentistry were added to internship rotations to increase the clinical exposure of students.



Teaching and Learning Methods

Selection of appropriate teaching and learning methods is of critical importance in a competency or outcomes-based curriculum. In selection of the teaching and learning methods, the following considerations were made and faculty should keep in mind these features in appraising and revising teaching and learning methods during implementation of the curriculum:

➤ Alignment.

The selected methods are closely aligned with the intended learning outcomes across the three learning domains (knowledge, skills and attitude) and the expected level of performance by the learner thereby maximizing the chances that learner will master the required knowledge, skills and attitude

➤ Variety.

One size does not fit all. What might be considered as the best teaching and learning method for a certain learning outcome may not be so for another learning outcome. In addition, using different methods is likely to stimulate and motivate learners. Hence, a variety of teaching and learning methods are suggested to be used in the curriculum.

➤ Collaborative

Where possible, methods that encourage collaboration and cooperation among learners are proposed.

The major teaching and learning methods suggested to be used in the implementation of the curriculum are described below when and where.

1. Interactive lecture: Lecture is an efficient way to integrate and present information from multiple sources on complex topics³. Additional advantage of lectures is that it gives students a chance to follow and model the way an expert thinks, reasons and asks questions⁴. Lecture is appropriate for teaching knowledge objectives. Recognizing the limitations of didactic lecture, interactive lecture is suggested to be used in this curriculum during the first four years(Pre-med to clerkship I). Lectures can be made interactive by enhancing them with engagement of learners mentally and physically using questions, brainstorming, discussion, think-pair-share, debate, role play, case study, providing opportunities for reading, talking, listening, writing and reflecting, and other learner activities.⁵ The use of multimedia will aid in the engagement of the learners⁶

2. Problem-based learning^{7, 8} (PBL): PBL is a method of learning in which learners first encounter a problem followed by a systematic, learner-centered inquiry and reflection process. As applied to education for the health professions, PBL is a method designed to help students learn the sciences basic to medicine at the same time they develop the reasoning process used by physicians and other health professionals in their clinical practice. The problem comes first without advance readings, lectures, or preparation and the problem serves as a stimulus for the need to know. Problem-based learning is designed to develop integrated, context-specific knowledge base; decision-making/critical thinking process and skills; self-directed, life-long learning skills; interpersonal, collaboration, and communication skills; constructive self and peer assessment skills; professional ethics and behavior. PBL is suggested to be used in the in the second and third years. During pre-clerkship years, written hypothetical cases will be used to facilitate PBL tutorials. Typically a PBL tutorial involves a group of 5 to 8 students discussing and analyzing a

³Daniel Richardson. Don't dump the didactic lecture; fix it. *Adv Physiol Educ* 2008;32: 23–24

⁴ANNA FYRENIUS, BJO'' RN BERGDAHL & CHARLOTTE SILE'N. Lectures in problem-based learning—Why, when and how? An example of interactive lecturing that stimulates meaningful learning. *Medical Teacher*, 2005; Vol. 27, No. 1, pp. 61–65

⁵Kevin McLaughlin & Henry Mandin. A schematic approach to diagnosing and resolving lecturalgia. *Medical Education* 2001;35:1135-1142

⁶ Ainsworth,S.(1999). The function of multiple representation. *Computational education*,33,pp131-152

⁷ Division of Educational Development & Research, Teacher & Educational Development, University of New Mexico School of Medicine. Faculty and student guide to PBL tutorials in phase I curriculum of the University of New Mexico School of Medicine. 2002.

⁸David and Mifflin, Barbara. Problem-based learning: Where are we now? *Medical Teacher* 2008;30:8,742-763

common patient problem in two meetings over two weeks, each meeting lasting 2-3 hours. In the first meeting, learners identify problems, generate hypotheses and explaining mechanisms. The days until the next meeting are time for independent self-study of learning issues identified. During the second meeting, students will discuss the learning issues and apply what they have learned to the problem. Based on the experiences of NMEI schools and Debre-Tabor universities in the implementation of PBL the time between sessions was short. The cases topics had tendencies to limit the learner towards single diagnosis their by decreasing engagement and exploration by the learner. And the PBL sessions had tendencies toward clinical decision making skill and less on integration of biomedical sciences and social and population health. Thus, the PBL cases topics were modified in such way that they can integrate biomedical sciences with clinical sciences, social and population health, professionalism and ethics. The case topics also promote student engagement to explore possible pathophysiologic mechanism involved in clinical and public health context by not limiting the problem towards a single diagnosis. In this format a single PBL case could be used to let the students explore different pathophysiologic conditions. This also enabled us to decrease the number of PBL cases per module and increase the time for self-directed learning. We suggested running the first PBL session once a week and then conclude the case on the second session in a week time. The students will have one week between PBL sessions.

3. Case study: Case studies present realistic scenarios/situations that focus on a specific issue or problem, which may be related to diagnosis or treatment of patients, interpersonal skills or any of a wide range of managerial or organizational problems. Learners typically read, study and react to the case study individually or in small groups. Case studies are important to teach higher order knowledge objectives (application, analysis and synthesis) and critical thinking skills/clinical decision making skills. Given the dense medical curriculum and need for efficient use of student and faculty time case based learning offers an alternative methods to tradition PBL teaching. Thus in clerkship modules case based learning is used more than PBL to prevent unfocused tangent and give more opportunity for focused clinical skill application.⁹

⁹ Srinivassan M,et al. Comparing Problem based learning with case –based learning : effects of a mjobr curricula shift at two instituions. Acad Med (2007) Jan;82 (1):74-82

4. Simulated practice (clinical skills lab)¹⁰: Simulated practice is the use of simulated person, device or set of conditions for instructional purpose. The learner is required to respond to the situation as he or she would under natural circumstances. Simulation takes various forms. Simulation can be static (like using anatomical models that closely resemble the human body or parts of it) or automated using advanced computer technology. Some are individual, prompting solitary performance, or interactive, involving groups of people. In medical education, simulation complements patient-based education and is best employed to prepare learners for real patient contact. It allows them to practice and acquire patient care skills in a controlled, safe and forgiving environment. Simulations are used to develop psychomotor, procedural and clinical decision-making skills. Simulation also aids development of communication and teamwork skills as well as the ability to respond to medical emergencies systematically. Simulated teaching facilitates learning under the right conditions including, but not limited to, learners receiving feedback on their performance, learners having the opportunity for repetitive practice and simulation being an integral part of the curriculum. Clinical skills lab is suggested to be used during pre-clerkship and clerkship years of the medical curriculum.

Role play^{11,12,13}: In a role play, learners play out different roles or parts-such as of a patient and provider-in a simulated situation. Role play addresses knowledge, skills and attitude objectives. Role plays promote learning through behavior modeling, observation, feedback, analysis and conceptualization. They are also often useful for exploring, discussing and influencing behaviors and attitudes of learners, as well as for helping learners develop skills such as history-taking, physical examination and counseling. It is also useful for teaching management and supervision skills.

¹⁰Issenberg, S. Barry MD, Mcgaghie, William C., Petrusa, Emil R., Gordon, David Lee and Scalese, RossJ. Features and uses of high-fidelity medical simulations that lead to effective learning: a BEME systematic review. Medical Teacher, 2005;27(1):10- 286.

¹¹Jhpiego Corporation. Training skills for healthcare providers: reference manual. 3rd edition. Baltimore: 2010.

¹²Jhpiego Corporation. In-service training techniques, timing, setting and media; findings from a systematic review of the literature. Maryland, 2012

¹³ Issenberg, S. Barry MD, Mcgaghie, William C., Petrusa, Emil R., Gordon, David Lee and Scalese, RossJ. Features and uses of high-fidelity medical simulations that lead to effective learning: a BEME systematic review. Medical Teacher, 2005;27(1):10- 28

5. Clinical practicum¹⁴: Clinical practicum or clinical teaching is the use of direct patient or client experiences to develop and practice knowledge, skills and attitude required for healthcare delivery or patient care under the supervision of a skilled clinical instructor or preceptor. These skills include generic skills (communication skills, mental and physical examination skills and basic clinical testing and procedural skills), problem-based clinical skills (skills related to patient complaints or diagnoses), discipline-specific clinical skills (such as sepsis in surgery) and continuum of care skills. Clinical learning opportunities include placements at a variety of clinical and community settings for outpatient emergency care, acute care (outpatient and inpatient), operation theatre, chronic care (outpatient and institutional), palliative and end of life care, wellness and preventive care, and population-based healthcare (community, public health). Outpatient departments are appropriate to practice interviewing, interpersonal and counseling skills as well as clinical skills. Inpatient departments are good to teach patient management, practice healthcare delivery skills including documentation of care plan and treatment given and demonstrate management of rarely seen conditions. Clinical teaching and learning uses a variety of techniques including observation, demonstration, role-modeling, practice, coaching, feedback, discussion and reflection. Clinical teaching starts in the first year with weekly hospital and primary health care visits and runs throughout the medical curriculum increasing in complexity, level of involvement and responsibility. During pre-clerkship, clinical experience must include assignment with nurse preceptors and students to master basic nursing procedural skills develop caring attitude and foster inter-professional collaborative practice competencies.

6. Community-based learning: Community-based education refers to learning activities that take place outside academic hospital in the community setting¹⁵. These settings include, but are not limited to, family homes, primary health care networks (health post, health center, and primary hospital), clinics, outreach sites, schools and prison facilities. Uses of community-based education include increasing the willingness and ability of doctors to work in rural and underserved communities thereby contributing to solution of inequity in health service delivery; enhancing learning (like PBL) by providing opportunities for students to learn in situations similar to those in later professional lives and opportunities to elaborate on previously acquired knowledge;

¹⁴Association of American Medical Colleges. Recommendations for clinical skills curricula for undergraduate medical education. 2005.

¹⁵Mohi Eldin M.A. Magzoub and Henk G. Schmidt. A Taxonomy of Community-based Medical Education. Acad. Med. 2000;75:699–707

equipping students with competencies that they would never learn adequately otherwise, e.g., leadership skills, ability to work in a team, the capability to interact with the community, the need for continuity of care, the effect illness has on a family and the early signs of disease and spectrum of health problems; offering an opportunity to learn and work with other health professionals; keeping the curriculum responsive to changing needs of the community; rendering opportunities for partnerships between the community, the university and the government. In the medical curriculum, there are three main modalities for community-based learning, namely, weekly community/primary health care visits, yearly community-based training program and team training program during internship. During year 1 to 5, every Friday students will have weekly community/primary healthcare visits in nearby urban and peri-urban areas. Depending on student number and logistics, students may be distributed among different community-learning sites. Similarly, in years 2 to 5, students will have a 4 weeks block community attachment at the end of each academic year in rural communities (community-based training program). On the sixth year, internship rotation includes a 4 weeks placement in a primary hospital or health center with other health professions students (team training program).

7. Laboratory practice: Students will have opportunities for demonstration, guided practice and coaching in labs to deepen their understanding and apply principles and methods of basic and clinical sciences. To this end, students will have two days per week of basic science lab session during premedical and pre-clerkship years and 1-2 days per week of side lab (clinical lab practice) during clerkship one and two.

8. Portfolio-based learning: Portfolio¹⁶¹⁷¹⁸ is collection of products collected by the student that provides evidence of learning and achievements related to a learning plan. Portfolio develops self-directed learning and reflective ability. It provides personal and professional educational evidence for student learning, contextualizes learning, links experience with personal interpretation, enhances interactions between students and teachers, allows students to receive

¹⁶ACGME and ABMS. Toolbox of assessment methods. A product of the joint initiative of the ACGME Outcome Project of the Accreditation Council for Graduate Medical Education (ACGME), and the American Board of Medical Specialties (ABMS). Version 1.1, September 2000

¹⁷M. FRIEDMAN BEN DAVID, M.H. DAVIS, R.M. HARDEN, P.W. HOWIE, J. KER & M.J. PIPPARD. AMEE Medical Education Guide No. 24: Portfolios as a method of student assessment. Medical Teacher, Vol. 23, No. 6, 2001

¹⁸HANKE DEKKER, ERIK DRIESSEN, EDITH TER BRAAK, FEDDE SCHEELE, JORIS SLAETS, THYS VAN DER MOLEN & JANKE COHEN-SCHOTANUS. Mentoring portfolio use in undergraduate and postgraduate medical education. Medical Teacher 2009; 31: 903–909

feedback, stimulates the use of reflective strategies and expands understanding of professional competence. The basic structure of the portfolio may include a title page (giving student's name, year of training and name of the mentor), contents page (listing what is in the portfolio with page references), a list of learning objectives (whose achievement the evidence in the portfolio claims to demonstrate), a short reflective overview (summarizing the learning that has taken place since the last portfolio review, and indicating which items of evidence relate to which learning objectives) and the evidence itself (probably grouped together into the areas contained in the learning objectives). Mentoring is crucial for portfolio-based learning, as it enhances the feedback process and stimulates students' reflections. Students will have individual mentors (preferably with medical background) from first year and will stay with one mentor until the point of graduation. The aims of the mentoring are to provide feedback, stimulate reflection, support students in compiling portfolio, monitor students competency development, support students in developing a better awareness and understanding of their strengths and weaknesses, support students in drawing up a learning plan for the coming period and motivate/inspire students, The Mentor will evaluate portfolio of the students at least two times a year and hold discussion to provide feedback.

Personal research and reflection exercise: In this methodology the student selects content area from list of topics provided (e.g. examine the impact of culture on the delivery of health care) then use journals, self- reflection, community based research, clinical experiences, discussions etc., and is expected to present the findings (in writing and /or orally). This will help the student apply literature review, self-reflection and critical thinking as a method of professional exploration and growth to enhance their research and communication skill and deepen and broaden their knowledge. In addition to its use as teaching/learning method in the first five years of the medical curriculum, PRRE will also be used as formative and summative assessment method.

9. Whole group session: During years 1 to 5, all students and faculty will meet on Friday afternoon for whole group session. The purpose of the session is to consolidate and reflect on the different learning activities covered during the week. The session is student-centered discussion that will be facilitated by one or more faculty. These may include but not limited to discussion on clinical encounters, social and population health issues, professional and ethical issues.

10. Journal club: A journal club¹⁹ is a group of individuals who meet regularly to discuss the clinical applicability of articles in current medical journals. Journal club is an increasingly popular way to promote the uptake of research evidence into practice. To make it effective, evidence suggests mentoring and brief training of students on how to judge quality of research as well as the use of structured critical appraisal instrument. Journal club is suggested to be implemented during internship after students have completed research methods module.

Assessment Methods

Assessment plays a central role in education process: it determines much of the work students undertake affects their approach to learning and is an indication of which aspects of the course are valued most highly. The purposes of assessment are to motivate students to learn, create learning opportunities, to give feedback to students and teachers, grading and quality assurance. There is a distinction between a formative assessment, which is mainly intended to help the student learn and a summative assessment, which is intended to identify how much has been learned. Formative assessment is most useful part way through the module and will involve giving students feedback which they can use to improve future performance. Faculty should conduct at least two formative assessments of each student during a given module or rotation. Summative assessment is used to make a pass/fail or, promotion decision; findings of formative assessment are not used to make pass/fail decisions, however²⁰. That being said both formative and summative assessments are equally important; however, psychometric rigor is required more from summative assessment strategies. The following principles^{21,22} are considered in selection of assessment strategies and faculty should keep in mind these principles in appraising and revising assessment methods during implementation. Validity and reliability are of utmost importance but it is also recommended to consider feasibility, cost and fairness.

¹⁹Harris J, Kearley K, Heneghan C, Meats E, Katherine Kearley-Shiers, NiaRoberts, Perera R. Are journal clubs effective in supporting evidence-based decision making? A systematic review. BEME Report. November 2009

²⁰ Chris Rust. Learning and Teaching Briefing Papers Series: Purposes and Principles of Assessment. Oxford Centre for Staff and learning Development. 2002

²¹ Steven M Downing & Thomas M Haladyna. Validity threats: overcoming interference with proposed interpretations of assessment data. Medical Education 2004; 38: 327–333

²² Steven M Downing. Reliability: on the reproducibility of assessment data. Medical Education 2004; 38: 1006–1012

Reliability: Reliability is the reproducibility or consistency or generalizability of assessment scores. An assessment result is said to be reliable if students will get the same score if they re-take the exam. Similarly, for essay type and performance assessment, assessment scores are reliable, if the same results are obtained with different raters. Reliability of assessments can be improved by increasing the number of questions (or cases in clinical performance examination), aiming for middle difficulty questions, writing clear and unambiguous questions and increasing the number of raters.

Validity: Validity is the ability of an assessment to measure what it is supposed to measure. Validity is not about the method refers to the evidence presented to support or refute the meaning or interpretation assigned to assessment results. Simply put, assessment results are valid if they accurately distinguish competent from incompetent students and if the student who gets “A” grade is actually an “A” student, a student who gets a “B” grade is actually a “B” student, a student who gets an “F” grade is actually an “F” student, etc. Examples of factors that affect validity in written assessment are too few written questions to sample the content adequately, preparing questions from some chapters, mismatch of assessment questions with content covered in the curriculum, poorly constructed questions, too difficult or too easy questions, rater subjectivity and cheating. For performance (clinical) assessment, too few cases or observations to generalize performance, unrepresentative cases, rater bias, flawed rating scales/checklists and indefensible pass/fail cut off points are threats to validity. Note that reliability is a necessary but not sufficient condition for validity.

These are the descriptions of the major assessment methods including when and where they would be used in the curriculum.

Direct observation of clinical skills (DOCs)

The purpose of DOCs or mini-clinical evaluation exercise²³ is to assess clinical skills while a student interacts with patients in different settings. Typically, it takes 15-20 minutes and the assessor follows the student with a checklist and gives feedback at the end. The DOCs offer students immediate and ongoing feedback about their observed clinical skill and performance (interviewing skills, physical examination skills, and professionalism, clinical judgment,

²³John J. Norcini; Linda L. Blank; F. Daniel Duffy; and Gregory S. Fortna. The Mini-CEX: A Method for Assessing Clinical Skills. *Ann Intern Med.* 2003; 138:476-481

counseling skills, organization/efficiency and overall clinical competence). This method will be used in all years of the training and there will be at least two DOCs to be performed by a student in each module or clinical rotation. This assessment method enables one to follow the progress of the student and will be used for formative assessment.

Objective structured examination

Objective structured clinical examination (OSCE) is a performance-based exam. During the exam, students are observed and evaluated as they go through a series of 8 or more stations. It allows assessment of multiple competencies. It is Objective, because examiners use a checklist for evaluating the trainees; structured, because every student sees the same problem and performs the same tasks in the same time frame; Clinical, because the tasks are representative of those faced in

real clinical situations. These increase the reliability and validity of the assessment. OSCE²⁴ is a standardized means to assess history taking, physical examination skill, communication skills, ability to summarize and document findings, ability to make a differential diagnosis or plan treatment, clinical judgment based on patient 's note and procedural skills. OSCE may use manikins and simulators, standardized patients and real patients. Standardized patients are healthy persons trained to simulate a medical condition in a standardized way. Health science students, health facility staff and faculty may serve as standardized patients. Objective structured practical exam (OSPE) is a variant of OSCE to assess students' knowledge and skill in a non-clinical setting. Both OSCE/OSPE will be part of the summative assessment and will be implemented during premedical, pre-clerkship I and II as well as clerkship I and II.

Structured long cases

Structured long case assessment presents the student with a complete and realistic clinical challenge thereby enabling the evaluator to see the complete picture of the student's ability in addressing the challenges. The use of multiple cases improves reliability of the examination, which is a major weakness in the traditional long case. Additional improvements to the traditional long

case are required to improve reliability. For example, observing the student-patient interaction,

using checklist and increasing the number of examiners²⁵. Feasibility should be considered here.

Written assessments:

May include different item formats such as multiple-choice questions, matching, true-false, essay and short answer. Written assessment methods will help to evaluate knowledge and understanding of basic, clinical, public health and psychosocial sciences and professionalism and ethics. Important point to remember is to ensure written exams assess higher order knowledge in addition to recall and comprehension. Written assessments would be parts of both as formative and summative assessment in premedical, pre-clerkship and clerkship.

Standardized oral exam

The standardized oral examination²⁶ is a type of performance assessment using realistic patient cases for questioning the examinee. One or two faculty serve as examiners and students can be tested on several different cases. Oral exam will be part of the summative assessment in qualification exams at the end of pre-clerkship and clerkship modules. In the case of pre-clerkship the examiner begins by presenting to the examinee a clinical problem in the form of a patient case scenario and asks the examinee questions that correlate, integrate the biomedical sciences with clinical context. In the case of clerkship, the questions probe the reasoning for requesting clinical findings, interpretation of findings, and treatment plans.

Logbook

Logbook documentation serves as evidence of scope of patient care and community experience to meet requirements or specific learning outcomes. Maintaining logbook will encourage students to make use of all possible learning opportunities for clinical/procedural skills and community skills to fulfill minimum requirement. Regular review of logbook can be used to help the student track what procedures or experiences must be sought to meet requirements. The logbook document should be counter signed by faculty. The number reported in a logbook may not necessarily indicate competence. Logbook will be part of the formative assessment throughout the medical curriculum.

Portfolio

Portfolio²⁷ is collection of papers and other forms of evidence that learning has taken place. It provides evidence for learning and progress towards learning objectives. Reflecting upon what has been learned is an important part of constructing portfolio. In addition to products of learning, the portfolio can include statement about what has been learnt, its application, remaining learning need, how they can be met. Portfolio helps to assess learning outcomes including those that are not easy to assess with other methods like personal growth, self-directed learning, reflective ability, self-assessment of personal growth and professionalism. Portfolio allows assessment of progress towards learning outcomes by using chronological work samples collected at different points in time. Portfolio will be part of the formative assessment throughout the duration of the medical training and can be used as a summative assessment during internship.

Global Rating

Global Rating²⁸ is assessment of general categories of ability (e.g. patient care skill, medical knowledge, interpersonal and communication skills, professionalism, etc.) retrospectively based on general impression over a period of time and derived from multiple source of information. The purpose is to evaluate knowledge, skill and attitude over a period of time at the end of a rotation thereby helping the evaluation of the student's effort across time. Global rating will be part of both formative and summative assessment of students throughout the duration of the medical

curriculum360⁰ Evaluation²⁹ which consists of measurement tools completed by multiple people in a student's sphere of influence. Evaluators usually are faculty, other members of the health care team, peers, patients, families and community members. 360⁰ evaluations can be used to assess interpersonal and communication skills, teamwork ability, management skills, decision-making professional behaviors and some aspects of patient care. It will be used as part of the summative assessment in Community based training program (CBTP), team training program (TTP), and internship. In clerkship I and II it is more feasible to use it only as formative assessment

PBL /case study tutorials progressive assessment

PBL/ case study³⁰ progressive assessment is a continuous assessment of students' performance during PBL/case study tutorials with regards to content, process and professionalism. To improve reliability of the score, this assessment strategy will be guided by a checklist. The checklist may assess competencies in four areas: knowledge of basic, clinical, public health and psychosocial sciences and information gathering skills, reasoning skills, participation and communication skills, cooperation and teambuilding skills. This assessment method will be part of both formative and summative assessment during pre-clerkship I and II and clerkship I and II. PBL/case study tutorial assessment at the end of each session and midway during a module will be used as a formative assessment method while end of module assessment of students' performance during PBL tutorials will be used as part of the summative assessment.

Grading and Promotion criteria

Grading System

Letter grades shall be given based on the points earned out of 100. The letter grading system has a **fixed scale** as described in the table below

Letter of Grade	Point scored	Description	Corresponding fixed Number grades
A ⁺	[90,100]	Outstanding	4.0
A	[85, 90)	Excellent	4.0
A ⁻	[80, 85)	Excellent	3.75
B ⁺	[75, 80)	Very Good	3.50
B	[70, 75)	Good	3.00
C ⁺	[65, 70)	Satisfactory	2.50
C	[60, 65)	Fair	2.00
D	[50, 60)	Unsatisfactory	1.0
F	<50	Fail	0

Promotion criteria

1. General Promotion criteria

- 1.1. A student is allowed to stay a maximum of two years for one academic year
- 1.2. Pass mark for any course or module or attachment will be at least a "C" grade.
- 1.3. Any student who scores "D" or below in any module or attachment will not be allowed to progress to the next academic year.
- 1.4. Any student who has "F" grade will repeat the course

- 1.5. Any student who has Incomplete “I” grade will be required to complete the specific module if his /her evidence is accepted by School Academic commission.
 - 1.5.1. If the student does not produce justifiable reason in two weeks the “I” grade will be changed to “F” grade with in six weeks
- 1.6. Any student who has No Grade (NG) will take exam if his /her evidence is accepted by School Academic commission.
- 1.7. Any student who fails to achieve a CGPA of at least 2.0 on a given academic year but greater than or equal to 1.75 shall take a remedial exam on module/s or attachments s/he scored D grade
- 1.8. Any student who scores year GPA of 1.75- 2.0 shall have the privilege of readmission

2. Promotion Criteria for Pre – clerkship

Irrespective of the cumulative GPA;

- 2.1. Any student who scored “D” in a module shall take a remedial exam before promoting to the next academic year.
- 2.2. Any student who scores “D” in two or more modules, he/she shall not sit for remedial exam, rather shall repeat the failed modules
- 2.3. Any student who scores “F” in one or more modules shall repeat the failed module.
- 2.4. If a student scores “D” after repeating the module, s/he shall take remedial exam.
- 2.5. If a student scores “F” after repeating the module s/he shall be dismissed.
- 2.6. If a student scores “D” or below after remedial as per article 3.4 above for grade “D”, s/he shall be dismissed.
- 2.7. Any student who scores “D” or “F” on CBTP attachment shall be allowed to work on specific task assigned by the department (remidial).
 - 2.7.1. If this student scores “D” on CBTP attachment after the assignment (remidial), s/he shall repeat the attachment.

3. Step one exam (pre-clerkship qualifying exam)

- 3.1. Any student who has score of D or below in any preclerkship modules shall not be eligible for pre-clerckship qualification exam.
- 3.2. Progression from pre-clerkship to clerkship will be determined through pre-clerkship internal qualifying examination as pass or fail. The assessment will have the following components
 - Final written examination (comprised from all preclerkship modules)- 50%
 - Final OSCE (comprised from all preclerkship modules) -20%
 - Final oral examination - 20% (could be external or internal examination)

- 3.3. Sixty percent of the total score (100) will be the pass mark
- 3.4. Any student who scores below 60%, shall take remedial exam (OSCE and Oral).
- 3.5. Any student who scores below 60% after remedial, shall repeat pre-clerkship-II modules and will sit for qualification exam
- 3.6. Any student who scores below 60% after repeating the year shall take remedial exam (OSCE and Oral).
- 3.7. Any student who scores below 60% after second remedial may be transferred to other disciplines in the university if the university legislation allows

4. Promotion Criteria for Clerkship

Irrespective of the cumulative GPA;

- 4.1. A student who scores one “D” in any of the following modules (Internal medicine, surgery, Pediatrics, OBGYN and SPH) shall take remedial exam for the subject s/he scored “D”.
 - 4.1.1. If a student scores “D” or below after remedial exam shall be required to repeat the attachment on which he scored “D”.
 - 4.1.2. If a student scores “D” after repeating the module shall take remedial exam.
 - 4.1.3. If a student scores “D” or below after the remedial exam shall be dismissed.
- 4.2. A student who scores two or more “D” or one or more “F” in any of the following modules (Internal medicine, surgery, Pediatrics, OBGYN, SPH) shall be required to repeat the modules he / she scored D.
 - 4.2.1. If a student scores “D” in one module after repeating the module shall take remedial exam.
 - 4.2.2. If a student scores two or more “Ds” or below after repeating the modules, s/he shall be dismissed.
- 4.3. A student who scores one or two “D” in any of the following modules (Clinical methods, Psychiatry, Dentistry, Radiology, ENT, Dermatology, emergency medicine and Ophthalmology) shall be allowed to take remedial exam.
 - 4.3.1. If he/she fails after remedial, he/she will be required to repeat the attachment anytime before qualification exam
 - 4.3.2. If a student scores “D” in one or two modules after repeating the module shall take remedial exam.
 - 4.3.3. If this student scores “D” or below after remedial the modules, s/he shall be dismissed.
- 4.4. A student who scores more than two “D”s or one or more “F” in any of the following modules (Clinical methods, Psychiatry, Dentistry, Radiology, ENT, Dermatology, emergency medicine and Ophthalmology) shall be required to repeat the modules anytime before qualification exam.
 - 4.4.1. If a student scores “D” in one or two modules after repeating the module shall take remedial exam.
 - 4.4.2. If a student scores “D” or below after remedial exam shall be dismissed.
- 4.5. A student who scores “D” in one major and one minor shall take remedial for both
 - 4.5.1. If fails in either or both modules after remedial shall repeat the modules

- 4.5.2. If a student scores “D” in either module after repating the module shall take remedial exam
- 4.5.3. If fails in the minor module only after remedial, s/he shall clear the minor before internship
- 4.5.4. If fails in the major module after remedial, s/he shall be dismissed irrespective of the CGPA
- 5.6 A student who scores “F” in one major and one minor shall repeat both.
 - 5.6.1 If a student scores “D” in either or both modules after repating the module shall take remedial exam
 - 5.6.2 If fails in the minor module only after remedial, s/he shall clear the minor before internship
 - 5.6.3 If fails in the major module after remedial, s/he shall be dismissed irrespective of the CGPA
- 5.7. A student who scores ‘D” or below in one major and two minors will be required to repeat the modules he failed.
 - 5.7.1. If a student scores “D” in any of the three modules after repating, shall take remedial exam
 - 5.7.2. If fails in the minor modules only after remedial, s/he shall clear the minor before internship
 - 5.7.3. If fails in the major module after remedial, s/he shall be dismissed irrespective of the CGPA
- 5.8. A student who failed (got “D” or below) only in one minor module he/she shall still be allowed to take qualification exam, but, MUST repeat the course and secure pass mark before joining internship.

6. Step 2 (Clerkship qualifying exam)

- 6.1. Progression from clerkship to internship will be determined through a clerkship qualifying examination assessment. The assessment for each major module shall have the following components.
 - C2 for major clerkship module including progressive assessment shall account for 40%
 - Written examination shall account for 30%
 - External examination shall account for 30% for each major attachment
- 6.2. Sixty percent shall be the pass mark for each major clerkship modules
 - 6.2.1. Any student who scores below 60% in one or two module/s, shall be required to attach a solid three months for each failed module.
 - 6.2.2. Any student who scores below 60% in more than two modules, shall be required to re-attach for C-2 “major” modules.
 - 6.2.3. Any student who scores below 60% in more than two modules after repeating the year, shall be dismissed

7. Internship

- 7.1. Internship is salaried post
- 7.2. Any student in internship is required to secure “C” (satisfactory) and above in each attachment
- 7.3. Any student who got “D” (unsatisfactory result) in any internship attachment, shall repeat the attachment for the same period of duration
- 7.4. Internship repeat attachment is not allowed for more than once provided that his/her internship stay do not exceed two academic years.
- 7.5. If a student required to repeat more than once the internship attachment, the respective school academic commission decides.

8. Graduation Requirement

- 8.1. A student should pass successfully in all modules obtaining a minimum grade of C
- 8.2. A cumulative Grade Point Average (CGPA) of 2.00 must be obtained
- 8.3. A Student should successfully pass step 2 qualifying examination
- 8.4. Successful completion of internship attachments

Module Sequence: Module Listing

Year I, Semester I (freshman)

Course Code	Course Name	Credit Hour	ECTS
FLEn 1011	Communicative English Language skills I	3	5
Biol-1011	General physics	3	5
Psyc 1011	General psychology	3	5
Math 1011	Mathematics and general science	3	5
LoCT 1011	Critical Thinking	3	5
SpSc 1011	Physical fitness	P/F	2 contact hrs
GeES 1011	Geography of Ethiopia and the horn	3	5
	Semester Total	18	32
	Mid-Year Vacation	2 weeks	

Year I, Semester II (Premedicine)

Course Code		Course	Credit Hour	ECTS	
FLEn 1012		Communication English- Basic Writing skill	3	5	
Anth1012		Social anthropology	2	4	
EmTe 1012		Introduction to Emerging Technologies in Medicine	3	5	
Econ 1012		Economics	3	5	
Chem 1012		Organic Chemistry	3	5	
Incl 1012		Inclusiveness	2	4	
GLOT 1012		Global trends	2	4	
MCIE 1012		Moral and Ethics education	2	4	
ENTP 1012		Entrepreneurship	3	5	
		Total	23	41	
Exam week			1 week		
End –Year Vacation			4 weeks		
	Module Code	Module Name	Duration	ECTS	SPH (Social and

Pre-Clerkship I					Population Health Module)
	INMD 2011	Introduction to medicine	20 weeks	33	SPH-I: Determinants of Health:- 16 weeks (5 ECTS) COMH 2011)
	Mid –Year Vacation		2 weeks		SPH-II:- Measurement of Health and Disease – 18 weeks (20 weeks 5 ECTS) (COMH 2022)
	MEHO 2022	Metabolic homeostasis	5 weeks	7	
	HEMA 2032	Hematology system	4 weeks	6	
	CVLS 2062	Cardiovascular system and Lymphatic system	8 weeks	12	
	RESP 2052	Respiratory system	5 weeks	7	
	MSKI 2042	Musculoskeletal and Integumentary system	6 weeks	9	
	COMH 2032	Community Based Training –I	4 weeks	5	
	End –Year Vacation		4 weeks		
		Total	58 weeks	79	Total ECTS 79 +10= 89

	Module Code	Module Name	Duration	ECTS	SPH (Social and Population Health Module)
Pre-Clerkship II	GITS 3071	Gastrointestinal system	7 weeks	10	SPH-III:- Health promotion & Disease prevention —20 weeks (5 ECTS) COMH 3041
	RENA 3081	Renal system	5 weeks	7	
	ENDO 3091	Endocrine system	5 weeks	7	
	Mid–Year Vacation		2 weeks		
	REPR 3102	Reproductive system	5 weeks	7	
	NERV 3112	Nervous System	8 weeks	12	
	INFD 3122	Infectious Diseases & Rural medicine	5 weeks	7	
	COMH 3052	Community Based Training – II	4 weeks	5	
		End of Pre- Clerkship qualifying exam	4 weeks		
		End year Vacation	4 weeks		
		Total	49 weeks	55	Total ECTS 55+5=60

Internship		Course	Duration	ECTS	SPH (Social and Population Health Module)
	INME 6011	Internal Medicine	10 weeks	19	SPH VI:- Research project (5 ECTS) COMH 6101
	SURG 6011	Surgery including Ortho	10 weeks	19	
	PEDI 6011	Pediatrics	10 weeks	19	
	OBGY 6011	Gynecology/ Obstetrics	10 weeks	19	
	EMME 6011	Emergency Medicine	5 weeks	10	100+5=105
	PSYC 6011	Psychiatry	5 weeks	5	
	DERM 6011	Dermatovenereology	2 weeks	4	
	COMH 6011	TTP	4	5	
		Research leave	2		
Total			56	100	
	Module Code	Module Name	Duration	ECTS	SPH (Social and Population Health Module)
Clerkship I	CLMD 4011	Clinical methods	6 weeks	10	SPH-IV:- Health Economics Management and Policy—24 weeks COMH 4061 (5 ECTS)
	RADI 4011	Radiology	2 weeks	3	
	Module Code	Module Name	Duration	ECTS	SPH (Social and Population Health Module)
	INME 4011	Internal Medicine I	10 weeks	17	
	SURG 4011	Surgery –I (including Ortho)	10 weeks	17	
		Mid-Year Vacation	2 weeks		
Clerkship II	EMME 5011	Emergency Medicine	6 weeks	10	SPH-V:- Research Methodology--19 weeks COMH 5081 (4 ECTS) 86+5= 91 ECTS
	PEDI 5011	Pediatrics I	3 weeks	5	
	OBGY 5011	Gynecology/ Obstetrics I	3 weeks	5	
	DERM 5011	Dermatology	4 weeks	7	
	COMH 4072	Community Based Training III	4 weeks	5	
	PSYC 5011	Psychiatry	4 weeks	7	
		End year Vacation	4 weeks		
	DENT 5011	Dentistry	2 weeks	3	
		Total	58 weeks	86	
	ENT 5011	ENT	2 weeks	3	
	Mid- year Vacation		2 weeks		
	INME 5011	Internal Medicine II	7 weeks	12	
	SURG 5011	Surgery –II	7 weeks	12	
	PEDI 5011	Pediatrics –II	7 weeks	12	
	OBGY 5011	Gynecology/Obstetrics –II	7 weeks	12	
	COMH 5092	Community Based Training –IV	4 weeks	5	
		Study Leave and qualifying exam	6 weeks		
		End year vacation	2 weeks		
		Total	62 weeks	86	86+4= 89 ECTS

PREMEDICINE MODULES

COURSE TITLE: COMMUNICATIVE ENGLISH- WRITING ENGLISH SKILL

COURSE CODE: FLEn 1012

Course Description

This course is intended to develop and improve students' language competence. Generally, this course will cover the specific language aspects described below.

Writing (summarizing a text, writing descriptive texts)

Vocabulary (working out meanings from context, synonyms, antonyms, collocations, definitions);

Grammar (relative clauses, modals, voice, conditionals, tense, reported speech).

Course objectives

Upon completing the course, students will be able to:

- Compose well organized paragraph of the different type
- Compose well organized essay
- Use various vocabulary learning strategies and techniques
- Write and present reports
- Read various materials and make their own notes
- Identify the structure of written discourses
- Attend their academic work at ease and with clarity.

Week	Main Topic/Sub topic/s/ Chapter	Student Activities
1 st & 2 nd	1. Grammar: Learning to use grammar for facilitating meaning • Using relative clauses • using active and passive constructions for descriptive writing • using conditional sentences 2. Writing: Writing a short text that narrates one aspect of cultural practices • Definition of a paragraph • Components of a paragraph • Characteristics of a good paragraph • Basic types of paragraph • Techniques of paragraph development • Writing a paragraph 3. Vocabulary Guessing contextual meanings of words	-write a story on cultural practices
3 rd & 4 th	Writing Effective Essay • Basic Characters of a Good Essay • The difference between a paragraph and an essay • The Structure of an Essay - The Introduction - The Body - Summary	-write descriptive texts -Write a short essay

5 th & 6 th	Writing Developing various texts about campus students health problems Describing products using passive and active voices	-write a paragraph that shows cause-effect
7 th	Writing ➤ Write an expository and descriptive paragraph	Write an expository and descriptive paragraph
8 th and 9 th	Writing ➤ Writing argumentative Paragraph Grammar: ➤ Reported speech	-write argumentative paragraph Use reported speech in writing
10 th & 11 th	Letter Writing • Introduction to Letter Writing • Planning a Letter • Types of Letter • Personal or Informal Letters • Business or Formal letters • Four Considerations of a Business Letter • Tutorial based on questions prepared on Reflection/self-assessment checklist	
12 th and 13 th	1. Reading: ➤ Understanding writer's style ➤ Reacting to a text 2. Vocabulary: ➤ Understanding meanings of words from context 3. Writing: ➤ Writing narrative paragraph	-Practice understanding writers style and reacting to a text -use contexts to understand meanings of words -Write narrative paragraphs
14 th & 15 th wk	Writing ➤ summarize, paraphrase or quote texts correctly;	summarize, paraphrase or quote texts correctly;
16 th		
Teaching methods	Classroom contact/Lecture, group work, interactive tutorial sessions (group and pair work/discussions and individual work (independent learning)).	
Course policy	Attendance: It is compulsory to come to class on time and every time. If you are going to miss more than three classes during the term, you should not take this course. Assignments: you must do your assignment on time. No late assignment will be accepted. Tests/Quizzes: you will have short quizzes and tests almost every week. If you miss the class or, are late to class, you will miss the quiz or test. No makeup tests or	

	quizzes will be given. You are expected to observe the rules and the regulations of the University as well. Cheating/plagiarism: you must do your own work and not copy and get answers from someone else. The only way to learn English is to do the work yourself. Also, please do not chew gum, eat, listen to recorders or CD players, wear sunglasses, or talk about personal problems. Please be sure to turn off pagers and cell phones before class and exam sessions
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Assessment

1. Continuous Assessment (50%)

- | | |
|-------------------------------|-----|
| ➤ Reading and Vocabulary-test | 10% |
| ➤ Writing assignments | 30% |
| ➤ Grammar-test | 10% |

2. Final Examination (50%)

COURSE TITLE: ICT AND EMERGING TECHNOLOGIES IN MEDICINE

COURSE CODE: EmTe-1012

Course Description:

This module is intended to develop basic awareness of information and communication technology, computer system, computer network and data communication, computer security and ethics. And also to develop student's knowledge and skill on techniques and applications pertinent to medicine and create a fundamental understanding of how the application of these technologies to medicine and human health contribute to health service quality by making informed decision. It is offered in pre-medicine

Course Objectives

At the end of this module, learners will be able to apply knowledge and skills of basic computer technologies in improving the health service delivery, education and research by making informed decision.

Supporting Objective:

At the end of the course students will be able to:-

- Describe some of the basic computer terminologies in medicine
- Explain the application of computer in medicine
- Identify computer system
- Describe computer Arithmetic & data representation in computer
- Describe the concept of Health Information Organizations and how they fit into the Nationwide Health Information Network
- Explain Integrating Healthcare Enterprise (IHE), Hospital Information System (HIS) and Electronic Medical Card (EMR)
- Explain health Information Technology Interoperability (HL7 and DICOM)
- Explain the importance of networks in the field of medicine
- Describe the evidence pyramid and levels of evidence
- apply the process of using evidence-based medicine to answer a medical question
- identify and use the most important online evidence based medicine resources
- State the difference between telehealth and telemedicine
- List the various types of telemedicine such as tele radiology, tele neurology
- explain the potential benefits of telemedicine to patients and clinicians
- Describe the importance of data security and privacy in medicine

Prerequisites: None

Teaching and learning methods

- Interactive lecture
- Discussion
- Demonstration

- Video show
- Computer lab practice

Teaching-Learning Materials

- AV aids (LCD and computer or writing board and marker or chalk)
- Computers with appropriate software
- Hand-outs of lecture materials

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- Edward H. Shortliffe Leslie E. Perreault. Medical Informatics: Computer Applications in Health Care and Biomedicine. Second Edition. 2001
- Sharon E Straus, W. scott Richardson, Paul Glasziou, R. Brain Haynes. Evidence-Based Medicine: How to Practice and Teach It. fourth edition. 2011

Assessment methods

Formative Assessments:

- Assignment
- Practice Exercise in computer laboratory
- Student presentation

Summative Assessment:

- Class activity and presentation (5%)
- Group assignment (5 %)
- Individual assignment (5 %)
- Quiz (10 %)
- Mid-term test (30%)
- Final Exam (40 %)

Schedule

Schedule Date	Learning Activity	Required Reading/ Assignment
Week 1 -2	Interactive lecture ➤ Introduction to Information and Communication Technology ➤ INTRODUCTION TO COMPUTER ○ Definition of Computer and Computer Science ○ Types of computers ○ Characteristics of computers ○ Drawbacks of Computers ○ Parts of Computer System ○ Logical organization of a computer system ○ Computer Hardware ○ Computer Software ➤ EMERGING TECHNOLOGIES	

	○ Human to Machine Interaction ○ Future trends in emerging technologies ○ AI in Health	
Computer skills lab ➤ Computer simulations/video showing the different computer hardware ➤ Practice: identification and Examination of computer hardware		
Week 3 - 4	Interactive lecture Data Representation and Number System ➤ Number System ➤ Data representation inside computers ➤ Computer arithmetic ➤ Computer coding system	
Computer skills lab ➤ Computer simulations showing the data representation		
Week 5 – 6	Interactive lecture ➤ Communications and its component ➤ Computer networks ➤ Importance network for medical field ➤ Network topology ➤ Introduction to Internet ➤ Search engines ➤ Internet of things	
Computer skills lab ➤ Computer simulation: showing the different network topology ➤ Practical work: using different search engines		
Week 7	Interactive lecture and discussion ➤ Application of computer ➤ Benefits of using computer system in medicine ➤ Augmented reality ➤ Other emerging technologies ○ Nanotechnology ○ Biotechnology ○ Blockchain technology ○ Cloud and quantum computing ○ Autonomic computing ○ Computer vision ○ Embed systems ○ Cyber security ○ Additive manufacturing (3D Printing)	

Computer Skill Lab ➤ Computer simulation: showing different videos usage of computer in Medicine, e-granary and Terasim.		
Week 8 and 9	Interactive lecture and discussion ➤ Introduction to HMIS (Health Management Information System) ➤ Health information organization and flow ➤ Integrating Healthcare Enterprise (IHE) ➤ Hospital Information System	

	(HIS) ➤ Electronic Medical Record (EMR) ➤ Health Information Technology Interoperability	
Computer skills lab ➤ Practical work: using different EMR especially Smart-Care.		
Week 10 – 11	Interactive lecture and discussion ➤ Introduction to Evidence Based Medicine ➤ Evidence pyramid and level ➤ Process of Evidence Based Medicine ➤ Common online/on-the-shelf evidence based resources	
Computer skills lab ➤ Practical work: using online Evidence Based Medicine resources		
Week 12 -13	Interactive presentation and discussion ➤ Introduction to data security and privacy in medicine ➤ definition of computer ethics & security ➤ Health Information Portability and Accountability Act (HIPAA) ➤ Backup ➤ Encryption ➤ Viruses & worms protection	
Week 14- 15	Interactive lecture and discussion ➤ Introduction to telemedicine and tele-education ➤ Types of telemedicine ➤ Importance of telemedicine and tele-education	
Computer skills lab ➤ Demonstration: demonstration how telemedicine and tele-education work		
Week 16	Review of the Computer Application to Medicine ➤ Discussion ➤ Computer skills lab ➤ Study break for exam ➤ Written exam	

COURSE TITLE: MORAL AND CIVICS

Course Title	Introduction to Civics and Ethics
Course Code	CESt1011
Credit Hours	2 Cr.Hrs (4 ECTS)
Academic Year	
Status of Course	Compulsory Common Course

Student Work Load		Lectures	Tutorial	Library and Group Work	Assign Report	Home Study	Total W. L.
		32hrs.	--	14hrs.	12hrs.	35 hrs.	93hrs
Course Description	This course is designed for undergraduate students with the aim of producing good citizens. It emphasizes on equipping learners with the necessary civic competence and active participation in public life. It will also help them to exercise their democratic rights and discharging their responsibilities effectively by familiarizing them with necessary civic knowledge and skills. In countries such as ours, where the process of cultivating modern constitutional and democratic values in the minds of citizens is experiencing serious challenges, largely because the country had no established civic culture and partly because these values and principles are not yet well-institutionalized, civics and ethical education remains to be imperative. To this end, the course introduces learners to the basics of civics and ethics, citizenship, morality and the goals of studying civics and ethics. It exposes students to the meanings, foundations, approaches, values and principles of ethics and civic virtue that learners must be equipped with both as citizens and professionals in their encounter with real life situations both to be morally matured and responsible while making decisions and taking actions. The course also elucidate the nature, purpose and forms of state and government, constitution, democracy and human rights, the nature of democratic citizenship, modes of cultivating civic-virtues in our citizens mainly within the context of Ethiopia.						
	Ethiopia is currently going through a twin process of hope and despair. On the one hand there are tremendous social, economic and political changes. On the other hand, significant challenges are affecting the process, the pace and magnitude of this change. For such changes to be successful however, it is imperative that citizens develop rational thinking, critical support and reasonable opposition to the growing culture of mob mentality. Moreover, citizens also need to go beyond their narrow individual interests and prioritize broad national interests. The prevalence of corruption, which has been spreading like a wild fire is also frustrating the productive capacities of citizens that could positively contribute to the development of the country. All the aforementioned national concerns have largely been overlooked by the common course syllabi currently under construction. Such glaring absence of citizenship and moral education from the curriculum could be considered as one of the gaps that need to be urgently addressed. In this regard, by encouraging civil discourse on contending national issues, prioritizing peace and inculcating honourable disposition, the course civic and ethical studies would prepare students to contribute to the overall peace, stability and prosperity of the nation at large, hence magnifying the relevance and urgency of this course.						
Course Purpose							

C o n	Upon a successful completion of this course, students will be able to: • Understand the subject matter of Civics and Ethics; • Cultivate certain moral values and civic virtues that enable them to be morally matured and competent in their professional and citizenry lives by practically exposing them to moral and civic debates/discussions and engagements. • Develop such values/ virtues as recognition, appreciation and tolerance towards diversity and also build culture of peace • Gain knowledge about the theoretical discourses and practices of state, government and citizenship, and their mutual interplay especially in the context of Ethiopia; • Develop individual and/or collective potential of becoming self-confident citizens who can effectively participate in their legal-political, socio-economic and cultural lives; • Understand the essences of such values and principles as democracy and human rights, multiculturalism and constitution and constitutionalism with especial reference to Ethiopia; • Develop analytical and reflective skill of identifying global or national level development, democracy/governance and peace related issues of civics and ethics and then be able to produce or evaluate policies and practices in a civically and ethically responsible manner.
COURSE CONTENTS AND SCHEDULES	
C o n t a i n s	Chapters, Sections and Sub-sections
6 h o u r s	Chapter One: Understanding Civics and Ethics 1.1 Defining Civics, Ethics, Morality and amorality 1.2 The Origin and Development of Civics and ethical education 1.3 The purpose of civics and ethical education 1.4 Citizen: Rights and responsibilities Competences of good citizen

Chapter Two: Approaches to Ethics**2.1 Normative ethics****2.1.1 Teleological Ethics (Consequentialist)**

- Hedonism
- Ethical and psychological Egoism: Epicureanism and Cyrenaicism
- Social Hedonism: Utilitarianism

2.1.2 Deontological Ethics (Non- Consequentialist)

- Performance of One's own Duty
- Devine-based Morality
- Kant's Categorical Imperative
- W.D. Ross's Prima Facie duty

2.1.3 Virtue Ethics and Civic Virtues

- Basic Principles of Civic Virtues
- How to be virtuous person?

2.2 Non-Normative Ethics

- Meta Ethics
- Absolutism/Objectivism
- Relativism/Subjectivism and Conventionalism
- Naturalism and Non-naturalism

2.3 Issues in Applied Ethics**2.3.1 Development Ethics**

- How should a society Develop?
- Who is morally responsible for Underdevelopment?
- A 'Just' Development

2.3.2 Environmental Ethics

- Ecosystem and the environmental pollutions
- Principles of Environmental Ethics

2.3.3 Professional Ethics

- Profession and Professionalism
- The scope of Professional Ethics
- Common Principles of Professional Ethics

06 hour	Chapter Three: Ethical Decision Making and Moral Judgments 3.1 Ethical Principles and Values of Moral Judgments ▪ The principle of equal consideration of interest ▪ Conflicting goals and ethical Justifications ▪ Ethical values and Justifiable exceptions 3.2 Why Should I act ethically?
1 2	Chapter Four: State, Government and Citizenship 4.1 Understanding State ▪ What is a state? ▪ Attributes of State ▪ State Structures 4.2 Understanding Government ▪ Major Function and Purpose of Government ▪ Types of Government: Limited and Unlimited ▪ Systems of Government 4.3 Understanding Citizenship ▪ What is Citizenship ▪ Inclusion and exclusion in Citizenship ▪ Ways of Acquiring Citizenship ▪ Ways of Losing Citizenship ▪ Citizenship in Ethiopian Context: Past and Present 4.4 State Formation and Nation-building in Ethiopian Context

1 2	Chapter Five: Constitution, Democracy and Human Rights 5.1 Constitution and Constitutionalism ▪ Peculiar features of Constitution ▪ Major Purpose and Functions of Constitution ▪ Classification of Constitutions ▪ The Constitutional Experience of Ethiopia: pre and post 1931 5.2 Democracy and Democratization ▪ Definitions and Forms of Democracy ▪ Views on Democracy: Substantive and Procedural Views ▪ Fundamental Values and Principles of Democracy ▪ Democratization and Its Waves ▪ Major actors in Democratization Process ▪ Democracy and Good Governance in Ethiopia 5.3 Human Rights ▪ Definitions and Nature of Human Rights ▪ Basic Characteristics of Human Rights ▪ Dimensions of Human Rights ▪ The Protection and Promotion of Human Rights - Human Rights Instruments: Documents - Oversight Mechanisms: Institutions
Teaching methodology : Lectures, Group discussions, debates & Reflections	
Recommended Mode of assessment	Quiz (5%) Tests (15%) Assignments (15%) Mid-Exam (25%) Final Examination (40%) Total: 100%
Instructor's Commitment: The course instructor is expected to provide timely lectures, demonstrate students to understand and practice the issues pertaining to central theme of the course, suggest available reading materials, and evaluate students' performance regularly.	
Course Policy: The policy, which administer this course, is in line with University's legislation (no? year?) available at (website). Meaningful participation during class, group work and presentation is	

important for the success of this course. Since each class builds on the one before it, attendance is mandatory. Concerning to academic integrity, learners are expected to exert high level of commitment, responsibility and academic honesty. Because the value of an academic degree depends upon the absolute integrity of the work done by the student for that degree, it is imperative that a student demonstrate a high standard of individual honor in his or her scholastic work. If you cheat on an exam you will receive a failing grade, and most likely will be dropped from the class. Academic dishonesty of any type by a student provides grounds for disciplinary action by the instructor or department. In written works, no material may be copied from another. The work that you submit must be your own, for both moral and legal reasons.

Recommended Reading Materials

- Alexander, Larry (eds.).(1998). *Constitutionalism: Philosophical Foundations*. *Cambridge*: Cambridge University Press.
- AssefaFisseha. (2006). *Federalism and Accommodation of Ethnic Diversity in Ethiopia: Comparative Study*. Utrecht: Wolf Legal Publishers.
- Charles F. Kettering Foundation. & Harwood Group.1991. *Citizens and politics: a view from Main Street America*. Dayton, Ohio: The Foundation.
- S. Oderberg and Timothy Chapel. (2004). *Human values , new essays on ethics and natural law* palgravemacmillan, Great Britain.
- Fasil Nahum. 1997. *Constitution for a Nation of Nations: The Ethiopian Prospect*. Lawrenceville,NJ: Red Sea Publishers.
- FDRE. (1995). *The Constitution of the Federal Democratic of Ethiopia*. Federal NegarritGazeta: Addis Abeba
- Francis Snare (1992). *The Nature of Moral Thinking*. Rutledge, U.S.A and Canada
- Frechette,S. (1981). *Environmental Ethics*. U.S.A.: The Boxwood Press.
- Goodin, Robert E. 2005. *Reflective Democracy*. Oxford University Press: New York.
- James Paul and Clapham .1972. *Ethiopian Constitutional Development: A source book*. Haile Selassie I university: Addis Ababa.
- Jeavons, T. (1991). *Learning for the common good: liberal education, civic education, and teaching about philanthropy*. Washington, DC: Association of American Colleges.
- John M.Rist *Real Ethics*. (2004).*Reconsidering the Foundations of Morality*Cambridge university press U.K and U.S.A
- Macedo, S. (2000). *Diversity and distrust: civic education in a multicultural democracy*. Cambridge, Mass: Harvard University Press.
- Melzer, A. M., Weinberger, J., &Zinman, M. R. (1998). *Multiculturalism and American Democracy*. Lawrence, Kansas: University Press of Kansas.
- Munitz, Milton K., (ed.) (1961). *A Modern Introduction to Ethics*, The Free Press of Clencoe
- Navia, Luis E. and Kelly, Eugene. (1980). *Ethics and the Search for Values*, Prometheus Books.
- Niemi, R. G., &Junn, J. (1998). *Civic education: what makes students learn*. New Haven: Yale University Press.
- Norman, Richard. (1985). *The Moral Photospheres: An introduction to Ethics*, Oxford, Clarendon Press.
- Nzongola, Ntalajia and Margaret C. 1998. *The State and Democracy in Africa*. *Asmara*: Africa World Press.
- Oppenheim, A. N. (1977). *Civic education and participation in democracy : the German case*.London ; Beverly Hills: Sage.
- Penrose, W. O. (1952). *Freedom is ourselves: Legal rights and duties of the citizen as a basis for civic education*. Newark: University of Delaware Press

COURSE TITLE: SOCIAL ANTHROPOLOGY

COURSE CODE: Anth 1012

Credit hour 2

Course Description:

This module which contain Sociology and Anthropology is designed for medical students to understand the subject matter of sociology by briefly covering some of the ideas of the founding fathers of sociology and the major sociological theories, cross-cultural survey of cultural diversity and similarity in the interpretation of health, illness and healing systems. It also exopected to introduce the anthropology of Ethiopian societies and cultures to first year students' of Higher Learning Institutions (HLIs). It covers basic concepts of anthropology such as culture, society and humanity. It also discusses themes including unity and diversity; kinship, marriage and family; indigenouse knowledge systems and local governance, identity, multiculturalism, conflict, conflict resolution and peacemaking system; intra and inter-ethnic relations of Ethiopian peoples. In addition, the course explores culture areas of Ethiopia such as plough culture, enset culture and pastoralism. The course further covers marginalized minority and vulnerable groups in terms of age, gender, occupation and ethnicity by taking ethnographic case studies into account and discuss ways of inclusive growth.

General Objective:

At the end of the course students will be able to assess & recognize the basic social components of societies in relation to health.

Specific Objectives: At the end of the course the student will be able to

1. Describe the definition of sociology
2. Discuss the importance of sociology in Medicine
3. Explain the relationship of Anthropology and sociology
4. Discuss the concept of family as a social unit and the status of an Individual in a family
5. Explain the dynamics of society and common social problems
6. Identify the socio-cultural and economic aspects of the community
7. Develop an understanding of the nature of anthropology and its broader scope in making sense of humanity in a global perspective;
8. Understand the cultural and biological diversity of humanity and unity in diversity across the world and in Ethiopia;
9. Analyze the problems of ethnocentrism against the backdrop of cultural relativism;
10. Realize the socially constructed nature of identities & social categories such as gender, ethnicity, race and sexuality;
11. Explore the various peoples and cultures of Ethiopia;
12. Understand the social, cultural, political, religious& economic life of different ethno-linguistic & cultural groups of Ethiopia;
13. Understand different forms marginalization and develop skills inclusiveness;
14. Appreciate the customary systems of governance and conflict resolution institutions of the various peoples of Ethiopia;
15. Know about values, norms and cultural practices that maintain society together;
16. Recognize the culture area of peoples of Ethiopia and the forms of interaction developed over time among themselves; and
17. Develop broader views and skills to deal with people from a wide variety of socio-economic and cultural backgrounds.

Schedule

WEEK	Topic	Activities/Tasks	
		Instructor	Students
1	<i>The Discipline of Sociology</i> 1.1. Definition and subject matter of sociology 1.2. Sociological imagination 1.3. Scope of sociology: Micro and Macro Sociology <i>Group assignment</i> 1.4. Sociology and Other Social Sciences 1.5. The Significance of Learning Sociology <i>The Development of Sociology: A Historical Review</i> 2.1. Early Origins and Development: Factors which contributed for the emergence and development of Sociology 2.2. Founders of Sociology 2.2.1 Auguste Comte 2.2.2 Herbert Spencer 2.2.3 Emile Durkheim 2.2.4 Karl Marx 2.2.5 Max Weber <i>Quiz 1</i>	✓ Present lecture ✓ Raise questions that require critical thinking and encourage class room discussion ✓ Briefly summarize the lecture towards the end of the session ✓ Present lecture ✓ Raise questions that require critical thinking and encourage class room discussion ✓ Briefly summarize the lecture towards the end of the session ✓ Present lecture ✓ Raise questions that require critical thinking and encourage class room discussion ✓ Briefly summarize the lecture towards the end of the session	✓ Actively listen to the lecture ✓ Take note of important points ✓ Actively participate in class discussion ✓ Actively listen to the lecture ✓ Take note of important points ✓ Actively participate in class discussion ✓ Actively listen to the lecture ✓ Take note of important points ✓ Actively participate in class discussion
2	<i>Theoretical Perspectives in Sociology</i> 3.1. Structural Functionalism 3.2. Conflict Perspective 3.3. Symbolic – Interactionism	✓ Present lecture ✓ Raise questions that require critical thinking and encourage class room discussion ✓ Briefly summarize the lecture towards the end of the session	✓ Actively listen to the lecture ✓ Take note of important points ✓ Actively participate in class discussion
3	<i>Culture</i> 7.1. The Concept of Culture 7.2. Components of Culture 7.3. Definition of basic cultural concepts	✓ Present lecture ✓ Raise questions that require critical thinking and encourage class	✓ Actively listen to the lecture ✓ Take note of important points ✓ Actively participate in

	<i>Quiz 2</i>	✓ room discussion ✓ Briefly summarize the lecture towards the end of the session	class discussion
4	<i>Socialization</i> 5.1. Socialization defined 5.2. Types of socialization 5.3. Agents of socialization	✓ Present lecture ✓ Raise questions that require critical thinking and encourage class room discussion ✓ Briefly summarize the lecture towards the end of the session	✓ Actively listen to the lecture ✓ Take note of important points ✓ Actively participate in class discussion
5	<i>Social Organization and Interaction</i> 6.1. Social structure: status and role 6.2. Groups and Institutions 6.3. Types of social groups 6.4. Social Values, Norms and Social Control <i>Quiz 3</i>	✓ Present lecture ✓ Raise questions that require critical thinking and encourage class room discussion ✓ Briefly summarize the lecture towards the end of the session	✓ Actively listen to the lecture ✓ Take note of important points ✓ Actively participate in class discussion
6	<i>Social Inequality and social processes</i> 7.1. Social stratification 7.2. Forms of social stratification 7.3. Social mobility 7.4. Social change 7.5. Social movements <i>Submission of assignment paper</i>	✓ Present lecture ✓ Raise questions that require critical thinking and encourage class room discussion ✓ Briefly summarize the lecture towards the end of the session	✓ Actively listen to the lecture ✓ Take note of important points ✓ Actively participate in class discussion
7	<i>Deviance and Crime</i> 8.1. Definition of deviance and crime 8.2. Major differences between deviance and crime 8.3. Types of crime 8.4. Theories of deviance <i>Quiz 4</i>	✓ Present lecture ✓ Raise questions that require critical thinking and encourage class room discussion ✓ Briefly summarize the lecture towards the end of the session	✓ Actively listen to the lecture ✓ Take note of important points ✓ Actively participate in class discussion
8	Paper Presentation	✓ Introduce presenters and moderate presentations ✓ Raise questions regarding the papers being	✓ Actively listen to presentations ✓ Present papers ✓ Answer questions raised during the course of their

		presented ✓ Provide clarifications	presentation ✓ Raise questions regarding other presentations
9	Introducing Anthropology and its Subjects 1.1. What is anthropology – a Mirror for Humanity? Sketching the subject matter, scope and concerns of anthropology Anthropological imagination: asking questions and seeing the world anthropologically. Q- What does it mean by using the anthropological lens when looking at the world? Defining Features of Anthropology- holism, relativism & comparative perspectives Methods of Research in anthropology: ethnography & ethnographic methods 1.2. Sub-fields of Anthropology: Four Mirrors for Understanding Humanity 1.3. The relation between anthropology and other disciplines		
10	Human Culture and Ties that Connect 2.1. Conceptualizing Culture: What Culture Is and What Culture Isn't? 2.2. Characteristics features of culture: what differentiates culture from other traditions? 2.3. Aspects of Culture – Material & Non-material (values, beliefs & norms) 2.4. Levels of culture: universality, generality and particularity (cultural diversity) 2.5. Ethnocentrism, Cultural relativism, and human rights Discussion- Debating cultural		

	relativism: Human rights law and the demonization of culture and anthropology along the way 2.6. Cultural Change: what is cultural change? Cultural Diffusion versus Cultural Assimilation Innovation Discussion - Contesting culture as sharply bounded versus unbounded ‘cultural flows’ or as ‘fields of discourse’ in the context of globalization. 2.7. Ties that Connect: Marriage, Family and Kinship Marriage -rules, functions and forms of Marriage Family -types and functions of Family Q. How families and marriage differ in different societies? Kinship System -types of kin groups and rules of descent Kinship and Gender Across Cultures  Sex and Gender: Mapping differences		
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	in cross cultural perspective  Gender –as power relations 2.8. Cultural practices, norms and values that maintain society together		
11	Human Diversity, Culture Areas, and Contact in Ethiopia 3.1. Human Beings & Being Human: What it is to be human? –(a bio-cultural animal?) 3.2. Origin of the Modern Human Species: Homo sapiens sapiens (that’s you!) Religious, biological & evolutionary (paleo-anthropological) explanations 3.3. The Kinds of Humanity: human physical variation Q. Why isn’t everyone the same? Q. Why do people worldwide have differences in their phenotypic attributes? 3.4. Human Races: the history of racial typing The Grand Illusion: Race, turns out, is arbitrary Q. What can we say for sure about human races?		
	3.5. Why is Everyone Different? Human Cultural Diversity - anthropological explanations Q. Why don’t others do things the way we/I do? 3.6. Culture areas and		

	cultural contacts in Ethiopia 3.6.1.1. Plough culture area 3.6.1.2. <i>Enset</i> culture area 3.6.1.3. Pastoral societies culture area 3.6.1.4. Historical and social interactions between culture areas		
12	Marginalized, Minorities, and Vulnerable Groups 4.1. Gender based marginalization 4.2. Occupational cast groups 4.3. Age based vulnerability (children and old age issues) 4.4. Religious and ethnic minorities 4.5. Human right approaches and inclusive growth, anthropological perspectives		
13	Theories of inter-ethnic relations and multiculturalism in Ethiopia 5.1. The Scales of Human Identity: Who am I?- Understanding ‘self’ & ‘other’ Q- What are the ways we tell for others who we are? 5.2. Ethnicity and Race: What’s in a name? 5.3. Ethnic Groups & Ethnic Identity Q. What is the basis of one’s ethnic identity? Q. Is ethnicity a fundamental aspect of human nature & self-consciousness, essentially unchanging and unchangeable		

	identity? Or Q. Is it, to whatever extent, socially constructed, strategically or tactically manipulable, and capable of change at both the individual and collective levels? 5.4. Race –the social construction of racial identity Q. Do the claims of some people/groups about superior & inferior racial groups have any scientific validity? 5.5. Primordialism; Instrumentalism; Social constructivism 5.6. Debates on inter-ethnic relations and identities		
14	Customary and local governance systems and peace making 6.1. Indigenous knowledge systems and local governance 6.2. Intra and inter-ethnic conflict resolution institutions Ethnographic cases: commonalities and shared practices (e.g., Oromo and Somali, Afar and Tigray; Gedeo and Oromo; Guraghe and Siltie; Amara and Tigray) 6.3. Customary/Local governance systems Ethnographic cases: Oromo Geda; Somali-Gurti; Gamo, Gofa, Wolayita-Woga; Guraghe-Sera 6.4. Legal pluralism: interrelations		

	between customary, religious and state legal systems	
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Teaching method- lecture, and Group discussion

Teaching materials-LCD, White Board, chalk and board, video show

Assessment and Evaluation Criteria:

Based on the progressive understandings of the course, students will be evaluated continuously through both non-graded assignments/activities, like (reading assignments) and graded assignments/activities and assessments including class discussion & participation, Test, Term Paper & presentation, Home Taken Exam/case studies and Final Exam.

Assessment:

Continuous assessment (class participation, Group and Individual assignment, quizzes)...60%

Final written examination ---40%

Course Requirements:

(Classroom, resources, and other inputs required to deliver the course will be listed)

Grading Policy:

(Legislation of the concerned university will be adopted by departments/schools handling the common course across the country)

Suggested readings:

1. ZerihunDoda,(2005), Introduction to sociology for health students , Debub University
2. Diana Kendal, Rick Linden, J. Lothain Murry (2001), sociology in our times: the essentials , 2nd ed., Nelson Thomson learning
3. Macaronis John J. (2008). SOCIOLOGY, 12th ed. Pearson prentice hall
4. Schafer, Richard. (2003). Sociology, New York: McGraw Hill, Inc.
5. Asmarom Legesse (2006). Oromo Democracy: an Indigenous African Political System. The Red Sea Press, Inc.
6. Cameron, M. Smith and Evan T. Davies (2008). Anthropology for Dummies. Wiley Publishing, Inc., Indianapolis, Indiana.
7. Clifored Geertz . (1973). The Interpretation of Cultures. A division of Harper Collins Publishers
8. Donald Donham . (1986). Marxist Modern. The Ethnographic History of Marxist Ethiopia.
9. Donald N. Levine. (1974). Greater Ethiopia: The Evolution of A Multiethnic Society. Chicago & London., University of Chicago.
10. Dunif-Hattis and Howard C. (1992). Anthropology: Understanding Human Adaptation. New York: Harper Collins, Inc
11. Eriksen, T. H. (2001). Small Places, larger Issues: An introduction to social and cultural anthropology. London: Pluto Press.
12. Eriksen, T. H. (2004). What is anthropology? London: Pluto Press.
13. Eriksen, T. Hylland. (2002). Ethnicity and Nationalism. London; Pluto Press.
14. Eriksen, T.H. and Nielsen, F.S. (2001). A History of Anthropology. London: Pluto Press.
15. Hallpike, Christopher R. (1972). The Konso of Ethiopia: A Study of the Value of a Cushitic People. Oxford: Clarendon Press.
16. Hamer, John. (1970). The Sidama Generational Class Cycles: A Political Gerontocracy. Africa 40,1 (Jan,1970): 50-70.
17. Haviland, WA, (1999). Cultural Anthropology (9th ed.). Fort Worth: Harcourt and Brace College Pub.
18. Kottak, C. P. (2004) – Anthropology: the Exploration of Human Diversity (10th ed.). McGraw Hill, New York.
19. Lavenda, R. and Emily S. (2015). Anthropology. What Does It Mean to Be Human?. (3rded.). Oxford. Oxford University Press.

20. Pankhurst, R. (2001). *Historic Images of Ethiopia*. Shamans Books. Addis Ababa, Ethiopia.
21. Richard Jenkins. (2006). *Rethinking Ethnicity*. London Sage Publication.
22. Rosman, A., Rubel, P.G. and Weisgrau, M. (2009). *The Tapestry of Culture: an Introduction to Social Anthropology*. Lanham: Rowman and Little field.
23. Scupin and DeCorse (1988). *Anthropology: A Global Perspective* (2nd ed.). New Jersey: Prentice Hall.
24. Shack, William S. (1966). *The Gurage: A People of the Enset Culture*. London: Oxford University Press.
25. Triulzi et al. (2002). *Remapping Ethiopia Easer African Studies*. Addis Ababa

COURSE TITLE: GLOBAL TRENDS

Course Code: GLOT 1012

Credit Hour= 2 Cr. Hr

Instructor's Contact Information	Name					
	Academic Position					
	Cell Phone					
	Email					
	Office No					
Course Title	Global Trends					
Course Code	IRGI 1021					
Credit Hours	02 Cr.H (4 ECTS)					
Status of Course	Compulsory Common Course					
Student Work Load	Lectures		Library and Group Work	Assign Report	Home Study	Total W. L.
	32hrs.	06	16hrs.	13hrs.	30	97hr
Course description	The course is designed to familiarize learners on the nature and development of international relations and global issues. It deals with nations, states, national interest, cooperation and conflict among states, and the role of state and non-state actors in the international system. Additionally, it explains the nature of international law, global political economy and the nexus between regionalism and globalization. It also critically examines the contemporary global issues and how the international community is trying to address them. It is organized to systematically examine international issues by employing different theories and providing concrete examples from different parts of the world. Last but not least, after providing rigorous understanding of how the international system functions, it will equip learners to consciously observe and critically understand the Ethiopia's Relations with the outside world. As the saying goes "Think globally acts locally!"					

Course Purpose	We live in an exciting yet challenging period in history. The world seems to dominantly progress in constantly opposing directions. On the one hand, because of globalization, the world is getting closer and becoming interconnected in ways never experienced before generating more wealth, scientific innovation, and cross-national cooperation. On the other hand, the challenges of war, terror, arms trade, money laundering, disease, poverty, environmental problems, human and drug trafficking still generate an aura of uncertainty for the present and future generations. As such, decisions made by states, multinational corporations, non-governmental organizations, and terrorists have a direct impact on our life. Thanks to global flow of information, there may be a multitude of individuals who know the events that are occurring in the world. But some still do not understand why events happened the way they have happened; and what consequences they may bring. In light of this, this course is designed to equip students with a necessary knowledge and skill which enable them to understand the political, economic and social dynamics of the global system, how it works, its actors, its influence and ways to cope up global issues from theoretical and practical point of view.
Course objectives and expected learning outcomes	At the end of the course, students will be able to: ✓ Understand nations, nationalism and states ✓ Explain the nature and historical development of international relations ✓ Gain basic knowledge of the major theories in the discipline of International relations and develop the ability to critically evaluate and apply such theories ✓ Elucidate national interest, foreign policy and diplomacy ✓ Explicate the nature and elements of international political economy and international law ✓ Examine the extent and degree of influence of state and non-state actors in the international system ✓ Examine the roles major international and regional institutions play in world Politics ✓ Critically evaluate the major contemporary global issues ✓ Assess the overriding foreign policy guidelines of Ethiopia in the past and present ✓ Explore Ethiopia's role in regional, continental and global institutions and affairs
Couse Contents and Schedule	
Contac Hours	Chapters, Sections and Sub-Sections
6 Hours	1. Chapter one: Understanding International relations 1.1. Conceptualizing Nations, Nationalism and States 1.2. The Nature and Evolution of International Relations 1.3. Actors of International Relations ✓ State Actors ✓ Non-State Actors 1.4. Levels of Analysis in the International Relations 1.5. Power, Anarchy and Sovereignty in the International System 1.6. The Structure of International System
6 Hours	2. Chapter Two: Contending Theories of International Relations 2.1. Realism and Neo-Realism 2.2. Liberalism and Neo-Liberalism 2.3. Marxism and Neo-Marxism 2.4. Critical Theory 2.5. Constructivism 2.6. Modernism and Post-Modernism

8 Hours	3. Chapter Three: Foreign Policy and Diplomacy 3.1. Conceptualizing National Interest, Foreign Policy and Diplomacy 3.2. National Interest and Foreign Policy ✓ Determinants of National Interest and Foreign Policy ✓ Objectives of Foreign Policy ✓ Foreign Policy Orientations ✓ Instruments of Foreign Policy 3.3. A Survey of Foreign Policy and Diplomacy of Ethiopia ✓ Foreign Policy of Ethiopia during the Reign of Emperor Menilik II ✓ Foreign Policy of Ethiopia during the Reign of Emperor Hailesillassie ✓ Foreign Policy of Ethiopia during the Derg Regime ✓ Foreign Policy of Ethiopia during the EPRDF
8 Hours	4. Chapter Four: The International Political Economy (IPE) 4.1. Meaning and Nature of IPE ✓ The Nexus between Politics (State) and Economics (Market) 4.2. Theoretical Perspectives on IPE ✓ Classical Mercantilism and Economic Nationalism ✓ Classical Liberalism and Adam Smith ✓ Comparative Advantage and David Ricardo ✓ Neoliberalism and Keynesianism ✓ Marxism and Dependency Theory ✓ Hegemonic Stability Theory ✓ Developmental State Model 4.3. The Political Economy of North-South, South-South: Conflict and Cooperation
8 Hours	5. Chapter Five: International Law 5.1. Meaning, Nature and Areas of International Law 5.2. Sources and Subjects of International Law 5.3. Law Making and Enforcement process at International and Domestic level 5.4. Formation, Recognition and Responsibility of State under International Law
6 Hours	6. Chapter Six: Regionalism and Globalization 6.1. The Concept, Nature and Development of Regionalism and Regional Integration ✓ The Old and New Regionalism 6.2. Major Theories of the Regional Integrations ✓ Functionalism ✓ Neo-functionalism ✓ Inter- governmentalism ✓ Supra-nationalism ✓ Selected Cases of Regional Integration (EU, AU...) 6.3. Definition and Evolution of Globalization ✓ Aspects of Globalization ✓ Actors of Globalization ✓ Pros and Cons of Globalization ✓ Ethiopia in a globalized World 6.4. Regionalization versus Globalization and State ✓ The Convergence, Divergence and Overlapping relations of Regionalization and Globalization ✓ The Hypocrisy of Sovereignty

6 Hours	7. Chapter Seven: Major Contemporary Global Issues 7.1. Conceptualizing Global Issues 7.2. Survey of Global Issues ✓ Security Issues ○ Terrorism, Religious Fundamentalism and political Extremism ○ Weapons of Mass Destruction and The Nuclear Power paradox ○ Illicit Human Trafficking, Drug Trafficking, Firearms Trafficking ✓ Environmental Issues ○ Climate Change and Global warming ✓ Technology Related Issues ○ Cyber Crime and Cyber Security ✓ Other Social, Economic and Political Issues ○ Human Rights ○ Migration and Refugee ○ Trade War ○ Aid, Debt Relief
Teaching methodology: Lectures, Group discussions, debates & Reflections	
Recommended Mode of Assessment	✓ Tests (20%) ✓ Assignment and Presentation (15%) ✓ Mid-Exam (25%) ✓ Final Examination (40%)
Instructor's Commitments: The course instructor is expected to provide timely lectures, demonstrate students to understand and analyze the issues pertaining to central theme of the course, suggest available reading materials, and evaluate students' performance regularly.	
Course Policy: Meaningful participation during class, group work and presentation is important for the success of this course. Since each class builds on the one before it, attendance is mandatory.	
Academic Integrity: The department expects from its students a high level of responsibility and academic honesty. Because the value of an academic degree depends upon the absolute integrity of the work done by the student for that degree, it is imperative that a student demonstrate a high standard of individual honor in his or her scholastic work. If you cheat on an exam you will receive a failing grade, and most likely will be dropped from the class. Academic dishonesty of any type by a student provides grounds for disciplinary action by the instructor or department. In written work, no material may be copied from another. The work that you submit must be your own, for both moral and legal reasons.	
Recommended Reading Materials	
1. Altinay, Hakan (2011) Global Civics: Responsibilities and Rights in an Interdependent World. The Brookings institution: Washington 2. Armstrong, David (ed.)(2009). Routledge Handbook of International Law. London: Routledge 3. Baylis, J. and Smith, S. (eds.) (1997).The Globalization of World Politics. Oxford: Oxford University Press. 4. Browlie, Ian (2003). Principles of Public International Law. (6th ed.). New York: Oxford University 5. Copson, Raymond w.(2007)The United States in Africa: Bush policy and beyond in association with International African Institute Royal African Society of Social Science Research Council, Zed Books: London 6. Crane, George T. and Abal Amawi (1997). The Theoretical evolution of International Political Economy: A Reader (2nd Edition). Oxford University Press: New York. 7. Crawford, Robert (2000) Idealism and Realism in International Relations: Beyond the Discipline. Routledge: USA 8. DeLombaerde, Philippe (ed.) (2008) Governing Regional Integration for Development: Monitoring	

- Experiences, Methods and Prospects. Ashgate Publishing Company: England
9. Demelo, Jaime and Arvind Panagariy (eds.) (1993) A New Dimensions in Regional Integration ,Centre for Economic Policy Research 1993, Cambridge University Press: USA
 10. Demelo, Jaime and Arvind Panagariy (eds.) (1993) A New Dimensions in Regional Integration, Centre For Economic Policy Research 1993, Cambridge University, Press: USA
 11. Der, James D. (2009) Critical Practices in International Theory: Selected Essays,Routledg, Abingdon, Oxon
 12. Farrell, Mary (ed.) (2005) Global Politics of Regionalism: Theory and Practice. Pluto Press: London
 13. Genest, Mark A. (1996). Conflict and Cooperation: Evolving Theories of International Relations. Fourth Worth: Harcourt Brace and Co.
 14. Goldestein, Joshua S. and John C. Pevhouse (2006). International Relations brief 3rd ed. Prisscilla Mc Greehon.
 15. Goldstein J. S. (2003) International Relations. 5th edition. Washington, D.C. Pearson Education Press, Inc
 16. Griffiths, Martin (Ed.) (2007). International Relations Theory for the Twenty-First Century:An introduction. New York: Routledge
 17. Griffiths, Martin and Terry O'Callaghan (2002) International Relations: The Key Concepts. Routledge: London
 18. Griffiths, Martin, et al. (2008). International Relations: The Key Concepts (Second Edition). New York: Routledge
 19. Griffiths, Martin, Steven C. and M. Scott (2009), Fifty Key Thinkers in International Relations (2nd edition)
 20. Hancock, Kathleen J. (2009) Regional Integration; Choosing Plutocracy, Palgrave Macmillan: United States
 21. Henderson, Conway W.(1998) International Relations: Conflict and Cooperation at the Turn of the 21 st Century. Guilford: McGraw-Hall.
 22. Hollis, Martin and Steve Smith (1990) Explaining and Understanding International Relations. United States: Oxford University Press.
 23. Holsti, K.J. (1995) International Politics: A Framework for Analysis. 7th ed. New Jersey: Prentice Hall.
 24. J.M. Biswaro. (2012) The Quest for Regional Integration in the Twenty First Century: Rhetoric versus Reality - A Comparative Study, Mkukina Nyota Publishers Ltd, Dares Salaam: Tanzania
 25. Macdonald, David B.et.al (ed.)(2007)The Ethics of Foreign Policy. Ashgate Publishing Limited: England
 26. Malm, Endreas and Shora Esmailian (2007). Iran on the Brink Rising Workers and Threats of War. Pluto Press
 27. Mintz, Alex and Karl De Rouen (2010) Understanding Foreign Policy Decision Making, Cambridge University Pres: Cambridge
 28. Rengger, N.J.(2000) International Relations, Political Theory and the Problem of Order: Beyond International Relations theory? Routledge: London
 29. Rourke, John T. and Mark A. Boyer (1998). World Politics: International Politics on the world Stag., brief. 2nd ed. Guilford: Dushkin/McGraw-Hall.
 30. Salmon, Trevor C. (Ed.) (2005). Issues in International Relations. New York: Routledge
 31. Steans, Jill and Lloynd Pettiford (2005). Introduction to International relations: Perspectives and Themes. 2nd ed. Harlow: Pearson Prentice Hall.
 32. Sutch, Peter and Juanita Elias (2007) International Relations, the basics. Taylor & Francis.
 33. Todaro, Michael P. and Stephen Smith (2003). Economic Development (8thed).
 34. Trevor S. Salmon and Mark F. Imber (ed) (2008). Issues in International Relations. Routledge Publishing, 2nd Edition
 35. Vinay Bhargava (2006). Introduction to Global Issues.
 36. Walts, Kenneth N. (2003) Progress in International Relations Theory. Belfer Center for Science and International Affairs John F. Kennedy School of Government, Harvard University Cambridge, Massachusetts

COURSE TITLE: INCLUSIVENESS

COURSE CODE: Incl 1012

Credit Hour = 2 Cr. Hr

Inclusion in Education and Development: A course for Higher Education

Students of all disciplines

1. Course information

1.1. Course Title: Inclusion in Education and Development

1.2. Course code: SNIE _____

1.3. Credit hours: 2

1.4. Contact hours: 48 Hours

2. Introduction

Development efforts of any organization need to include and benefit people with disabilities through providing education, creating employability, promoting prosperity, reducing poverty and enhancing stability. Unfortunately, this has not been the practice for the majority of people with disabilities due to unfavorable attitude, negligence and exclusion from all development endeavors. It is obvious that people with disabilities are the large stand most disadvantaged minority in the world. They are about 15 percent of the global population (about one billion people), and 17.6 million in Ethiopia, with most extended families including someone with a disability (World Health Organization and World Bank and 2011). Exclusion practices of this large number of persons with disabilities in Ethiopia seem undermines their potential/ability to contribute to poverty reduction and economic growth within their household, their community and the country. It is clear that it is not impairment, but, the exclusion practices that has contributed for poverty aggravation for persons with disabilities. Exclusion practices of persons with disabilities have a long history, affecting the life of people with disabilities and the society at large. In the past and even today people have been discriminated due to their disabilities.

Inclusions promote effective developments through full participation of all members of a population and people with disabilities, where both are agents of development and beneficiaries. Through identifying and removing barriers, people with disabilities participate and benefit from the developments. Genuine inclusion of people with disabilities allow them actively participate in development processes and eliminate dependence syndrome, leads to broader benefits for families and communities, reduces the impacts of poverty, and positively contributes to a country's economic growth. All stages of development processes of any organization should be inclusive through creating equal access to education, health care services, work and employment, social protection and all development center of human being.

3. Course Description

Special needs education refers to people with divers' disabilities, gifted and talented, and divers' population being at risk of education and development. As per the institutional reform that is focusing on enhancing development for all population, the field of inclusive education is taking center stage in institutional planning and improvement. This course introduces the process of achieving inclusion with all appropriate accessibility and established support system at institutional level.

In this course, the higher education students will learn how to assess, understand and address the needs of persons with disabilities and provide relevant support or seek extra support form experts. He/she also learns how to adapt and implementing services for an inclusive environment that aimed to develop holistic development such as affective, cognitive and psychosocial skills of the population with disabilities. Identification and removal/management of environmental barriers would find a crucial place in the course. The students learn how to give more attention and support for students with; hearing impairments, visual impairment, deaf-Blind, autism, physical and health impairments, intellectually challenged, emotional and

behavior disorders, learning difficulty, communication disorders, gifted and talented student, and those at risk due to different reason (population who are environmentally and culturally deprived, abused, torched, abandoned, and orphaned and vulnerable..etc). All University students will be given the chance to study the specific developmental characteristics of each group of students with disabilities and come up with appropriate intervention strategies in inclusive settings of their respective professional environment and any development settings where all citizens are equally benefited.

4. Learning outcome of the course

The goal of this course is to provide the tools and strategies that help to create a convenient environment that accommodates population with divers' disabilities and potential. This course encourages exploring the benefits of collaborating with colleagues to design and implement inclusion an all sphere of life. It also guides the discovery of ways to modify environment as well as services and practices to meet the needs of all persons with disabilities in inclusive environment.

As a result of reviewing various reading materials, completing the assignments, engaging in related discussions, and strongly workings on activities, towards the completion of the course, the University students of all fields in Ethiopia will be able to:

1. Aware the needs of people with special needs, their potential and include all aspects of developmental needs
2. Identify population with special needs, their potentials and the learning and working styles of all population with special needs in their environment.
3. Demonstrate desirable attitude towards all population with special needs in their learning, working and living environment
4. Apply various assessment strategies for evidence-based planning to meet their needs
5. Attempt to adapt environments they are working and living in according to the need and potential of the population with special needs
6. Develop an accommodative and inclusive attitude help to think for the wellbeing and development of population with special needs.
7. Identify and select appropriate support and services method that addresses the life needs of population with special needs individually and on group bases.
8. Collaborate with experts and relevant others for the life success of all persons with disabilities in all environments.
9. Create and maintain successful inclusive environment

5. Detailed Syllabus components

1. Unit 1. Understanding students with diverse needs/special needs

Time allotted: 10 contact hours

1.1. Unit objectives

At the end of completing this unit, the students will be able to:

- 1.1.1. Brief historical trends of special needs population and their holistic development
- 1.1.2. Describe the effect of negative attitude on educational and life success of people with special needs
- 1.1.3. Describe the nature of difficulties, preventable causes, identification, and assessment, of students with various impairments that affect their daily learning.
- 1.1.4. Identify students with special needs whose daily life and functioning is challenged and those students who are at risk.
- 1.1.5. Describe the need and characteristics of gifted and talented population
- 1.1.6. Depict the condition of student at risk because of different reasons (environmentally, culturally and linguistically deprived, abused, torched, abandoned, and orphaned and vulnerable student) who need special attention in educational setting.

1.2. Unit Contents and sub contents

- 1.2.1. Overview of students with various special needs; sensory, physical, below average intelligence, emotional and behavioral challenges, learning difficulties, communication

difficulties, gifted and talented, students at risk, their developmental characteristics and preventable causes.

1.2.2.Theoretical models such as the right model, medical model and social model of special needs and its relevance.

1.2.3.Educational of students with various special needs.

1.2.4.Attitudinal barriers and means to enhance awareness among the community members and political leaders.

1.2.5.The relationship between Poverty and Disability

1.2.6.Planning effective development of population with special needs in inclusive settings

1.3. General Approach/Methods/Strategies

This section is flexible to involve the instructor's creativity in identifying selecting and adapting the instructional method to the context of the learner. Some general approaches are listed below. The instructor can select among this and add his own that he/she feels appropriate

1.3.1.Interactive lectures

1.3.2.Collaborative learning

1.3.3.Independent/self-learning

1.3.4.Institutional/school visits and practicum

1.3.5.Individual and group assignments and presentation

1.3.6.Seminars

1.3.7.Expert consultancy

1.3.8.Group participation

1.3.9.Guest practitioners

1.3.10. Others...

1.4. Detailed activities

1.4.1.Students' Activities

1.4.1.1. Reading the materials delineated for this purpose

1.4.1.2. Discus and/or report the diversities of learners in the classroom and report

1.4.1.3. Conduct mini case studies of students with divers needs

1.4.1.4. Visit schools that integrate students with special needs and report the teaching and learning experience at school.

1.4.1.5. Find out gifted and talented students and gather facts on the teaching and learning experience at school.

1.4.1.6. Find out evidence the effective instructional approach from existing experience and literature.

1.4.2.Instructors Activities

1.4.2.1. Facilitate students learning by posing questions and guiding their activities

1.4.2.2. Provide some examples of students with special needs and their success.

1.4.2.3. Provide group case study questions

1.4.2.4. Support learners in all aspects in relation to their learning

1.4.2.5. Facilitate dialogue among learners to find out effective instructional approach that address divers learning needs.

1.5. Assessment Strategies and Techniques

1.5.1.Recording attendance

1.5.2.Recording group and individual participation

1.5.3.Visit reports and presentation

1.5.4.Assignment (individual and group)

1.5.5.Test at the end of the unit

2. Unit 2. Understanding Inclusion

Time allotted: 8 hours

2.1. Unit Objectives

Upon accomplishing this unit, the teacher candidates will be able to perform the following activities.

2.1.1.Organize and implement inclusion for people with varying special needs

- 2.1.2. Demonstrate understanding of the principles of an inclusive environment, the rationale for inclusion, and its effect on education, and development.
- 2.1.3. Define terms associated with inclusion and its practices
- 2.1.4. Recognize what an inclusive environment looks and sounds like
- 2.1.5. Respect rights of students with special needs along with the disability convention ratified by the Ethiopian Government
- 2.1.6. Identify the benefits and challenges of inclusion
- 2.1.7. Modify environment to meaningfully accommodate population with special needs in all environment

2.2. Unit Contents and sub contents

- 2.2.1. Definition of inclusive environment and the support system
- 2.2.2. Elements of Inclusive environment
- 2.2.3. Characteristics of inclusive environment
- 2.2.4. Special needs population's right in the inclusive environment
- 2.2.5. Benefits and Challenges of Inclusion
- 2.2.6. Strategies in addressing individual needs.
- 2.2.7. Policies, legislations, strategies, legal framework and other related documents.

2.3. General Approach/Methods/Strategies

This section is flexible to involve the instructor's creativity in identifying selecting and adapting the instructional method to the context of the learner. Some general approaches are listed below. The instructor can select among this and add his own that he/she feels appropriate.

- 2.3.1. Interactive lectures
- 2.3.2. Group task
- 2.3.3. Collaborative/cooperative learning
- 2.3.4. Independent/self-learning
- 2.3.5. Institutional visits and practicum
- 2.3.6. Individual and group assignments and presentation
- 2.3.7. Seminars
- 2.3.8. Expert consultancy
- 2.3.9. Other...

2.4. Detailed activities

- 2.4.1. Students' Activities
 - 2.4.1.1. Discuss; define what inclusive environment and the support system for population with divers needs
 - 2.4.1.2. Discus and list elements and characteristics of inclusive environment
 - 2.4.1.3. List special populations' right to development and discuss how to implement Inclusive environment
 - 2.4.1.4. Explore the benefits and challenges of inclusion, individually and in-group
- 2.4.2. Instructors Activities
 - 2.4.2.1. Facilitate students learning by posing questions and guiding their activities
 - 2.4.2.2. Create participatory learning classroom.
 - 2.4.2.3. Provide activity guide for the students
 - 2.4.2.4. Encourage all students to work on the activities in cooperative manner
 - 2.4.2.5. Provide special support for those with difficulties or special needs
 - 2.4.2.6. Provide feedback on the students various activities
 - 2.4.2.7. Stabilize the learning by summarizing core points
 - 2.4.2.8. Demonstrate the importance and application of adapted technology
 - 2.4.2.9. Introduce Web sites devoted to technology use in the inclusive classroom.
 - 2.4.2.10. Introduce software and multimedia applications.

2.5. Assessment Strategies and Techniques

- 2.5.1. Recording attendance

- 2.5.2. Recording group and individual participation
- 2.5.3. Recording the product of students
- 2.5.4. Test at the end of the unit
- 2.5.5. Assignment and visit reports

3. Unit 3: Identification of population with special needs

Time allotted: 5 hours

3.1. Unit objectives

Upon completing this unit, the students will be able to:

- 3.1.1. Learn and engage in developing identification tools that would be applicable in the environment
- 3.1.2. Identify different needs among population with special needs and use various strategies that support their developmental needs.
- 3.1.3. Demonstrate the process of identifying students who need special support and the options available for serving these students' educational needs

3.2. Unit Contents and sub contents

- 3.2.1. Development of checklists for identification of various difficulties.
- 3.2.2. Procedure of identification
- 3.2.3. Identifying learners needs, potentials and difficulties in learning

3.3. General Approach/Methods/Strategies

This section is flexible to involve the instructor's creativity in identifying selecting and adapting the instructional method to the context of the learner. Some general approaches are listed below. The instructor can select among this and add his own that he/she feels appropriate.

- 3.3.1. Interactive lectures
- 3.3.2. Group activities
- 3.3.3. Collaborative learning
- 3.3.4. Independent/self-learning
- 3.3.5. Institutional/school visits and practicum
- 3.3.6. Individual and group assignments and presentation
- 3.3.7. Seminars
- 3.3.8. Expert consultancy
- 3.3.9. Others...

3.4. Detailed activities

- 3.4.1. Students' Activities
 - 3.4.1.1. Read materials on how to identifying learning styles of learners, from literature and experiences.
 - 3.4.1.2. Discuss how to identify various support strategies
 - 3.4.1.3. Discuss on various instruments on how to identify students need, potentials and difficulties
 - 3.4.1.4. Develop identification checklist for specific difficulty.
- 3.4.2. Instructors Activities
 - 3.4.2.1. Facilitate students learning by posing questions and guiding their activities
 - 3.4.2.2. Provide activity guide for the students
 - 3.4.2.3. Encourage all students to work on the activities in cooperative manner
 - 3.4.2.4. Provide special support for those with who need it
 - 3.4.2.5. Provide feedback on the students various activities

3.5. Assessment Strategies and Techniques

- 3.5.1. Recording attendance
- 3.5.2. Recording group and individual participation
- 3.5.3. Recording the product of students
- 3.5.4. Test at the end of the unit
- 3.5.5. Assignment
- 3.5.6. Portfolio

4. Unit 4. Assessment in special needs

Time allotted: 5 hours

4.1. Unit objectives

- 4.1.1. At the end of this unit, the students will be able to:
- 4.1.2. Adapt assessments for students with special needs
- 4.1.3. Understand potential challenges of using standard assessment tools to measure the progress of students with special needs
- 4.1.4. Modify and create assessments that accurately evaluate the skills and progress of all students, including those with special needs
- 4.1.5. Use ongoing as well as summative assessments
- 4.1.6. Use portfolios to assess ethically and appropriately what each student knows and able to do in inclusive classroom.
- 4.1.7. Design an assessment that addresses an equity issue
- 4.1.8. Assess, design and decide the most appropriate educational programming for student/youth with sensory impairments, physical and health impairments, intellectually challenged, emotional and behavior disorders, learning difficulty, communication disorders, and students at risk and gifted and talented students.
- 4.1.9. Assess and design on elimination of social and environmental barriers that would facilitate inclusive education

4.2. Unit Contents and sub contents

- 4.2.1. Strategy and procedure to develop assessment instrument.
- 4.2.2. Relevant components of assessment instrument.
- 4.2.3. Progressive assessments
- 4.2.4. Portfolios
- 4.2.5. Implication of assessment

4.3. General Approach/Methods/Strategies

This section is flexible to involve the instructor's creativity in identifying selecting and adapting the instructional method to the context of the learner. Some general approaches are listed below. The instructor can select among this and add his own that he/she feels appropriate.

- 4.3.1. Assignment
- 4.3.2. Group and individual presentations
- 4.3.3. Collaborative learning
- 4.3.4. Independent/self-learning
- 4.3.5. Institutional/school visits and practicum
- 4.3.6. Individual and group assignments and presentation
- 4.3.7. Seminars
- 4.3.8. Expert consultancy
- 4.3.9. Others....

4.4. Detailed activities

- 4.4.1. Students' Activities
 - 4.4.1.1. Identify and discuss some basic assessments techniques for students with special needs and their challenges
 - 4.4.1.2. Exercise how to assessments progresses of all students
 - 4.4.1.3. Discuss how to use portfolios for all students
 - 4.4.1.4. Practice to develop assessment tools
- 4.4.2. Instructors Activities
 - 4.4.2.1. Facilitate students learning by posing questions and guiding their activities
 - 4.4.2.2. Provide activity guide for the students
 - 4.4.2.3. Encourage all students to work on the activities in cooperative manner
 - 4.4.2.4. Provide special support for those with difficulties or special need
 - 4.4.2.5. Provide feedback on the students various activities

4.5. Assessment Strategies and Techniques

- 4.5.1. Recording attendance
- 4.5.2. Recording group and individual participation
- 4.5.3. Recording the product students
- 4.5.4. Test at the end of the unit

5. Unit 5: Differentiated services for populations of special needs

Time allotted: 5 hours

5.1. Unit Objectives

At the end of this unit, the students will be able to involve effectively in the following activities:

- 5.1.1. Demonstrate understanding of the individualized services plan for population with special needs as a means of ensuring that these population receive services opportunities tailored to their needs
- 5.1.2. Describe the purpose of an individualized services plan
- 5.1.3. Identify the components of an individualized services plan
- 5.1.4. Develop strategies for providing remediation to population with special needs
- 5.1.5. Identify applicable technologies and software that will be useful for persons with various
- 5.1.6. Use the internet and other technology tools to enhance services and developments for populations of persons with various special needs
- 5.1.7. Evaluate technology applications for population with special needs
- 5.1.8. Explain the need for interdisciplinary individualized services plan teams, and describe the role and responsibility of each team member
- 5.1.9. Develop group intervention and describe its approach

5.2. Unit Contents and sub contents

- 5.2.1. Strategies of mediation to students with special needs
- 5.2.2. Content-specific resources for students
- 5.2.3. Instructional technology
- 5.2.4. Individualized service plan
- 5.2.5. Interdisciplinary individualized services plan teams
- 5.2.6. Curriculum enrichment
- 5.2.7. The role and responsibilities of a general education teacher in the
- 5.2.8. Implementation of the individualized services p
- 5.2.9. Planning group intervention

5.3. General Approach/Methods/Strategies

This section is flexible to involve the instructor's creativity in identifying selecting and adapting the instructional method to the context of the learner. Some general approaches are listed below. The instructor can select among this and add his own that he/she feels appropriate.

- 5.3.1. Interactive lectures
- 5.3.2. Group activities
- 5.3.3. Collaborative learning
- 5.3.4. Independent/self-learning
- 5.3.5. Institutional/school visits and practicum
- 5.3.6. Individual and group assignments and presentation
- 5.3.7. Seminars
- 5.3.8. Guest lecture
- 5.3.9. Others...

5.4. Detailed activities

- 5.4.1. Students' Activities
 - 5.4.1.1. Define what individualized services plan mean
 - 5.4.1.2. Discuss the purposes of individualized services plan
 - 5.4.1.3. List the components of individualized services plan
 - 5.4.1.4. Discuss how individualized services plan could be realized in the school
 - 5.4.1.5. Design and present single case individualized services plan
 - 5.4.1.6. Develop group intervention plan

5.4.2. Instructors Activities

- 5.4.2.1. Facilitate students learning by posing questions and guiding their activities
- 5.4.2.2. Provide activity guide for the students
- 5.4.2.3. Encourage all students to work on the activities in cooperative manner
- 5.4.2.4. Give assignments
- 5.4.2.5. Provide special support for those with who need it
- 5.4.2.6. Provide feedback on the students various activities
- 5.4.2.7. Design and present the sample for good individualized services plan and group educational intervention.

5.5. Assessment Strategies and Techniques

- 5.5.1. Recording attendance
- 5.5.2. Recording group and individual participation
- 5.5.3. Recording the product of the students
- 5.5.4. Test at the end of the unit
- 5.5.5. Assignment

6. Unit 6. Promoting Positive Behaviors Institution-wide

Time allotted: 5 hours

6.1. Unit objectives

- 6.1.1. Upon the accomplishing this unit, the Higher education students will be able to perform the following activities.
- 6.1.2. Implement strategies for managing an inclusive environment effectively
- 6.1.3. Describe behavior management modifications in an inclusive environment
- 6.1.4. Use strategies to increase desirable behaviors while decreasing undesirable behaviors
- 6.1.5. Develop effective techniques for responding to inappropriate behavior both in and out of the classroom
- 6.1.6. Build positive social relationships between all populations with special needs.
- 6.1.7. Demonstrate understanding of the importance of collaboration in an inclusive environment

6.2. Unit Contents and sub contents

- 6.2.1. Behavior management modifications
- 6.2.2. Classroom management for inclusive environment
- 6.2.3. Social relationships and collaboration in an inclusive environment

6.3. General Approach/Methods/Strategies

This section is flexible to involve the instructor's creativity in identifying selecting and adapting the instructional method to the context of the learner. Some general approaches are listed below. The instructor can select among this and add his own that he/she feels appropriate.

- 6.3.1. Interactive lectures
- 6.3.2. Collaborative learning
- 6.3.3. Independent/self-learning
- 6.3.4. Institutional/school visits and practicum
- 6.3.5. Individual and group assignments and presentation
- 6.3.6. Seminars
- 6.3.7. Expert consultancy
- 6.3.8. Discussion
- 6.3.9. Others....

6.4. Detailed activities

- 6.4.1. Students' Activities
 - 6.4.1.1. Discuss how to learn ways to create an inclusive community.
 - 6.4.1.2. Explore how to modify the current environment.
 - 6.4.1.3. Discuss techniques for relationship building in the inclusive environment.
 - 6.4.1.4. Instructors Activities
 - 6.4.1.5. Facilitate students learning by posing questions and guiding their activities
 - 6.4.1.6. Encourage interactive approach.

- 6.4.1.7. Provide activity guide for the students
- 6.4.1.8. Encourage all students to work on the activities in cooperative manner
- 6.4.1.9. Provide special support for those with who need it
- 6.4.1.10. Provide feedback on the students various activities

6.5. Assessment Strategies and Techniques

- 6.5.1. Recording attendance
- 6.5.2. Recording group and individual participation
- 6.5.3. Recording the product of students
- 6.5.4. Test at the end of the unit

7. Unit 7: Resources for the Inclusive environment

Time allotted: 5 hours

7.1. Unit objectives

- 7.1.1. At the end of this unit, the students will be able to accomplish the following tasks:
- 7.1.2. Apply constructivist techniques to create a conducive climate to diverse populations' success.
- 7.1.3. Find out existing resource that enhances success of inclusive environment.
- 7.1.4. Make adaptations based on the nature of the disabilities
- 7.1.5. Adapt communication for people with special needs education, such as, Braille, augmentative communication and Sign Language

7.2. Unit Contents and sub contents

- 7.2.1. Modification of environment and materials
- 7.2.2. Adapting learning and working process according to the needs
- 7.2.3. Identifying human material and other resources that help inclusive environmental activities.
- 7.2.4. Accessing adapted technologies
- 7.2.5. Accessing communication through various means such as Sign Language

7.3. General Approach/Methods/Strategies

This section is flexible to involve the instructor's creativity in identifying selecting and adapting the instructional method to the context of the learner. Some general approaches are listed below. The instructor can select among this and add his own that he/she feels appropriate.

- 7.3.1. Assignment
- 7.3.2. Interactive lectures
- 7.3.3. Visits and reporting
- 7.3.4. Collaborative learning
- 7.3.5. Independent/self-learning
- 7.3.6. Institutional/school visits and practicum
- 7.3.7. Individual and group assignments and presentation
- 7.3.8. Seminars
- 7.3.9. Expert consultancy
- 7.3.10. Others...

7.4. Detailed activities

- 7.4.1. Students' Activities
 - 7.4.1.1. Review some high school curriculum in group and exercise how to adapt for learners
 - 7.4.1.2. Adapting instructional techniques and present them in the classroom
- 7.4.2. Instructors Activities
 - 7.4.2.1. Facilitate students learning by posing questions and guiding their activities
 - 7.4.2.2. Provide activity guide for the students
 - 7.4.2.3. Encourage all students to work on the activities in cooperative manner
 - 7.4.2.4. Provide special support for those with difficulties and special needs
 - 7.4.2.5. Provide feedback on the students various activities

7.5. Assessment Strategies and Techniques

- 7.5.1. Recording attendance
- 7.5.2. Recording group and individual participation

7.5.3. Recording the product of students

7.5.4. Test at the end of the unit

8. Unit 8: Collaborative Partnerships with stakeholders

Time allotted: 5 hours

8.1. Unit objectives

At the end of this unit the students will be able to:

8.1.1. Identify key elements of successful collaboration

8.1.2. Describe the benefits and challenges of collaboration for various stockholders for the success of inclusive education

8.1.3. Explain the process of cop-planning, and develop strategies for effective co-planning and team learning and working

8.1.4. Identify characteristics of successful stockholders partnerships,

8.1.5. Design and plan strategies for community involvement

8.2. Unit Contents and sub contents

8.2.1. Collaboration to successfully move towards inclusion

8.2.2. Planning Inclusive development in all sectors

8.2.3. Implementing inclusive

8.2.4. Individualized support as per the law, policies and directive

8.2.5. Evaluation and monitoring

8.3. General Approach/Methods/Strategies

This section is flexible to involve the instructor's creativity in identifying selecting and adapting the instructional method to the context of the learner. Some general approaches are listed below. The instructor can select among this and add his own that he/she feels appropriate.

8.3.1. Interactive lectures

8.3.2. Collaborative learning

8.3.3. Independent/self-learning

8.3.4. Institutional/school visits and practicum

8.3.5. Individual and group assignments and presentation

8.3.6. Seminars

8.3.7. Individual and group presentations

8.3.8. Expert consultancy

8.3.9. Others...

8.4. Detailed activities

8.4.1. Students' Activities

8.4.1.1. Discuss and sort out key elements of successful team or co teaching.

8.4.1.2. Discuss the benefits and challenges of collaboration in the move towards inclusive education.

8.4.1.3. Identify and list resources for successful inclusion

8.4.2. Instructors Activities

8.4.2.1. Facilitate students learning by posing questions and guiding their activities

8.4.2.2. Provide activity guide for the students

8.4.2.3. Encourage all students to work on the activities in cooperative manner

8.4.2.4. Provide special support for those with who need it

8.4.2.5. Provide feedback on the students various activities

8.5. Assessment Strategies and Techniques

8.5.1. Recording attendance

8.5.2. Recording group and individual participation

8.5.3. Recording the product of students

8.5.4. Test at the end of the unit

9. Unit 9. Responsibilities

9.1. General Responsibilities of Instructors

Profile of teacher educator teaching this course must be the right professional in Special needs education. In the past, it was observed that non-special needs educators used to teach similar course. In order to produce quality teachers, this course should be offered only by teacher educator, MEd/or MA or PhD in special needs education. To meet the learning out come aforementioned and enhance teachers' quality, the special needs teacher educator will have the following major responsibilities.

- 9.1.1. Advise students on all the aspects of the course
- 9.1.2. Provide the students with the syllabus and other materials well ahead of the delivery of it
- 9.1.3. Conduct the interactive lectures as per the plan
- 9.1.4. Facilitate students' individual assignments, group assignments, field works, practicum, seminars, presentations, and collaborative learning
- 9.1.5. Periodically assess the students' work
- 9.1.6. Provide the students with timely feedbacks on their graded and ungraded academic works
- 9.1.7. Follow on students' progress and communicate to the students
- 9.1.8. Keep student records on the whole work of the students
- 9.1.9. Design and execute students' consultation program

9.2. General Responsibilities of Students

This course is designed for would teachers after completion of Bachelor degree in various fields. For successful completion of this course the teacher candidates would have the following responsibilities

- 9.2.1. Students are expected to actively and fully attend and participate all the in class and outclass learning activities. Missing a single class will cost students 2 points.
- 9.2.2. Carry out individual assignments, group assignments, field works, and practicum as per the details and deadlines
- 9.2.3. Students are expected to read given materials before class
- 9.2.4. Students are expected to read selected books and ten articles
- 9.2.5. Actively participate in the planning, organizing and conducting of all the seminars and presentations
- 9.2.6. Reflect on feedbacks and initiate actions on them
- 9.2.7. Passing the exams successfully

10. Unit 10. General Course Assessment and Evaluation Methods learning

Dear teacher candidates, for each contents you will complete getting started activities, read selected materials complete course works and group assignments. Assessment of the students would be a continuous process. The following scheme of evaluation would be used:

- 10.1. Individual assignments 20% (optional, depending on the class size and teacher educators teaching load)
- 10.2. Group assignment: 20%
- 10.3. Overall performance (punctuality, attendance, participation and collaboration): 10%. This is based on concrete records of punctuality, attendance and fruitful participation, that is measured by teacher educator
- 10.4. Written examination (could be more than one time): 50 to 70%

11. References

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3. Blackwell: Singapore
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5. Berit H. Johanson and Alemayehu Teklemariam (2006). Towards Special Needs Education as a University Discipline: An Important step on the way to Education for All. In When All Means All. Hakapaino Oy: Helsinki

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7. MOE (2007). School Improvement Program
8. MOE (2010). Special Needs Program strategies implementation guide.
9. MOE (Ministry of Education). (2006). Special Needs Education Program Strategy. Addis Ababa
10. Understanding and responding to children's need in inclusive classroom (2010). www.european-agency.org
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12. ዓለማየሁ ትክለማርያም (2011). አካቶ ትምህርት ለምን፣ምን፣ለነማን እንዴት፣ አዲስ አበባ፡-ፋርኢስት ኢትዮጵያ

Annexes

Instructional Materials and Equipment for Demonstration in the learning and teaching process

- LCD and/or Smart Board
- E - Chart
- Various magnifying lenses
- Slate and styles
- Perkins Braille writer
- White Cane
- Blind folder
- Tuning fork
- Audiometer
- Hearing aids (various type)
- Sign language books and videos
- Various instructional videos related this unit
- Braille atlases
- Molded plastic, dissected and dissected relief maps
- Relief globs
- Land form model
- Abacus
- Raised clock faces
- Geometric area and volume aids
- Write forms for matched planes and volumes
- Braille rulers
- Raised-line check books
- Signature guide
- Longhand-writing kit
- Script letter – sheets and boards
- Talking calculator
- Closed-circuit television
- Computer software for various students with special needs; for example Jawse for blind and sign language software for deaf
- And others additional

Course Title: Entrepreneurship and Business Development

Course Code: MGMT 102

Course Status: Common

Course is coordinated by: Management Department

ECTS/Credit: 5/3

Course Description: This interdisciplinary course is designed to introduce students the meaning and concept of entrepreneurship, creativity, innovation and their manageable processes that can be applied across careers and work settings. It focuses on building entrepreneurial attitude and behavior that will lead to creative solution within community and organizational environments. The Course topics include the history of entrepreneurship, the role of entrepreneurs in the globalized economy and the identification of entrepreneurial opportunities. The development of a business ideas, products and services, marketing and developing new ventures , the examination of feasibility studies and the social and ethical implications of entrepreneurship are incorporated. Besides, issues related to starting and financing a new venture are included. Finally, managing growth, transition and sustainability of the venture are considered. And forms of business organizations, legal and regulatory frameworks of governing the whole system are also encompassed in the course syllabus.

Pre-requisite: None

Course Objectives

Upon the completion of this course, students will be able to:

- Define entrepreneurship within the context of society
- Identify business opportunities
- Prepare business plan
- Distinguish forms of business ownership
- Comprehend intellectual property rights in business practices
- Define basic marketing concepts
- Formulate context-based marketing strategies
- Identify and evaluate sources of financing new ventures
- Manage business growth and transition
- Practice ethical business with all stakeholders

Expected Learning outcomes

As the intension of the course entrepreneurship is preparing University students for self employment, the curriculum is designed focusing on changing the behavior of students. It is designed in such a way that graduates will be more of “job creators than job seekers”. Much should be done on the behavioral aspects than the technical aspects of entrepreneurship.

Students are expected to develop the basic competencies that successful entrepreneurs should process

Schedule

Week	Topics and subtopics	Course Objectives and Competences to be Acquired
Week 1 & 2	1.1. Definition and philosophy of Entrepreneurship Vs Entrepreneurs 1.1.1. Historical origin of entrepreneurship 1.2. Type of Entrepreneurs 1.3. Role within the economy 1.4. Entrepreneurial Competence and Environment 1.4.1. Entrepreneurial Mindset 1.4.2. Demographic Factors 1.4.3. Entrepreneurial Environment 1.5. Entrepreneurship, creativity and innovation	After completing this chapter, students will be able to: - Define the term entrepreneurship and entrepreneur - Identify types of entrepreneur - Recognize the role of entrepreneurship in the economy - Analyze the entrepreneurial competencies - Differentiate the term creativity and innovation
	Activities	
	Teacher's Activity	Students' Activity
	- Introducing objectives to the students - Asking brain storming questions - Giving brief introduction to the sub topics - Giving class room and home based works - Checking, evaluating, and giving feedback to student's work - Summarizing the chapter	- Define the term entrepreneurship and entrepreneur - Discuss the role of entrepreneurship within the economy - Explain the entrepreneurial competences
	Delivery Methods	Interactive Lecture, group discussion and reflection
	Assessment	
Week 3, 4&5	Quiz	5%
	Business Planning 1. Opportunity Identification and Evaluation 2. Business Idea Development 2.1 Business Idea Identification 2.2. Sources of Business Ideas 2.3. Methods for generating Business Ideas 3. The Concept of Business Planning	After completing this chapter, students will be able to: - Identify opportunity in the environment - Evaluate the opportunities in the

	4. Business Feasibility 5. The Business plan 6. Developing a business plan	• environment • Generate business idea • Explain the concept of business planning • Identify components of business plan • Develop business plan
	Activities	
	Teacher's Activity	Students' Activity
	• Introducing objectives to the students • Asking brain storming questions • Giving brief introduction to the sub topics • Giving class room and home based works • Checking, evaluating, and giving feedback to student's work • Summarizing the chapter	• Discuss business opportunities in the environment • Generate business idea • Synthesize the components of business plan • Develop business plan
	Delivery Methods	Interactive Lecture, group discussion and reflection
	Assessment	
	• Group Project/ Business Plan Development • Presentation	15% 5%
Week 6 & 7	Business Formation 1. The Concept of Business development 2. Forms of Business (a short formation) 3. Definition and Importance of SMEs 4. Setting up small scale business 5. Roles of SMEs 6. Business failure and success factors. 7. Problems of small scale business in Ethiopia 8. Organizational structure and entrepreneurial team formation	After completing this chapter, students will be able to: • Explain the concept of business development • Identify the forms of business ownership • Define SMEs • Analyze the importance of SMEs • Set Up small scale business • List role of SMEs • Distinguish the failure and success factors of SMEs • Identify the problem of small scale business in Ethiopia

		Develop organizational culture
	Activities	
	Teacher's Activity	Students' Activity
	- Introducing objectives to the students - Asking brain storming questions - Giving brief introduction to the sub topics - Giving class room and home based works - Checking, evaluating, and giving feedback to student's work - Summarizing the chapter	- Discuss the concept of business development - Brainstorm the importance of SMEs - Discuss the failure and success factors of SMEs
	Delivery Methods	Interactive Lecture, group discussion and reflection
	Assessment	
	Individual assessment	10%
Week 8 & 9	Product or Services Development - The Concept of product or service technology - Product or service development Process - Legal and regulatory frameworks - Intellectual Property Protection/Product or service protection - Patent - Trademarks - Copyrighting	After completing this chapter, students will be able to: - Describe the concept of product and services - List product or service development process - Discuss the intellectual property protection
	Activities	
	Teacher's Activity	Students' Activity
	- Introducing objectives to the students - Asking brainstorming questions - Giving brief introduction to the sub topics - Giving class room and home based works - Checking, evaluating, and giving feedback to student's work - Summarizing the chapter	- Describe the concept of product and services - Discuss the failure and success factors of SMEs - Analyze Product or service process - Recognize legal and regulatory frameworks

		Describe intellectual property protection
	Delivery Methods	Interactive Lecture, group discussion and reflection
	Assessment	
Week11&12	Marketing - The Concept and philosophy marketing - Marketing Mix and Strategies - Marketing Information System - Marketing intelligence - Marketing research - Competitive analysis - Selling and Customer Service	After completing this chapter, students will be able to: - Define marketing - Identify Marketing mix and strategies - Analyze components of marketing information system - Explain competitive environment - Explain competitive environment
	Activities	
	Teacher's Activity	Students' Activity
	- Introducing objectives to the students - Asking brainstorming questions - Giving brief introduction to the sub topics - Giving class room and home based works - Checking, evaluating, and giving feedback to student's work - Summarizing the chapter	- Define marketing concept - Discuss marketing mix strategies - Differentiate components of marketing information system - Explain competitive environment
	Delivery Methods	Interactive Lecture, group discussion and reflection
	Assessment	
	Test	15%
Week 13&14	Financing the new venture - Overview of Business Financing - Source of financing - Equity financing - Debt financing - Trade credit - Lease financing - Traditional Financing (Equib/Edir, etc...) - Crowd Funding	After completing this chapter, students will be able to: - Know business financing - Identify the sources of finance - Understand with traditional

		financing techniques • Familiarize with crowd funding • Know Ethiopian micro finance system
	5. Micro finance in Ethiopia	
	Activities	
	Teacher's Activity	Students' Activity
	• Introducing objectives to the students • Asking brainstorming questions • Giving brief introduction to the sub topics • Giving class room and home based works • Checking, evaluating, and giving feedback to student's work • Summarizing the chapter	• Discuss business financing • Identify the sources of finance • Explore traditional financing techniques • Aware about crowd funding • Examine Ethiopian micro finance system
Week 15&16	Delivery Methods	Interactive Lecture, group discussion and reflection
	Assessment	
	Managing Growth and Transition 1. Managing business growth 2. New venture expansion strategies 3. Business Ethics and Social responsibility	After completing this chapter, students will be able to: • Know how to manage business growth • Understand business expansion strategies • Know & Internalize business ethics & social responsibilities
	Activity	
	Teacher's Activity	Students' Activity
	• Introducing objectives to the students • Asking brainstorming questions • Giving brief introduction to the sub topics • Giving class room and home based works • Checking, evaluating, and giving feedback to student's work • Summarizing the chapter	• Discuss business growth & its management • Identify new venture expansion strategies • Examine business ethics and social responsibility system

	Delivery Methods	Interactive Lecture, group discussion and reflection
	Assessment	
Course Teaching Learning Methods	Listen to a lecture and take notes on the lesson treated, take part in reading assignment, Group Discussion, Individual reflection, Debate among groups, Case study discussion and analysis, Assignment presentation	
Assessment Methods	• Quiz • Test • Individual Assignment • Business Plan Preparation • Presentation • Final Exam	5% 15% 10% 15% 5% 50%
Commitment of Instructors and Learners	Preparedness: You must come to class prepared by bringing with you the appropriate materials like handouts, worksheets and exercises given, text books and completed assignments. Complete the individual and group assignments and other activities on time. You must plan your own learning through reading various course related materials and chapters in books. You are expected to work much individually to meet the requirement of the course. You have to use your time for group work and home study effectively. Participation: Make active participation during discussions (you must participate in class). You are not participating if you are simply talking to a friend, doing homework, daydreaming, or not doing what the rest of the class is doing. If you are working in a group or with a partner, you must talk to your group members or partner and be a part of the group. Always be ready and willing to give constructive feedback to partners'/group members and to listen to their comments on your work. Medium: Use only English, which is the medium of instruction, especially in the class room	
Policy of University	Attendance: It is compulsory to come to class on time and every time. If learners are going to 85% during the term, they should not take this course Assignments: Learners must do their individual and group projects and submit on time. Any assignments will be submitted on and before the specified deadline. Tests/Quizzes: Learners have short quizzes and tests almost every unit. If they miss the class or, are late for class, they will miss the quiz or test and no makeup test or quizzes will be given for late	

	arrivals. Therefore, learners are expected to comply with the rules and the regulations of the University as well. Cheating: Learners must do their own work and not copy and get answers from someone else. When learners are in class, there are strictly forbidden from chewing gum, consuming any addictive substances, listening to recorders or CD players, or being involved in acts that interrupt the normal teaching-learning process. Besides, learners are required to switch off their cell phones before class and exam sessions. Learners who attempt to disobey these rules and regulations will be subject to disciplinary measures accordingly to the Senate Legislations of the University.	
Reference & Texts	Recommended Text book Hirsh RobertD. and D. and Peters MichaelP. “Entrepreneurship” Fifth Edition, Tata McGraw Hill Edition, 2002. Further References • Justin G. Longenecker and Carlos W. Moore, Small Business Management 12th edition, College Division South Western Publishing Co. Dallas, 2003 • Holt David H. “Entrepreneurship – New venture Creation “Eastern Economy Edition • DonaldF. Kutatko and RichardM.Hodgetts, “Entrepreneurship: A Cotemporary Approach” Fourth Edition. • Hailay Gebretinsae, Entrepreneurship and Small Business Management, 2nd Edition. approach “. Fourth Edition, the Dryden Press, 1998	

Course title: Introduction to Economics

Course Code= Econ 1012

I. Instructor Contact Instructor's Name _____ Mob.No _____

Information	Meeting day (s): _____		Meeting time: _____			
	Building location: _____ Floor No _____ Room No _____					
	Email: _____		Consultation Hours: _____			
II. Course Information	Lecture	Discussion with Peers & instructor	Group & Individual Assignment	Home Study/ Self-study/	Assessment	Total number of hours per semester
	48	30	15	40	25	158
Course is offered to	All first year Students					
Degree Program	B.A./BSc./BED/LLB/MD/DVM					
Course Title	Economics					
Course status	Common Course					
Course is coordinated	By Economics Department/School					
Course code	Econ-1103					
ECTS /Credit Hours	5/3					
	Academic Year: _____			Semester: I/II		

III. Course description

This course provides a general introduction to economics combining elements of micro and macro fundamentals. The first part of the course focuses on theories of consumers' and producers' behavior. Besides the course will also cover the neoclassical theory of product and/or service pricing for perfectly competitive market and provide brief introduction to monopoly, monopolistic competition, and oligopoly market structures. The second major part of the course will discuss elements of macroeconomics such as macroeconomic goals, national income account and its

measurement, macroeconomic problems and policy instruments. In offering the course, the real contexts Ethiopia will be thoroughly considered.

IV. General objective

The course will introduce students to the fundamental economic concepts and principles.

V. Specific objectives of the course

This course is aimed at:

- Describing the major economic agents and their respective roles and objectives,
- Introducing the concepts of demand and supply and their interactions.
- Introducing students to the neoclassical theory of consumer preferences and utility maximization approaches,
- Discuss short- run behaviour of production and the related cost structure,
- Introduce the different market structures and their real world applications, and
- Equipping students with macroeconomic goals, national income accounting, economic problems and policy instruments in light Ethiopian context.

VI. Expected learning outcomes

After completing introduction to economics, students will be able to:

- Describe the major economic agents and their corresponding roles and objectives;
- Understand the concepts of demand and supply and their interactions;
- Explain the objective functions of consumers and producers' behaviour in the short run, .
- Differentiate the various types of market structures,
- Understand the fundamental macroeconomic concepts, problems and policy instruments in the context of Ethiopia.

VII. Units and contents

Lecture	Topic & Sub Topics of the Course
Chapter One: Introduction	
<i>hour 6 S</i>	1. Introduction 1.1. DeFinition and Meaning of Economics 1.2. Rationale of Economics 1.3. Scope and methods of economic analysis 1.3.1. Micro and macro economics 1.3.2. Positive and normative economics 1.3.3. Inductive and deductive reasoning in economics. Scarcity, choice, opportunity cost and production 1.4. possibilities frontier 1.5. Basic economic questions,

- 1.6. Economic systems
- 1.7. Decision making units and the circular flow model

Chapter Two: Theory of Demand and Supply

8 hours

- 2. Theory of Demand and Supply
 - 2.1. Theory of Demand
 - 2.1.1. Demand function, demand schedule and demand curve
 - 2.1.2. Determinants of Demand
 - 2.1.3. Elasticity of Demand
 - 2.2. Theory of Supply
 - 2.2.1. Supply function, supply schedule and supply curve
 - 2.2.2. Determinants of supply
 - 2.2.3. Elasticity of supply
 - 2.3. Market equilibrium

Chapter Three: Theory of Consumers' Behaviour

9 hours

- 3. Theory of Consumers' Behaviour
 - 3.1. Consumer preferences
 - 3.2. The concept of utility
 - 3.3. Approaches of measuring Utility
 - 3.3.1. The cardinal utility approach
 - 3.3.1.1. Assumptions of cardinal utility theory
 - 3.3.1.2. Total and marginal utility
 - 3.3.1.3. Law of diminishing marginal utility (LDMU)
 - 3.3.1.4. Equilibrium of the consumer
 - 3.3.2. The ordinal utility approach
 - 3.3.2.1. Assumptions of ordinal utility approach
 - 3.3.2.2. Indifference curve and map
 - 3.3.2.3. Properties of indifference curves
 - 3.3.2.4. The marginal rate of substitution (MRS)
 - 3.3.2.5. The budget line or the price line
 - 3.3.2.6. Equilibrium of the consumer

Chapter Four : The Theory of Production and Costs

8 hours	4. Theory of Production and Costs 4.1. Theory of production in the short run 4.1.1. Definition of production 4.1.2 . Production function 4.1.3 . Total, average, marginal product 4.1.4. The law of variable proportions 4.1.5. Stages of production 4.2. Theory of costs in the short run
	4.2.1. Definition and types of costs 4.2.2. Total, average, marginal costs in the short run 4.2.3. Relationship between short-run production and cost curves

Chapter Five: Market structure

6 hours	5. Market structure 5.1. The concept of market in physical and digital space 5.2. Perfectly Competitive market 5.2.1 . Assumptions 5.2.2. Short run equilibrium of the firm 5.2.3 . Short run equilibrium of the industry 5.3. Monopoly market 5.3.1. Definition and Characteristics 5.3.2. Sources of Monopoly 5.4. Monopolistically competitive market 5.4.1. Definition and characteristics 5.5. Oligopolistic market 5.5.1. Definition and characteristics
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Chapter Six: Fundamentals of macroeconomics (with stylized facts from Ethiopia)

	6. Fundamentals of macroeconomics 6.1. Goals of Macroeconomics 6.2. The National Income Accounting
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11 hours	6.2.1. Approaches to measure national income (GDP)
	6.2.2. Other income accounts (GNP, NNP, NI, PI and DI)
	6.3. Nominal versus Real GDP
	6.4. The GDP deflator and the Consumer Price Index(CPI)
	6.5. The Business Cycle
	6.6. Macroeconomic Problems
	6.6.1
	. Unemployment
	6.6.2
	. Inflation
	6.6.3. Trade deficit and budget deficit
	6.7. Macroeconomic Policy Instruments
	6.7.1
	. Monetary policy
	6.7.2
	. Fiscal policy

VIII. Course teaching methodology

The course will involve deploying different teaching methods that attempt to make the teaching-learning process as effective as possible. For most part of the course, delivery method will be arranged as to make the process student-centered. There shall be full and active participation from students and they are strongly encouraged to ask questions, to reflect on brain-storming queries, and be involved actively and attentively in take-home assignments and peer discussions that appear during the semester both within and outside class-room sessions.

While there is no limit to the imagination and flexibility of the instructor, the course delivery techniques will generally involve the following items:

- Lecture
- Brain-storming sessions
- Group discussions
- Individual and group assignments

IX. Assessment Methodology

Students will be evaluated using different mechanisms and their weights as indicated in the table below.

Table1. General assessment profile

Assessment method	Weight
Assignment (individual and/or group)	20%
Tests/ quizzes	30%

Final Exam	50 %
Total	100%

X. Course policy

- **Attendance:** it is compulsory to come to class on time and every time. If students are going to miss **85% of the class** during the term, they shall not be allowed to sit the final exam,
- **Assignments:** students must do their individual and group assignments and submit on time. Assignments shall be submitted on or before the due date as specified by the instructor,
- **Tests/Quizzes:** instructors should give short quizzes and tests as appropriate.
- **Cheating:** students must do their own work and should not copy answers from someone else.
- **Acts and mannerisms:** When students are in class, they are strictly forbidden from chewing gum, consuming any addictive substances, listening to recorders or CD players, or being involved in acts that interrupt the normal teaching-learning process. Besides, students are required to switch off their cell phones before class and exam sessions. Students who attempt to disobey these rules and regulations will be subject to disciplinary measures accordingly to the Senate Legislations of the University.

XI. Commitments of instructor & students

- **Preparedness:** students must come to class prepared by bringing the appropriate materials like handouts, worksheets, exercises given, text books and assignments. Students must plan their own learning through reading various course related materials and chapters in books. They are expected to work a lot individually to meet the requirement of the course. They have to use their time for group work and home study effectively.
- **Participation:** students are expected make active participation during class sessions.
- **Coordination:** instructors shall play a pivotal role in facilitating the teaching and learning processes both in the class room and outside the class rooms.

XII. Readings and texts

1. A. Koutsoyiannis, *Modern Microeconomics*
2. D.N.Dwivedi, 1997, *Micro Economic Theory*, 3rd edition., Vikas Publishing
3. R.S. Pindyck & D.L. Rubinfeld, *Microeconomics*.
4. Hal R. Varian, *Intermediate Microeconomics: A Modern Approach*, 6th edition.
5. C.L.Cole, *Micro Economics: A Contemporary Approach*.
6. Ferguson & Gould's, 1989, *Microeconomic Theory*, 6th edition.
7. N. Gregory Mankiw, 2007, *Macroeconomics*, 4th edition.
8. P. Aghion and P. Howitt, 2009, *The Economics of Growth*, The MIT Press.
9. A. B. Abel and B.S. Bernanke, 2017, *Macroeconomics*, 9th edition, Pearson.

10. Ayele Kuris, *Introduction to Economics*, 2001.
11. Begg, Fisher & Dornbusch, 2005, *Macroeconomics*, 8th Ed.
12. Liberman, Marc and Hill, Robert E, 2005, *Introduction to Economics* 2nd Ed.
13. Richard E. Carmichael, 2006, *Economics for Everyone: An introduction to Economics*.

Course title: Organic Chemistry

Course Code = Chem 1012

CREDIT HRS.: 3

Course content:

1. Functional groups in organic chemistry
 - 1.1. Classes of organic compounds
 - 1.2. Alkanes
 - 1.3. Alkenes or Olefines
 - 1.4. Alkynes
 - 1.5. Aromatic compounds
 - 1.6. Carbonyl groups
 - 1.7. Hydroxyl groups
 - 1.8. Carboxyl groups
 - 1.9. Amino groups
 - 1.10. Ether groups
2. Stereochemistry
 - 2.1. Introduction
 - 2.2. Optical isomerism
 - 2.3. Concept of chirality's
 - 2.4. Optical activity
 - 2.5. Representation of enantiomers
 - 2.6. Properties of enantiomers
 - 2.7. Symmetry and dissymmetry
 - 2.8. Meso compounds
 - 2.9. Diastereoisomers
 - 2.10. Asymmetric synthesis
 - 2.11. Racemisation
 - 2.12. Resolution of Racemic mixture
 - 2.13. Configuration
 - 2.14. D and L notation for stereo isomers
 - 2.15. R and S sequence of specification of configuration
 - 2.16. Fischer projection
 - 2.17. R and S configuration from planar formula
 - 2.18. Geometrical isomerism (restricted rotation)
 - 2.19. Syn---Anti isomerism
 - 2.20. The E and Z nomenclature of geometrical isomers
 - 2.21. Cis and trans isomerism in cyclic compounds

3. Structure elucidation and molecular spectroscopy
 - 3.1. Introduction
 - 3.2. Combustion analysis
 - 3.3. Molecular weight and molecular formula
 - 3.4. The structure of methane
 - 3.5. Tetrahedral structure
 - 3.6. Citronellal-sites of unsaturated
 - 3.7. some simple reactions of citronellal
 - 3.8. Ozonization
 - 3.9. Ozonization of citronellal
 - 3.10. Molecular spectroscopy
 - 3.11. UV spectroscopy
 - 3.12. IR spectroscopy
4. Substitution reaction
 - 4.1. Introduction
 - 4.2. SN2 mechanism
 - 4.3. SN1 mechanism
 - 4.4. Miscellaneous comments
 - 4.5. Alcohol
 - 4.6. Ethers
 - 4.7. Carboxylic acids
 - 4.8. Alkanes, alkenes, and Alkynes
 - 4.9. Amines
 - 4.10. Epoxide ring opening
 - 4.11. Malonic ester and acetoacetic ester synthesis
5. Elimination reaction
 - 5.1. Introduction
 - 5.2. Mechanism
 - 5.3. E1 versus E2
 - 5.4. Elimination versus substitution
 - 5.5. Saytzeff and Hoffmann rules
 - 5.6. Dehydration of alcohols
 - 5.7. Dehydrohalogenation of alkyl halides
 - 5.8. Elimination of vicinal dihalides
 - 5.9. Hoffmann elimination
 - 5.10. Acetate pyrolysis
 - 5.11. Cope reaction
6. Addition reaction
 - 6.1. Introduction
 - 6.2. Mechanism
 - 6.3. Reactivity
 - 6.4. Markovnikov rule
 - 6.5. Michael addition
 - 6.6. Radical reaction
 - 6.7. Addition of halogen
 - 6.8. Addition of hydrogen halides
 - 6.9. Addition of hypohalous acids
 - 6.10. Hydration of alkenes

7. Aromatic Chemistry
 - 7.1. Introduction
 - 7.2. Aromaticity
 - 7.3. Aromatic substitution
 - 7.4. Mechanism
 - 7.5. Directing effects
 - 7.6. Halogenations
 - 7.7. Sulphonation
 - 7.8. Nitration
 - 7.9. Friedal-Crafts alkynylation
 - 7.10. Frieal-Crafts acylation
 - 7.11. Diazotization of amines
 - 7.12. Reaction of aromatic side chains
8. Carbonyl reactions
 - 8.1. Carbonyl addition-Elimination
 - 8.2. Cyanohydrins
 - 8.3. Witting reaction
 - 8.4. Acetal and ketal reaction
 - 8.5. Halogenation
 - 8.6. Alkylation
 - 8.7. Aldol condensation
 - 8.8. Claisen-Schmidts condensation
 - 8.9. Mannich condensation
 - 8.10. Perikin condensation
 - 8.11. Claisen condensation
9. Rearrangement reactions
 - 9.1. Introduction
 - 9.2. Rearrangement to an electron deficient center
 - 9.3. Pinacol-Pinacolone rearrangement
 - 9.4. Rearrangement to an electron deficient oxygen. The Baeyrs villager oxidation
 - 9.5. Rearrangement to an electron-deficient nitrogen. Beckmann rearrangement
 - 9.6. Hoffmann rearrangement
10. Biological molecules
 - 10.1. Glucose-an introduction to carbohydrate chemistry
 - 10.2. Disaccharides and polysaccharides
 - 10.3. Amino acids, peptides and proteins
 - 10.4. The structure and properties of a-amino acids
 - 10.5. Analysis of a-amino acids
 - 10.6. Synthesis of A-amino acids
 - 10.7. Peptides and proteins
 - 10.8. Peptide synthesis
 - 10.9. Natural products
 - 10.10. Terpenoids and steroids
 - 10.11. Bio synthesis
11. Techniques in experimental organic chemistry:
 - Re crystallization
 - Melting point determination

Simple and fractional distillation
Steam distillation
Functional group identification
Organic compounds in three dimensions using models
Preparation of aspirin, soap, esters, qualitative organic analysis and chromatography

Pre-Clerkship One

MODULE TITLE: INTRODUCTION TO MEDICINE

MODULE CODE: INMD-2011

NOMINAL DURATION: 20 WEEKS

Module Description: This introductory module is intended to provide students with an overview of the medical profession including ethical principles and laws, the medical curriculum, introductory concepts and principles of biomedical sciences and the art of medical interviewing, physical examination and first aid. Students will also develop relevant competencies in health promotion and disease prevention, communication, ethics & professionalism, management of information, critical thinking, practice-based improvement and research and inter-professional collaborative practice. This module is given during the first semester of the the second year.

Module Design

This module is designed by integrating overview of the medical profession. Building on in-depth understanding of the principle of medical profession and biomedical sciences as a foundation, this module aims to provide the analytical and cognitive skills necessary for a successful transition from basic medical profession ethical principles and laws, medical curriculum concepts and principles of biomedical sciences to the care of patients.

The professional competency development component will support the student in acquiring a practical understanding of how to approach patients and develop the basic skill of examining patients. Social and Population Health (SPH) module will run in parallel and will be made to correlate with the content being taught.

Pre-requisite

Successful completion of pre-med courses

Module Objectives

By the end of this module, students will be able to apply knowledge of the ethical, legal methods and values underpinning the medical profession, basic Biological principles, and develop foundational skills in patient-doctor interaction. Students will also be able to describe essential components of the curriculum and take active responsibility for their own learning.

Supporting Objectives

As a Medical Expert

- Describe the basic organization and functions of cells and intracellular organelles
- Describe the benefit of homeostasis for the survival of organisms
- Describe the structure, classification and functions of bio molecules
- Describe the classification, mechanism of action and kinetics of enzymes
- Describe the cell cycle, and the molecular processes necessary for cell division and cancer biomarker

- Describe the different patterns of inheritance
- Describe types of tissue in the body
- Describe the nature and organization of the major systems of the human body
- Describe cellular responses to injury and cellular adaptations.
- Describe the basic principles of carcinogenesis, and the functional properties of tumors.
- Describe the general overview on principle and classification of micro-organisms
- Describe overview of basic immunology and immune response against microbial infections.
- Describe basic principles of pharmacology used for the preventive, diagnosis and treatment purpose
- Apply basic general principles of pharmacokinetics and pharmaco-dynamics of drugs used for the treatment of infectious disease, toxic substances and anti-neoplastic agents in a simulated setting
- Describe intrauterine development of the human embryo including the common congenital anomalies from conception to birth
- Describe components of medical history
- Demonstrate medical interviewing skills
- Describe components of complete physical examination
- Demonstrate physical examination skills
- Demonstrate the ability to give first aid
- Recognize the role of a physician in health promotion and disease prevention

Communicator

- Demonstrate clear, sensitive and effective communication skills in interacting with a simulated patient by listening, sharing and responding
- Communicate effectively with healthy clients and client's family

Collaborator

- Work in team with her/his class mate
- Provide support to her/his classmate
- Shares his/her knowledge

Leader

- Takes initiative in his/her learning
- Takes initiative to take responsibilities without being asked
- Demonstrate positive behavior in leading a group
- Understand self and identifying their own learning needs and plan learning activities to address their need

Scholar

- Describe the core competencies, teaching/learning strategies and assessment methods in the medical curriculum
- Attempts to critique and challenge situations
- Demonstrate asking critical questions
- Participate in research undertaking

Professional

- Demonstrate independent learning
- Describe moral and ethical principles and legal responsibilities underlying the medical profession
- Apply ethical principles in daily practice

- Recognize their obligation to promote, protect, and enhance essential elements of the medical profession for the benefit of patients, the profession and society at large
- Identify professional values which include excellence, altruism, responsibility, compassion, empathy, accountability, honesty, integrity, and a commitment to scientific methods
- Participate in providing care in compassionate and respectful manner

Exhibit the ability to effectively plan and efficiently manage one's own time Duration: 16 weeks

Teaching and learning methods

- Interactive lecture
 - Assignment
 - Discussion and debate
 - Case studies
 - Role play
 - Clinical simulation
 - Demonstration
- Eg. Video show
- Observation in clinical setting
 - Community visit
 - Personal research and reflection exercise/ Portfolio
 - Whole group session
 - Problem-Based learning

Educational Resources

- Basic sciences lab
- Clinical skills lab
- Health Facility
- Videos of patient examination
- Anatomical charts and mannequins
- Histological and pathological slides and specimens
- Computer-based simulations

References

- Tortora G.J and B. Derrickson. Principles of Anatomy and physiology. 12th edition. 2008
- Drake RL., Vogl W., and Mitchell AW. Gray's anatomy for students. 2007
- Keith L. Moore, T.V.N. Persaud and Mark G Turchia. The developing human. Clinically oriented embryology. 9th edition. 2013
- Junqueira's basic histology: text and atlas. 13th edition. 2013
- Guyton and Hall. Textbook of medical physiology. 13th edition. 2013.
- Vanders. Human physiology. 13th edition. 2014
- Ashis Banerjee. Clinical physiology. An examination primer. 2005.
- Lehninger A.L., principles of biochemistry, 5th edition 2013
- Voet. D and voet .J . G. Biochemistry , 5th edition, 2012
- Vasudevan DM, Text book of biochemistry for medical students, Jaypee, 6th edition 2011
- John Baynes and Marek Dominiczak. Medical biochemistry. 4th edition. 2014
- Alan Gaw. Clinical biochemistry 5th edition. 2013

- Eric Arthur Newsholme and Tony R. Leech. Functional biochemistry in health and disease. 2010.
- Richard N Mitchell, Ramzi S. Cotran. Robbins basic pathology. 8th edition. 2007.
- David A Levison, Robin Reid, Alistair D Burt, David J Harrison and Stewart Fleming. **Muir's Textbook of Pathology, Fourteenth Edition**
- **Levinson W. Review of Medical Microbiology and immunology.**12th edition.2010
- Jawetz, Melnick, & Adelberg's. Medical Microbiology, 24th Ed, 2007
- Murray P., Rosenthal S., Foller M, Medical Microbiology, 8th edition, 2010
- Kuby, Goldsby. Immunology. 5thed, 2007.
- Bertram G. Katzung. Basic and clinical pharmacology. 12th edition. 2011
- Goodman and Gilman's. The pharmacological basis of therapeutics. 12th edition. 2011
- Lynn S. Bickley, Peter G. Szilagy. Bates' guide to physical examination and history taking. — 10th ed. 2009
- Michael Swash and Michael Glynn. Hutchison's clinical methods. An integrated approach to clinical practice. 22nd edition. 2007
- People to People. The Manual of Ethiopian Medical History. 2013
- FMHACA. Scope of Practice for Medical Practitioners in Ethiopia. 2013
- EMA. Medical ethics for doctors in Ethiopia.Ethiopian Medical Association. 2010
- Gwen Van Servellen. Communication skill for the Health Care Professional: Concepts, Practice. Jones and Bartlett Publisher. 2009
- Competency-based professionalism in anesthesiology: Continuing Professional Development (DOI 10.1007/s12630-012-9747-z)
- Richard E. Ashcroft. Principles of health care ethics, 2nd Edn, 2007
- Raanan Gillon. Principles of health care ethics, 1994
- EFMOH compassionate and respectful care training manual for health work force, 2017
- EFMHCA Regulation 299/213 (<http://www.fmhaca.gov.et/>)

Learners Guide

- Handout
- Books
- Text books
- E-books

Learning Facilities / Infrastructure

- Class rooms
- Skill lab
- Dissection Room
- ICT center
- Clinical set up
- Library

Portable and stationary plants, machines and equipment

- Flip charts
- Video cassette
- CD-ROM
- Computers
- DVD Player

- LCD projector Overhead Projector
- White board
- WMA. Medical ethics manual. 2nd edition. World Medical Association. 2009

Methods of assessment

Progressive

- Written progress /quiz/ - 25%
- Clinical skill lab - 5%
- Basic science lab - 15%
- Hospital visit - 5%
 - Professionalism mini-evaluation exercise (P-MEX)- Direct observation of student's professional behaviors
 - Review of students' reflective portfolio

End module

- Written - 30%
- OSCE/OSPE - 20%

Module Schedule

¹ Students are required to maintain in their portfolio learning experiences, learning outcomes achieved, evidence of learning and reflections longitudinally and their mentors would provide feedback occasionally. Clinical instructors should also ask students to submit reflective portfolio twice during the attachment for formative assessment and feedback.

²⁴ACGME and ABMS. Toolbox of assessment methods. A product of the joint initiative of the ACGME Outcome Project of the Accreditation Council for Graduate Medical Education (ACGME), and the American Board of Medical Specialties (ABMS). Version 1.1, September 2000.

²⁵John Norcini. The death of the long case? BMJ 2002;324:408–9

²⁶ACGME and ABMS. Toolbox of assessment methods. A product of the joint initiative of the ACGME Outcome Project of the Accreditation Council for Graduate Medical Education (ACGME), and the American Board of Medical Specialties (ABMS). Version 1.1, September 2000

²⁷ M. FRIEDMAN BEN DAVID, M.H. DAVIS, R.M. HARDEN, P.W. HOWIE, J. KER & M.J. PIPPARD. AMEE Medical Education Guide No. 24: Portfolios as a method of student assessment. Medical Teacher, Vol. 23, No. 6, 2001

^{28,29}ACGME and ABMS. Toolbox of assessment methods. A product of the joint initiative of the ACGME Outcome Project of the Accreditation Council for Graduate Medical Education (ACGME), and the American Board of Medical Specialties (ABMS). Version 1.1, September 2000.

²⁹

³⁰Division of Educational Development & Research, Teacher & Educational Development, University of New Mexico School of Medicine. Faculty and student guide to PBL tutorials in phase I curriculum of the University of New Mexico School of Medicine. 2002.

Date	Learning Activity	Required Reading/ Assignment
Week 1	Interactive lecture Day One Introduction to the curriculum (4 and half hours) ▪ Structure and design (30 min) ▪ Education strategies (30 min) ▪ Core competencies (1 hour) ▪ Teaching and learning methods (1 hours) ▪ Assessment methods (1 hours) ▪ Introduction to the module (30 min) Day two ▪ History of medicine (4 hours) • Global perspectives • Ethiopian medical history Day Three Introduction to Anatomy (4 hours) ▪ Definition of Anatomy and historical background ▪ Subdivisions of Anatomy ▪ Approaches to studying Anatomy ▪ Levels of Structural Complexity of the body ▪ The Language of Anatomy ▪ Body Planes and Sections ▪ Body Cavities ▪ Regions of the body	'Reflective portfolio
	Basic science lab • Tour of the different basic science labs and introduction to staff	
	Clinical skills lab • Tour of the clinical skills lab and introduction to staff	
	Hospital visit • Tour of the hospital and introduction to staff	
	Primary health care or community visit • Tour of primary healthcare facilities and introduction to staff • Introduction to the community and assignment of individual homes	

	Whole group session Faculty facilitated discussion and reflection on the week's learning activities	
Week 2	Interactive lecture Day One Introduction to Physiology (4 hours) ○ Definition and approach to study physiology ○ Functional organization of the human body and control of the internal environment	Reflective portfolio
	Day Two ○ Cell physiology (1hr) ○ Body fluid compartment and composition (1 hr) ○ Plasma membrane and membrane transport mechanism (2hr)	
	Day Three Introduction to Biochemistry (4hr) • Cells • Cellular organelles and their biochemical roles • Membranous organelles and their biochemical roles • Cell signaling (1hr) • Non-membranous organelles and their biochemical roles	
	Basic science lab • Anatomical charts and atlas • Computer-simulations of the cell structure and function • Computer simulations of transport of substances through the cell membrane • Computer-simulations of homeostasis	
	Clinical skills lab • Video show of doctor-patient encounter and reflections on the communication skills	
	Hospital visit • Observation of patient-doctor encounter at OPDs	
	Primary health care or community visit • Observation of patient-doctor encounter at OPDs • Home visits trying to understand societal views about physicians and physician-patient communication	
	Whole group session • Faculty facilitated discussion and reflection on the week's learning activities	

Week 3	Interactive lecture Day one • Introduction to Histology (4hr) ○ Overview of histology ○ Epithelial Tissue ○ Connective Tissue ○ Epithelial Membranes Day Two • Anatomy of the integumentary system (2 hours) • Introduction to medical ethics (2hours) ○ What is medical ethics? ○ Why study medical ethics? ○ Medical ethics, medical professionalism, human rights and law	
	Day Three • Overview of the anatomy, histology and embryology of the skeletal system (2hrs) ▪ Functions of bone ▪ Gross structure of bone ▪ Histology of bone ▪ Growth and development of bones ▪ Cartilages ▪ Introduction to the skeleton ▪ Introduction to joints ▪ Terms of movement • Overview of the anatomy and histology of the muscular system (2hours) ▪ Muscle Tissue • Skeletal muscle • Cardiac muscle • Smooth muscle ▪ Introduction to skeletal muscles of the Body	Reflective portfolio
	Basic science lab • Bone types and Skeletal • Histological microscopic slides of epithelial and connective tissue, bone, muscle and nervous tissues • Histology slides and atlas • Histological microscopic slides of Skin • Computer simulations of the structures and functions of the integumentary system • Computer-simulations of the structures and functions of the musculoskeletal system	
	Clinical skills lab	

	• Simulated practice: Medical interviewing (art and ethics) Hospital visit • Observation of patient-doctor encounter at OPDs • Talking to medical doctors Primary health care or community visit • Observation of patient-doctor encounter at OPDs • Talking to medical doctors • Home visits trying to practice medical interviewing Whole group session • Faculty facilitated discussion and reflection on the week's learning activities • Establishing Rapport & Empathy (2 hrs of introductory ppt and role plays)	
Week 4	Interactive lecture Day one ▪ Overview of the anatomy and histology of the Circulatory System (2hours) ▪ Introduction ▪ Functions of the circulatory system ▪ Structure of Blood Vessels(2hours) Day two Introduction to respiratory system (2 hours) Introduction to renal system (2 hours) Day three Introduction to reproductive system (2 hours) Introduction to digestive system (2 hours)	Reflective portfolio
	Basic science lab • Anatomical charts and atlas • Histological microscopic slides of blood vessels • Computer-simulations of the structures and functions of the GI system • Computer simulations of the kidney and urinary syst • Computer-simulations cell reproduction • Computer simulations of structure and functions of the respiratory syste	
	Clinical skills lab • Simulated practice: Medical interviewing (art and ethics)	
	Hospital visit • Observation of patient-doctor encounter at OPDs • Talking to medical doctors Primary health care or community visit • Observation of patient-doctor encounter at OPDs • Talking to medical doctors	

	• Home visits trying to practice medical interviewing	
	Whole group session • Faculty facilitated discussion and reflection on the week's learning activities • Basic Communication skills & Non Verbal Communication(2 hrs of introductory ppt and role plays)	
Week 5	Interactive lecture Day one • Overview of the anatomy and histology of the Nervous system (4 hours) ▪ Functions of the nervous system ▪ Organization of the nervous system ▪ Nervous tissue Day Two Introduction to Physiology of nervous system(4 hours) • Functional organization of the nervous system • Nerve tissues, Membrane Potentials and synaptic transmission • Autonomic nervous system Day three ▪ Introduction to Endocrine system	
	Basic science lab • Anatomical charts and atlas • Histology slides and atlas • Computer simulations of the endocrine system • Computer simulations of the nervous system	
	Clinical skills lab • Simulated practice: Medical interviewing	
	Hospital visit • Observation of patient-doctor encounter at OPDs • Talking to medical doctors	
	Primary health care or community visit • Observation of patient-doctor encounter at OPDs • Talking to medical doctors • Home visits trying to practice medical interviewing	
	Whole group session • Faculty facilitated discussion and reflection on the week's learning activities • Ethical practice in healthcare (1hrs of case studies)	

Week 6	Interactive lecture Day one • Genetic control of protein synthesis, cell function and cell reproduction (2hours) • Principal features of medical ethics (2hr) • What is special about medicine? • What is special about medical ethics? • Who decides what is ethical? • Do medical ethics change? • Do medical ethics differ from one country to the other? • The role of the World Medical Association (WMA) and the Ethiopian Medical Association (EMA) • How does WMA and EMA decide what is ethical? • How do individuals decide what is ethical? Day two ▪ Muscle physiology (4 hours) ○ Functional structure of the skeletal muscle ○ Contraction and excitation of skeletal muscle (neuromuscular transmission, excitation contraction coupling and contraction of skeletal muscle) ▪ Muscle mechanics Day three ▪ General Embryology (4hours) ▪ Introduction ▪ Gametogenesis ▪ First Week of Development ▪ Second Week of Development ▪ Third Week of Development Basic science lab • Anatomical charts and atlas • Histology slides and atlas • Embryology charts and atlas • Computer simulations of the membrane potentials and action potentials • Computer simulations of contraction and excitation of skeletal and smooth muscle	
	Clinical skills lab • Simulated practice: Physical examination (skill and professional behavior) Hospital visit • Observation of patient-doctor encounter at OPDs • Talking to medical doctors Primary health care or community visit • Observation of patient-doctor encounter at OPDs • Talking to medical doctors • Home visits trying to practice medical interviewing	

	Whole group session - Faculty facilitated discussion and reflection on the week's learning activities Introduction to professionalism & Life-long learning (1 hr of introductory ppt, action plan, small group discussion)	
Week 7	Interactive lecture Day one General Embryology continued. (4 hours) - Third to Eighth Week - Third month to birth - Multiple pregnancy - Fetal membranes and placenta Day two General Embryology continued. (4 hours) - Body cavities, mesenteries, and diaphragm - Development of pharyngeal apparatus - Birth defects Day three - Physicians and patients (2 hours) - What is special about physician-patient relationship? - Respect and equal treatment - Communication and consent - Decision making for incompetent patients - Confidentiality - Beginning of life issues - End of life issues Basic science lab - Embryology charts and atlas - Computer simulation of embryology	
	Clinical skills lab - Simulated practice: Physical examination (skill and professional behavior) Hospital visit - Observation of patient-doctor encounter at OPDs - Assisting with medical interviewing Primary health care or community visit - Observation of patient-doctor encounter at OPDs - Assisting with medical interviewing - Home visits trying to practice physical examination Whole group session - Faculty facilitated discussion and reflection on the week's learning activities	
Week 8	Interactive lecture Day one	

	- Chemical composition in the cell (4hr) - Carbohydrates: Structure, function, classification and biomedical importance - Lipid: Structure, function, classification and biomedical importance importance Day two - Chemical composition in the cell..... (4 hr) - Proteins: Structure, function, classification and biomedical - Nucleic acids: structure, classification, function Day three Basic Enzymology (4hrs) - Nomenclature - Classification - Properties and kinetics - Mechanism of action - Biomedical importance	
	Basic science lab - Computer-simulations of genetic control of protein synthesis - Qualitative and quantitative analysis of carbohydrates, lipids, proteins and aminoacids. - Computer simulations of enzyme actions and regulations - Computer simulations of food digestion and absorption - Computer simulations of nutrient metabolism	
	Clinical skills lab - Simulated practice: Physical examination (Skills and professional behavior) Hospital visit - Observation of patient-doctor encounter at OPDs - Assisting with medical interviewing Primary health care or community visit - Observation of patient-doctor encounter at OPDs - Assisting with medical interviewing - Home visits trying to practice medical interviewing Whole group session - Faculty facilitated discussion and reflection on the week's learning activities	
Week 9	Interactive lecture Day one Pathology (4 hrs) - Introduction to pathology - Cellular injury, death and adaptation Day Two	

	Acute and chronic inflammation (4 hours) Day three Tissue repair: regeneration, healing and fibrosis (4 hours)	
	Basic science lab • Pathology slides and atlas • Computer-simulations of cell injury, death and adaptation • Computer simulations of acute and chronic inflammation • Computer simulations of tissue repair	
	Clinical skills lab • Simulated practice: Physical examination (Skills and professional behavior)	
	Hospital visit • Observation of patient-doctor encounter at OPDs • Assisting with interviewing patients	
	Primary health care or community visit • Observation of patient-doctor encounter at OPDs • Assisting with interviewing patients • Home visits trying to practice first aid skills	
	Whole group session Faculty facilitated discussion and reflection on the week's learning activities	
Week 10	Interactive lecture Day one Hemodynamic disorders, thrombosis and shock (4 hours) Day two Introduction to Genetic and pediatrics diseases (2 hours) Physicians and society(2hrs) ▪ What is special about physician-society relationship? ▪ Dual loyalty ▪ Resource allocation ▪ Public health ▪ Global health Day three • Cell cycle and its regulation (2hrs) • Phases of the cell cycle: Inter-phase and mitosis - Regulation of cell cycle • Free radicals and their roles in diseases (2 hours) ▪ Basic science lab • Computers simulations of hemodynamic disorders	

	• Computer simulation of cell cycle and regulation	
	Clinical skills lab • Simulated practice: Physical examination (Skills and professional behavior)	
	Hospital visit • Observation of patient-doctor encounter at OPDs • Assisting with interviewing patients	
	Primary health care or community visit • Observation of patient-doctor encounter at OPDs • Assisting with interviewing patients • Home visits trying to practice first aid skills	
	Whole group session • Faculty facilitated discussion and reflection on the week's learning activities • Legal aspects in health care practice (1 hr of introductory lecture, small discussion)	
Week 11	Interactive lecture Day one Introduction to Basic Molecular Genetics (2hrs) ▪ DNA replication and repair ▪ Transcription ▪ Translation ▪ Introduction to recombinant DNA technology and gene therapy Physicians and colleagues(2hours) ▪ Challenges to medical authority ▪ Relationship with physician colleagues, teachers and students ▪ Reporting unsafe or unethical practices ▪ Relationships with other health professionals ▪ Cooperation ▪ Conflict resolution Day two • Neoplasia (4hr) Day three • Neoplasia (4hr) continued	
	Basic science lab • Computer simulation of molecular genetics • Computer simulations of carcinogenesis • Computer simulations of mutation •	
	Clinical skills lab • Simulated practice: Physical examination (Skills and professional behavior)	

	Hospital visit • Observation of patient-doctor encounter at OPDs • Assisting with interviewing patients	
	Primary health care or community visit • Observation of patient-doctor encounter at OPDs • Assisting with interviewing patients • Home visits to traditional medicine practitioners	
	Whole group session • Faculty facilitated discussion and reflection on the week's learning activities • Compassionate & Respectful care (2 hrs of introductory lecture, case studies, roleplay)	
Week 12	Day one • Neoplasia (4hr) continued Day two • Introduction to microbiology (definition, biological principles and classification of microorganisms) • The microbial cell structure(2hrs) Day three • Introduction to microbial genetics (2hrs) • Classification of bacteria (1hrs) The growth, survival and death of microorganisms(1hrs)	
	Basic science lab ▪ Microbial charts and atlas ▪ Basic microbiology equipments and reagents ▪ Demonstration of basic staining techniques ▪ Computer simulation of carcinogenesis	
	Clinical skills lab • Simulated practice: Physical examination (Skills and professional behavior)	
	Hospital visit • Observation of patient-doctor encounter at OPDs • Assisting with interviewing patients	
Week 13	Day one ▪ Cultivation of microorganisms(1hr) ▪ Normal microbial flora(1hr) ▪ Medical research (2hrs) ➤ Importance of medical research ➤ Ethical requirements ➤ Ethics review committee approval ➤ Scientific merit ➤ Social value ➤ Risks and benefits ➤ Informed consent, confidentiality ➤ Conflict of roles ➤ Honest reporting	

	➤ Whistle blowing ➤ Unresolved issues Day two Overview of virology (4hrs) Day three Overview mycology(4hrs)	
	Basic Science Lab ▪ Preparing cultivation media for microorganisms ▪ Anti microbial susceptibility testing ▪ Computer simulations of properties of viruses ▪ Computer simulations of properties of fungi	
	Clinical skills lab • Simulated practice: Physical examination (Skills and professional behavior)	
	Hospital visit • Observation of patient-doctor encounter at OPDs • Assisting with interviewing patients	
	Primary health care or community visit • Observation of patient-doctor encounter at OPDs • Assisting with interviewing patients • Home visits to traditional medicine practitioners	
	Whole group session • Faculty facilitated discussion and reflection on the week's learning activities • Leadership: Self-assessment and Self-management (1 hr of introductory ppt, check list, small group discussion)	
Week 14	Interactive lecture Day one ▪ Principles of infection prevention (2 hrs) ▪ Principle of disinfection and Sterilization (2 hrs) Day two ▪ Overview of Parasitology (4hrs) Day Three ▪ Overview of Immune system (4hours)	
	Basic science lab ▪ Ova-parasite identification ▪ Haemoparasite and flagellates identification ▪ Computer simulations of properties of parasites ▪ Computer-simulations of microorganism-host interactions and adaptive immune response	
	Clinical skills lab • Simulated practice: First aid skills	
	Hospital visit	

	• Observation of patient-doctor encounter at OPDs • Assisting with interviewing patients	
	Primary health care or community visit • Observation of patient-doctor encounter at OPDs • Assisting with interviewing patients • Home visits to traditional medicine practitioners	
	Whole group session • Faculty facilitated discussion and reflection on the week's learning activities • Leadership: Self-assessment and Self-management (1 hr of introductory ppt, check list, small group discussion)	
Week 15	Day one ▪ Overview of Immune system (4hours) continued Day two ▪ Diseases of the immune system, autoimmune diseases and immunodeficiency states (4hrs) Day three ▪ Principles of diagnostic medical microbiology(2hours) ▪ Introduction to pharmacology (2 hours) ▪ Definition and scope of pharmacology ▪ Source, nomenclature and dosage forms of drugs Basic science lab • Computer-simulations of drug absorption, distribution and metabolism • Clinical skills lab • Simulated practice: Physical examination (Skills and professional behavior) Hospital visit • Observation of patient-doctor encounter at OPDs • Assisting with interviewing patients Primary health care or community visit • Observation of patient-doctor encounter at OPDs • Assisting with interviewing patients • Home visits trying to practice physical examination Whole group session • Faculty facilitated discussion and reflection on the week's learning activities	
Week 16	Day one Pharmacokinetics (4hrs) Day two • Pharmacodynamics (4 hours) Day three Introduction to autonomic nervous system in pharmacology (4 hrs) Basic science lab • Animal handling and routes of drug administration	

	• Computer-simulations of molecular mechanisms of drug action • Computer simulation of autonomic nervous system	
	Clinical skills lab • Simulated practice: Physical examination (Skills and professional behavior)	
	Hospital visit • Observation of patient-doctor encounter at OPDs • Assisting with interviewing patients	
	Primary health care or community visit • Observation of patient-doctor encounter at OPDs • Assisting with interviewing patients • Home visits trying to practice physical examination	
	Whole group session • Faculty facilitated discussion and reflection on the week's learning activities	
Week 17	Day one •—General principle of drug toxicity and poisoning (4hrs) Day two Common Ethiopian Traditional medicine medical practices (4hrs) Day three ▪ Cancer biomarkers (1hr) ▪ Pharmacology of antineoplastic agents(3hrs)	
	Basic science lab • Computer simulations of membrane transport and drug response • Computer simulations of membrane transport and drug response	
	Clinical skills lab • Simulated practice: Physical examination (Skills and professional behavior)	
	Hospital visit • Observation of patient-doctor encounter at OPDs • Assisting with interviewing patients	
	Primary health care or community visit • Observation of patient-doctor encounter at OPDs • Assisting with interviewing patients • Home visits trying to practice physical examination	
	Whole group session • Faculty facilitated discussion and reflection on the week's learning activities	
Week 18	Day one General principle of Antimicrobial agents (4hours)	

	▪ Definition of Antimicrobial agents ▪ General classification Antimicrobial agents Day two (4hours) ▪ General MOA of Antimicrobial agents (4 hours) Day three (4hrs) ▪ First aid (4hours) ▪				
	Basic Science Lab • Computer simulation of mechanism MOA of Antimicrobial agents				
	Skills lab First aid demonstration				
	Primary health care or community visit • Drug information visit				
	Whole group session • Faculty facilitated discussion and reflection on the week's learning activities				
Week 19	Day one (4hrs) ▪ General mechanism of resistance of Antimicrobial agents and number system (Theory and practice) Day two ▪ Drug interaction ▪ Adverse drug reaction ▪ Drug discovery and development Day three Independent learning				
	Basic Science Lab • Computer simulation of mechanism MOA of Antimicrobial agents				
	Skills lab First aid demonstration				
	Primary health care or community visit • Drug information visit				
	Whole group session				
	Monday	Tuesday	Wednesday	Thursday	Friday
	Faculty facilitated discussion and reflection on the week's learning activities				
Week 20	Day One: Interactive Lecture	Day two: Interactive Lecture	Day three: Study break	Day four: Interactive Lecture	Day five: Primary Health Care/Community Visit
8-10 am	Introduction to medicine (Written exam)	Introduction to medicine (OSPE/OSCE)	Independent study	Clinical skills lab/hospital visit	
10 am-12 pm	Interactive Lecture	Interactive Lecture			
1:30-5:30 pm	Biomedical lab	Biomedical lab	Biomedical lab	Clinical skills lab/hospital visit	Whole Group Session

MODULE TITLE: METABOLIC HOMEOSTASIS

MODULE CODE: MEHO 2022

NOMINAL DURATION: 5 WEEKS

Module Description

This module deals with the functions of the different biomolecules with a focus on metabolic processes. The sequences of biological information flow are also discussed in great detail. Examples of disease processes and therapeutic approaches are included to help students appreciate the biochemical processes that underlie different disease conditions.

In addition to portraying how life functions at the molecular level, MEHO 2022 would also help students to better understand the contents of subsequent preclinical modules. A lot of the lessons learned here lend themselves to clinical medicine; it is hoped that students would try to make associations along the course of their medical study.

Module Design

This module is designed by integrating the normal and abnormal metabolism and homeostatic mechanisms of the body function. Building on in-depth understanding of the metabolic process as a foundation, this module aims to provide the analytical and cognitive skills necessary for a successful transition from the major metabolic reactions, physiology, pathology and pharmacology to the care of patients with diseases of the metabolism.

The Professional competency development component will support the student in acquiring a practical understanding of how to approach a patient with known or suspected abnormal metabolic disorders and develop the basic skill of examining patients with different metabolic disorders. Social and Population Health (SPH) module will run in parallel and will be made to correlate with the content being taught.

Pre-requisite

Successful completion of the Introduction to Medicine Module

Module objective

By the end of this module, students will be able to apply knowledge of basic sciences for understanding the metabolic homeostasis and disorder associated with it.

Supporting objectives

As a Medical Expert

- Apply knowledge of the normal metabolic process and functions of the metabolic pathway in the body homeostasis in analyzing clinical problems
- Apply principles of physiologic mechanisms and processes of the metabolic pathway in analyzing clinical problems
- Apply principles of human anabolic, catabolic and amphibolic processes in analyzing clinical problems including metabolic anomalies
- Identify the various causes of a disease (genetic, metabolic, and autoimmune) of the human body metabolic system and analyze the metabolomics in clinical problems
- Analyze the pharmacokinetics and pharmacodynamics of drugs used in the treatment of disorders of the metabolic homeostasis
- Apply important determinants and risk factors affecting metabolic disorders to protect and promote individual and population health
- Demonstrate taking history from the patient with metabolic disorder in a simulated setting
- Demonstrate pertinent physical examination of patients with metabolic disorder in a simulated setting

- Perform basic investigations in the diagnosis of metabolic disorders
- Demonstrate an initial assessment of a clinical problem related to metabolic homeostasis in a simulated setting
- Discuss important life style, genetic, demographic, environment, health system, social, economic, psychological and cultural determinants of metabolic disorders
- Contribute to a culture that promotes patient safety

As a Communicator

- Demonstrate communication with a patient with metabolic disorder by using a patient centered approach that encourages patient trust and autonomy and is characterized by empathy, respect, and compassion
- Optimize the physical environment for patient comfort, dignity, privacy, engagement, and safety
- Recognize when the values, biases, or perspectives of patients, physicians, or other health care professionals may have an impact on the quality of care, and modify the approach to the patient accordingly
- Respond to patients' non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients
- Manage disagreements and emotionally charged conversations
- Use patient centered interviewing skills to effectively gather relevant biomedical and psychosocial information
- Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information
- Provide a clear structure for and manage the flow of an entire patient encounter
- Seek and synthesize relevant information from other sources, including the patient's family, with the patient's consent
- Skillfully share information and explanations that are clear, accurate, timely, and adapted to patient's and his or her family's level of understanding and need in a simulated setting

As a Collaborator

- Establish and maintain healthy relationships with physicians and other colleagues in the health care professions to support relationship centered collaborative care
- Recognize one's own professional roles and limits
- Share knowledge and information to other colleagues
- Show respect toward other physicians and members of the health care team
- Respect and value diversity among other colleagues

As a Manager

- Develop their leadership skills
- Design and organize elements of health care delivery
- Set priorities and manage time to balance practice and personal life
- Promote occupational health as physicians for sustainable practice

As a Health Advocate

- Work with patients with metabolic disorders and their families to address the determinants of health that affect them and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors
- Incorporate metabolic disorders prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health

- Work with a community or population to identify the determinants of metabolic disorders
- Participate in a process to improve health in the community or population they serve

As a Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize knowledge gaps in biomedical, clinical and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Integrate other intrinsic roles into their learning and practice of medicine
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in practice
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance

- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Duration: 4 weeks

Teaching and learning methods

- Interactive lecture
- Assignment
- Discussion and debate
- Case studies
- Role play
- Clinical simulation
- Demonstration
- Video show
- Observation in clinical setting
- Lab quiz and report
- Community visit
- Personal research and reflection exercise/ Portfolio
- Problem based learning
- Whole group session
- Seminar

Educational Resources

- Basic sciences lab
- Clinical skills lab
- Health Facility
- Videos of patient examination
- Anatomical charts and mannequins
- Histological and pathological slides and specimens
- Computer-based simulations

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Learners Guide

- Handout
- Books
- Text books
- E-books

Learning Facilities/Infrastructure

- Class rooms
- Skill lab
- ICT center
- Clinical set up
- Library

Portable and stationary plants, machines and equipment

- Flip charts
- Video cassette
- CD-ROM
- Computers
- DVD Player
- LCD projector Overhead Projector
- White board

Methods of assessment

Progressive

- Written progress /quiz/-15%
- Clinical skill lab- 5%
- Bio-lab- 15%
- Hospital visit-5%
- PBL/Case study/Seminar-10%

END-module

- Written -30%
- OCSE/OSPE- 20%

Module Schedule

Week	Learning activity	Reading assignment
Week one	Day One <u>Interactive lecture (4 hours)</u> • Introduction to the module (30 minute) • Introduction to Bioenergetics (1.5 hrs.) • Biochemical Catalysis and Enzymes (2 hrs.) <u>Problem-based learning 1st session (2hours)</u> • A case of lactose intolerance Day Two <u>Interactive Lecture (4hours)</u>	Portfolio

	• Mechanisms of enzyme catalysis, enzyme kinetics with emphasis on the importance of V_{max} and K_m, the effects of pH and temperature; enzyme inhibition (2hrs) • Enzyme regulation and the clinical application of enzymes, Coenzymes, cofactors and their functions (2hrs) <u>Integrated Basic science lab (3hours)</u> • Estimation of the levels of the following enzymes: alanine transaminase, aspartate transaminase, alkaline phosphatase Day Three <u>Interactive lecture (2hours)</u> Carbohydrate Metabolism: • Enzymes and transporters involved in the digestion and absorption of carbohydrates (2hr) • Glycolysis and its clinical significance in different cells (2hr) <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Approach to a patient with flatulence, abdominal cramp and diarrhea (history and physical examination) • Computer simulation showing metabolism Day Four <u>Interactive lecture (4 hours)</u> • Hexose monophosphate pathway and glucuronic acid pathway, and their clinical significance in different cells (2hrs) • Regulation of the pentose phosphate pathway; the glucuronic acid pathway and its functions; Metabolism and clinical significance of other hexose sugars (2hr) <u>Integrated Basic science lab (3hours)</u> • Blood/urine glucose estimation <u>Problem-based learning 2ndsession (2hours)</u> • Previous PLB session will be continued Day Five <u>Primary health care or community visit (4hours)</u> • Community visits – home visits to identify individuals with hypoglycemia, their risk factors and determinants, perceptions of causes, local remedies and provide advice on health promotion and disease prevention <u>Hospital visit</u> • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
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Week 2	Day One <u>Interactive lecture (4 hours)</u> • Glycogen storage sites, synthesis and degradation; regulation of glycogen metabolism (1 hr.) • The significance of gluconeogenesis and responsible organs, reactions and regulation and the Cori cycle. The Pyruvate Dehydrogenase Reaction (1 hr.) • Electron Transport and Oxidative Phosphorylation: The link between redox potential and free energy, components of the respiratory chain, coupling of electron transport and oxidative phosphorylation, the energy yield of reduced coenzymes, inhibitors and uncouplers (2 hr.) <u>Problem-based learning 2nd session (2hours)</u> • Case of glycogen storage disease <u>Integrated Basic science lab (3hours)</u> • General carbohydrate test/Molish test • Computer-simulations of liver metabolism Day Two <u>Interactive Lecture (4hours)</u> • Dietary lipid utilization by cells and fatty acid oxidation (2hr) • Fatty acid biosynthesis (1hr) • Lipogenesis, and lipolysis with their clinical importance (1hr) <u>Integrated Basic science lab (3hours)</u> • Computer simulation of lipid metabolism in adipose tissue Day Three <u>Interactive lecture (4 hours)</u> • Ketone body production and utilization (1hr) • The synthesis and degradation of phospholipids, sphingolipids and glycolipids, and their clinical significance (1.5 hr.) • Lipoprotein metabolism and related disorders (1.5 hr.) <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Approach to a patient with obesity (history taking and physical examination) Day Four <u>Interactive lecture</u> • SPH (2hours) <u>Integrated Basic science lab(3hours)</u> • Determination of lipid profile TG,VLDL,LDL,HDL,TC <u>Problem-based learning 1st session (2hours)</u> • A case of lactose intolerance Day Five <u>Primary health care or community visit, Hospital visit (4</u>	
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	<u>hours)</u> <u>Hospital visit</u> • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Primary health care or community visit</u> • General OPD • Home visits trying to identify individuals with metabolic syndrome, their risk factors and determinants, perceptions of causes and local remedies and provide advice on health promotion and disease prevention <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
Week 3	Day One <u>Interactive lecture</u> <u>Protein metabolism (4hours)</u> • Proteolytic enzymes, protein turnover, nitrogen balance: lysosomal and the ubiquitin/proteasome pathway • Amino acid metabolism (catabolism) and their clinical significance • TCA Cycle and its regulations <u>Problem-based learning (2hours)</u> • Case of rickets <u>Integrated Basic science lab (3hours)</u> • General protein estimation (Lowry and Biuret methods) Day Two <u>Interactive Lecture (4hours)</u> • Amino Acid Derived Nitrogenous Compounds metabolism and their clinical significance; Inborn Errors of Amino Acid Metabolism (2hr) • Ammonia metabolism (The urea cycle) and its clinical significance and related metabolic inborn errors (2hr). <u>Integrated Basic science lab (3hours)</u> • Measurement of ammonia in the urine • Computer simulation of urea cycle Day Three <u>Interactive lecture (4hours)</u> • Dyslipidemia, Familial hypercholesterolemia, Severe hypertriglyceridemia, Lipid storage disease (4hr) <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Approach to a patient with hyperammonemia (history taking and physical examination) Day Four <u>Interactive lecture</u> • SPH (2hours)	

	<u>Interactive lecture</u> Metabolism of micronutrients (4hours) • Metabolic roles of vitamins (water soluble and insoluble vitamins) (2hr) • Phosphate and calcium metabolism (2hr) <u>Integrated Basic science lab (3hours)</u> • Determination of kidney function test (Creatinine and urea) Day Five <u>Hospital visit, Primary health care or community visit (4 hours)</u> <u>Hospital visit</u> • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Primary health care or community visit</u> • General OPD • Home visits trying to identify individuals with diabetes mellitus, their risk factors and determinants, perceptions of causes and local remedies and provide advice on health promotion and disease prevention <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
Week 4	Day One <u>Interactive lecture (4hours)</u> Nucleotide Metabolism (4hrs.): • Purine metabolism: de novo and salvage pathways, degradation; • Pyrimidine metabolism; • Inborn errors of nucleotide metabolism <u>Problem-based learning (2hours)</u> • Previous PLB session will be continued <u>Integrated Basic science lab (3hours)</u> • Determination of serum uric acid Day Two <u>Interactive Lecture (4hours)</u> • Pathophysiology of Hyperuricemia and gout (1 hr.) • Pharmacology of drugs used for treatment of gout and hyperlipidemia (3 hrs.) <u>Integrated Basic science lab (3hours)</u> • Computational pyrimidine and purine synthesis Day Three <u>Interactive lecture (4hours)</u> • The packaging of DNA in chromosomes, the nucleoid; Prokaryotic and eukaryotic DNA replication • Mutations and DNA repair mechanisms; Prokaryotic and eukaryotic gene transcription, Translation	

	- Recombinant DNA and cloning, gene therapy, DNA amplification, DNA fingerprinting <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> - Simulated practice: Approach to a patient with joint pain (history taking and physical examination) - Computer simulation showing DNA structural models Day Four <u>Interactive lecture</u> - SPH (2hours) <u>Integrated Basic science lab (3hours)</u> - Phenylketonuria Day Five <u>Hospital visit, Primary health care or community visit (4 hours)</u> <u>Hospital visit</u> - General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Primary health care or community visit</u> - General OPD - Home visits trying to identify individuals with genetic disorder, their risk factors and determinants, perceptions of causes and local remedies and provide advice on health promotion and disease prevention <u>Whole group session (3hours)</u> Faculty facilitated discussion and reflection on the week's learning activities	
Week 5	Day One Review of the metabolic homeostasis - Discussion (2hrs) - Basic science lab (3hrs) - Clinical skills lab (3hrs) Day Two : Study leave Day Three: Study leave Day Four: Written MCQ's Day Five: OSCE	

MODULE TITLE: HAEMATOLOGY MODULE

MODULE CODE: HEMA 2032

NOMINAL DURATION: 4 WEEKS

Module Description

This system-based module is intended to equip medical students with profound knowledge of basic sciences and development of early clinical skills relevant to understanding of the hematology system and related clinical problems. Students will also develop relevant competencies with the Medical expertise, Scholar, Collaborator, Communicator, Manager, and Health Advocate Professional. This module is given on the first semester during the second year of the pre-clerkship period.

Module Design

This module is designed by integrating normal and abnormal structure and function of the haematology system. Building on in-depth understanding of the haematology system as a foundation, this module aims to provide the analytical and cognitive skills necessary for a successful transition from basic haematology structure, function and pathology to the care of patients with diseases of the haematology system.

The Professional competency development component will support the student in acquiring a practical understanding of how to approach a patient with known or suspected haematology disease and develop the basic skill of examining patients with different haematology system disorders. Social and Population Health (SPH) module will run in parallel and will be made to correlate with the content being taught.

Pre-requisite

Successful completion of the Introduction to Medicine Module

Module objective

By the end of this module, students will be able to apply knowledge of basic sciences for understanding the hematology system and analyzing disorder of the hematology system.

Supporting objectives

As a Medical Expert

- Apply knowledge of the normal structures and functions of the hematology system in analyzing clinical problems
- Apply principles of physiologic mechanisms and processes of the hematology system in analyzing clinical problems
- Apply principles of human life cycle, growth, development and aging of the hematology system in analyzing clinical problems including developmental anomalies
- Identify the various causes of a disease (genetic, developmental, metabolic, toxic, infectious, autoimmune, neoplastic and degenerative) of the hematology system and analyze the pathogenesis in clinical problems
- Analyze the pharmacokinetics and pharmacodynamics of drugs used in the treatment of disorders of the hematology system
- Apply important determinants and risk factors affecting hematology diseases to protect and promote individual and population health
- Demonstrate taking history from the patient with hematology system disorder in a simulated setting
- Demonstrate physical examination pertinent to with hematology system disorder in a simulated setting
- Perform basic investigations in the diagnosis of hematology system disorders

- Demonstrate making an initial assessment of a clinical problem related to hematology system in a simulated setting
- Discuss important life style, genetic, demographic, environment, health system, social, economic, psychological and cultural determinants of hematology system disorders
- Contribute to a culture that promotes patient safety

As a Communicator

- Demonstrate communication with a patient with hematology system disorder by using a patient centered approach that encourages patient trust and autonomy and is characterized by empathy, respect, and compassion
- Optimize the physical environment for patient comfort, dignity, privacy, engagement, and safety
- Recognize when the values, biases, or perspectives of patients, physicians, or other health care professionals may have an impact on the quality of care, and modify the approach to the patient accordingly
- Respond to patients' non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients
- Manage disagreements and emotionally charged conversations
- Use patient centered interviewing skills to effectively gather relevant biomedical and psychosocial information
- Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information
- Provide a clear structure for and manage the flow of an entire patient encounter
- Seek and synthesize relevant information from other sources, including the patient's family, with the patient's consent
- Skillfully share information and explanations that are clear, accurate, timely, and adapted to patient's and his or her family's level of understanding and need in a simulated setting

As a Collaborator

- Establish and maintain healthy relationships with physicians and other colleagues in the health care professions to support relationship centered collaborative care
- Recognize one's own professional roles and limits
- Share knowledge and information to other colleagues
- Show respect toward other physicians and members of the health care team
- Respect and value diversity among other colleagues

As a Manager

- Develop their leadership skills
- Design and organize elements of health care delivery
- Set priorities and manage time to balance practice and personal life
- Promote occupational health as physicians for sustainable practice

As a Health Advocate

- Work with patients with hematology system disorders and their families to address the determinants of health that affect them and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors
- Incorporate hematology system disease prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health

- Work with a community or population to identify the determinants of hematology system health that affect them
- Participate in a process to improve hematology system health in the community or population they serve

As a Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize knowledge gaps in biomedical, clinical and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Integrate other intrinsic roles into their learning and practice of medicine
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in practice
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting

- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Duration: 4 weeks

Teaching and learning methods

- Interactive lecture
- Assignment
- Discussion and debate
- Case studies
- Role play
- Clinical simulation
- Demonstration
- Video show
- Observation in clinical setting
- Community visit
- Personal research and reflection exercise/ Portfolio
- Whole group session

Educational Resources

- Basic sciences lab
- Clinical skills lab
- Health Facility
- Videos of patient examination
- Anatomical charts and mannequins
- Histological and pathological slides and specimens
- Computer-based simulations

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- Goldman. Cecil Medicine. 23rd edition. 2007
- Courtney M. Townsend Jr. et al. Sabiston textbook of surgery: the biological basis of modern surgical practice. 19th edition 2012

Learners Guide

- Handout
- Books
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Learning Facilities/Infrastructure

- Class rooms
- Skill lab
- Dissection Room
- ICT center
- Clinical set up
- Library

Portable and stationary plants, machines and equipment

- Flip charts
- Video cassette
- CD-ROM
- Computers
- DVD Player
- LCD projector Overhead Projector
- White board

Methods of assessment

Progressive

- Written progress /quiz/-15%
- Clinical skill lab- 5%
- Bio-lab- 15%
- Hospital visit-5%

- PBL-10%
- END-module
- Written -30%
 - OCSE/OSPE- 20%

Module Schedule

Week	Learning activity	Reading assignment
Week one	Day One <u>Interactive lecture (2hours)</u> • Introduction to the module (1hr) • Microscopic structure and functions of the red blood cell(1hr) <u>Problem-based learning 1st session (2hours)</u> • A case of bleeding tendency and repeated infection <u>Integrated Basic science lab (3hours)</u> • Hemoglobin/haematocrit determination Day Two <u>Interactive Lecture (4hours)</u> • Haematocrit, Functions and production of RBC (2hrs) • Heme synthesis, breakdown and associated diseases(2hrs) <u>Integrated Basic science lab (3hours)</u> • Blood cell morphology • Blood grouping Day Three <u>Interactive lecture (2hours)</u> • Disorders of the red blood cells(anaemia and polycythaemia) (2hrs) <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Approach to a patient with generalized weakness (history taking and physical examination) • Computer simulation showing haematopoiesis Day Four <u>Interactive lecture (SPH) (2hours)</u> <u>Interactive lecture (2hours)</u> • Parasitic causes of haematological infections (malaria, hookworm, leishmania, etc)(2hrs) <u>Integrated Basic science lab (3hours)</u> • Pathologic slide on blood cell morphology • Determination of serum bilirubin and urobilinogen Day Five	Portfolio

	<u>Primary health care or community visit (4hours)</u> <u>Primary health care or community visit</u> • General OPD • Community visits – home visits to identify individuals with anaemia, their risk factors and determinants, perceptions of causes, local remedies and provide advice on health promotion and disease prevention <u>Hospital visit</u> • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
Week 2	Day One <u>Interactive lecture (2hours)</u> • Microbial causes of haematological infections (Rickettsia, Brucellosis, Borrelia and causes of septicaemia and bacteremia)(2hrs) <u>Problem-based learning 2nd session (2hours)</u> • Previous PLB session will be continued <u>Integrated Basic science lab (3hours)</u> • Blood film and Haemo-parasite identification • Blood culture and antibiotic susceptibility Day Two <u>Interactive Lecture (4hours)</u> • Pharmacology of anti-anaemic drugs, hematopoietic growth factors (2hrs) • Microscopic structure of WBC (1hr) • Production and functions of white blood cells(1hr) <u>Integrated Basic science lab (3hours)</u> • White blood cell count and differential • Haemoglobin and haematocrit determination Day Three <u>Interactive lecture (2hours)</u> • Disorders of white blood cells(non-neoplastic and neoplastic proliferation of WBC <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Approach to a patient with a white blood cell disorder (history taking and physical examination) Day Four <u>Interactive lecture (SPH)(2hours)</u> <u>Interactive lecture (2hours)</u>	

	• Microscopic structure and formation of platelets(1hr) • functions of platelets,Hemostasis and coagulation(1hr) <u>Integrated Basic science lab(3hours)</u> • Previous integrated lab session will be continued Day Five <u>Primary health care or community visit, Hospital visit (4 hours)</u> <u>Hospital visit</u> • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Primary health care or community visit</u> • General OPD • Home visits trying to identify individuals with immune deficiencies, their risk factors and determinants, perceptions of causes and local remedies and provide advice on health promotion and disease prevention <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
Week 3	Day One <u>Interactive lecture (2hours)</u> • Bleeding disorder (DIC, thrombocytopenia, coagulation disorder)(2hrs) <u>Problem-based learning (2hours)</u> Previous PLB session will be continued <u>Integrated Basic science lab (3hours)</u> • Cell morphology • Determination of blood coagulation time(bleeding time , prothrombin time , activated partial thromboplastin time) Day Two <u>Interactive Lecture (4hours)</u> • Pharmacology of coagulant, anti-coagulant, anti-platelets and thrombolytic agents (4hrs) <u>Integrated Basic science lab (3hours)</u> • Previous integrated lab session will be continued Day Three <u>Interactive lecture (2hours)</u> • Plasma proteins and their clinical application <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Approach to a patient	

	with a bleeding disorder (history taking and physical examination) • Computer simulations of haemostasis Day Four <u>Interactive lecture (SPH)(2hours)</u> <u>Interactive lecture (2hours)</u> • Haematopoiesis <u>Integrated Basic science lab (3hours)</u> • Determination of Serum albumin and globulin Day Five <u>Hospital visit, Primary health care or community visit (4 hours)</u> <u>Hospital visit</u> • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Primary health care or community visit</u> • General OPD • Home visits trying to identify individuals with a bleeding disorder, their risk factors and determinants, perceptions of causes and local remedies and provide advice on health promotion and disease prevention <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
Week 4	Day One • Review of the hematology system • Discussion (2hrs) • Basic science lab (3hrs) • Clinical skills lab (3hrs) Day Two : Written MCQ's Day Three: OSCE Day Four: Interactive lecture (SPH) (2hrs) Day Five: Primary health care or community visit (4hour) Whole group session (3hrs)	

Time Table

Time	Monday	Tuesday	Wednesday	Thursday	Friday
8 -10 am	Interactive	Interactive	Interactive	Interactive	Primary Health

	Lecture	Lecture	Lecture	Lecture	Care/ Community Visit
10 -12 am	PBL	Interactive Lecture		Interactive Lecture	
1:30- 5:30 pm	Biomedical lab	Biomedical lab	Clinical skills lab	Biomedical lab	Whole Group Session

MODULE TITLE: CARDIOVASCULAR AND LYMPHATIC SYSTEMS

MODULE CODE: CVLS 2062

NOMINAL DURATION: 8 WEEKS

Module Description:

This system-based module is intended to equip medical students with profound knowledge of basic sciences and development of early clinical skills relevant to understanding of the cardiovascular and lymphatic system and related clinical problems. Students will also develop relevant competencies with the Medical expertise, Scholar, Collaborator, Communicator, Manager, Health Advocate and Professional. This module is given on the second semester during the second year of the pre-clerkship period of the undergraduate medical education program.

Module Design:

This module is designed by integrating normal and abnormal structure and function of the cardiovascular and lymphatic system. Building on in-depth understanding of the cardiovascular and lymphatic system as a foundation, this module aims to provide the analytical and cognitive skills necessary for a successful transition from basic cardiovascular structure, function and pathology to the care of patients with heart and lymphatic disease.

The Professional competency development component will support the student in acquiring a practical understanding of how to approach a patient with known or suspected cardiac disease and develop the basic skill of examining the cardiovascular system. Social and Population Health (SPH) module will run in parallel and will be made to correlate with the subject being taught.

Pre-requisite

Successful completion of Introduction to Medicine module

Module Objective

By the end of this module, students will be able to apply knowledge of basic sciences for understanding the cardiovascular and lymphatic system and analyzing disorder of the cardiovascular and lymphatic system.

Supporting objectives

As a Medical Expert

- Apply knowledge of the normal structures and functions of the cardiovascular and lymphatic system in analyzing clinical problems
- Apply principles of physiologic mechanisms and processes of the cardiovascular and lymphatic system in analyzing clinical problems
- Apply principles of human life cycle, growth, development and aging of the cardiovascular and lymphatic system in analyzing clinical problems including developmental anomalies
- Identify the various causes of a disease (genetic, developmental, metabolic, toxic, infectious, autoimmune, neoplastic, degenerative and traumatic) of the cardiovascular and lymphatic system and analyze the pathogenesis in clinical problems
- Analyze the pharmacokinetics and pharmacodynamics of drugs used in the treatment of disorders of the cardiovascular and lymphatic system
- Apply important determinants and risk factors affecting cardiovascular and lymphatic diseases to protect and promote individual and population health
- Demonstrate taking history from the patient with cardiovascular and lymphatic system disorder in a simulated setting.
- Demonstrate physical examination pertinent to with cardiovascular and lymphatic system disorder in a simulated setting.

- Perform basic investigations in the diagnosis of cardiovascular and lymphatic system disorders
- Demonstrate making an initial assessment of a clinical problem related to cardiovascular and lymphatic system in a simulated setting.
- Discuss important life style, genetic, demographic, environment, health system, social, economic, psychological and cultural determinants of cardiovascular and lymphatic system disorders.
- Contribute to a culture that promotes patient safety

As a Communicator

- Demonstrate communication with a patient with cardiovascular and lymphatic system disorder disusing a patient-centered approach that encourages patient trust and autonomy and is characterized by empathy, respect, and compassion
- Optimize the physical environment for patient comfort, dignity, privacy, engagement, and safety
- Recognize when the values, biases, or perspectives of patients, physicians, or other health care professionals may have an impact on the quality of care, and modify the approach to the patient accordingly
- Respond to patients' non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients
- Manage disagreements and emotionally charged conversations
- Use patient-centered interviewing skills to effectively gather relevant biomedical and psychosocial information
- Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information
- Provide a clear structure for and manage the flow of an entire patient encounter
- Seek and synthesize relevant information from other sources, including the patient's family, with the patient's consent.
- Skillfully share information and explanations that are clear, accurate, timely, and adapted to patient's and his or her family's level of understanding and need in a simulated setting

As a Collaborator

- Establish and maintain healthy relationships with physicians and other colleagues in the health care professions to support relationship-centered collaborative care
- Recognize one's own professional roles and limits
- Share knowledge and information to other colleagues
- Show respect toward other physicians and members of the health care team
- Respect and value diversity among other colleagues

As a Manager

- Develop their leadership skills
- Design and organize elements of health care delivery
- Set priorities and manage time to balance practice and personal life
- Promote occupational health as physicians for sustainable practice

As a Health Advocate

- Work with patients with cardiovascular and lymphatic system disorders and their families to address the determinants of health that affect them and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors

- Incorporate cardiovascular and lymphatic system disease prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of cardiovascular and lymphatic system health that affect them
- Participate in a process to improve cardiovascular and lymphatic system health in the community or population they serve

As a Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize knowledge gaps in biomedical, clinical and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Integrate other intrinsic roles into their learning and practice of medicine
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in practice
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice

- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Duration: 8 weeks

Teaching and learning methods

- Interactive lecture
- Assignment
- Discussion and debate
- Case studies
- Role play
- Clinical simulation
- Demonstration
- Video show
- Observation in clinical setting
- Community visit
- Personal research and reflection exercise/ Portfolio
- Whole group session

Educational Resources

- Basic sciences lab
- Clinical skills lab
- Health Facility
- Videos of patient examination
- Anatomical charts and mannequins
- Histological and pathological slides and specimens
- Computer-based simulations

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Portable and stationary plants, machines and equipment

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- CD-ROM
- Computers
- DVD Player
- LCD projector/Overhead Projector
- White board

Methods of assessment

Progressive

Written progress /quiz/-15%

- Clinical skill lab- 5%
- Biomedical-lab- 15%
- Hospital visit-5%
- PBL-10%

End module

- Written -30%
- OCSE/OSPE- 20%

Module Schedule

Week	Learning Activity	Required Reading / Assignment
Week 1	Day One <u>Interactive lecture (2hours)</u> • Introduction to the module (1hr) • Introduction to CVS and overview of thoracic cavity (1hr) <u>Problem-based learning 1st session (2hours)</u> • A case of chest pain Day Two <u>Interactive lecture (4hours)</u> • Pericardium (1hr) • Gross and microscopic structures of the heart (3hrs) <u>Integrated Basic science lab (3hours)</u> • Anatomical charts and models • Dissection of Pericardium and heart • Tissue slides and specimens of heart Day Three <u>Interactive lecture (2hours)</u> • Anterior and superior mediastinum (1hr) • The heart muscle: The heart as a pump and function of heart valves (1hr) <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Approach to a patient with heart disease (history taking and physical examination of the CVS) • Audio recordings of normal heart sounds • Chest x-ray of the normal heart Day Four <u>Interactive lecture – SPH (2hours)</u> <u>Interactive lecture (2hours)</u> • Cardiac energetic(1hr) • Biomarkers of cardiac injury(1hr) <u>Integrated basic science Lab (3hours)</u>	Portfolio

	• Dissection of superior mediastinum • Computer-simulations of the heart as a pump • Demonstration of properties of cardiac muscle in a lab animal • Exercise stress test/heart function test Day Five <u>Primary health care, hospital or community visit (4hours)</u> • General OPD • Home visits trying to identify common cardiac diseases, their risk factors and determinants and local remedies and provide advice on promotion of health and prevention of disease Hospital visit • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
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Week 2	Day One <u>Interactive lecture (2hours)</u> • Electrophysiology of the heart (2hrs) <u>Problem-based learning 2nd session (2hours)</u> • Previous PLB session will be continued <u>Integrated Basic science lab (3hours)</u> • Determination of cardiac biomarkers ✓ AST, CK, LDH, Troponin and Mb Day Two <u>Interactive lecture (4hours)</u> • The normal electrocardiogram (ECG)(2hrs) • The Cardiac cycle (2hrs) <u>Integrated Basic science lab (3hours)</u> • ECG measurement and interpretation • Cardiac biomarkers of injury, AST, CK, LDH, Troponin, Myoglobin • Tissue slides, pathology slides and specimens Day Three <u>Interactive lecture (2hours)</u> • Lipoproteins metabolism and hyperlipidaemia (2hrs) <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Approach to a patient with heart disease (history taking and physical examination of the CVS) • Chest x-rays or computer simulations of abnormal heart conditions Day Four <u>Interactive lecture – SPH (2hours)</u> <u>Interactive lecture (2hours)</u> • Microbes affecting the heart and the pericardium (Valvular heart disease, infective endocarditis) (2hrs)...continues <u>Integrated basic science Lab (3hours)</u> • Determination of lipid profile ✓ TAG,VLDL,LDL,HDL,TC Day Five <u>Primary health care, hospital or community visit (4hours)</u> • General OPD • Home visits trying to identify common cardiac diseases, their risk factors and determinants and local remedies and provide advice on promotion of health and prevention of disease <u>Hospital visit</u> • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Whole group session (3hours)</u>	
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	Faculty facilitated discussion and reflection on the week's learning activities	
Week 3	Day One <u>Interactive lecture (2hours)</u> - Microbes affecting the heart and the pericardium (Valvular heart disease, infective endocarditis) (2hrs) <u>Problem based learning 1st session (2hours)</u> - A case of leg edema <u>Integrated basic science Lab (3hours)</u> - Detection of anti-streptolysin O and S Day Two <u>Interactive lecture (4hours)</u> - Pathophysiology of common infectious and autoimmune diseases of the heart (heart failure, ischemic heart diseases, hypertensive heart disease, valvular heart diseases, congenital heart disease, cardiomyopathies, cardiac tumors, pericarditis, endocarditis) <u>Integrated basic science Lab (3hours)</u> - Audio recordings or computer simulations of the dynamics of heart valves and sounds in valvular and congenital heart diseases - Computer-simulations of drug actions on the heart Day Three <u>Interactive lecture (2hours)</u> - Overview of Pharmacology of autonomic drugs (1hr) - Pharmacology of sympathomimetics drugs (1hr) <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> - Simulated practice: Approach to a patient with heart disease (history taking and physical examination of the CVS) - Chest x-rays or computer simulations of abnormal heart conditions Day Four <u>Interactive lecture – SPH (2hours)</u> <u>Interactive lecture (2hours)</u> - Pharmacology of sympathomimetics drugs (2hrs) <u>Integrated basic science Lab (2hours)</u> - Previous integrated lab session continues Day Five <u>Primary health care, hospital or community visit (4hours)</u> - General OPD - Home visits trying to identify common cardiac diseases, their risk factors and determinants and local remedies and provide advice on promotion of health and prevention of disease <u>Hospital visit</u>	

	• General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
Week 4	Day One <u>Interactive lecture (2hours)</u> • Pharmacology of adrenergic blockers (2hrs) <u>Problem-based learning 2nd session (2hours)</u> • Previous PLB session will be continued <u>Integrated Basic science lab (3Hours)</u> • Computer simulations of sympathomimetics drugs Day Two <u>Interactive lecture (4hours)</u> • Pharmacology of sympathomimetics drugs • Pharmacology of drugs used for treatment of heart failure (2 hrs) <u>Integrated Basic science lab (3Hours)</u> • Previous lab session continues Day Three <u>Interactive lecture (2hours)</u> • Pharmacology of drugs used for treatment of ischemic heart disease (2hrs) <u>Professional competency development</u> <u>Clinical skills lab</u> • Measuring blood pressure in different positions and after exercise • Measuring arterial pulse after exercise, temporary arterial occlusion and applying temperature Day Four <u>Interactive lecture – SPH (2hours)</u> <u>Interactive lecture (2hours)</u> • Major blood vessels of the body(1hr) • Histology of blood vessels (1hr) <u>Integrated basic science Lab (2hours)</u> • Anatomical charts and models • Dissection of blood vessels • Tissue slides and specimens • Computer simulations of blood circulation Day Five <u>Hospital visit</u> • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Primary health care or community visit</u> • General OPD • Home visits trying to identify common diseases of blood	

	vessels (including hypertension), their risk factors and determinants, local remedies and provide advice on promotion of health and prevention of disease <u>Whole group session</u> Faculty facilitated discussion and reflection on the week's learning activities	
Week 5	Day One <u>Interactive lecture (2hours)</u> - Development of CVS (2hrs) <u>Problem-based learning 1st session (2hours)</u> - A case of sever blood loss <u>Integrated Basic science lab (3hours)</u> - Previous lab session will be continues Day Two <u>Interactive lecture (4hours)</u> - Regulation Of Cardiac Output (1hr) - The Arterial and the Venous system (1hr) - The local, humoral and nervous regulation of the circulation (1hr) - Regulation of Arterial blood Pressure(1hr) <u>Integrated Basic science lab (3hours)</u> - Computer simulations of blood pressure control - Computer simulations of pathogenesis of hypertensive vascular disease Day Three <u>Interactive lecture (2hours)</u> - Physiology of microcirculation(1hr) - Dynamics of the peripheral circulation(1hr) <u>Professional Competency development (4hours)</u> <u>Clinical skills lab</u> - Measuring blood pressure in different positions and after exercise - Measuring arterial pulse after exercise, temporary arterial occlusion and applying temperature Day Four <u>Interactive lecture – SPH (2hours)</u> <u>Interactive lecture (2hours)</u> - Pathophysiology of hypertensive vascular disease(1hr) - Pharmacology of drugs used for treatment of hypertension and hypotension (1hr) <u>Integrated basic science Lab (2hours)</u> - Pathology tissue slides and specimen - Computer simulations of effect of anti-hypertensive agents on blood pressure Day Five	

	<u>Hospital visit</u> • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Primary health care or community visit</u> • General OPD • Home visits trying to identify common diseases of blood vessels (including hypertension), their risk factors and determinants, local remedies and provide advice on promotion of health and prevention of disease	
Week 6	Day One <u>Interactive lecture (2hours)</u> • Mechanism of atherosclerosis (LDL/HDL) • Dyslipidaemia • Familial hypercholesterolemia • Severe hypertriglyceridemia <u>Problem-based learning 2nd session (2hours)</u> • Previous PLB session will be continued <u>Integrated Basic science lab (3hours)</u> • Previous lab session continues Day Two <u>Interactive lecture (4hours)</u> • Pharmacology of agents used in hyperlipidemia(2hrs) • Pathophysiology of venous disorders (varicose vein, thrombophlebitis)(2hrs) <u>Integrated Basic science lab (3hours)</u> • Pathology slides and specimens • Computer simulations of atherosclerosis Day Three <u>Interactive lecture (2hours)</u> • Anatomy and histology of lymphatic system • Lymphatic vessels and lymph flow • Lymphoid tissue <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Examination of the vascular system Day Four <u>Interactive lecture – SPH (2hours)</u> <u>Interactive lecture (2hours)</u> • Lymphoid organs(thymus, bone marrow, lymph nodes, spleen, tonsils, MALT)(2hrs) <u>Integrated basic science Lab (2hours)</u> • Anatomical charts and models • Discussion of lymphoid organs • Histological slide of lymphoid organs	

	Day Five <u>Hospital visit</u> • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Primary health care or community visit</u> • General OPD • Home visits trying to identify common diseases of blood vessels (including hypertension), their risk factors and determinants, local remedies and provide advice on promotion of health and prevention of disease	
Week 7	Day One <u>Interactive lecture (2hours)</u> • Development of lymphatic system(2hrs) <u>Problem-based learning session (2hours)</u> • Previous PLB session will be continued <u>Integrated Basic science lab (3hours)</u> • Previous lab session continues Day Two <u>Interactive lecture (4hours)</u> • Physiology of lymphatic system (2hrs) • Pathology of lymphatic system (2hrs) <u>Integrated Basic science lab (3hours)</u> • Previous lab session continues Day Three <u>Interactive lecture (2hours)</u> • Pharmacology of anti-arrhythmic drugs <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Approach to a patient with heart disease (history taking and physical examination of the CVS) • Audio recordings of normal heart sounds • Chest x-ray of the normal heart DayFour <u>Interactive lecture – SPH (2hours)</u> <u>Interactive lecture (2hours)</u> • Pharmacology of drugs used for treatment of infective endocarditis and pericarditis (2hrs) <u>Integrated basic science Lab (2hours)</u> • Previous lab session continues Day Five <u>Primary health care, hospital or community visit (4hours)</u> • General OPD • Home visits trying to identify common cardiac diseases, their risk factors and determinants and local remedies and provide	

	advice on promotion of health and prevention of disease Hospital visit • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
Week 8	Exam week Day One • Review of the cardiovascular and lymphatic systems • Discussion (2hours) Basic science lab (3hours) and clinical skills lab (3hours) Day Two : MCQ's Day Three: OSCE Day Four: Interactive lecture (SPH) (2hrs) Day Five: Primary health care or community visit (4hrs) Whole group session (3hrs)	

Time Table

Time	Monday	Tuesday	Wednesday	Thursday	Friday
8-10 am	Interactive Lecture	Interactive Lecture	Interactive Lecture	Interactive Lecture	Primary Health Care/Community Visit
10-12 am	PBL	Interactive Lecture		Interactive Lecture	
1:30-5:30pm	Biomedical lab	Biomedical lab	Clinical skills lab	Biomedical lab	Whole Group Session

MODULE TITLE: RESPIRATORY SYSTEM

MODULE CODE: RESP 2052

NOMINAL DURATION: 5 WEEKS

Module Description:

This system-based module is intended to equip medical students with profound knowledge of basic sciences and development of early clinical skills relevant to understanding of the respiratory system and related clinical problems. Students will also develop relevant competencies with the Medical expertise, Scholar, Collaborator, Communicator, Manager, and Health Advocate Professional. This module is given on the first semester during the second year of the pre-clerkship period of the undergraduate medical education program.

Module Design

This course is designed by integrating normal and abnormal structure and function of the respiratory system. Building on in-depth understanding of the respiratory system as a foundation, this module aims to provide the analytical and cognitive skills necessary for a successful transition from basic respiratory structure, function and pathology to the care of patients with respiratory/lung disease. The Professional competency development component will support the student in acquiring a practical understanding of how to approach a patient with known or suspected respiratory disease and develop the basic skill of examining the respiratory system. Social and Population Health (SPH) module will run in parallel and will be made to correlate with the content being thought.

Prerequisite

Successful completion of the Introduction to Medicine Module

Module objective

By the end of this module, students will be able to apply knowledge of basic sciences for understanding the respiratory system and analyzing disorder of the respiratory system.

Supporting objectives

As a Medical Expert

- Apply knowledge of the normal structures and functions of the respiratory system in analyzing clinical problems
- Apply principles of physiologic mechanisms and processes of the respiratory system in analyzing clinical problems
- Apply principles of human life cycle, growth, development and aging of the respiratory system in analyzing clinical problems including developmental anomalies
- Identify the various causes of a disease (genetic, developmental, metabolic, toxic, infectious, autoimmune, neoplastic and degenerative) of the respiratory system and analyze the pathogenesis in clinical problems
- Analyze the pharmacokinetics and pharmacodynamics of drugs used in the treatment of disorders of the respiratory system
- Apply important determinants and risk factors affecting respiratory system diseases to protect and promote individual and population health

- Demonstrate taking history from the patient with respiratory system disorder in a simulated setting
- Demonstrate physical examination pertinent to with respiratory system disorder in a simulated setting
- Perform basic investigations in the diagnosis of respiratory system disorders
- Demonstrate making an initial assessment of a clinical problem related to respiratory system in a simulated setting
- Discuss important life style, genetic, demographic, environment, health system, social, economic, psychological and cultural determinants of respiratory system disorders.
- Contribute to a culture that promotes patient safety

As a communicator

- Demonstrate communication with a patient with respiratory system disorder by using a patient-centered approach that encourages patient trust and autonomy and is characterized by empathy, respect, and compassion
- Optimize the physical environment for patient comfort, dignity, privacy, engagement, and safety
- Recognize when the values, biases, or perspectives of patients, physicians, or other health care professionals may have an impact on the quality of care, and modify the approach to the patient accordingly
- Respond to patients' non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients
- Manage disagreements and emotionally charged conversations
- Use patient-centered interviewing skills to effectively gather relevant biomedical and psychosocial information
- Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information
- Provide a clear structure for and manage the flow of an entire patient encounter
- Seek and synthesize relevant information from other sources, including the patient's family, with the patient's consent
- Skillfully share information and explanations that are clear, accurate, timely, and adapted to patient's and his or her family's level of understanding and need in a simulated setting

As a Collaborator

- Establish and maintain healthy relationships with physicians and other colleagues in the health care professions to support relationship-centered collaborative care
- Recognize one's own professional roles and limits
- Share knowledge and information to other colleagues
- Show respect toward other physicians and members of the health care team
- Respect and value diversity among other colleagues

As a Manager

- Develop their leadership skills
- Design and organize elements of health care delivery
- Set priorities and manage time to balance practice and personal life
- Promote occupational health as physicians for sustainable practice

As a Health Advocate

- Work with patients with respiratory system disorders and their families to address the determinants of health that affect them and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors
- Incorporate respiratory system disease prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of respiratory system health that affect them
- Participate in a process to improve respiratory system health in the community or population they serve

As a Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize knowledge gaps in biomedical, clinical and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Integrate other intrinsic roles into their learning and practice of medicine
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in practice
- Exhibit professional behaviors in the use of technology enabled communication

- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Duration: 5 weeks

Teaching and learning methods

- Interactive lecture
- PBL
- Case study
- Video show
- Demonstration
- Role play
- Observation in clinical setting
- Basic sciences lab
- Clinical skills lab
- Community visit
- Personal research and reflection exercise
- Portfolio
- Computer lab practice
- Whole group session

Educational Resources

- Basic sciences lab
- Clinical skills lab
- Health Facility
- Videos of patient examination
- Anatomical charts and mannequins
- Histological and pathological slides and specimens
- Computer-based simulations

References

- Tortora G.J and B. Derrickson. Principles of Anatomy and physiology. 12th edition. 2008
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- Courtney M. Townsend Jr.et al. Sabiston textbook of surgery: the biological basis of modern surgical practice. 19th edition 2012

Learners Guide

- Handout
- Books
- Text books
- E-books

Learning Facilities / Infrastructure

- Class rooms
- Skill lab
- Dissection Room
- ICT center
- Clinical set up
- Library

Portable and stationary plants, machines and equipment

- Flip charts
- Video cassette
- CD-ROM
- Computers
- DVD Player
- LCD projector Overhead Projector
- White board

Methods of assessment

Progressive

- Written progress /quiz/-15%
- Clinical skill lab- 5%
- Biomedical lab- 15%
- Hospital visit-5%
- PBL-10%

END module

- Written -30%
- OCSE/OSPE- 20%

Module schedule

Week	Activity	Required assignment
Week 1	Day One <u>Interactive lecture (2 hours)</u> • Introduction to the module • Introduction to the respiratory system <u>PBL case- 1st session (2 hours)</u> • A case of shortness of breath with cough <u>Integrated basic science lab (3 hours)</u> • Anatomical charts and models Day Two : <u>Interactive lecture (4 hours)</u> • Thoracic wall • Diaphragm • Movements of the thoracic wall and diaphragm during breathing <u>Integrated basic science lab (3hours)</u> • Dissection of the thoracic wall and mediastinum • Tissue slides of the respiratory system • Computer-simulations of chest movement during breathing Day Three: <u>Interactive lecture (2 hours)</u> • Gross and microscopic structures and functions of the upper respiratory tract ✓ Nose and Paranasal sinuses <u>Professional competency development (4 hours)</u> • Approach to a patient with respiratory problem (History taking and physical examination) Day Four <u>Interactive lecture (2 hours)</u> • Over view of Anatomy of the neck • Anatomy of Pharynx <u>Interactive lecture (SPH) (2 hourss)</u> <u>Integrated Basic science Lab (3 hours)</u> Previous integrated lab session will be continued Day Five • Primary health care or community visit (4hrs) • Whole group session (3hrs) • Faculty facilitated discussion and reflection on the week's learning activities	
Week 2	Day One <u>Interactive lecture (2hours)</u> • Gross and microscopic structures of the larynx, trachea,	

	bronchi and bronchioles • Posterior mediastinum <u>Problem-based learning 2nd session (2 hours)</u> • Previous PLB session will be continued Day Two <u>Interactive lecture (4 hours)</u> • Pathophysiology of common diseases of the upper respiratory tract (common cold, acute pharyngitis, acute epiglottitis, otitis media, laryngitis, Nasopharyngeal carcinoma and laryngeal tumors)(2hrs) • Gross and microscopic structures and function of the lung and pleura (1hrs) • Development of Respiratory system (1hrs) <u>Integrated basic science lab (3 hours)</u> • Anatomical charts and models • Dissection of the upper respiratory tract • Tissue slides, pathology slides and specimens • Computer-simulations of clearance of mucus and cilia movement Day Three <u>Interactive lecture (2 hours)</u> • Mechanics of Pulmonary Ventilation (mechanics of breathing) <u>Professional competency development (4 hours)</u> • Simulated practice: Approach to a patient with cough (history taking and physical examination including inspection, auscultation, percussion) • Computer-based simulations of normal and abnormal breathing sounds including wheezing, crepitation Day Four <u>Interactive lecture (2 hours)</u> • Pulmonary volumes and capacities <u>Interactive lecture (SPH) (2 hours)</u> <u>Integrated Basic science Lab (3 hours)</u> • Previous integrated lab session will be continued Day Five <u>Primary health care or community visit (4 hours)</u> <u>Whole group session (3 hours)</u> Faculty facilitated discussion and reflection on the week's learning activities	
Week 3	Day One <u>Interactive lecture (2 hours)</u> • Pathophysiology of common diseases of the lungs and pleura (atelectasis, restrictive and obstructive pulmonary diseases) <u>Day one PBL case- first session(2 hours) :</u> • A case of chest trauma	

	<u>Integrated Basic science Lab (3 hours)</u> • Anatomical charts and models • Clinical measurements of spirometry Day Two <u>Interactive lecture (4 hours)</u> • Pathophysiology of common diseases of the lungs and pleura (atelectasis , restrictive and obstructive pulmonary diseases) Continued....(2hrs) • Pulmonary circulation ,edema and pleural fluids(2hrs) <u>Integrated Basic science lab (3 hours)</u> • Dissection of the pleura and lungs • Tissue slides, pathology slides and specimens • Computer-simulations of pulmonary ventilation, pulmonary circulation, gas exchange, transport of oxygen and carbon dioxide, regulation of respiration Day Three <u>Interactive lecture (2 hours)</u> • Physical principles of gas exchange and GasTransport (1hr) • Regulation of respiration(1hr) <u>Professional competency development (4 hours)</u> • Simulated practice: Approach to a patient with shortness of breath (history taking and physical examination focusing on percussion and auscultation to pick abnormalities) • Interpretation of chest x-rays of different chest pathologies Day Four <u>Interactive lecture (2hours)</u> • Hemoglobin (Structure, Oxygen and Carbon Dioxide transport, (1hrs) • Lung surfactants(1hrs) <u>Interactive lecture (SPH) (2 hour)</u> <u>Integrated basic Science Lab (3hours)</u> • Previous integrated lab session will be continued Day Five <u>Primary health care or community visit (4 hour)</u> <u>Whole group session (3 hours)</u> • Faculty facilitated discussion and	
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	reflection on the week's learning activities	
Week 4	Day One <u>Interactive lecture (2hours)</u> • Anti-inflammatory drugs used for asthma • Pharmacotherapy of cough and nasal decongestant (2hrs) <u>Problem-based learning 2nd session (2hours)</u> • Previous PLB session will be continued <u>Integrated basic Science Lab (3hours)</u> • Anatomical charts and models • Tissue slides, pathology slides and specimens Day Two <u>Interactive lecture (4 hours)</u> • Bacterial causes of the respiratory infections (4hrs) <u>Integrated Basic science lab(3 hours)</u> • Effect of drugs on bronchial smooth muscle • Sputum culture and antibiotic susceptibility • Ziehl-nelson staining • Basic principles of fungal culture Day Three <u>Interactive lecture (2 hours)</u> Pathophysiology of common diseases of the lungs and pleura (lung neoplasms and pleural lesions) <u>Professional competency development (4 hours)</u> • Simulated practice: Approach to a patient with Hemoptysis (history taking and physical examination focusing on percussion and auscultation to pick abnormalities) • Interpretation of chest x-rays of different chest pathologies Day Four <u>Interactive lecture (2 hours)</u> • Viral and fungal causes of the respiratory infections (2hrs) <u>Interactive lecture (SPH) (2 hours)</u> <u>Integrated basic science Lab (3 hours)</u> • Previous integrated lab session will be continued Day Five <u>Primary health care or community visit (4 hours)</u> <u>Whole group session (3 hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
Week 5	Day One	

	Review of the respiratory system • Discussion (2hrs) • Basic science lab (3hrs) • Clinical skills lab (3rs) Day Two: written exam Day Three: OSCE Day Four: Interactive lecture (SPH) (2hrs) Day Five: Primary health care or community visit (4hrs) Whole group session (3hrs)	
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Time Table

Time	Monday	Tuesday	Wednesday	Thursday	Friday
8 -10 am	Interactive Lecture	Interactive Lecture	Interactive Lecture	Interactive Lecture	Primary Health Care/ Community Visit
10 -12 am	PBL	Interactive Lecture		Interactive Lecture	
1:30-5:30 Pm		Biomedical lab	Clinical skills lab	Biomedical lab	Whole Group Session

MODULE TITLE: MUSCULOSKELETAL AND INTEGUMENTARY SYSTEM

MODULE CODE: MSKI 2042

NOMINAL DURATION: 6 WEEKS

Module Description: This system-based module is intended to equip medical students with profound knowledge of basic sciences and development of early clinical skills relevant to understanding of the musculoskeletal and integumentary system and related clinical problems. Students will also develop relevant competencies with the Medical expertise, Scholar, Collaborator, Communicator, Manager, Health Advocate, and Professional. This module is given on the second semester during the second year of the pre-clerkship period of the undergraduate medical education program.

Module Design

This module is designed by integrating normal and abnormal structure and function of the musculoskeletal and integumentary system. Building on in-depth understanding of the musculoskeletal and integumentary system as a foundation, this module aims to provide the analytical and cognitive skills necessary for a successful transition from basic musculoskeletal

and integumentary structure, function and pathology to the care of patients with diseases of the musculoskeletal and integumentary system.

The Professional competency development component will support the student in acquiring a practical understanding of how to approach a patient with known or suspected disorders of musculoskeletal and integumentary and develop the basic skill of examining musculoskeletal and integumentary system .Social and Population Health (SPH) module will run in parallel and will be made to correlate with the content being taught.

Prerequisite

Successful completion of Introduction to Medicine Module

Module objective

By the end of this module, students will be able to apply knowledge of basic sciences for understanding the musculoskeletal and integumentary system and analyzing disorder of the musculoskeletal and integumentary system.

Supporting objectives

As a Medical Expert

- Apply knowledge of the normal structures and functions of the musculoskeletal and integumentary system in analyzing clinical problems
- Apply principles of physiologic mechanisms and processes of the musculoskeletal and integumentary system in analyzing clinical problems
- Apply principles of human life cycle, growth, development and aging of the musculoskeletal and integumentary system in analyzing clinical problems including developmental anomalies
- Identify the various causes of a disease (genetic, developmental, metabolic, toxic, infectious, autoimmune, neoplastic, degenerative and traumatic) of the musculoskeletal and integumentary system and analyze the pathogenesis in clinical problems
- Analyze the pharmacokinetics and pharmacodynamics of drugs used in the treatment of disorders of the musculoskeletal and integumentary system
- Apply important determinants and risk factors affecting musculoskeletal and integumentary diseases to protect and promote individual and population health
- Demonstrate taking history from the patient with musculoskeletal and integumentary system disorder in a simulated setting
- Demonstrate physical examination pertinent to with musculoskeletal and integumentary system disorder in a simulated setting
- Perform basic investigations in the diagnosis of musculoskeletal and integumentary system disorders
- Demonstrate making an initial assessment of a clinical problem related to musculoskeletal and integumentary system in a simulated setting

- Discuss important life style, genetic, demographic, environment, health system, social, economic, psychological and cultural determinants of musculoskeletal and integumentary system disorders
- Contribute to a culture that promotes patient safety

As a communicator

- Demonstrate communication with a patient with musculoskeletal and integumentary system disorder disusing a patient-centered approach that encourages patient trust and autonomy and is characterized by empathy, respect, and compassion
- Optimize the physical environment for patient comfort, dignity, privacy, engagement, and safety
- Recognize when the values, biases, or perspectives of patients, physicians, or other health care professionals may have an impact on the quality of care, and modify the approach to the patient accordingly
- Respond to patients' non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients
- Manage disagreements and emotionally charged conversations
- Use patient-centered interviewing skills to effectively gather relevant biomedical and psychosocial information
- Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information
- Provide a clear structure for and manage the flow of an entire patient encounter
- Seek and synthesize relevant information from other sources, including the patient's family, with the patient's consent
- Skillfully share information and explanations that are clear, accurate, timely, and adapted to patient's and his or her family's level of understanding and need in a simulated setting

As a Collaborator

- Establish and maintain healthy relationships with physicians and other colleagues in the health care professions to support relationship-centered collaborative care
- Recognize one's own professional roles and limits
- Share knowledge and information to other colleagues
- Show respect toward other physicians and members of the health care team
- Respect and value diversity among other colleagues

As a Manager

- Develop their leadership skills
- Design and organize elements of health care delivery
- Set priorities and manage time to balance practice and personal life
- Promote occupational health as physicians for sustainable practice

As a Health Advocate

- Work with patients with musculoskeletal and integumentary system disorders and their families to address the determinants of health that affect them and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors
- Incorporate musculoskeletal and integumentary system disease prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of musculoskeletal and integumentary system health that affect them
- Participate in a process to improve musculoskeletal and integumentary system health in the community or population they serve

As a Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize knowledge gaps in biomedical, clinical and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them

- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Integrate other intrinsic roles into their learning and practice of medicine
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in practice
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Duration: 6 weeks

Teaching and learning methods

- Interactive lecture
- Assignment
- Discussion and debate
- Case studies
- Role play
- Clinical simulation
- Demonstration
- Video show
- Observation in clinical setting
- Community visit
- Personal research and reflection exercise/ Portfolio
- Whole group session

Educational Resources

- Basic sciences lab
- Clinical skills lab
- Health Facility
- Videos of patient examination

- Anatomical charts and mannequins
- Histological and pathological slides and specimens
- Computer-based simulations

References

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- Skill lab
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- ICT center
- Clinical set up
- Library

Portable and stationary plants, machines and equipment

- Flip charts
- Video cassette
- CD-ROM
- Computers
- DVD Player
- LCD projector Overhead Projector
- White board

Methods of assessment

Progressive

- Written progress /quiz/-15%
- Clinical skill lab- 5%
- Bio-lab- 15%
- Hospital visit-5%
- PBL-10%

END module

- Written -30%
- OCSE/OSPE- 20%

Module Schedule

Week	Learning activity	Reading assignment
Week One	Day One <u>Interactive lecture (2hours)</u> • Introduction to the module • Introduction to the musculoskeletal system • Gross and microscopic structure and functions of the skin • Development of the Integumentary system <u>Problem-based learning (2hours)</u> • A case of skin lesion <u>Integrated Basic science lab (3hours)</u> • Atlas (anatomy, histology) or charts showing gross and microscopic structure of the skin Day Two <u>Interactive Lecture (4hours)</u> • Collagen biosynthesis, metabolism and interrelated diseases (2hrs) • Pathology of the skin (inflammatory dermatoses, infectious dermatoses) (1hr) • Microbial causes of skin infection and muscle (1hr) <u>Integrated Basic science lab (3hours)</u> • Microscopic structure of the skin • Dissection of the skin • Histology and pathology slides Day Three • <u>Interactive lecture (2hours)</u> • Dermatologic pharmacology(2hr) <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Examination of the integumentary and musculo-skeletal system (history taking and physical examination) • Computer simulations showing the different skin structures Day Four <u>Interactive lecture (SPH) (2hours)</u> <u>Interactive lecture (2hours)</u> • Pathology of common diseases of muscle (muscle atrophy, dystrophy, myopathy, disease of the neuromuscular junction, skeletal muscle tumour) (2hr) <u>Integrated basic science Lab (3hours)</u> • Previous integrated lab session will be continued • Histology of muscle tissue	

	Day Five <u>Hospital visit, Primary health care or community visit (4hours)</u> <u>Hospital visit</u> • General OPD, Medical OPD, Pediatrics OPD, surgical OPD, Dermatology clinic <u>Primary health care or community visit</u> • General OPD • Home visits trying to identify diseases of the skin, their risk factors and determinants, local perceptions of causes, local remedies and provide advice on promotion and prevention • Home visits to meet with traditional healer <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
Week 2	Day One <u>Interactive lecture (2hours)</u> • Histology of muscle tissue (1hr) • Physiology of muscle (neuromuscular transmission, excitation contraction coupling and contraction of muscle) (1hr) <u>Problem-based learning 2nd session (2hours)</u> • A case of skin lesion <u>Integrated Basic science lab (3hours)</u> • KOH fungal staining Day Two <u>Interactive Lecture (4hours)</u> • Pharmacology of agents acting at the neuromuscular junction (1.5hr) • Muscle metabolism and exercise (1hr) • Drugs acting on muscle spasm disorder (1.5hr) <u>Integrated Basic science lab (3hours)</u> • Anatomic atlas showing muscle structure • Histological and pathology slides and specimens • Dissection of the muscle Day Three <u>Interactive lecture (4hours)</u> • Gross and microscopic structure and functions of bone(1hr) • Calcium and bone metabolism (1hr) • Pathology of common diseases of the bone (congenital diseases of the bone, acquired diseases of bone development, fracture, bone tumour)... (2hr) <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u>	

	• Simulated practice: Approach to a patient with muscle wasting (history taking and physical examination) • Computer simulation to study muscles origin, insertion and action Day Four <u>Interactive lecture (SPH)(2hours)</u> <u>Interactive lecture (2hours)</u> • Gross and microscopic structure and functions of joints and supporting structure (1hr) • Microbial causes of bone infection and arthritis (1hr) <u>Integrated basic science Lab (3hours)</u> • Previous integrated lab session will be continued Day Five <u>Hospital visit, Primary health care or community visit (4hours)</u> <u>Hospital visit</u> • General OPD, Medical OPD, Paediatrics OPD, Surgical OPD <u>Primary health care or community visit</u> • General OPD • Home visits trying to identify diseases of muscle, their risk factors and determinants, home remedies and provide advice on promotion of health and prevention of disease <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
Week 3	Day One <u>Interactive lecture (2hours)</u> • Pathology of common diseases of the bone (congenital diseases of the bone, acquired diseases of bone development, fracture, bone tumour)... (2hr) <u>Problem-based learning 1st session (2hours)</u> • A case of fracture <u>Integrated Basic science lab (3hours)</u> • Synovial fluid analysis • Rheumatoid factor detection Day Two <u>Interactive Lecture (4hours)</u> • Pathology of the common diseases of joints (arthritis, joint tumour)(2hr) • Pharmacology of agents affecting the bone (1hr) • Pharmacology of agents used for the treatment of common joint problems(1hr) <u>Integrated Basic science lab (3hours)</u> • Anatomical charts and models of bones • Dissection of Axioappendicular muscles, axilla and arm	

	• Histology and pathology specimens and slides Day Three <u>Interactive lecture (2hours)</u> • Anatomy of upper limb ○ Introduction to the upper limb ○ Bones of the upper limb ○ Superficial structures of upper limb <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Approach to a patient with disease of bone growth (history taking and physical examination) • Computer simulations of bone growth and healing Day Four <u>Interactive lecture (SPH)(2hr)</u> <u>Interactive lecture (2hr)</u> • Anatomy of upper limb (continue) ○ Cubital fossa ○ Forearm <u>Integrated basic science Lab (3hours)</u> • Previous integrated lab session will be continued • Dissection of fore arm Day Five <u>Hospital visit , Primary health care or community visit (4hours)</u> <u>Hospital visit</u> General OPD, Medical OPD, Paediatrics OPD, Surgical OPD <u>Primary health care or community visit</u> • General OPD • Home visits trying to identify diseases of the bone including fracture, home remedies used and provide advice on promotion of health and prevention of disease • Home visits to meet with bone setters <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
Week 4	Day One <u>Interactive lecture (2hours)</u> • Anatomy of upper limb (continue) ○ Axio - appendicular Muscles ○ Axilla ○ Brachial plexus ○ Arm <u>Problem-based learning 2nd session (2hrs)</u> ➤ A case of fracture	

	<u>Integrated Basic science lab (3hours)</u> • Dissection of hand Day Two <u>Interactive Lecture (4hours)</u> • Hand • Surface anatomy the upper limb • Joints of the upper limb • Anatomy of Lower limb ○ Bones of the lower limb ○ Superficial structures of the lower limb <u>Integrated Basic science lab (3hours)</u> • Anatomical models and charts of joints • Dissection of anterior and medial thigh, anterior leg and dorsum of foot Day Three <u>Interactive lecture (2hours)</u> • Thigh and gluteal region • Lumbar and Sacral plexuses • Popliteal fossa • Leg <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Approach to a patient with rheumatism (history taking and physical examination) • Computer simulations of major joints Day Four <u>Interactive lecture (SPH)(2hours)</u> <u>Interactive lecture (2hours)</u> • Anatomy of Lower limb (continue) ○ Foot ○ Joints of lower limb <u>Integrated basic science Lab (3hours)</u> • Previous integrated lab session will be continued • Dissection of gluteal region, posterior thigh, posterior leg and sole of the foot Day Five <u>Hospital visit</u> • General OPD, Medical OPD, Paediatrics OPD, surgical OPD <u>Primary health care or community visit</u> • General OPD • Home visits trying to identify diseases of joints, their risk factors and determinants, perceived causes, local remedies	
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	and provide advice on promotion of oral health and prevention of disease • Home visits to meet with traditional healers <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
Week 5	Day One Arches of foot • Development of the skeletal system • Development of the muscular system • Development of limbs Day two • Review of the musculoskeletal and integumentary system • Discussion (2hrs) • Basic science lab (3hrs) • Clinical skills lab (3hrs)	
Week 6	• Day Two: Written exam Day Three: OSCE Day Four: Interactive lecture (SPH)(2hrs) Day Five:	

Time Table

Time	Monday	Tuesday	Wednesday	Thursday	Friday
8-10 am	Interactive Lecture	Interactive Lecture	Interactive Lecture	Interactive Lecture	Primary Health Care/ Community Visit
10-12 am	PBL	Interactive Lecture	-----	Interactive Lecture	
1:30-5:30pm	Biomedical lab	Biomedical lab	Clinical skills lab	Biomedical lab	Whole Group Session

PRE- CLERKSHIP II MODULES

MODULE TITLE: GASTROINTESTINAL MODULE

MODULE CODE: GAST 3071

NOMINAL DURATION: 7 WEEKS

Module Description: This system-based module is intended to equip medical students with profound knowledge of basic sciences and development of early clinical skills relevant to understanding of the gastrointestinal system and related clinical problems. Students will also develop relevant competencies with the Medical expertise, Scholar, Collaborator, Communicator, Manager, and Health Advocate Professional. This module is given on the second semester during the second year of the pre-clerkship period of the undergraduate medical education program.

MODULE DESIGN

This module is designed by integrating normal and abnormal structure and function of the gastrointestinal system. Building on in-depth understanding of the gastrointestinal system as a foundation, this module aims to provide the analytical and cognitive skills necessary for a successful transition from basic gastrointestinal structure, function and pathology to the care of patients with diseases of gastrointestinal system.

The Professional competency development component will support the student in acquiring a practical understanding of how to approach a patient with known or suspected disease of gastrointestinal system and develop the basic skill of examining the gastrointestinal system. Social and Population Health (SPH) module will run in parallel and will be made to correlate with the subject being taught.

Prerequisite

Successful completion of Introduction to Medicine Module.

Module objective

By the end of this module, students will be able to apply knowledge of basic sciences for understanding the gastrointestinal system and analyzing disorder of the gastrointestinal system.

Supporting objectives

As a Medical Expert

- Apply knowledge of the normal structures and functions of the gastrointestinal system in analyzing clinical problems
- Apply principles of physiologic mechanisms and processes of the gastrointestinal system in analyzing clinical problems
- Apply principles of human life cycle, growth, development and aging of the gastrointestinal system in analyzing clinical problems including developmental anomalies
- Identify the various causes of a disease (genetic, developmental, metabolic, toxic, infectious, autoimmune, neoplastic and degenerative) of the gastrointestinal system and analyze the pathogenesis in clinical problems

- Analyze the pharmacokinetics and pharmacodynamics of drugs used in the treatment of disorders of the gastrointestinal system
- Apply important determinants and risk factors affecting gastrointestinal diseases to protect and promote individual and population health
- Demonstrate taking history from the patient with gastrointestinal system disorder in a simulated setting.
- Demonstrate physical examination pertinent to with gastrointestinal system disorder in a simulated setting.
- Perform basic investigations in the diagnosis of gastrointestinal system disorders
- Demonstrate making an initial assessment of a clinical problem related to gastrointestinal system in a simulated setting.
- Discuss important life style, genetic, demographic, environment, health system, social, economic, psychological and cultural determinants of gastrointestinal system disorders.
- Contribute to a culture that promotes patient safety

As a Communicator

- Demonstrate communication with a patient with gastrointestinal disorder by using a patient-centered approach that encourages patient trust and autonomy and is characterized by empathy, respect, and compassion
- Optimize the physical environment for patient comfort, dignity, privacy, engagement, and safety
- Recognize when the values, biases, or perspectives of patients, physicians, or other health care professionals may have an impact on the quality of care, and modify the approach to the patient accordingly
- Respond to patients' non-verbal communication and use appropriate non-verbal behaviours to enhance communication with patients
- Manage disagreements and emotionally charged conversations
- Use patient-centered interviewing skills to effectively gather relevant biomedical and psychosocial information
- Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information
- Provide a clear structure for and manage the flow of an entire patient encounter
- Seek and synthesize relevant information from other sources, including the patient's family, with the patient's consent.
- Skilfully share information and explanations that are clear, accurate, timely, and adapted to patient's and his or her family's level of understanding and need in a simulated setting

As a Collaborator

- Establish and maintain healthy relationships with physicians and other colleagues in the health care professions to support relationship-centered collaborative care
- Recognize one's own professional roles and limits
- Share knowledge and information to other colleagues
- Show respect toward other physicians and members of the health care team
- Respect and value diversity among other colleagues

As a Manager

- Develop their leadership skills
- Design and organize elements of health care delivery

- Set priorities and manage time to balance practice and personal life
- Promote occupational health as physicians for sustainable practice

As a Health Advocate

- Work with patients with gastrointestinal system disorders and their families to address the determinants of health that affect them and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors
- Incorporate gastrointestinal system disease prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of gastrointestinal system health that affect them
- Participate in a process to improve gastrointestinal system health in the community or population they serve

As a Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize knowledge gaps in biomedical, clinical and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Integrate other intrinsic roles into their learning and practice of medicine

- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in practice
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Duration: 7 weeks

Teaching and learning methods

- Interactive lecture
- Assignment
- Discussion and debate
- Case studies
- Role play
- Clinical simulation
- Demonstration
- Video show
- Observation in clinical setting
- Community visit
- Personal research and reflection exercise/ Portfolio
- Whole group session

Educational Resources

- Basic sciences lab
- Clinical skills lab
- Health Facility
- Videos of patient examination
- Anatomical charts and mannequins
- Histological and pathological slides and specimens
- Computer-based simulations

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- Lynn S. Bickley, Peter G. Szilagyi. Bates' guide to physical examination and history taking. 10th ed. 2009
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- Courtney M. Townsend Jr. [et al.]. Sabiston textbook of surgery: the biological basis of modern surgical practice. 19th edition 2012

Learners Guide

- Handout
- Books
- Text books
- E-books

Learning Facilities / Infrastructure

- Class rooms
- Skill lab
- Dissection Room
- ICT centre
- Clinical set up
- Library

Portable and stationary plants, machines and equipment

- Flip charts
- Video cassette
- CD-ROM
- Computers
- DVD Player
- LCD projector Overhead Projector
- White board

Methods of assessment

Progressive

- Written progress /quiz/-15%
- Clinical skill lab- 5%
- Bio-lab- 15%
- Hospital visit-5%
- PBL-10%

END module

- Written -30%
- OCSE/OSPE- 20%

Module Schedule

Week	Activity	Required reading/ Assignment
Week one	Day One <u>Interactive lecture (4 hours)</u> • Introduction to the module(30min.) • Introduction to the digestive system • Overview of the abdomen • Development of the digestive system (1hr) <u>Problem-based learning 1st session (2hours)</u> • A case of diarrhea <u>Integrated Basic science lab (3hours)</u> • Anatomical charts and models Day Two <u>Interactive Lecture (4hours)</u> • Anterolateral abdominal wall • Gross and microscopic structure of the oral cavity, salivary glands, oropharynx and esophagus <u>Integrated Basic science lab (3hours)</u> • Anatomical charts and models • Dissection of oral cavity and esophagus • Dissection of anterolateral abdominal wall • Tissue slides, pathology slides and specimens • Computer-simulations of mastication and deglutition process Day Three <u>Interactive lecture (4 hours)</u> • Peritoneum and peritoneal cavity (2hrs) • Regulation of GIT functions (hormonal and neuronal)(2hrs) <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Examination of the oral and GI system (history taking and physical examination) Day Four <u>Interactive lecture (4hours)</u> • Functions of the oral cavity and esophagus(2hr) • Microbes and parasites affecting the oral cavity and esophagus) (2hr) <u>Interactive lecture SPH (2hours)</u> <u>Integrated basic science Lab (3hours)</u> • Dissection of peritoneum, peritoneal cavity • Previous lab session continues	

	Day Five <u>Hospital visit, Primary health care or community visit (4hours)</u> • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD • Home visits trying to identify common diseases of oral cavity, their risk factors and determinants and local remedies, local remedies and provide advice on promotion of oral health and prevention of disease • Home visits to meet with traditional tooth extractors <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
Week 2	Day One <u>Interactive lecture (3hours)</u> • Pathophysiology of common diseases of the oral cavity and esophagus <u>Problem-based learning 2nd session (2hours)</u> • Previous PLB session will be continued <u>Integrated Basic science lab (3hours)</u> • Laboratory diagnosis of H. pylori test Day Two <u>Interactive Lecture (4hours)</u> • Gross and microscopic structure of the stomach, small intestine and pancreas <u>Integrated Basic science lab (3hours)</u> • Anatomical charts and models • Dissection of stomach, duodenum and pancreases • Tissue slides, pathology slides and specimens Day Three <u>Interactive lecture (3hours)</u> • Gross and microscopic structure of the stomach, small intestine and pancreas (3hrs) <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Approach to a patient with dysphagia (history taking and physical examination) Day Four <u>Interactive lecture (3hours)</u> • Movements and secretions of stomach and small intestine(3hrs) <u>Interactive lecture – SPH (2hours)</u> <u>Integrated basic science Lab (3hours)</u> • Previous lab session continues • Dissection of Small intestine Day Five	

	<u>Hospital visit, Primary health care or community visit (4hours)</u> • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD • Home visits trying to identify common diseases of esophagus, stomach and small intestine; their risk factors and determinants and local remedies and provide advice on promotion of health and prevention of disease • Home visits to meet with traditional tooth extractors <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
Week 3	Day One <u>Interactive lecture (4hours)</u> • Gross and microscopic structures of the liver and hepatobiliary system <u>Problem-based learning 1st session (2hours)</u> • A case of jaundice <u>Integrated basic science Lab (3hours)</u> • Tissue slides, pathology slides and specimens Day Two <u>Interactive Lecture (4hours)</u> • Pancreatic digestive enzymes • Functions of the Biliary Tree • Bile Secretion by Liver <u>Integrated Basic science lab (3hours)</u> • Anatomical charts and models • Dissection of liver and biliary tract • Tissue slides, pathology slides and specimens • Computer-simulations of food digestion and absorption in the stomach and small intestine and hormonal and neural control mechanisms Day Three <u>Interactive lecture (4hours)</u> • Pathophysiology of common diseases of the stomach <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Approach to a patient with epigastric pain (history taking and physical examination) • Simulated practice: Nasogastric tube insertion Day Four <u>Interactive lecture (4hours)</u> • Microbial causes of stomach, and small intestine ○ Microbial cause of Diarrheal ○ Microbial cause of food born ○ Opportunistic causes of soil transmitted	

	<u>Interactive lecture SPH (2hours)</u> <u>Integrated basic science Lab (2hours)</u> • Widal test, stool culture and antibiotic susceptibility • Modified Zeihl nelson and stool examination for opportunistic soil transmitted helminthes • Previous lab session continues Day Five <u>Hospital visit , Primary health care or community visit (4hours)</u> • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD • Home visits trying to identify common diseases of esophagus, stomach ,small intestine and pancreas ; their risk factors and determinants and local remedies and provide advice on promotion of health and prevention of disease • Home visits to meet with traditional tooth extractors <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
Week 4	Day One <u>Interactive lecture (3hours)</u> • Digestion and absorption in the GIT (3hrs) <u>Problem-based learning 2nd session (2hours)</u> • Previous PLB session will be continued <u>Integrated Basic science lab (3hours)</u> • Anatomical charts and models • Histology of the liver and hepatobiliary system • Gamma glutamate transferase Day Two <u>Interactive Lecture (4hours)</u> • Gross and microscopic structures and functions of the large intestine (3hrs) <u>Integrated Basic science lab (3hours)</u> • Anatomical charts and models • Tissue slides and specimens • Histology of large intestine Day Three <u>Interactive lecture (3hours)</u> • Movement and absorption in the large intestine and defecation reflex <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u>	

	- Simulated practice: Approach to a patient with abdominal distension (history taking and physical examination) Day Four <u>Interactive lecture (4hours)</u> - Pathophysiology of small and large intestine <u>Interactive lecture – SPH (2hours)</u> <u>Integrated basic science Lab (3hours)</u> - Pathology slides and specimens - Previous lab session continues Day Five <u>Hospital visit, Primary health care or community visit (4hours)</u> - General OPD, Medical OPD, Pediatrics OPD, Surgical OPD - Home visits trying to identify diseases of the lower GI, their risk factors and determinants, local remedies and provide advice on promotion of health and prevention of disease <u>Whole group session (3hours)</u> - Faculty facilitated discussion and reflection on the week's learning activities	
Week 5	Day One <u>Interactive lecture (4hours)</u> Microbial infection of liver - Hepatitis - EBV - CMV <u>Problem-based learning 1st session (2hours)</u> - A case of failure to pass stool <u>Integrated Basic science lab (3hours)</u> - Serologic tests for hepatitis Day Two <u>Interactive Lecture (4hours)</u> - Pathophysiology of common diseases of the liver and hepatobiliary system (4hr) <u>Integrated Basic science lab (3hours)</u> - Dissection of the large intestine - Tissue slides, pathology slides and specimens Day Three <u>Interactive lecture (4hours)</u> - Pharmacology of drugs used for treatment of GERDs and PUD <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u>	

	- Simulated practice: Approach to a patient with jaundice (history taking and physical examination) Day Four <u>Interactive lecture (3hours)</u> - Digestion and absorption of nucleotides (1hr) - Integrated organs metabolism (2hr) <u>Interactive lecture – SPH (2hours)</u> <u>Integrated basic science Lab (2hours)</u> - Previous lab session continues - Computer simulation of integrated metabolism Day Five <u>Hospital visit, Primary health care or community visit (4hours)</u> - General OPD, Medical OPD, Pediatrics OPD, Surgical OPD - Home visits trying to identify diseases of the hepatobiliary system, their risk factors and determinants, local remedies and provide advice on promotion of health and prevention of disease <u>Whole group session (3hours)</u> - Faculty facilitated discussion and reflection on the week's learning activities	
Week 6	Day One <u>Interactive lecture (3hours)</u> - Pathophysiology of common bile duct and pancreas <u>Problem-based learning 2nd session (2hours)</u> - Previous PLB session will be continued <u>Integrated Basic science lab (3hours)</u> - Pathology slides and specimens Day Two <u>Interactive Lecture (4hours)</u> - Microbes and parasites affecting the large intestine (4hr) <u>Integrated Basic science lab (3hours)</u> - Tissue slides, pathology slides and specimens - Computer-simulations of water absorption in the large intestine - Colonoscopy, sigmoidoscopy Day Three <u>Interactive lecture (3hours)</u> - Pharmacology of drugs used for treatment of Constipation, nausea and vomiting (3hr) <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> - Simulated practice: Approach to a patient intestinal obstruction (history taking and physical examination including rectal examination)	

	• Interpretation of basic results of investigations Day Four <u>Interactive lecture (3hours)</u> • Pharmacology of drugs used for treatment of diarrheal disease <u>Interactive lecture – SPH (2hours)</u> <u>Integrated basic science Lab (2hours)</u> • Ova/parasite identification of intestinal helminthes • Previous lab session continues Day Five <u>Hospital visit , Primary health care or community visit (4hours)</u> • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD • Home visits trying to identify diseases of large intestine, their risk factors and determinants and local remedies and provide advice on promotion of health and prevention of disease <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
Week 7	Exam week Day One: Review of the Gastro intestinal system • Discussion (2hrs) • Basic science lab (3rs) • Clinical skills lab (3hrs) Day Two: Written exam Day Three: OSCE Day Four: Study break for SPH II exam Day Five: SPH II exam Whole group session (3hrs)	

Time Table

Time	Monday	Tuesday	Wednesday	Thursday	Friday
8-10 am	Interactive Lecture	Interactive Lecture	Interactive Lecture	Interactive Lecture	Primary Health Care/Community Visit
10-12 am	PBL	Interactive Lecture	-----	Interactive Lecture	Community Visit
1:30-5:30pm	Biomedical lab	Biomedical lab	Clinical skills lab	Biomedical lab	Whole Group Session

MODULE TITLE: RENAL SYSTEM

MODULE CODE: RENA 3081

NOMINAL DURATION: 5 WEEKS

Module Description: This system-based module is intended to equip medical students with profound knowledge of basic sciences and development of early clinical skills relevant to understanding of the kidney and urinary system and related clinical problems. Students will also develop relevant competencies with the Medical expertise, Scholar, Collaborator, Communicator, Manager, Health Advocate, and Professional. This module is given on the first semester during the second year of the pre-clerkship period of the undergraduate medical education program.

Module Design

This module is designed by integrating normal and abnormal structure and function of the renal system. Building on in-depth understanding of the kidney and urinary system as a foundation, this module aims to provide the analytical and cognitive skills necessary for a successful transition from basic structure of kidney and urinary system, function and pathology to the care of patients with diseases of kidney and urinary system.

The Professional competency development component will support the student in acquiring a practical understanding of how to approach a patient with known or suspected kidney and urinary system and develop the basic skill of examining the kidney and urinary system disorders. Social and Population Health (SPH) module will run in parallel and will be made to correlate with the content being thought.

Prerequisite

Successful completion of the Introduction to Medicine Module

Module Objective

By the end of this module, students will be able to apply knowledge of basic sciences for understanding the kidney and urinary system and analyzing disorder of the kidney and urinary system.

Supporting objectives

As a Medical Expert

- Apply knowledge of the normal structures and functions of the kidney and urinary system in analyzing clinical problems
- Apply principles of physiologic mechanisms and processes of the kidney and urinary system in analyzing clinical problems
- Apply principles of human life cycle, growth, development and aging of the kidney and urinary system in analyzing clinical problems including developmental anomalies
- Identify the various causes of a disease (genetic, developmental, metabolic, toxic, infectious, autoimmune, neoplastic, degenerative and traumatic) of the kidney and urinary system and analyze the pathogenesis in clinical problems
- Analyze the pharmacokinetics and pharmacodynamics of drugs used in the treatment of disorders of the kidney and urinary system
- Apply important determinants and risk factors affecting kidney and urinary diseases to protect and promote individual and population health
- Demonstrate taking history from the patient with kidney and urinary system disorder in a simulated setting.
- Demonstrate physical examination pertinent to with kidney and urinary system disorder in a simulated setting.

- Perform basic investigations in the diagnosis of kidney and urinary system disorders
- Demonstrate making an initial assessment of a clinical problem related to kidney and urinary system in a simulated setting.
- Discuss important life style, genetic, demographic, environment, health system, social, economic, psychological and cultural determinants of kidney and urinary system disorders.
- Contribute to a culture that promotes patient safety

As a communicator

- Demonstrate communication with a patient with kidney and urinary system disorder disusing a patient-centered approach that encourages patient trust and autonomy and is characterized by empathy, respect, and compassion
- Optimize the physical environment for patient comfort, dignity, privacy, engagement, and safety
- Recognize when the values, biases, or perspectives of patients, physicians, or other health care professionals may have an impact on the quality of care, and modify the approach to the patient accordingly
- Respond to patients' non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients
- Manage disagreements and emotionally charged conversations
- Use patient-centered interviewing skills to effectively gather relevant biomedical and psychosocial information
- Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information
- Provide a clear structure for and manage the flow of an entire patient encounter
- Seek and synthesize relevant information from other sources, including the patient's family, with the patient's consent.
- Skillfully share information and explanations that are clear, accurate, timely, and adapted to patient's and his or her family's level of understanding and need in a simulated setting

As a Collaborator

- Establish and maintain healthy relationships with physicians and other colleagues in the health care professions to support relationship-centered collaborative care
- Recognize one's own professional roles and limits
- Share knowledge and information to other colleagues
- Show respect toward other physicians and members of the health care team
- Respect and value diversity among other colleagues

As a Manager

- Develop their leadership skills
- Design and organize elements of health care delivery
- Set priorities and manage time to balance practice and personal life
- Promote occupational health as physicians for sustainable practice

As a Health Advocate

- Work with patients with kidney and urinary system disorders and their families to address the determinants of health that affect them and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors

- Incorporate kidney and urinary system disease prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of kidney and urinary system health that affect them
- Participate in a process to improve kidney and urinary system health in the community or population they serve

As a Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize knowledge gaps in biomedical, clinical and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Integrate other intrinsic roles into their learning and practice of medicine
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in practice
- Exhibit professional behaviors in the use of technology enabled communication

- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Duration: 5 weeks

Teaching and learning methods

- Interactive lecture
- Assignment
- Discussion and debate
- Case studies
- Role play
- Clinical simulation
- Demonstration
- Video show
- Observation in clinical setting
- Community visit
- Personal research and reflection exercise/ Portfolio
- Whole group session

Educational Resources

- Basic sciences lab
- Clinical skills lab
- Health Facility
- Videos of patient examination
- Anatomical charts and mannequins
- Histological and pathological slides and specimens
- Computer-based simulations

References

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- Goldman. Cecil Medicine. 23rd edition. 2007
- Courtney M. Townsend Jr, et al. Sabiston textbook of surgery: the biological basis of modern surgical practice. 19th edition 2012

Learners Guide

- Handout
- Books
- Text books
- E-books

Learning Facilities / Infrastructure

- Class rooms
- Skill lab
- Dissection Room
- ICT center
- Clinical set up
- Library

Portable and stationary plants, machines and equipment

- Flip charts
- Video cassette
- CD-ROM
- Computers
- DVD Player
- LCD projector Overhead Projector

- White board

Methods of assessment

Progressive

- Written progress /quiz/-15%
- Clinical skill lab- 5%
- Bio-lab- 15%
- Hospital visit-5%
- PBL-10%

END module

- Written -30%
- OCSE/OSPE- 20%

Module Schedule

Week	Activity	Required assignment
Week One	Day One <u>Interactive lecture (2hours)</u> • Introduction to the module (30minute) • Introduction to the urinary system <u>PBL case- 1st session (2hours)</u> • A case centralized obesity <u>Integrated lab basic science lab(3hours)</u> • Atlas (anatomy, histology) or charts showing gross and microscopic structure of the renal and urinary system Day Two <u>Interactive lecture (4hours)</u> • Gross and microscopic structure and functions of the renal system(2hrs) • Functions of the kidneys and Urine formation by kidneys–glomerular filtration, renal blood flow (2hrs) <u>Integrated lab basic science lab (3hours)</u> • Dissection of the kidney, ureter and bladder • Histology and pathology slides and specimens Day Three: <u>Interactive lecture (2hours)</u> • Development of the urinary system • Posterior abdominal wall <u>Professional competency development (4hours)</u> • Simulated practice: Approach to a patient with renal failure (history taking and physical examination) Day Four <u>Interactive lecture (2hours)</u> • Tubular processing of glomerular filtrate	

	• Reabsorption and Secretion in the Proximal Tubule • Reabsorption and Secretion in the Loop of Henle , Distal Nephron and collecting duct <u>Interactive lecture (SPH) (2hours)</u> <u>Integrated basic science Lab (3hours)</u> • Previous integrated lab session will be continued • Dissection of posterior abdominal wall and retroperitoneal structures Day Five • Primary health care or community visit (4hrs) <u>Hospital visit</u> • General OPD, Medical OPD, Pediatrics OPD, surgical OPD, Renal follow up clinic <u>Primary health care or community visit</u> • General OPD • Community visits – home visits to trying to find patients with renal and urinary problems, identify risk factors and determinants, traditional remedies and provide advice Whole group session (3hrs) • Faculty facilitated discussion and reflection on the week’s learning activities	
Week Two	Day One <u>Interactive lecture (2hours)</u> • Urine concentration and dilution • Countercurrent mechanisms <u>Problem-based learning 2nd session (2hours)</u> • Previous PLB session will be continued <u>Integrated basic science lab (3hours)</u> • Qualitative and quantitative analysis of urine Day Two <u>Interactive lecture (4hours)</u> • Biochemical tests for renal function (2hrs) • Role of kidney in regulating extracellular fluid osmolality and sodium concentration (2hrs) <u>Integrated basic science lab (3hours)</u> • Video showing micturition cycle (filling phase and storage phase, initiation phase and voiding phases) and urodynamic assessment • Computer simulation of urine formation Day Three <u>Interactive lecture (2hours)</u> • Renal regulation of potassium, calcium, phosphate and magnesium	

	• Micturation reflex <u>Professional competency development (4hours)</u> • Simulated practice: Approach to a patient with renal/ureteric colic (history taking and physical examination) Day Four <u>Interactive lecture (2hours)</u> • Regulation of acid-base balance <u>Interactive lecture (SPH) (2hours)</u> <u>Integrated basic science Lab (3hours)</u> • Computer simulation of diuretic agents Day Five <u>Primary health care or community visit (4hours)</u> <u>Hospital visit</u> • General OPD, Medical OPD, Pediatrics OPD, surgical OPD, Renal follow up clinic <u>Primary health care or community visit</u> • General OPD • Community: home visits to trying to find patients with renal and urinary problems, identify risk factors and determinants, traditional remedies and provide advice <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
Week Three	Day One <u>Interactive lecture (2hours)</u> • Pathophysiology of common kidney diseases (glomerular diseases, diseases affecting the tubules and interstitium) <u>Day one PBL case- first session (2hours) :</u> • A case of polyuria <u>Integrated Basic science lab (3hours)</u> • Tissue slides, pathology slides and specimens Day Two <u>Interactive lecture (4hours)</u> • Pathophysiology of common kidney diseases (glomerular diseases, diseases affecting the tubules and interstitium) • Pathophysiology of diseases of the blood vessels • Pathophysiology of cystic diseases of the kidney • Pathophysiology of tumors <u>Integrated Basic science lab (3hours)</u> • Renal function test	

	• Blood urea, serum creatinine, urinary protein Day Three <u>Interactive lecture (2hours)</u> • Pathophysiology of urinary outflow obstruction • Metabolic disorders of the urinary System(AA metabolism and related metabolism) <u>Professional competency development(4hours)</u> • Demonstration and practice of catheterization in a simulated setting Day Four <u>Interactive lecture (2hours)</u> Pathophysiology of renal failure. <u>Interactive lecture (SPH)(2hours)</u> <u>Integrated Basic science lab (3hours)</u> • Previous integrated lab session will be continued • Tissue slides, pathology slides and specimens Day Five <u>Primary health care or community visit (4hour)</u> <u>Hospital visit</u> • General OPD, Medical OPD, Pediatrics OPD, surgical OPD, Renal follow up clinic <u>Primary health care or community visit</u> • General OPD • Community: home visits to trying to find patients with renal and urinary problems, identify risk factors and determinants, traditional remedies and provide advice <u>Whole group session (3 hour)</u> Faculty facilitated discussion and reflection on the week's learning activities	
Week Four	Day One <u>Interactive lecture (2 hours)</u> • Microbial causes of of the kidney and urinary system infection (urethritis,pyelonephritis and cystitis) <u>Problem-based learning 2nd session (2hours)</u> • Previous PLB session will be continued <u>Integrated Basic science lab (3hours)</u> • Physical,chemical and microscopic analysis of Urine Day Two <u>Interactive lecture (2hours)</u> • Pharmacology of drugs used for treatment of urinary tract infections • Microbial causes of the kidney and urinary system infection (urethritis,pyelonephritis and cystitis)	

	<u>Integrated Basic science lab (3hours)</u> • Computer simulation of renal clearance Day Three <u>Interactive lecture (2hours)</u> • Pharmacology of drugs used for treatment of renal failure <u>Professional competency development (4hours)</u> • Simulated practice: Approach to a patient with renal/urinary infection (history taking and physical examination) Day Four <u>Interactive lecture (2hours)</u> • Pharmacology of diuretic agents <u>Interactive lecture (SPH)(2hours)</u> <u>Integrated basic science Lab (3hours)</u> • Urine culturing and antibiotic susceptibility test Day Five <u>Primary health care or community visit (4hours)</u> <u>Hospital visit</u> • General OPD, Medical OPD, Pediatrics OPD, surgical OPD, Renal follow up clinic <u>Primary health care or community visit</u> • General OPD • Community: home visits to trying to find patients with renal and urinary problems, identify risk factors and determinants, traditional remedies and provide advice <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
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Week Five	Day One - Review of the urinary system - Discussion (2hrs) - Basic science lab (3hrs) - Clinical skills lab (3hrs) Day Two: Written exam Day Three: OSCE Day Four: Interactive lecture (SPH) (2hrs) Day Five: Primary health care or community visit (4hrs) Whole group session (3hrs)	
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Time Table

Time	Monday	Tuesday	Wednesday	Thursday	Friday
8 -10 am	Interactive Lecture	Interactive Lecture	Interactive Lecture	Interactive Lecture	Primary Health Care/ Community Visit
10-12 am	PBL	Interactive Lecture		Interactive Lecture	
1:30- 5:30pm	Biomedical lab	Biomedical lab	Clinical skills lab	Biomedical lab	Whole Group Session

MODULE TITLE: ENDOCRINE SYSTEM

MODULE CODE: ENDO 3091

Nominal duration: 5wks

Module Description: This system-based module is intended to equip medical students with profound knowledge of basic sciences and development of early clinical skills relevant to understanding of the endocrine system and related clinical problems. Students will also develop relevant competencies with the Medical expertise, Scholar, Collaborator, Communicator, Manager, and Health Advocate Professional. This module is given on the first semester during the second year of the pre-clerkship period of the undergraduate medical education program.

Module design

This module is designed by integrating normal and abnormal structure and function of the endocrine system. Building on in-depth understanding of the endocrine system as a foundation, this module aims to provide the analytical and cognitive skills necessary for a successful transition from basic endocrine system structure, function and pathology to the care of patients with endocrine system disease.

The Professional competency development component will support the student in acquiring a practical understanding of how to approach a patient with known or suspected endocrine system disease and develop the basic skill of examining patients with clinical problem of the endocrine system. Social and Population Health (SPH) module will run in parallel and will be made to correlate with the content being taught.

Pre-requisite

Successful completion of the Introduction to Medicine Module

Module Objective

By the end of this module, students will be able to apply knowledge of basic sciences for understanding and analyzing disorder of the endocrine system.

Supporting Objectives**As a Medical Expert**

- Apply knowledge of the normal structures and functions of the endocrine system in analyzing clinical problems
- Apply principles of physiologic mechanisms and processes of the endocrine system in analyzing clinical problems
- Apply principles of human life cycle, growth, development and aging of the endocrine system in analyzing clinical problems including developmental anomalies
- Identify the various causes of a disease (genetic, developmental, metabolic, toxic, infectious, autoimmune, neoplastic and degenerative) of the endocrine system and analyze the pathogenesis in clinical problems
- Analyze the pharmacokinetics and pharmacodynamics of drugs used in the treatment of disorders of the endocrine system
- Apply important determinants and risk factors affecting endocrine diseases to protect and promote individual and population health
- Demonstrate taking history from the patient with endocrine system disorder in a simulated setting
- Demonstrate physical examination pertinent to with endocrine system disorder in a simulated setting
- Perform basic investigations in the diagnosis of endocrine system disorders
- Demonstrate making an initial assessment of a clinical problem related to endocrine system in a simulated setting
- Discuss important life style, genetic, demographic, environment, health system, social, economic, psychological and cultural determinants of endocrine system disorders.
- Contribute to a culture that promotes patient safety

As a communicator

- Demonstrate communication with a patient with endocrine system disorder by using a patient-centered approach that encourages patient trust and autonomy and is characterized by empathy, respect, and compassion
- Optimize the physical environment for patient comfort, dignity, privacy, engagement, and safety
- Recognize when the values, biases, or perspectives of patients, physicians, or other health care professionals may have an impact on the quality of care, and modify the approach to the patient accordingly
- Respond to patients' non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients
- Manage disagreements and emotionally charged conversations
- Use patient-centered interviewing skills to effectively gather relevant biomedical and psychosocial information
- Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information

- Provide a clear structure for and manage the flow of an entire patient encounter
- Seek and synthesize relevant information from other sources, including the patient's family, with the patient's consent
- Skilfully share information and explanations that are clear, accurate, timely, and adapted to patient's and his or her family's level of understanding and need in a simulated setting

As a Collaborator

- Establish and maintain healthy relationships with physicians and other colleagues in the health care professions to support relationship-centered collaborative care
- Recognize one's own professional roles and limits
- Share knowledge and information to other colleagues
- Show respect toward other physicians and members of the health care team
- Respect and value diversity among other colleagues

As a Manager

- Develop their leadership skills
- Design and organize elements of health care delivery
- Set priorities and manage time to balance practice and personal life
- Promote occupational health as physicians for sustainable practice

As a Health Advocate

- Work with patients with endocrine system disorders and their families to address the determinants of health that affect them and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors
- Incorporate endocrine system disease prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of endocrine system health that affect them
- Participate in a process to improve endocrine system health in the community or population they serve

As a Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize knowledge gaps in biomedical, clinical and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources

- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Integrate other intrinsic roles into their learning and practice of medicine
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
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- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Duration: 5 weeks

Teaching and learning methods

- Interactive lecture
- Assignment
- Discussion and debate
- Case studies
- Role play
- Clinical simulation
- Demonstration
- Video show
- Observation in clinical setting
- Community visit
- Personal research and reflection exercise/ Portfolio
- Whole group session

Educational Resources

- Basic sciences lab
- Clinical skills lab
- Health Facility

- Videos of patient examination
- Anatomical charts and mannequins
- Histological and pathological slides and specimens
- Computer-based simulations

References

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- Skill lab
- Dissection Room
- ICT center
- Clinical set up
- Library

Portable and stationary plants, machines and equipment

- Flip charts
- Video cassette
- CD-ROM
- Computers
- DVD Player
- LCD projector Overhead Projector
- White board

Methods of assessment**Progressive**

- Written progress /quiz/-15%
- Clinical skill lab- 5%
- Bio-lab- 15%
- Hospital visit-5%
- PBL-10%

END module

- Written -30%
- OCSE/OSPE- 20%

Module Schedule

Week	Activity	Required assignment
Week One	Day One <u>Interactive lecture (2hours)</u> • Introduction to the module (30minute) • Introduction to the Endocrine system • Gross and microscopic structure and functions of the pituitary gland (1.5hr) <u>PBL case- 1st session (2hours)</u> A case of pituitary adenoma <u>Integrated basic science lab (3hours)</u> • Anatomical atlas or charts showing gross structure of the anterior and pituitary glands and thyroid glands • Dissection of pituitary gland Day Two <u>Interactive lecture (4hours)</u> • General principles of endocrinology • The Hypothalamic-Pituitary Control System <u>Integrated basic science lab (3hours)</u> • Computer simulation of the pituitary axis	

	- Histology and pathology slides of endocrine system Day Three: <u>Interactive lecture (2hours)</u> - Gross and microscopic anatomy of thyroid and parathyroid glands <u>Professional competency development (4hours)</u> - Simulated practice: Approach to a patient with anterior neck mass (history taking and physical examination) Day Four <u>Interactive lecture (2hours)</u> - Parathyroid Glands and Calcitonin - Calcium & phosphate metabolism <u>Interactive lecture (SPH)(2hours)</u> <u>Integrated basic science lab (3hours)</u> - Previous integrated lab session will be continued - Determination of Ca, calcitonin, PTH and phosphate. Day Five <u>Hospital visit</u> General OPD, Medical OPD, Pediatrics OPD, surgical OPD <u>Primary health care or community visit</u> - General OPD - Home visits trying to identify a woman with goiter, their risk factors and determinants, local remedies and provide advice on health promotion <u>Whole group session (3hours)</u> - Faculty facilitated discussion and reflection on the week's learning activities	
Week Two	Day One <u>Interactive lecture (2hours)</u> - Pharmacology of gonadotropins, GnRH receptor antagonist, prolactin, Vasopressin and Oxytocin drugs <u>Day one PBL case- first session (2hours) :</u> A case of Cushing's syndrome <u>Integrated basic science lab (3 hours)</u> - Anatomical charts and atlas of the thyroid, parathyroid glands and adrenal glands - Dissection of thyroid and parathyroid glands Day Two : <u>Interactive lecture (4hours)</u>	

	• Pathophysiology of the common diseases of the pituitary gland <u>Integrated basic science lab (3hours)</u> • Histology and pathology slides and specimens of thyroid and parathyroid gland • Hormone assay: TSH, T3 and T4 determination Day Three: <u>Interactive lecture (2hours)</u> • Pathophysiology of the common diseases of the thyroid and parathyroid glands(2hrs) <u>Professional competency development (4hours)</u> • Simulated practice: Approach to a patient with anterior neck mass (history taking and physical examination) Day Four <u>Interactive lecture (SPH)(2hours)</u> • Gross and microscopic structure and functions of the adrenal gland <u>Integrated basic science lab (3hours)</u> • Previous integrated lab session will be continued Day Five <u>Hospital visit</u> • General OPD, Medical OPD, Pediatrics OPD, surgical OPD <u>Primary health care or community visit</u> • General OPD • Home visits trying to identify a woman with goiter, their risk factors and determinants, local remedies and provide advice on health promotion <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
Week 3	Day One <u>Interactive lecture (2hours)</u> • Mechanism of action of hormones (signal transduction) • Intracellular Signal Transduction • Signaling via G-Protein-linked Cell-Surface Receptors • Signaling via Enzyme-linked Cell-Surface Receptors <u>Day one PBL case- first session (2hours) :</u> a case of diabetes mellitus <u>Integrated basic science lab (3hours)</u> • Determination of glycated Hb Day Two	

	<u>Interactive lecture (4hours)</u> • Pharmacology of thyroid and anti-thyroid drugs • Agents affecting mineral ion homeostasis • Pharmacology of drugs acting on adrenal cortex (Adrenocorticosteroids, synthetic corticosteroids, mineralocorticoids) <u>Integrated basic science lab (3hours)</u> • Computer simulations of blood glucose homeostasis • Testing blood glucose Day Three: <u>Interactive lecture (2hours)</u> • Histology of Endocrine Pancreas (1hr) • Blood glucose regulation(2hrs) <u>Professional competency development (4hours)</u> • Simulated practice: Approach to a patient with adrenal insufficiency (history taking and physical examination) Day Four <u>Interactive lecture (2hrs)</u> • Pathophysiology of endocrine pancreas and parathyroid • Metabolic derangements in diabetes mellitus (2hrs) <u>Interactive lecture (SPH) (2hours)</u> <u>Integrated basic science lab (3hours)</u> • Determination pituitary hormones (ACTH,TSH,GH) Day Five <u>Primary health care or community visit (4hour)</u> <u>Whole group session (3hours)</u> Faculty facilitated discussion and reflection on the week's learning activities	
Week 4	Day One <u>Interactive lecture (2hours)</u> • Anatomy and histology Pineal gland • Endocrine tissues in other organs • Development of the Endocrine system <u>Day one PBL case- first session (2hours)</u> a case of diabetes mellitus <u>Integrated basic science lab (3hours)</u> • Histology and pathology slides and specimens Day Two <u>Interactive lecture (4hours)</u> • Pharmacology of insulin and oral anti-diabetic agents <u>Integrated basic science lab (3hours)</u> • Computer simulations of blood glucose homeostasis • Testing blood glucose Day Three: <u>Interactive lecture (2hours)</u> • Pathophysiology of adrenal gland <u>Professional competency development (4hours)</u>	

	- Simulated practice: Approach to a patient with adrenal insufficiency (history taking and physical examination) Day Four <u>Interactive lecture (2hours)</u> - Physiology of sex hormones <u>Interactive lecture (SPH) (2hours)</u> <u>Integrated basic science lab (3hours)</u> - Previous integrated lab session will be continued - Determination of steroid hormones (Testosterone, estrogen and progesterone) Day Five <u>Primary health care or community visit (4hours)</u> <u>Whole group session (3hours)</u> - Faculty facilitated discussion and reflection on the week's learning activities	
Week 5	Day One - Review of the Endocrine system system - Discussion (2hrs) - Basic science lab (3hrs) - Clinical skills lab (3hrs) Day Two: Written exam Day Three: OSCE Day Four: Study break for SPH III exam Day Five: SPH III exam Whole group session(3hrs)	

Time Table

Time	Monday	Tuesday	Wednesday	Thursday	Friday
8 -10 am	Interactive Lecture	Interactive Lecture	Interactive Lecture	Interactive Lecture	Primary Health
10 -12 am	PBL	Interactive Lecture		Interactive Lecture	Care/Community Visit
1:30-:30pm	Biomedical lab	Biomedical lab	Clinical skills lab	Biomedical lab	Whole Group Session

MODULE TITLE: REPRODUCTIVE SYSTEM

MODULE CODE: REPR 3102

NOMINAL DURATION: 5 WEEKS

Module Description: This system-based module is intended to equip medical students with profound knowledge of basic sciences and development of early clinical skills relevant to

understanding of the reproductive system related clinical problems. Students will also develop relevant competencies with the Medical expertise, Scholar, Collaborator, Communicator, Manager, Health Advocate, and Professional. This module is given on the first semester during the second year of the pre-clerkship period of the undergraduate medical education program.

MODULE DESIGN

This module is designed by integrating normal and abnormal structure and function of the reproductive system. Building on in-depth understanding of the reproductive system as a foundation, this module aims to provide the analytical and cognitive skills necessary for a successful transition from basic reproductive structure, function and pathology to the care of patients with diseases of reproductive system.

The Professional competency development component will support the student in acquiring a practical understanding of how to approach a patient with known or suspected disease of reproductive system and develop the basic skill of examining the reproductive system. Social and Population Health (SPH) module will run in parallel and will be made to correlate with the subject being taught.

Prerequisite

Successful completion of the Introduction to Medicine Module

Module Objective

By the end of this module, students will be able to apply knowledge of basic sciences for understanding the reproductive system and analyzing disorder of the reproductive system.

Supporting objectives

As a Medical Expert

- Apply knowledge of the normal structures and functions of the reproductive system in analyzing clinical problems
- Apply principles of physiologic mechanisms and processes of the reproductive system in analyzing clinical problems
- Apply principles of human life cycle, growth, development and aging of the reproductive system in analyzing clinical problems including developmental anomalies
- Identify the various causes of a disease (genetic, developmental, metabolic, toxic, infectious, autoimmune, neoplastic, degenerative and traumatic) of reproductive system and analyze the pathogenesis in clinical problems
- Analyze the pharmacokinetics and pharmacodynamics of drugs used in the treatment of disorders of the reproductive system
- Apply important determinants and risk factors affecting reproductive system diseases to protect and promote individual and population health
- Demonstrate taking history from the patient with reproductive system disorder in a simulated setting
- Demonstrate physical examination pertinent to with reproductive system disorder in a simulated setting
- Perform basic investigations in the diagnosis of reproductive system disorders
- Demonstrate making an initial assessment of a clinical problem related to reproductive system in a simulated setting
- Discuss important life style, genetic, demographic, environment, health system, social, economic, psychological and cultural determinants of reproductive system disorders
- Contribute to a culture that promotes patient safety

As a communicator

- Demonstrate communication with a patient with reproductive system disorder disusing a patient-centered approach that encourages patient trust and autonomy and is characterized by empathy, respect, and compassion
- Optimize the physical environment for patient comfort, dignity, privacy, engagement, and safety
- Recognize when the values, biases, or perspectives of patients, physicians, or other health care professionals may have an impact on the quality of care, and modify the approach to the patient accordingly
- Respond to patients' non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients
- Manage disagreements and emotionally charged conversations
- Use patient-centered interviewing skills to effectively gather relevant biomedical and psychosocial information
- Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information
- Provide a clear structure for and manage the flow of an entire patient encounter
- Seek and synthesize relevant information from other sources, including the patient's family, with the patient's consent.
- Skillfully share information and explanations that are clear, accurate, timely, and adapted to patient's and his or her family's level of understanding and need in a simulated setting

As a Collaborator

- Establish and maintain healthy relationships with physicians and other colleagues in the health care professions to support relationship-centered collaborative care
- Recognize one's own professional roles and limits
- Share knowledge and information to other colleagues
- Show respect toward other physicians and members of the health care team
- Respect and value diversity among other colleagues

As a Manager

- Develop their leadership skills
- Design and organize elements of health care delivery
- Set priorities and manage time to balance practice and personal life
- Promote occupational health as physicians for sustainable practice

As a Health Advocate

- Work with patients with reproductive system disorders and their families to address the determinants of health that affect them and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors
- Incorporate reproductive system disease prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of reproductive system health that affect them
- Participate in a process to improve reproductive system health in the community or population they serve

As a Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice

- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize knowledge gaps in biomedical, clinical and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Integrate other intrinsic roles into their learning and practice of medicine
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in practice
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Duration: 5 weeks

Teaching and learning methods

- Interactive lecture
- Assignment

- Discussion and debate
- Case studies
- Role play
- Clinical simulation
- Demonstration
- Video show
- Observation in clinical setting
- Community visit
- Personal research and reflection exercise/ Portfolio
- Whole group session

Educational Resources

- Basic sciences lab
- Clinical skills lab
- Health Facility
- Videos of patient examination
- Anatomical charts and mannequins
- Histological and pathological slides and specimens
- Computer-based simulations

References

- Tortora G.J and B. Derrickson. Principles of Anatomy and physiology. 12th edition. 2008
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- Keith L. Moore, Arthur F. Dalley, Anne M.R. Agur. Clinically oriented anatomy. 6th edition. 2010
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- Junqueira's basic histology: text and atlas. 13th edition. 2013
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- Eric Arthur Newsholme and Tony R. Leech. Functional biochemistry in health and disease. 2010.
- Richard N Mitchell, Ramzi S. Cotran. Robbins basic pathology. 8th edition. 2007.
- David A Levison, Robin Reid, Alistair D Burt, David J Harrison and Stewart Fleming. Muir's Textbook of Pathology, Fourteenth Edition
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- Murray P., Rosenthal S., Foller M, Medical Microbiology, 8th edition, 2010
- Abhay R. Satoskar [et al. Medical parasitology. 2009
- Color atlas of parasitology.
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- Goldman. Cecil Medicine. 23rd edition. 2007
- Courtney M. Townsend Jr. [et al.]. Sabiston textbook of surgery: the biological basis of modern surgical practice. 19th edition 2012

Learners Guide

- Handout
- Books
- Text books
- E-books

Learning Facilities / Infrastructure

- Class rooms
- Skill lab
- Dissection Room
- ICT center
- Clinical set up
- Library

Portable and stationary plants, machines and equipment

- Flip charts
- Video cassette
- CD-ROM
- Computers
- DVD Player
- LCD projector Overhead Projector
- White board

Methods of assessment

Progressive

- Written progress /quiz/-15%
- Clinical skill lab- 5%
- Bio-lab- 15%
- Hospital visit-5%
- PBL-10%

END module

- Written -30%
- OCSE/OSPE- 20%

Module schedule

Week	Learning activity	Reading assignment
Week One	Day 1 <u>Interactive lecture (2hours)</u> • Introduction to the module(30minute)	Portfolio

	• Introduction to the reproductive system • Overview of pelvis and perineum • Gross and microscopic structures and functions of the male reproductive organs <u>Problem-based learning 1st session (2hours)</u> • A case of amenorrhea <u>Integrated basic science lab (3hours)</u> • Pelvic bone • Atlas (anatomy, histology) or charts showing gross and microscopic structure of the male reproductive organ Day Two <u>Interactive Lecture (4hours)</u> • Sex Determination and Differentiation • Spermatogenesis • Male Sexual Act <u>Integrated Basic science lab (3hours)</u> • Dissection of male and perineum • Histology and pathology slides and specimens • Sperm analysis and count Day Three <u>Interactive lecture (2hours)</u> • Testosterone and other sex hormones(1hr) • Pathophysiology of common problems of the male sexual and reproductive system (1hr) <u>Professional competency development (4 hours)Clinical skills lab</u> • Simulated practice: Approach to a patient with erectile dysfunction (history taking and physical examination) Day Four <u>Interactive lecture (SPH) (2hours)</u> <u>Interactive lecture (2hours)</u> • Pharmacology of androgens and other anabolic steroids(1hr) • Gross and microscopic structures and functions of the female reproductive organs(1hr) • Uterus and uterine tubes <u>Integrated basic science Lab (3hours)</u> • Computer simulations/video showing spermatogenesis, congenital anomalies of male genital tract • Dissection of female pelvis and perineum Day Five <u>Hospital visit , Primary health care or community visit (4hours)</u> • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
	Day One	

Week 2	<u>Interactive lecture (2hours)</u> - Gross and microscopic structures and functions of the female reproductive organs <u>Problem-based learning 2nd session (2hours)</u> - Previous PLB session will be continued Day Two <u>Interactive Lecture (4hours)</u> - Gross and microscopic structures and functions of the female reproductive organs (2hrs) - External genitalia - Development of the reproductive system (2hrs) <u>Integrated Basic science lab (3hours)</u> - Anatomical charts of the female reproductive system and anomalies - Dissection of female pelvis and perineum..... - Computer simulations of the embryological and developmental changes of the female reproductive system Day Three <u>Interactive lecture (2hours)</u> - Functions of female reproductive system - Monthly ovarian cycle - Functions of the ovarian hormones <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> - Simulated practice: Approach to a patient with uterine/vaginal bleeding (history taking and physical examination) - Computer simulations of the monthly ovarian cycle and regulation of the monthly rhythm Day Four <u>Interactive lecture (SPH) (2hours)</u> <u>Interactive lecture (2hours)</u> - The endometrial cycle & menstruation - Hormonal Control of the Menstrual Cycle - Female sexual act - Puberty & menarche <u>Integrated basic science Lab (3hours)</u> - Slides–histology of female reproductive organs; pathology specimens of female genital tract disease including malignancies - Pap smear and/or visual inspection with acetic acid Day Five <u>Hospital visit , Primary health care or community visit (4hours)</u> - General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Whole group session (3hours)</u>	
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	Faculty facilitated discussion and reflection on the week's learning activities	
Week 3	Day One <u>Interactive lecture (2hours)</u> - Common pathologies of the female reproductive organs(2hrs) <u>Problem-based learning 1st session (2hours)</u> - A case of infertility <u>Integrated Basic science lab (3hours)</u> - Basic principles of immune chromatography test - Culturing of vaginal swab and anti biotic susceptibility Day Two <u>Interactive Lecture (4hours)</u> - Classes of pathogenic microorganisms according to their microscopic appearance staining characteristics ,and mechanisms of pathogenesis in the reproductive system(2hrs) - Sexual and reproductive tract infections (2hrs) <u>Integrated Basic science lab (3 hours)</u> - Slides–histology of female reproductive organ; pathology specimens of women genital tract disease - Gram staining and KOH of vaginal/urethral discharge Day Three <u>Interactive lecture (2hours)</u> - Pharmacology of agents used for treatment of sexual and reproductive tract infections (2hrs) <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> - Simulated practice: Approach to a client who has vaginal discharge (history taking and physical examination) Day Four <u>Interactive lecture (SPH) (2hrs)</u> <u>Interactive lecture (2hrs)</u> - Physiology of fertilization, pregnancy, childbirth and lactation(2hrs) (continues) <u>Integrated Basic science lab (3hours)</u> - Treponemal and non treponemal syphilitic test Day Five <u>Hospital visit , Primary health care or community visit (4hours)</u> - General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Whole group session (3hours)</u> - Faculty facilitated discussion and reflection on the week's learning activities	
Week 4	Day One <u>Interactive lecture (2hours)</u> Physiology of fertilization, pregnancy, childbirth and lactation(2hrs) <u>Problem-based learning 2nd session (2hours)</u>	

	• Previous PLB session will be continued <u>Integrated Basic science lab (3hours)</u> • Pregnancy test (HCG) Day Two <u>Interactive Lecture (4hours)</u> • Fetal physiology (2hrs) ▪ Pathophysiology of common pregnancy, childbirth and lactation problems(2hrs) <u>Integrated Basic science lab (3hours)</u> • Computer simulations and video of pregnancy and child birth Day Three <u>Interactive lecture (2hours)</u> • Pharmacology of uterotonics, abortifacients, uterine relaxants, drugs affecting milk production and drug use in pregnancy(2 hrs) <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Approach to a woman who is pregnant (history taking and physical examination) Day Four <u>Interactive lecture (SPH)</u> <u>Interactive lecture (2hours)</u> • Pharmacology of contraceptives(2hrs) <u>Integrated Basic science lab (3hours)</u> • Computer simulation of contraceptive of drugs Day Five <u>Hospital visit , Primary health care or community visit (4hours)</u> • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
Week 5	Day One Review of the reproductive system • Discussion (2 hours) • Basic science lab (3hours) • Clinical skills lab (3 hours) Day Two :MCQ exam Day Three: OSCE Day Four: Interactive lecture (SPH)(2hrs) Day Five:Primary health care or community visit (4hrs) <u>Whole group session (3hours)</u>	

Time Table

Time	Monday	Tuesday	Wednesday	Thursday	Friday
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8-10 am	Interactive Lecture	Interactive Lecture	Interactive Lecture	Interactive Lecture	Primary Health Care/Community Visit
10-12 am	PBL	Interactive Lecture		Interactive Lecture	
1:30-5:30pm	Biomedical lab	Biomedical lab	Clinical skills lab	Biomedical lab	Whole Group Session

MODULE TITLE: NERVOUS SYSTEM

MODULE CODE: NERV 3112

NOMINAL DURATION: 8 WEEKS

Module Description: This system-based module is intended to equip medical students with profound knowledge of basic sciences and development of early clinical skills relevant to understanding of the nervous system and related clinical problems. Students will also develop relevant competencies with the Medical expertise, Scholar, Collaborator, Communicator, Manager, Health Advocate, and Professional. This module is given on the second semester during the second year of the pre-clerkship period of the undergraduate medical education program.

Module Design: This module is designed by integrating normal and abnormal structure and function of the nervous system. Building on in-depth understanding of the nervous system as a foundation, this module aims to provide the analytical and cognitive skills necessary for a successful transition from basic nervous structure, function and pathology to the care of patients with clinical problems of the nervous system.

The Professional competency development component will support the student in acquiring a practical understanding of how to approach a patient with known or suspected disease of the nervous system and develop the basic skill of examining the nervous system. Social and Population Health (SPH) module will run in parallel and will be made to correlate with the subject being taught.

Prerequisite

Successful completion of the introduction to medicine module

Module Objective

By the end of this module, students will be able to apply knowledge of basic sciences for understanding the nervous system and analyzing disorder of the nervous system.

Supporting Objectives

As a Medical Expert

- Apply knowledge of the normal structures and functions of the nervous system in analyzing clinical problems
- Apply principles of physiologic mechanisms and processes of the nervous system in analyzing clinical problems
- Apply principles of human life cycle, growth, development and aging of the nervous system in analyzing clinical problems including developmental anomalies
- Identify the various causes of a disease (genetic, developmental, metabolic, toxic, infectious, autoimmune, neoplastic, degenerative and traumatic) of the nervous system and analyze the pathogenesis in clinical problems

- Analyze the pharmacokinetics and pharmaco-dynamics of drugs used in the treatment of disorders of the nervous system
- Apply important determinants and risk factors affecting nervous system diseases to protect and promote individual and population health
- Demonstrate taking history from the patient with nervous system disorder in a simulated setting
- Demonstrate physical examination pertinent to with nervous system disorder in a simulated setting
- Perform basic investigations in the diagnosis of nervous system disorders
- Demonstrate making an initial assessment of a clinical problem related to nervous system in a simulated setting
- Discuss important life style, genetic, demographic, environment, health system, social, economic, psychological and cultural determinants of nervous system disorders
- Contribute to a culture that promotes patient safety

As a Communicator

- Demonstrate communication with a patient with nervous system disorder disusing a patient-centered approach that encourages patient trust and autonomy and is characterized by empathy, respect, and compassion
- Optimize the physical environment for patient comfort, dignity, privacy, engagement, and safety
- Recognize when the values, biases, or perspectives of patients, physicians, or other health care professionals may have an impact on the quality of care, and modify the approach to the patient accordingly
- Respond to patients' non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients
- Manage disagreements and emotionally charged conversations
- Use patient-centered interviewing skills to effectively gather relevant biomedical and psychosocial information
- Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information
- Provide a clear structure for and manage the flow of an entire patient encounter
- Seek and synthesize relevant information from other sources, including the patient's family, with the patient's consent
- Skillfully share information and explanations that are clear, accurate, timely, and adapted to patient's and his or her family's level of understanding and need in a simulated setting

As a Collaborator

- Establish and maintain healthy relationships with physicians and other colleagues in the health care professions to support relationship-centered collaborative care
- Recognize one's own professional roles and limits
- Share knowledge and information to other colleagues
- Show respect toward other physicians and members of the health care team
- Respect and value diversity among other colleagues

As a Manager

- Develop their leadership skills
- Design and organize elements of health care delivery

- Set priorities and manage time to balance practice and personal life
- Promote occupational health as physicians for sustainable practice

As a Health Advocate

- Work with patients with nervous system disorders and their families to address the determinants of health that affect them and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors
- Incorporate nervous system disease prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of nervous system health that affect them
- Participate in a process to improve nervous system health in the community or population they serve

As a Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize knowledge gaps in biomedical, clinical and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Integrate other intrinsic roles into their learning and practice of medicine
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality

- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in practice
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Duration: 8 weeks

Teaching and learning methods

- Interactive lecture
- Assignment
- Discussion and debate
- Case studies
- Role play
- Clinical simulation
- Demonstration
- Video show
- Observation in clinical setting
- Community visit
- Personal research and reflection exercise/ Portfolio
- Whole group session

Educational Resources

- Basic sciences lab
- Clinical skills lab
- Health Facility
- Videos of patient examination
- Anatomical charts and mannequins
- Histological and pathological slides and specimens
- Computer-based simulations

References

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Learning Facilities / Infrastructure

- Class rooms
- Skill lab
- Dissection Room
- ICT center
- Clinical set up
- Library

Portable and stationary plants, machines and equipment

- Flip charts
- Video cassette
- CD-ROM
- Computers

- DVD Player
- LCD projector Overhead Projector
- White board

Methods of assessment

Progressive

- Written progress /quiz/-15%
- Clinical skill lab- 5%
- Biomedical-lab- 15%
- Hospital visit-5%
- PBL-10%

END module

- Written -30%
- OCSE/OSPE- 20%

Module Schedule

Week	Learning Activity	Required Reading/ Assignment
Week one	Day One <u>Interactive lecture (2hours)</u> • Introduction to the module(30minute) • Introduction to the nervous system <u>Problem-based learning 1st session (2hours)</u> • A case of quadriparasis/paraparsis <u>Integrated Basic science lab (3hours)</u> • Anatomical charts and models of the CNS Day Two <u>Interactive lecture (4hours)</u> • Anatomy of the Head ▪ Introduction ▪ Skull ▪ Meningitis ▪ Scalp <u>Integrated Basic science lab (3hours)</u> • Dissection of the scalp and meanings • Computer-simulations of the CNS Day Three <u>Interactive lecture (2hours)</u> • Anatomy of the Head ▪ Face <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Examination of the nervous system (history taking and physical examination) Day Four <u>Interactive lecture (2hours)</u>	

	- Anatomy of the Head - Parotid region - Infratemporal fossa <u>Interactive lecture – SPH (2hours)</u> <u>Integrated basic science Lab (3hours)</u> - Previous lab session continues - Dissection of face, parotid and infratemporal fossa Day Five <u>Primary health care, hospital or community visit (4hours)</u> <u>Primary health care or community visit</u> - General OPD - Home visits trying to identify patients with neurological disorders, their risk factors and determinants, perception of causes, local remedies and provide advice on promotion and prevention of disease - Home visits to meet with traditional healers <u>Hospital visit</u> - General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Whole group session (3hours)</u> - Faculty facilitated discussion and reflection on the week's learning activities	
Week two	Day One <u>Interactive lecture (2hours)</u> - Anatomy of the head - Pterygopalatine fossa - Anatomy of the brain <u>Problem-based learning 2nd session (2hours)</u> - Previous PLB session will be continued <u>Integrated Basic science lab (3hours)</u> - Dissection of head Day Two <u>Interactive lecture (4hours)</u> - Introduction to neuro anatomy - Nervous tissue - Anatomy of Brain <u>Integrated Basic science lab (3hours)</u> - Computer simulations of neurotransmission - Dissection of pterygopalatine fossa - Tissue and pathology slides and specimen - CSF analysis Day Three <u>Interactive lecture (2hours)</u> - Basic design of the nervous system - Electrochemical properties and communication of neurons - Cerebral blood flow, Cerebrospinal Fluid and Blood-brain	

	Barrier <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Approach to a patient with loss of consciousness (history taking and physical examination) Day Four <u>Interactive lecture (4hours)</u> • Anatomy of spinal cord • Functions of the spinal cord (1hr) • Sensory receptors & its neural circuit (1hr) <u>Interactive lecture – SPH (2hours)</u> <u>Integrated basic science Lab (3hours)</u> • Dissection of brain and spinal cord Day Five <u>Primary health care, hospital or community visit (4hours)</u> • General OPD • Home visits trying to identify common neurological diseases, their risk factors and determinants, perceptions of causes, local remedies and provide advice on promotion of health and prevention of disease <u>Hospital visit</u> • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
Week 3	Day One <u>Interactive lecture (2hours)</u> • Sensory receptors & its neural circuit.... (1hr) • Overview of microbial causes of CNS infection(1hr) <u>Problem based learning 1st session (2hours)</u> • A case of unconscious patient and hemiplegic <u>Integrated Basic science lab (3hours)</u> • Dissection of brain and spinal cord Day Two <u>Interactive lecture (4hours)</u> • Pathophysiology of common brain disorders (e.g., Cerebrovascular disease, CNS trauma, tumor, primary disease of the myelin, acquired metabolic and toxic disturbance, degenerative disease and dementia) <u>Integrated basic science Lab (3hours)</u> • Computer simulation of autonomic nervous system • Autonomic function testing Day Three <u>Interactive lecture (2hours)</u> • Metabolism in brain, Neurochemistry ,	

	Neurotransmitters(2hrs) <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Approach to a patient with organophosphate poisoning (history taking and physical examination) Day Four <u>Interactive lecture (2hours)</u> • Introduction to CNS Pharmacology • Sedative hypnotics <u>Interactive lecture – SPH (2hours)</u> <u>Integrated basic science Lab (3hours)</u> • Previous integrated lab session continues • Nicotine toxicity Day Five <u>Primary health care, hospital or community visit (4hours)</u> • General OPD • Home visits trying to identify common neurologic disorders, their risk factors and determinants, local perceptions of causes, local remedies and provide advice on promotion of health and prevention of disease <u>Hospital visit</u> • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
Week Four	Day One <u>Interactive lecture (2hours)</u> • Sensory functions of Thalamus & cerebral Cortex(1hr) • Gross and microscopic structure of orbit and eye • Vision (1hr) • Central neurophysiology of vision and the visual cortex <u>Problem-based learning 2nd session (2hours)</u> • Previous PLB session will be continued <u>Integrated Basic science lab (3hours)</u> • Anatomy of meninges and blood supply of brain • Cerebral hemisphere Day Two <u>Interactive lecture (4hours)</u> • Gross and microscopic structure of ear(1hr) • Auditory and vestibular physiology (1hr) • Central auditory pathways and auditory cortex • The Pharmacology of drugs used for Parkinsonism, Alzheimer's disease (2hrs) <u>Integrated Basic science lab (3Hours)</u> • Anatomical charts and models	

	- Dissection vestibular system of the spinal cord and orbit - Tissue slides, pathology slides and specimens - Computer-simulations of spinal cord transection Day Three <u>Interactive lecture (2hours)</u> - Pharmacology of sedative/hypnotics drugs <u>Professional competency development</u> <u>Clinical skills lab</u> - Simulated practice: Approach to a patient with pain (history taking and physical examination) Day Four <u>Interactive lecture (2hours)</u> - Cranial nerves <u>Interactive lecture – SPH (2hours)</u> <u>Integrated basic science Lab (3hours)</u> - Previous integrated lab session continues - Cerebellum Day Five <u>Hospital visit</u> - General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Primary health care or community visit</u> - General OPD - Home visits trying to identify neurological diseases, their risk factors and determinants, perceptions of causes, local remedies and provide advice on promotion of health and prevention of disease <u>Whole group session(3hours)</u> - Faculty facilitated discussion and reflection on the week's learning activities	
Week 5	Day One <u>Interactive lecture (2hours)</u> - Common brain function disturbance (sleep disorder, psychiatric disease and depression) (2hrs) - Pharmacology of anti-seizure drugs (2hrs) <u>Problem-based learning 1st session (2hours)</u> - A case of hearing difficulty and vision loss <u>Integrated Basic science lab (3hours)</u> - Brain stem Day Two <u>Interactive lecture (4hours)</u> - Pharmacology of antipsychotic and antidepressants <u>Integrated Basic science lab (3hours)</u> - Anatomical charts and models of peripheral nerves - Dissection of the peripheral nerves - Computer-simulations of drugs actions on CNS (analgesics,	

	antipsychotics, anti-depressants, anti-seizures Day Three <u>Interactive lecture (2hours)</u> • Histology of taste buds • Histology of olfactory epithelium • Physiology of the Chemical senses(taste & smell) • Pain physiology <u>Professional Competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Approach to a patient with psychosis (history taking and physical examination) Day Four <u>Interactive lecture (2hours)</u> • Bacterial and parasitic causes of CNS infection(2hrs) <u>Interactive lecture – SPH (2hours)</u> <u>Integrated basic science Lab (2hours)</u> • Previous integrated lab session continues Day Five <u>Hospital visit</u> • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Primary health care or community visit</u> • General OPD • Home visits trying to identify individuals with pain, their risk factors and determinants, local perceptions of causes, local remedies and provide advice on promotion of health and prevention of disease	
Week 6	Day One <u>Interactive lecture (2hours)</u> • Pharmacology of pain medication (NSAIDs and Opioid analgesics)(1hr) • Pharmacology (1hr) <u>Problem-based learning 2nd session (2hours)</u> • Previous PLB session will be continued <u>Integrated Basic science lab (3hours)</u> • Diencephalon Day Two <u>Interactive lecture (2hours)</u> • Viral and fungal causes of CNS infection(2hrs) <u>Integrated Basic science lab (3hours)</u> • Anatomical charts and models • Dissection of the eye, ear, tongue and nose • Computer-simulations of seeing, hearing, tasting and smelling functions Day Three <u>Interactive lecture (2hours)</u>	

	• Motor and integrative neurophysiology ▪ Motor function of the spinal cord: Spinal reflexes <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Visual acuity testing, Hearing testing Day Four <u>Interactive lecture (2hours)</u> • Gross and microscopic structure of eye and ear (1hr) • Histology of taste buds and olfactory epithelium(1hr) <u>Interactive lecture – SPH (2hours)</u> <u>Integrated basic science Lab (2hours)</u> • Previous integrated lab session continues Day Five <u>Hospital visit</u> • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Primary health care or community visit</u> • General OPD • Home visits trying to identify individuals with difficulty in seeing/hearing/tasting/smelling, their risk factors and determinants, perceptions of causes, local remedies and provide advice on promotion of health and prevention of disease	
Week 7	Day One <u>Interactive lecture (2hours)</u> • Cortical and brain stem control of motor function • Basal ganglia and its motor function • Cerebellum & its motor functions <u>Problem-based learning 2nd session (2hours)</u> • Previous PLB session will be continued <u>Integrated Basic science lab (3hours)</u> • Computer simulation of CNS drugs Day Two <u>Interactive lecture (4hours)</u> • Integration of the many parts of motor control • Sympathetic and parasympathetic systems functions • Basic principles of Learning and Memory • The Limbic System and the Hypothalamus • Physiology and chemistry of sleep and brain waves • Body Temperature Regulation <u>Integrated Basic science lab (3hours)</u> • Anatomical charts and models • Dissection of the eye, ear, tongue and nose • Computer-simulations of seeing, hearing, tasting and smelling functions Day Three	

	<u>Interactive lecture (2hours)</u> - Pharmacology of General anesthesia <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> - Simulated practice: Visual acuity testing, Hearing testing Day Four <u>Interactive lecture (2hours)</u> - Development of head and neck - Development of the nervous system - Development of eye and ear <u>Interactive lecture – SPH (2hours)</u> <u>Integrated basic science Lab (2hours)</u> - CSF culture and antibiotic susceptibility <u>PBL second session continues (2hours)</u> Day Five <u>Hospital visit</u> - General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Primary health care or community visit</u> - General OPD - Home visits trying to identify individuals with difficulty in seeing/hearing/tasting/smelling, their risk factors and determinants, perceptions of causes, local remedies and provide advice on promotion of health and prevention of disease	
Week 8	Exam week Day One - Review of the nervous system - Discussion (2hrs) - Basic science lab (3hrs) - Clinical skills lab (3hrs) Day Two: Written exam Day Three: OSCE Day Four: Interactive lecture (SPH)(2hrs) Day Five: Primary health care or community visit (4hrs) <u>Whole group session (3hours)</u>	

Time Table

Time	Monday	Tuesday	Wednesday	Thursday	Friday
8-10 am	Interactive Lecture	Interactive Lecture	Interactive Lecture	Interactive Lecture	Primary Health Care/Community Visit
10-12 am	Interactive Lecture	Interactive Lecture		Interactive Lecture	
1:30-5:30pm	PBL	Biomedical lab	Clinical skills lab	Biomedical lab	Whole Group

					Session
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MODULE TITLE: INFECTIOUS DISEASES AND RURAL MEDICINE

MODULE CODE: INFD 3122

NOMINAL DURATION: 5 weeks

Module Description: This system-based module is intended to equip medical students with profound knowledge of basic sciences and development of early clinical skills relevant to understanding of the infectious diseases and related clinical problems. Students will also develop relevant competencies with the Medical expertise, Scholar, Collaborator, Communicator, Manager, Health Advocate, and Professional. This module is given on the second semester during the second year of the pre-clerkship period of the undergraduate medical education program.

Module Design

This module is designed by integrating normal and abnormal structure and function of the micro organisms. Building on in-depth understanding of the infectious diseases as a foundation, this module aims to provide the analytical and cognitive skills necessary for a successful transition from basic infectious diseases pathology to the care of patients with diseases of the infection.

The Professional competency development component will support the student in acquiring a practical understanding of how to approach a patient with known or suspected infectious diseases and develop the basic skill of examining patients with different infectious diseases disorders. Social and Population Health (SPH) module will run in parallel and will be made to correlate with the content being taught.

Prerequisite

Successful completion of the Introduction to Medicine Module

Module Objective

By the end of this module, students will be able to apply knowledge of basic sciences for understanding the infectious diseases and analyzing disorder of the infection.

Supporting objectives

As a Medical Expert

- Apply knowledge of the normal structures and functions of the infectious diseases in analyzing clinical problems
- Apply principles of physiologic mechanisms and processes of the infectious diseases in analyzing clinical problems
- Apply principles of human life cycle, growth, development and aging of the infectious diseases in analyzing clinical problems including developmental anomalies
- Identify the various causes of a disease (genetic, developmental, metabolic, toxic, infectious, autoimmune, neoplastic, degenerative and traumatic) of the micro organisms and analyze the pathogenesis in clinical problems
- Analyze the pharmacokinetics and pharmacodynamics of drugs used in the treatment of disorders of the infectious diseases
- Apply important determinants and risk factors of infectious diseases to protect and promote individual and population health
- Demonstrate taking history from the patient with infectious diseases in a simulated setting
- Demonstrate physical examination pertinent to with infectious diseases in a simulated setting
- Perform basic investigations in the diagnosis of infectious diseases
- Demonstrate making an initial assessment of a clinical problem related to infectious diseases in a simulated setting

- Discuss important life style, genetic, demographic, environment, health system, social, economic, psychological and cultural determinants of infectious diseases
- Contribute to a culture that promotes patient safety

As a Communicator

- Demonstrate communication with a patient with infectious diseases using a patient-centered approach that encourages patient trust and autonomy and is characterized by empathy, respect, and compassion
- Optimize the physical environment for patient comfort, dignity, privacy, engagement, and safety
- Recognize when the values, biases, or perspectives of patients, physicians, or other health care professionals may have an impact on the quality of care, and modify the approach to the patient accordingly
- Respond to patients' non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients
- Manage disagreements and emotionally charged conversations
- Use patient-centered interviewing skills to effectively gather relevant biomedical and psychosocial information
- Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information
- Provide a clear structure for and manage the flow of an entire patient encounter
- Seek and synthesize relevant information from other sources, including the patient's family, with the patient's consent.
- Skillfully share information and explanations that are clear, accurate, timely, and adapted to patient's and his or her family's level of understanding and need in a simulated setting

As a Collaborator

- Establish and maintain healthy relationships with physicians and other colleagues in the health care professions to support relationship-centered collaborative care
- Recognize one's own professional roles and limits
- Share knowledge and information to other colleagues
- Show respect toward other physicians and members of the health care team
- Respect and value diversity among other colleagues

As a Manager

- Develop their leadership skills
- Design and organize elements of health care delivery
- Set priorities and manage time to balance practice and personal life
- Promote occupational health as physicians for sustainable practice

As a Health Advocate

- Work with patients with infectious diseases and their families to address the determinants of health that affect them and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors
- Incorporate infectious diseases prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of infectious diseases health that affect them

- Participate in a process to improve infectious diseases health in the community or population they serve

As a Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize knowledge gaps in biomedical, clinical and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Integrate other intrinsic roles into their learning and practice of medicine
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in practice
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions

- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Duration: 5 weeks

Teaching and learning methods

- Interactive lecture
- Assignment
- Discussion and debate
- Case studies
- Role play
- Clinical simulation
- Demonstration
- Video show
- Observation in clinical setting
- Community visit
- Personal research and reflection exercise/ Portfolio
- Whole group session

Educational Resources

- Basic sciences lab
- Clinical skills lab
- Health Facility
- Videos of patient examination
- Anatomical charts and mannequins
- Histological and pathological slides and specimens
- Computer-based simulations

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Learners Guide

- Handout
- Books
- Text books
- E-books

Learning Facilities / Infrastructure

- Class rooms
- Skill lab
- Dissection Room
- ICT center
- Clinical set up
- Library

Portable and stationary plants, machines and equipment

- Flip charts
- Video cassette
- CD-ROM
- Computers
- DVD Player
- LCD projector Overhead Projector
- White board

Methods of assessment

Progressive

- Written progress /quiz/-15%
- Clinical skill lab- 5%
- Biomedical-lab- 15%
- Hospital visit-5%
- PBL-10%

END module

- Written -30%
- OCSE/OSPE- 20%

Module schedule

Week	Learning Activity	Required Reading/ Assignment
Week One	Day One <u>Interactive lecture (2hours)</u> • Introduction to the module(30 minute) • Basics of infectious diseases (pathology) ▪ Host-pathogen interaction ▪ Molecular mechanisms of pathogenesis ▪ Laboratory diagnosis ▪ Prevention and control <u>Problem-based learning 1st session (2hours)</u> • A case of HIV/AIDS and TB <u>Integrated Basic science lab (3hours)</u> • Pathological charts and models • Tissue slides, pathology slides and specimen Day Two <u>Interactive lecture (4hours)</u> • Overview of antimicrobials (antibacterial, antiviral, antifungal and antiparasitic)(2hrs) • General Classification, mechanism of action and resistance mechanisms • Diarrheal diseases (2hrs) ▪ Common etiologies, risk factors and epidemiology ▪ Pathogenesis, alterations in structures and functions, and clinical presentations ▪ Laboratory diagnosis ▪ Prevention and treatment <u>Integrated Basic science lab (3hours)</u> • Laboratory safety and hazards • Sterilization and disinfection techniques • Computer-simulations of the pathophysiology Day Three <u>Interactive lecture (2hours)</u> • Pharmacology of common ant-parasitic and anti-helminthic	

	drugs <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Approach to a patient with acute infectious diarrhea with dehydration(history taking and physical) Day Four <u>Interactive lecture (2hours)</u> • Overview of Common Zoonotic and vector borne diseases(2hr) ▪ Definition ▪ Causes ▪ Mode of transmission ▪ Prevention and Treatment <u>Interactive lecture – SPH (2hours)</u> <u>Integrated basic science Lab (3 hours)</u> ✓ Basic principles of autoclaves ✓ Stool microscopy and culture Day Five <u>Primary health care, hospital or community visit (4hours)</u> <u>Primary health care or community visit</u> • General OPD • Home visits trying to identify diarrheal diseases, their risk factors and determinants, perceptions of causes and local remedies and provide advice on promotion of health and prevention of disease <u>Hospital visit</u> • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Whole group session (3hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
Week two	Day One <u>Interactive lecture (2hours)</u> • Neglected tropical and infectious disease <u>Problem-based learning 2nd session (2hours)</u> • Previous PLB session will be continued <u>Integrated Basic science lab (3hours)</u> • Computer-simulations of pathophysiology of HIV/AIDS • HIV rapid test • Elisa test Day Two <u>Interactive lecture (4 hours)</u> • HIV/AIDS(2 hrs) • Etiology, risk factors and epidemiology • Pathogenesis, alterations in structures and functions ,and clinical presentations of HIV/AIDS	

	• Laboratory diagnosis of HIV/AIDS • Biological basis of prevention and control methods • Pharmacology of drugs used for the treatment of HIV/AIDS (2hrs) • Classification, mechanism of action and common side effects <u>Integrated Basic science lab (3hours)</u> • Parasitological concentration techniques Day Three <u>Interactive lecture (2hours)</u> • Tuberculosis(pathology& microbiology)(2hrs) ▪ Etiologies, risk factors and epidemiology ▪ Pathogenesis, alterations in structures and functions, and clinical presentations in different systems ▪ Laboratory diagnosis ▪ Biological basis of prevention and control methods <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Approach to a patient with hospital acquired infection (history taking and physical examination) • Approach to a patient with HIV infection(history taking and physical examination) Day Four <u>Interactive lecture (2hours)</u> • Pharmacology of drugs used for treatment of tuberculosis(1hr) ▪ Relapsing fever and brucellosis(microbiology) (1hr) ▪ Etiologies, risk factors and epidemiology ▪ Pathogenesis, alterations in structures and functions, and clinical presentations in different systems <u>Interactive lecture – SPH (2hours)</u> <u>Integrated basic science Lab (3hours)</u> • Antibiotic Susceptibility test Day Five <u>Primary health care, hospital or community visit (4 hours)</u> • General OPD • Home visits trying to identify common sexually transmitted infections, their risk factors, determinants, perceptions of causes and local remedies and provide advice on promotion of health and prevention of disease <u>Hospital visit</u> • General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Whole group session (3 hours)</u> • Faculty facilitated discussion and reflection on the week's learning activities	
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Week 3	Day One <u>Interactive lecture (2hours)</u> • Malaria(microbiology)(2hrs) ▪ Common etiologies, risk factors and epidemiology ▪ Pathogenesis,alterations in structures and functions and clinical presentations ▪ Laboratory diagnosis ▪ Treatment and prevention methods <u>Problem based learning 1st session (2hours)</u> • A case of a child with acute febrile illness <u>Integrated Basic science lab (3hours)</u> • Tissue slides, pathology slides and specimens • Computer-simulations of pathophysiology of TB • Computer-simulations of pathophysiology of TB • AFB sputum staining and mycobacteria identification • Blood film preparation and parasite detection Day Two <u>Interactive lecture (4hours)</u> • Leishmaniasis (2hrs) ▪ Etiologies, risk factors and epidemiology ▪ Pathogenesis, alterations in structures and functions, and clinical presentations ▪ Laboratory diagnosis ▪ Biological basis of prevention and control methods • Anti-malarial and leishmaniasis drugs(1hr) • Schistosomiasis (1hr) ▪ Etiologies, risk factors and epidemiology ▪ Pathogenesis, alterations in structures and functions, and clinical presentations in different systems <u>Integrated basic science Lab (3hours)</u> • Blood film examination Day Three <u>Interactive lecture (2hours)</u> • Microorganisms causing outbreak ▪ Bioterrorism agents(influenza, SARS, Tularemia, Anthrax) <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> • Simulated practice: Approach to a patient with chronic cough (history taking and physical examination including inspection, auscultation, percussion) • Simulated practice: Approach to a patient with acute febrile illness (historytakingand physical examination) • Interpreting results of chest x-rays of TB patients Day Four <u>Interactive lecture (2hours)</u>	
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	- Emerging viral infections(Ebola, yellow fever, dengue,Zika) <u>Interactive lecture – SPH (2hours)</u> <u>Integrated basic science Lab (3hours)</u> - Fungal culture Day Five <u>Primary health care, hospital or community visit (4hours)</u> - General OPD - Home visiting to identify tuberculosis and malaria cases, their risk factors and determinants, perceptions of causes and local remedies and provide advice on promotion of health and prevention of disease <u>Hospital visit</u> - General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Whole group session (3 hours)</u> - Faculty facilitated discussion and reflection on the week's learning activities	
Week 4	Day One <u>Interactive lecture (2hours)</u> - Dermatophytes (2hrs) - Etiologies, risk factors and epidemiology - Pathogenesis, alterations in structures and functions, and clinical presentations - Laboratory diagnosis - Biological basis of prevention and control methods <u>Problem-based learning 2nd session (2hours)</u> - Previous PLB session will be continued <u>Integrated Basic science lab (3hours)</u> - A case of hepato-splenomegally - Computer-simulations of pathophysiology of leishmaniasis - Computer-simulations of pathophysiology of measles Day Two <u>Interactive lecture (4hours)</u> - Microbial causes of opportunistic infections - Microbial causes of ocular infection <u>Integrated Basic science lab (3 Hours)</u> - TB detection Day Three <u>Interactive lecture (2hours)</u> - Microbiological Diagnostic techniques and antimicrobial susceptibility testing <u>Professional competency development (4hours)</u> <u>Clinical skills lab</u> - Simulated practice: Approach to a child with fever(history taking and physical examination) - Simulated practice: Approach to a patient with visceral	

	- Leishmaniasis (history taking and physical examination) Day Four <u>Interactive lecture (2hours)</u> - Microbiological Diagnostic techniques and antimicrobial susceptibility testing (Continued of previous class) <u>Interactive lecture – SPH (2hours)</u> <u>Integrated basic science Lab (3hours)</u> - Previous lab will be continued Day Five <u>Hospital visit</u> - General OPD, Medical OPD, Pediatrics OPD, Surgical OPD <u>Primary health care or community visit</u> - General OPD - Home visits trying to identify cases of measles and leishmaniasis, their risk factors and determinants, perceptions of causes, local remedies and provide advice on promotion of health and prevention of disease <u>Whole group session(3hr)</u> - Faculty facilitated discussion and reflection on the week's learning activities	
Week 5	Day One: Review of infectious diseases module Day Two: MCQ Day Three: OSCE Day Four :SPH Day Five: Community visit	

Time Table

Time	Monday	Tuesday	Wednesday	Thursday	Friday
8-10 am	Interactive Lecture	Interactive Lecture	Interactive Lecture	Interactive Lecture	Primary Health Care/ Community Visit
10-12 am	PBL	Interactive Lecture		Interactive Lecture	
1:30-5:30pm	Biomedical lab	Biomedical lab	Clinical skills lab	Biomedical lab	Whole Group Session

CLERKSHIP I MODULES

Typical Week in Clerkship I

Time	Monday	Tuesday	Wednesday	Thursday	Friday
8 -10 am	Classroom teaching ³¹	Classroom teaching	Classroom teaching	Classroom teaching (SPH)	Primary Health units/Community attachment.
10am-12 pm	Clinical Practicum ³² , Round ³³	Clinical practicum	Clinical Practicum, Round	Clinical practicum	*Last week is for exam
1:30-4:30 pm	PBL ³⁴ /case study	Bed side teaching	Skills lab ³⁵	Bed side teaching	Whole group sessions
4:30 pm -	Independent study/preceptor ship ³⁶	Independent study/preceptor ship		Independent study/preceptor ship	

³¹ Students are required to maintain in their portfolio learning experiences, learning outcomes achieved, evidence of learning and reflections longitudinally and their mentors would provide feedback occasionally. Clinical instructors should also ask students to submit reflective portfolio twice during the attachment for formative assessment and feedback.

³² This includes mixed methods like (Interactive lectures, small group activities, written clinical simulations, seminars etc.) Interactive lectures should be case based and emphasize patient approach to enable students apply the concepts in a real clinical context of the country. They should utilize appropriate multimedia for teaching and group learning activities to maximize engagement of students.

³³ Clinical practicum includes all clinical learning activities that students engage in inpatient, outpatient, and/or Operation theatre settings supervised and/or unsupervised (clerking patients, observing, assisting, and/or performing procedures, patient chart review including investigation results)

³⁴ Clinical rounds should be aimed at guiding students identify their learning outcomes based on the patients they clerked.

³⁵ PBL should focus on problems that can build clinical decision making skills that may not be addressed in clinical teaching sessions. These might include rare patient conditions, professional, legal and ethical issues, medical errors, public health issues.

³⁶ Teaching in the skill lab (communication, examination, and procedural skills)

CLINICAL I MODULES

MODULE TITLE: CLINICAL METHODS

MODULE CODE: CLMD 4011

NOMINAL DURATION: 6 WEEKS

Module Description

The aim of the module is to introduce the student to the knowledge, attitude, and skills required to communicate with a patient effectively and elicit the health problems, explain and demonstrate common manifestations of diseases and their correlation with the underlying pathophysiological mechanism and to understand the role of clinical laboratory in health care system; differentiate different laboratory equipment; select, evaluate and interpret laboratory test results. And also, to introduce the institutional organization of patient care and appropriate documentation of information.

It is also designed to introduce students to ethical issues in health care policy, planning and service delivery in the global context. Through the application to practical decision-making at all levels of the healthcare system of a broad range of ethical principles and moral theories, the students will gain an understanding of 1) the definitions of healthcare ethics and bioethics, including comparative interpretations; 2) the complexities of healthcare ethics and bioethics in the global context; 3) the protocol and systems in place to ensure adherence to ethical principles; and 4) their specific roles and responsibilities to ensure such adherence.

The subject will be delivered in the form of lectures, basic labs, clinical labs, ward rounds and bedside teaching. Student assessment will be done using progressive assessment, written and practical examination.

Pre-requisite

Successful completion of Step I qualification exam

Module objective;

Course objective:

At the end of this course, the student will be able to correlate the role of history taking, physical examination, clinical laboratory in order to reach the diagnosis. And also, to understand medical ethics and professionalism in health care system.

Specific objectives

- Upon the completion of this course students would be able to:
 - Aware the structural organization health care system
 - Apply history taking, physical examination and clinical laboratories to reach the diagnosis.
 - Identify different laboratory equipment's and glass/plastic wares
 - Describe and practice the collection, handling, shipment and safe disposal of laboratory specimens.

- Perform common laboratory tests, select, evaluate and interpret different laboratory test results
- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions

Duration: 6 weeks

Teaching and learning methods

- Interactive lecture
- Assignment
- Discussion and debate
- Case studies
- Role play
- Clinical simulation
- Demonstration
- Video show
- Observation in clinical setting
- Community visit
- Personal research and reflection exercise/ Portfolio
- Whole group session

Educational Resources

- Basic sciences lab
- Clinical skills lab
- Health Facility
- Videos of patient examination
- Computer-based simulation.

References

1. Hutchison's clinical methods 24th edition, 2018
2. Barbara bates' guide to physical examination, 13th ed. 2021
3. Medical ethics for doctor in Ethiopia, 1st ed. 2010
4. Linne Jean Jergenson, Basic techniques of medical laboratory 4th ed. 2000
5. WHO, Manual of basic techniques for a health laboratory 2nd ed. 2003
6. Chees brough M. District Laboratory manual for tropical countries, 2018 (Vol I & vol II).
7. Seyoum B. Introduction to medical laboratory technology students lecture note series 2002.
8. Medical ethics for the boards, 3rd ed. 2016

Date	Learning Activity	Required Reading/ Assignment
Week 1	Interactive lecture Day One (4hrs) Introduction to the module ▪ Basic principles of history taking ▪ Physical examination ▪ General Appearance ▪ Vital Signs Day two(4hrs) ▪ Introduction to Medical ethics ▪ Historical background ▪ Ethics in medicine ▪ Hippocratic oath ▪ Medical ethics in Ethiopia Day Three ▪ INTRODUCTION clinical labs (2hrs) ▪ Definition and organization of Medical Laboratory ▪ Role of Clinical laboratory ▪ Selection, evaluation and interpretation of laboratory tests ▪ LABORATORY EQUIPMENTS AND WARES (2 hrs) ▪ Microscope ▪ Laboratory centrifuges ▪ Glass wares Day four ▪ Physical Examination • HEENT & Lympho-glandular system Day five • Medical Ethics continued ▪ Code of Medical ethics ▪ General Code of Medical ethics ▪ Doctor-patient and Doctor-community Relationships ▪ The doctor as a Professional ▪ Medical Secrecy	
	Basic science lab • Tour of the bio medical laboratory	

	Demonstration of how to use different types of microscope and accessories.	
	Clinical skills lab - Tour of the clinical skills lab and introduction to staff - Exercising history taking and physical examination (V/S taking) through role play - Simulation for HEENT and Lympho-glandular system - Role play for HEENT and Lympho-glandular system	
	Hospital visit - Exercising history taking and physical examination (Vital Sign taking) under supervision - Practicing HEENT and Lympho-glandular system Examination - Observing Doctor-patient and Doctor-community Relationships	
	Primary health care or community visit Optional for hospital visit	
	Whole group session Faculty facilitated discussion and reflection on the week's learning activities	
Week 2	Interactive lecture Day One - Physical Examination continued - Musculoskeletal System - Integumentary system	
	Day Two Medical ethics continued - Patients' Consent. - Torture and Punishment. - Certificates, Prescriptions and Signatures.	
	Day Three BASIC HAEMATOLOGICAL TESTS (2hrs) - Blood collection - Complete blood count - Erythrocyte sedimentation rate - Blood film preparation and staining - Hemostasis - Hematological disorders (anemia, leukemia) BASIC CLINICAL CHEMISTRY TESTS (2hrs) - Liver Function Test	

	○ Renal function test ○ Carbohydrate metabolism disorders ○ Plasma protein ○ Body fluid analysis	
	● Day Four(4hrs) ● Physical Examination continued ○ Respiratory system	
	● Day Five ● Medical ethics continued ○ Undisclosed Gains. ○ Advertisement and Publicity ○ The Doctor and his Professional Colleagues ○ Supervisory Role of the Doctor	
	Basic science lab ● Blood collection (venous and capillary) ● HCT, ESR, Hgb and differential count ● Common clinical chemistry tests (Glucose, Liver function tests, renal function tests)	
	Clinical skills lab .3. Simulation of Musculoskeletal, integumentary and RS examination ● Role play for musculoskeletal and integumentary system ● Role play for respiratory	
	Hospital visit ● Exercising musculoskeletal and integumentary system and respiratory examination under supervision ● Practicing how to take informed consent ● Counselling and disclosure ● Observing how to write proper prescription	
	Primary health care or community visit ● Optional	
	Whole group session ● Faculty facilitated discussion and reflection on the week's learning activities	
Week 3	Interactive lecture Day one ● Physical Examination continued ○ Cardiovascular System	
	Day Two	

	• Medical ethics continued • Mind and behavior control • Abortion. • Family Planning. • Artificial insemination.	
	Day Three • BASIC SERLOGICAL TESTS (2hrs) • Introduction to serology • Serology of Syphilis • Pregnancy test • Serological tests for febrile diseases • HIV tests • IMMUNOHAEMATOLOGICAL TESTS (2hrs) ○ Blood grouping ○ Cross match ○ Laboratory tests of blood transfusion reaction	
	Day four Physical Examination continued • Abdomen • Genitourinary system	
	Day five Medical Ethics continued • Severely handicapped children. • Care of children • Death • Euthanasia • Sexually transmitted diseases • HIV Infection and Doctors	
	Basic science lab • Blood grouping and Cross match • RPR/VDRL, HIV rapid tests and Pregnancy tests	
	Clinical skills lab • Simulation of Cardio-vascular, abdomen and genitourinary system • Role play for Cardio-vascular, abdomen and genitourinary system	
	Hospital visit	

	- Exercising Cardio-vascular, abdomen and genitourinary system physical examination - How to handle/communicate/inform/educate client with abortion /family planning/client with HIV disease	
	Primary health care or community visit - Observation of patient-doctor encounter at OPDs - Talking to medical doctors Home visits trying to practice medical interviewing	
	Whole group session Faculty facilitated discussion and reflection on the week's learning activities	
Week 4	Interactive lecture Day one - Physical Examination continued - Nervous system one	
	Day two Medical Ethics continued - Research ethics	
	Day three BASIC MICROBIOLOGICAL TEST (2hrs) - Specimen Collection and Handling - Examination method - Wet mount examination - Staining - Culture and biochemical - Molecular and serological methods URINALYSIS (2hrs) - Specimen collection - Examination methods - Physical - Chemical - Microscopic	
	Day four - Physical Examination continued (4hrs) - Nervous system two	
	Day five Medical Ethics continued - World medical association declaration of Helsinki	
	Basic science lab	

	• Smear preparation and Staining (blood, gram stain, AFB) • Chemical and microscopic examination of urine • Culture and biochemical tests	
	Clinical skills lab • Simulation of nervous system examination • Role play for nervous system examination	
	Hospital visit ▪ Exercising nervous system examination	
	Primary health care or community visit Optional	
	Whole group session ▪ Faculty facilitated discussion and reflection on the week's learning activities	
Week 5	Interactive lecture Day one ▪ PARASITOLOGICAL TESTS (2hrs) ▪ Types of parasitological specimen and their collection method ▪ Diagnostic method ▪ Macroscopic examination ▪ Microscopic examination ▪ Wet mount ▪ Concentration techniques ▪ Staining	
	Day two & day three • Unique features of OB/GYN physical examination	
	Day four and five ▪ Unique features of Pediatrics physical examination	
	Basic science lab ▪ Microscopic examination of intestinal and hemo-parasites	
	Clinical skills lab • Practice: Approach to a pediatric patient (history taking and physical examination in newborn and children) • Simulated practice: history taking and physical examination of a OB/GYN	
	Hospital visit ▪ Exercising hx and physical examination in pediatrics and OB/GYN.	
	Primary health care or community visit Optional	
	Whole group session	

	Faculty facilitated discussion and reflection on the week's learning activities	
Week 6	Day One- three: Study break Day four: Clinical methods (Written exam) Day five: Clinical methods (OSPE/OSCE)	

MODULE TITLE: RADIOLOGY

MODULE CODE: RADI 4011

NOMINAL DURATION: Clerkship I : 2 weeks

MODULE PREREQUISITE:

Completed Pre-clerkship courses successfully

COURSE DESCRIPTION:

Radiology attachment during Clerkship is intended to build upon the student's knowledge, skill and attitude required in the diagnostic radiology including assisting in radiological procedures. To enhance skill development, this attachment is delivered integrated Professional competency development (PCD)

COURSE DESIGN:

This will be delivered in Clerkship I:

The teaching will focus on consolidating the teachings across the different clerkship attachment building the ability of the student in interpretation of diagnostic radiological technique. This will prepare students to have the experience in using Radiology and Imaging investigations for patient care.

COMPETENCIES/MODULE OBJECTIVES

As a Medical Expert

- Demonstrate a commitment to high-quality care of their patients
- Apply knowledge of the clinical and biomedical sciences relevant to radiology
- Perform appropriately timed clinical assessments with recommendations that are well organized and properly documented in written and/or oral form
- Carry out professional duties in the face of multiple, competing demands
- Recognize and respond to the complexity, uncertainty, and ambiguity inherent in medical practice
- Identify and prioritize issues to be addressed in a patient encounter
- Perform basic investigations and interpret their results for the purpose of diagnosis and management, disease prevention, and health promotion
- Diagnose and establish a patient-centered management plan based on available evidences
- Provide clinical care within the practical context of patient physician relationship taking to account patient's culture and adapting to economic limitations, and encounter time constraints
- Establish goals of care in collaboration with patients and their families, which may include slowing disease progression, achieving cure, improving function, treating symptoms, and palliation
- Determine the most appropriate procedure(s) for the purpose of assessment and/or management

- Prioritize procedures, taking into account clinical urgency, potential for deterioration, and available resources
- Perform procedures in a skillful and safe manner, adapting to unanticipated findings or changing clinical circumstances

As a communicator

- Demonstrate communication with a patient, families, peers and preceptors using a patient-centered approach that encourages patient trust and autonomy and is characterized by empathy, respect, and compassion
- Respond to patients' non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients
- Manage disagreements and emotionally charged conversations
- Use patient-centered interviewing skills to effectively gather relevant biomedical and psychosocial information
- Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information
- Provide a clear structure for and manage the flow of an entire patient encounter
- Seek and synthesize relevant information from other sources, including the patient's family, with the patient's consent.
- Skillfully share information and explanations that are clear, accurate, timely, and adapted to patient's and his or her family's level of understanding and need in a simulated setting

As a Collaborator

- Establish and maintain healthy relationships with physicians and other colleagues in the health care professions to support relationship-centered collaborative care
- Recognize one's own professional roles and limits
- Share knowledge and information to other colleagues
- Show respect toward other colleagues and members of the health care team
- Respect and value diversity among other colleagues

As a Leader

- Develop their leadership skills
- Design and organize elements of health care delivery
- Set priorities and manage time to balance practice and personal life
- Promote occupational health as a medical student for sustainable practice

As a Health Advocate

- Work with pediatric patients and their families to address the determinants of health that affect them and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors
- Incorporate childhood illnesses prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of childhood disease
- Participate in a process to improve childhood health and child rights in the community

As a Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance

- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling on learning
- Promote a safe learning environment
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance
- Recognize knowledge gaps in biomedical, clinical and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in practice
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Course Content

Assessment types

Summative assessment

- Progressive (continuous) assessment (Quizzes, Concept application exercise Tutorial-based evaluation)----20%
- Written ----50%
- OSCE ----30%

•Formative assessment

- This will not contribute to the total grade of the student but provides continuous feed back to the student and it can be applied in the form of providing feed backs on assignments completed, students' professional behavior and etc.

Methods of assessment

1. DOCS (Direct Observation of Clinical Skills)

2. OSCE (Objective Structured Clinical Examination)
3. OSPE (Objective Structured Practical Exam)
4. PRRE (Personal Research and Reflection Exercise)
5. Project Work
6. Clinical examination (Long case and Short cases)
7. Oral exam (Viva)
8. Written exam (MCQ, Short Essay, Matching, True- False with reasoning)
9. Log books and portfolios
10. Global rating
11. 360° Assessment (Self, Peer, faculty, health facility staff and community member)

Module schedule

Schedule	Learning Activity	Required Reading/Assignment
Week One Day 1	Interactive Lecture • Introduction to radiology – principles of ordering a radiological investigation	Reflective portfolio
	Clinical practicum/case discussion • Student placement to radiology clinic	
	³⁷ Problem-based learning tutorial • Case: Chest pathology	
Day 2	Interactive Lecture • X-Ray Reading ○ Chest X ray	Reflective portfolio
	Clinical practicum Student placement to the radiology clinic	
	Clinical skills lab • Radiology atlas showing chest abnormalities • Chest x-rays with different pathologies	
Day 3	Interactive Lecture • X-Ray Reading ○ Abdominal X-ray	Reflective portfolio
	Clinical practicum Student placement to radiology clinic	
	PBL • Case: Chest pathologies	
Day 4	Interactive Lecture • X-Ray Reading ○ X ray of the upper and the lower limbs.	Reflective portfolio
	Clinical practicum Student placement to radiology clinic.	
	Clinical skill lab • Abdominal x-rays	

³⁷ Preceptorship is

	• X-rays showing fracture of upper and lower extremities • Radiology atlas showing fracture of upper and lower extremities	
Day 5	Primary health care/community visit • Health center • Home visit	
	Whole group discussion Faculty facilitates discussion and reflection on the week's learning activities	
Week Two Day 1	Interactive Lecture • X-Ray Reading ○ Spinal cord X ray	
	Clinical practicum Student placement to radiology clinic. Case discussion on topic of the week	
	Problem based learning Case: Paraplegia	
Day 2	Interactive Lecture • X-Ray reading ○ Special X rays	Reflective portfolio
	Clinical practicum Student placement to radiology clinic	
	Clinical skill lab Spinal x-rays Study radiology atlas of spinal injury	
Day 3	Interactive lecture • Basic Ultrasound	Reflective portfolio
	Clinical practicum Student placement to the radiology clinic	
	Problem based learning Case:	
Day 4	Interactive lecture • Introduction to CT and MRI scanning	Reflective portfolio
	Clinical practicum • Student placement to the radiology clinic	
	Clinical skills lab • Videos of ultrasound • CT and MRI images	
Day 5	Primary health care/community visit • Health center • Home visit	Reflective portfolio
Week Three Day 1	Whole group discussion Faculty facilitates discussion and reflection on the week's learning activities	Reflective portfolio
	Clinical practicum	

	• Student placement to the radiology clinic	
Day 2	Clinical practicum ○ Student placement to radiology clinic. Case discussion on topic of the week	Reflective portfolio
Day 3	Study break	
Day 4	Examination: Written exam	
Day 5	Examination: OSCE	

MODULE TITLE: INTERNAL MEDICINE I

MODULE CODE: INME 4011

Module duration: 10 weeks

Module Description: This first clerkship in internal medicine is intended to enable medical students to develop clinical knowledge, skills and attitude required for the understanding, diagnosis and management of medical disorders with emphasis on problems common in Ethiopia. They will also develop essential knowledge and skills in the provision of internal medical care. During this clerkship, students will have opportunities to review and relate knowledge of biomedical sciences for the understanding of patient problems they encounter in clinical settings. Students will also develop relevant competencies in health promotion and disease prevention, communication, collaboration, leadership, health advocacy, scholarship professionalism.

Module Objective

Upon completion of this module, students will be able to assess patients with medical disorders and formulate differential diagnoses and suggest the most likely diagnosis.

Supporting Objectives

1. Medical expert

- Apply knowledge of normal and abnormal structures and functions of the human body system in all body systems in analyzing medical problems.
- Apply knowledge of the human life cycle and effects of growth, development of the body systems in analyzing medical conditions.
- Apply knowledge of physiological mechanisms and processes of the human body system in analyzing medical conditions
- Apply knowledge of the various causes and pathogenesis of medical conditions in analyzing patient problems
- Analyze important determinants and risk factors of common medical problems.
- Apply knowledge of principles of drug pharmacokinetics and pharmacodynamics in the treatment of common medical conditions
- Demonstrate proper history taking skills from a patient with a medical problem.

- Demonstrate proper physical examination skills of a patient with a medical problem
- Attempt to identify abnormal physical findings in a patient with medical disorders
- Attempt to make an initial assessment of a patient presenting with signs and symptoms of an medical disorder
- Propose a plan of investigation for a patient with a medical disorder in partnership with the patient, obtaining informed consent as an essential part of this process
- Show the ability to perform bedside tests and procedures useful for the workup of common medical conditions
- Attempt interpreting the results of commonly used investigations
- Demonstrate the ability to formulate a likely diagnosis in a patient with a medical disorder
- Demonstrate clinical judgments and decisions, based on the available evidence in analyzing problems of an medical patient
- Attempt to outline a plan for management of a patient with medical disorder according to established principles and best evidence
- Explain the characteristics, indication, unwanted effects and management of adverse effects of drugs used in the treatment of medical conditions
- Demonstrate a willingness to adhere to infection prevention principles in approaching patients (clients) with medical conditions

2. Communicators

- Demonstrate communication with clients seeking medical care using a patient-centered approach that encourages patient trust and autonomy and is characterized by empathy, respect, and compassion
- Optimize the physical environment for clients seeking medical care comfort, dignity, privacy, engagement, and safety
- Recognize when the values, biases, or perspectives of clients seeking medical care, physicians, or other health care professionals may have an impact on the quality of care, and modify the approach to the patient accordingly
- Respond to patients' non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients
- Use patient-centered interviewing skills to effectively gather relevant biomedical and psychosocial information
- Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information
- Seek and synthesize relevant information from other sources, including the patient's/client's family, with the patient's consent.
- Demonstrate sharing information that are clear, accurate, timely, and adapted to patient's/client's and his or her family's level of understanding and need in a simulated setting
- Effectively communicate with ill patient and patient's family, and health care team

3. Collaborators

- Show the willingness to work with other professionals like nurses, lab technicians, etc to maintain a climate of mutual respect and shared values
- Establish and maintain healthy relationships with physicians and other colleagues in the health care professions to support relationship-centered collaborative care
- Recognize one's own professional roles and limits

- Share knowledge and information to other colleagues
- Show respect toward other physicians and members of the health care team
- Respect and value diversity among other colleagues

4. Leaders

- Demonstrate leadership in professional practice
- Manage their practice and career of themselves, other health care providers and support staff
- Demonstrate the ability to effectively plan and efficiently manage one's own time and activities to cope with uncertainty, and the ability to adapt to change
- Promote occupational health as physicians for sustainable practice
- Demonstrate the ability to build and manage team

5. Health Advocate

- Demonstrate responding to the needs of patients and the community as a whole by advocating for policy and system-level change
- Demonstrate responding to the individual patient health needs and rights by advocating with the community within and beyond the clinical environment
- Demonstrate working patients and their families to address the determinants of health that affect the community and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors
- Lead a healthy life style to be role models to promote health through for example participating in screening for TB, HIV, diabetes, hypertension etc.

6. Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice
- Ensure patient safety is maintained while learning
- Provide effective feedback to enhance learning and performance
- Analyze global and national causes and trends in morbidity and mortality (e.g. Infectious diseases TB, HIV, malaria,... Non-infectious diseases DM, Hypertension, etc)
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry
- Contribute to the teaching of medical and other health sciences students

7. Professional

- Integrate other intrinsic roles into their learning and practice of medical conditions
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in medicals care
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients/clients seeking medical care, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice in medical conditions
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the provision of health services
- Participate in peer assessment and standard-setting
- Demonstrate a culture that recognizes, supports, and responds effectively to colleagues in need
- Demonstrate the ability to provide care in compassionate and respectful manner
- Implement professional development plan that promote lifelong learning
- Apply preventive ethics in daily clinical practice

Prerequisites: Pre-clerkship modules

Duration: 10 weeks

Teaching and learning methods

- Interactive lecture
 - Case study, Role play
- Simulation based practice
- Clinical practicum/ rounds
- Bedside
 - Whole group sessions, Facilitated discussion
- PHC/community visit
- Preceptorship
- Reflective Portfolio

Teaching and learning materials

- Videos of patient examinations and clinical procedures
- Imaging studies of different medical conditions
- Mannequins for diagnostic and therapeutic procedures in Internal medicine

Methods of assessment

Formative

- Progressive assessment on engagement in interactive presentations, case study, whole group session
- Direct observation of clinical skills and feedback in clinical practicum, round, bedside and skill lab
- Global rating of performance or 360 degree evaluation in clinical practicum and

- PHC/community visit
- Logbook
- Portfolio/Personal research and reflection exercise
- Professionalism mini-evaluation exercise (P-MEX)- Direct observation of student's professional behaviors

Summative

- Progressive
 - Case study³⁸-5%
 - Clinical practicum/round, bedside³⁹-15%
 - Professionalism mini-evaluation exercise (P-MEX)- 5%
 - Skill lab⁴⁰ -15%
 - Logbook⁴¹-10%
- End Module
 - Written-30%
 - OSCE-20%

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- Keith L. Moor, Arthur F. Dally, Anne M.R. Agur. Clinically oriented anatomy. 6th edition. 2010
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- Harrison's text book of medicine, 18th edition
- Kumar and Clark, Clinical Medicine 8th edition
- Davidson's Principles and Practice of Medicine, 21st edition
- EFMOH, National compassionate and respectful care training participant manual for health work force, 2017
- EFMOH, National compassionate and respectful care facilitators guide, 2017

Schedule

³⁸ As much as possible, PBL cases during clerkship should preferably be based on real cases observed during clinical practicum

³⁹ These should be assessed based on structured checklist/global rating system

⁴⁰ These should be assessed based on structured checklist/global rating system

⁴¹ These should be assessed based on structured checklist/global rating system

Schedule	Learning Activity	Required Reading/ Assignment
Week 1	Interactive Lecture (10 hrs) - Cardinal manifestations and presentations of common diseases - General patient approach(HX, P.E, DDX) •(Headache , Cough, Chest pain, Hemoptysis, Coma, Fever)-Local names for cardinal symptoms	Reflective portfolio
	Clinical practicum (6 hrs) Student placement at outpatient and inpatient departments (medical OPD and medical ward)	
	PBL/ Case study Approach to a patient with fever/Headache/cough/coma/Syncope	
	Bedside Teaching (4 hrs) Patient with cough/fever/chest pain	
	Clinical skills lab (2 hrs) Practice: Hx and P/E. Approach to a patient with fever/Headache/cough/coma/Syncope	
	Primary health care or community visit (2hrs lecture/week and 4 hrs attachment/week) - Medical OPD in health centers - Home visits to patients with cardinal manifestations of common diseases. (fever/Headache/cough/coma/Syncope)	
	Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities	
	Seminar (2 hrs)	
Week 2	Interactive Lecture (6 hrs) - Infectious diseases: Viral infections (including rabies and HIV/AIDS) - Bacterial infections (streptococcal, staphylococcal, meningococcal, salmonella, tetanus, salmonella, brucella, bacillary dysentery, anthrax, mycoplasma, chlamydia) - Mycobacterial infection (TB, Leprosy)	Reflective portfolio
	Clinical practicum (6 hrs) Student placement at outpatient and inpatient departments (medical OPD, TB and leprosy clinic, medical ward)	
	PBL/Case Study HIV/AIDS, TB considering privacy and confidentiality and the role of professionals in caring as well as the public health aspect	

	• A case of Meningitis • A case of Tetanus	
	Bedside teaching (4 hrs) • Case with systemic bacterial infection, HIV, TB, leprosy , Meningitis	
	Clinical skills lab (2 hrs) • complete blood count, HIV rapid test • Video showing manifestations of opportunistic infections • Practice: history taking and physical examination of a patient with TB • Simulation – pleural tap Side lab • Gram’s stain, AFB, ESR • VDRL, stool exam, skin snip	
	Primary health care or community visit(2hrs lecture/week and 4 hrs attachment/week) • Associations of PLHIV and organization working on care and support for PLHIV in the community • Health center OPD • TB centers and rehabilitation centers • Home visits to patients on TB treatment and people with chronic cough in the community	
	Whole group session (3 hrs) • Faculty facilitated discussion and reflection on the week’s learning activities	
Week 3	Interactive Lecture (6hrs) • Spirochetes (relapsing fever, syphilis), rickettsia, fungi (aspergillosis, cryptococcosis, candida) • Protozoa(malaria, leishmaniasis, toxoplasmosis, amoebiasis, giardiasis, trypanosomiasis) a • Helminthes (schistosomiasis, hydatid cyst, strongyloidosis, ascariasis, hook worm, filariasis) • Approach to a patient with anemia,coagulation disorders • Acute and chronic leukemia, Hodgkins and non-Hodgkins lymphoma, Hemorrhagic disease, thrombotic disease Clinical practicum (6 hrs) Student placement at outpatient and inpatient departments	Reflective portfolio
	PBL/Case Study A case of Acute febrile illness (malaria), Intestinal parasitosis A case of hematologic malignancy, anemia	
	Bedside teaching (4 hrs) • Case: acute febrile illness, Intestinal parasitosis (hook work, amoebiasis), A case of hematologic malignancy, anemia.	

	Clinical skills lab (2 hrs) - Practice: history taking and physical examination from febrile patient Side lab - Hemoglobin/hematocrit - AFB, ESR - Slides of common parasites – protozoa and helminthes - Blood film, and peripheral morphology, VDRL, stool exam, skin snip	
	Primary health care or community visit(2hrs lecture/week and 4 hrs attachment/week) - Health center OPD - Health education about hygiene, helminthes, malaria and relapsing fever, anemia - Visit households, prisons, observe and assess risks for Mosquitoes, RF, anemia	
	Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities	
Week 4	Interactive Lecture (6 hrs) - Congestive heart failure (pathophysiology and management), cardiogenic shock - Disease of the myocardium and endocardium: infective endocarditis, rheumatic heart disease - Pericardial disease - Arrhythmia, ischemic heart disease, atherosclerosis, pulmonary hypertension - Hypertension	Reflective portfolio
	Clinical practicum (6 hrs) Student placement at outpatient and inpatient departments (medical OPD, medical ward)	
	PBL/ Case Study congestive heart failure, Acute coronary syndrome	
	Bedside teaching (4 hrs) Case: congestive heart failure Case: Emergency hypertension and its complications	
	Clinical skill lab (2 hrs) Practice: Examination of the cardiovascular system (taking history and physical examination on a patient with heart disease) ECG measurement Audio of abnormal heart sounds Simulation of a failed heart – animation IV line opening Approach to a patient with suspected arrhythmia or cardiac arrest (Practice taking history from a patient with heart disease) Practice cardio pulmonary resuscitation in a simulator	

	Abnormal findings on a chest x-ray, ECG, video, audio for heart sounds Side lab Cardiac enzymes, phlebotomy Primary health care/community(2hrs lecture/week and 4 hrs attachment/week) • Health center OPD and ward • Patients with heart disease Whole group discussion (3 hrs) • Faculty facilitated discussion and reflection on the week's learning activities	
Week 5	Interactive Lecture(6 hrs) • Respiratory diseases: Bronchial asthma, chronic obstructive lung disease, disease of the pleura • Lung cancer, interstitial lung disease • Occupational and environmental lung disease Clinical practicum (6hrs) Student placement at outpatient and inpatient departments (medical OPD, medical ward) PBL/Case Study A case of COPD Bedside teaching (4 hrs) Case: COPD, bronchial asthma Clinical skill lab (2 hrs) Approach to a patient with respiratory disease(Practice taking history and physical examination from a patient with respiratory problem) Pulse oximetry, spirometry Abnormal chest x-rays Primary health care/community visit(2hrs lecture/week and 4 hrs attachment/week) • Health center • Home visit to patient with bronchial asthma and COPD • Health education on prevention causes of lung disease(smoking, indoor pollution) Whole group discussion (3 hrs) • Faculty facilitated discussion and reflection on the week's learning activities • Ethical case analysis and ethics consultation & Preventive ethics(2 hrs): case study & introductive lecture	Reflective portfolio
Week 6	Interactive Lecture (6 hrs) • Acid peptic disease • GI bleed • Diseases of the liver – viral hepatitis, chronic hepatitis, cirrhosis	Reflective portfolio

	• Diseases of the biliary tract- cholecystitis • Pancreatitis • Diseases of rectum and anus • Gastrointestinal diseases: Disorder of GI motility, Functional GI disorder (inflammatory bowel disease, dyspepsia), Diseases of the esophagus • Malabsorption disorders	
	Clinical practicum (6 hrs) Student placement at outpatient and inpatient departments (medical OPD, medical ward)	
	PBL/ Case Study Peptic Ulcer disease, Chronic liver disease, Hepatitis, IBD	
	Bedside teaching (4hrs) Case: peptic ulcer disease, CLD, Hepatitis,	
	Clinical skill lab (2 hrs) • Approach to a patient with GI disease (Practice taking history and physical examination from a patient with GI disorder) • Insertion of NG tube • Peritoneal tap • Endoscopy videos Side lab • Liver function tests • Peritoneal fluid analysis including gram stain	
	Primary health care/community visit (2hrs lecture/week and 4 hrs attachment/week) • Health center • Visits to patients with chronic liver disease • Education on risk reduction (eg. Alcohol, HBV vaccination)	
	Whole group discussion (3 hrs) • Faculty facilitated discussion and reflection on the week's learning activities • Factors affecting provision of CRC, CRC fatigue, Disrespectful characteristics of CRC & nurturing problem solving skills on health care teams (2 hours introductive lecture & case study)	
Week 7	Interactive Lecture (6 hrs) ▪ Acute renal failure, chronic renal failure ▪ Glomerular disease – nephritic disease and nephrotic syndrome ▪ Acid base and electrolyte disorder, collagen vascular disease(SLE, Vasculitides, RA)	Reflective portfolio
	Clinical practicum (6 hrs) Student placement at outpatient and inpatient departments (medical OPD, medical ward)	

	PBL/ Case Study Renal Failure, Nephrotic syndrome	
	Bedside teaching (4hrs) Case: nephric disease and nephrotic syndrome	
	Clinical skill lab (2 hrs) - Approach to a patient with renal disease (Practice taking history and physical examination) - Practice catheterization in simulator Side lab - Renal function tests - Urine analysis (dipstick, microscopy)	
	Primary health care/community visit (2hrs lecture/week and 4 hrs attachment/week) - Health center - Patients with renal disease - Health education on prevention of renal failure(eg. Control HTN, DM)	
	Whole group discussion (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities	
Week 8	Interactive lecture (6 hrs) - Diabetes and disorders of endocrine pancreas - Diseases of the hypothalamus, pituitary - Diseases of thyroid and parathyroid glands - Diseases of adrenal gland - Disorders of Lipid metabolism and obesity	Reflective portfolio
	Clinical practicum (6 hrs) Student placement at outpatient and inpatient departments (medical OPD, medical ward)	
	PBL/Case study diabetic ketoacidosis, metabolic Syndrome	
	Bed side teaching (4 hrs) Case: diabetes mellitus Case: thyrotoxicosis, Cushing disease, or adrenal insufficiency	
	Clinical skill lab (2 hrs) Practice taking history and physical examination from a diabetic patient Slides and videos of DFU, Cushing, Acromegaly patients. Patients with Joint deformities Side lab Random or fasting blood glucose Lab: hormonal assay – TSH, T3, T4, Cortisol	
	Primary health care/community visit (2hrs lecture/week and 4 hrs attachment/week) - Health center - Home visits of patients with diabetes mellitus	

	Health education on prevention of Obesity and sedentary life	
	Whole group discussion (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities	
Week 9	Interactive Lecture (6 hrs/week) - Seizure disorders - Headache - Cerebrovascular accident (Ischemic Cerebrovascular disease and hemorrhagic Cerebrovascular disease) - Paraplegia - Myopathies and neuromuscular disorders - Meningitis, brain abscess, encephalitis, poliomyelitis - Infections of the CNS - Cranial nerve disorders, Parkinsonism and movement disorders - Brain tumors	Reflective portfolio
	Clinical practicum (6 hrs/week) Student placement at outpatient and inpatient departments and seizure clinic	
	PBL/ Case Study A case of paraplegia(Stroke), quadriplegia and GBS, ICSOL,	
	Bedside teaching (4hrs/week) Case with status epileptics, meningitis Case: Cerebrovascular accident or stroke	
	Side lab or clinical skills lab (2 hrs/week) - Approach to the patient with neurological disease (Practice: history taking and physical examination of a patient with neurological disease) - Skill lab: EEG, Video on type of seizure, video showing brain lesion, movement disorders	
	Primary health care or community visit (2hrs lecture/week and 4 hrs attachment/week) - Health center OPD - Home visits to patients with epilepsy and education on reduction of stigma and traditional beliefs and practices	
	Whole group session (3 hrs/week) on the week's learning activities	
Week 10	Monday-Tuesday Faculty facilitated discussion and reflection: Study break for exam	
	Wednesday: Written exam	
	Thursday and Friday: Standardized oral exam and structured long cases	

MODULE TITLE: SURGERY CLERKSHIP I

MODULE CODE: SURG 4011

DURATION: 10 weeks

MODULE DESCRIPTION:

This first clerkship in surgery is intended to enable medical students to develop their clinical knowledge, skills and attitude required for the understanding and diagnosis of surgical disorders with emphasis given to common life-saving procedures and interventions. During this clerkship, students will have opportunities to review and relate knowledge of biomedical sciences for the understanding of patient problems they encounter in clinical settings. Students will also develop relevant competencies in health promotion and disease prevention, communication, professionalism, management of information, critical thinking, practice-based improvement and research and inter-professional collaborative practice.

MODULE DESIGN:

The teaching will focus on building the ability of the student in history taking, physical examination in the clinical setting, formulating differential diagnosis, reaching to appropriate diagnosis, listing relevant investigation and constructing appropriate management plan.

MODULE PREREQUISITE:

- Students have completed Pre -clerkship courses successfully

MODULE OBJECTIVES

- By the end of this module, students will be able to assess patients with surgical and orthopedic disorders and formulate differential diagnoses.

Supporting Objectives

AS A MEDICAL EXPERT

- Demonstrate a commitment to high-quality surgical care of their patients
- Integrate the other six roles into their practice of medicine
- Apply knowledge of the clinical and biomedical sciences relevant to surgery and orthopedics
- Apply concepts and principles of public health sciences to protect and promote individual and population health
- Perform appropriately timed surgical clinical assessments with recommendations that are well organized and properly documented in written and/or oral form
- Carry out professional duties in the face of multiple, competing demands
- Recognize and respond to the complexity, uncertainty, and ambiguity inherent in surgical practice
- Identify and prioritize issues to be addressed in a surgical patient encounter
- Elicit a history, perform a complete physical exam, make a differential diagnosis, select appropriate investigations, perform basic investigations and interpret their results for the purpose of diagnosis and management, disease prevention, and health promotion
- Diagnose and establish a patient-centered management plan based on available evidence
- Recognize and respond to adverse events and near misses
- Contribute to a culture that promotes patient safety
- Adopt strategies that promote patient safety and mitigate negative human and system factors
- Demonstrate the ability to detect post op complications
- Demonstrate the ability to identify preoperative risk factors

- Formulate assessments and plans with attention to acuity of illness, risk benefit estimations, and patient preferences
- Describe common breast conditions and outline the evaluation of breast complains
- Describe gastrointestinal malignancies, including risk factors, signs and symptoms and initial evaluation.
- Formulate the differential diagnosis of Acute abdomen, based on data gathered through medical interview, physical examination, and investigation
- Describe common benign disorders of the thyroid, including risk factors, sign symptoms and initial evaluation,
- Demonstrate knowledge of perioperative care and familiarity with surgical procedures.

AS A COMMUNICATOR

- Develop Competence in medical interview and physical examination of surgical and orthopedic patients and incorporate ethical, social, and diversity perspectives to provide culturally competent health care.
- Deliver oral presentations and write chart entries that are accurate, organized, well reasoned and appropriately thorough.
- Communicate using a patient-centered approach that encourages patient trust and autonomy and is characterized by empathy, respect, and compassion
- Optimize the physical environment for patient comfort, dignity, privacy, engagement, and safety
- Recognize when the values, biases, or perspectives of patients, physicians, or other health care professionals may have an impact on the quality of care, and modify the approach to the patient accordingly
- Respond to patients' non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients
- Manage disagreements and emotionally charged conversations
- Adapt to the unique needs and preferences of each patient and to his or her clinical condition and circumstances
- Use patient-centered interviewing skills to effectively gather relevant biomedical and psychosocial information
- Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information
- Provide a clear structure for and manage the flow of an entire patient encounter
- Seek and synthesize relevant information from other sources, including the patient's family, with the patient's consent.
- Share health care information and plans with patients and their families
- Skillfully share information and explanations that are clear, accurate, timely, and adapted to patient's and his or her family's level of understanding and need
- Communicate appropriately in difficult circumstances, such as when breaking bad news, and when discussing sensitive issues, such as alcohol consumption, smoking or obesity
- Communicate appropriately with difficult/ violent patients and people with mental illness
- Facilitate discussions with patients and their families in a way that is respectful, non-judgmental, and culturally safe
- Assist patients and their families to identify, access and make use of information and communication technologies to support their care and manage their health

- Use appropriate communication skills and strategies to help patients and their families make informed decisions regarding their health
- Document clinical encounters in an accurate, complete, timely, and accessible manner, in compliance with legal and regulatory requirements
- Communicate effectively using a written health record, electronic medical record, or other digital technology
- **Effectively communication with ill patient and patient's family, and health care team**

AS A COLLABORATOR

- **Demonstrate a clear understanding of health care team goals, primary roles of team members, and work effectively to help the team achieve its goals**
- Establish and maintain healthy relationships with fellow students and other colleagues in the health care professions to support relationship-centered collaborative care
- Negotiate overlapping and shared responsibilities with fellow students and other colleagues in the health care professions
- Recognize one's own professional roles and limits
- Share knowledge and information to other colleagues
- Show respect toward fellow students and members of the health care team
- Respect and value diversity among other colleagues
- Implement strategies to resolve conflicts in a manner that supports a collaborative culture

AS A Leader

- Analyze adverse events and near misses to enhance systems of care
- Allocate health care resources considering justice, efficiency, and effectiveness for optimal patient care
- Apply evidence and management processes to achieve cost-appropriate care
- Develop their leadership skills
- Facilitate change in health care to enhance services and outcomes
- Set priorities and manage time to balance practice and personal life
- Implement processes to ensure personal practice improvement
- Promote occupational health for sustainable practice
- **Demonstrate the ability to build and manage team**

AS A Health Advocate

- Demonstrate ability to work with patients to address determinants of health that affect them and their access to needed health services or resources
- Demonstrate ability to work with patients and their families to increase their opportunities to adopt healthy behaviors
- Demonstrate the ability to incorporate disease prevention, health promotion, and health surveillance into interactions with individual patients
- Demonstrate the ability to lead a healthy life style to be role models to promote population health
- Demonstrate ability to work with a community or population to identify the determinants of health that affect them
- Demonstrate ability to improve clinical practice by applying a process of continuous quality improvement to disease prevention, health promotion, and health surveillance activities
- Demonstrate ability to participate in a process to improve health in the community or population they serve

AS A SCHOLAR

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance using various internal and external data sources
- Demonstrate the ability to engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling and the impact of the hidden curriculum on learners
- Demonstrate the ability to promote a safe learning environment
- Demonstrate the ability to ensure patient safety is maintained when learners are involved
- Demonstrate the ability to collaboratively identify the learning needs of others and prioritize learning outcomes
- Demonstrate effective teaching to facilitate learning
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize practice uncertainty and knowledge gaps in clinical and other professional encounters and generate focused questions that can address them
- Demonstrate proficiency in identifying, selecting, and navigating pre-appraised resources
- Demonstrate the ability to integrate evidence into decision-making in their practice
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion and critique of their relevance to professional practice
- Evaluate the applicability (external validity or generalizability) of evidence from a wide range of information resources
- Discuss the barriers to and facilitators of applying study findings to professional practice
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

AS A PROFESSIONAL

- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of practice
- Recognize and respond to ethical issues encountered in practice
- Recognize and manage conflicts of interest
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the Profession
- Fulfill and adhere to the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting

- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Manage personal and professional demands for a sustainable practice throughout the physician life cycle
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need
- **Efficiently identify knowledge gaps, utilize appropriate information sources and critically evaluate information and data.**
- **Behave with accountability and dependability**
- **Implement professional development plan that promote lifelong learning**
- **Apply preventive ethics in daily clinical practice**
- **Demonstrate the ability to provide care in compassionate and respectful manner**

COURSE CONTENT

Unit One: Approach to the surgical patient

1. History

Rationale: History is essential as part of evaluation for primary health care and preventive medicine.

Students will be able to:

- A. take Complete a comprehensive medical interview
- B. Demonstrate interpersonal communication skills that build trust and address contextual factors (race, ethnicity, spirituality/ religion, disability)
- C. Communicate results by well organized written and oral reports
- D. Perform risk assessment for malignancies, Emergency versus elective

2. Examination

Rationale: an accurate examination compliments the history, provides additional information, and helps determine diagnosis and guide management. It also provides an opportunity o educate and reassure the patient

The student will be able to:

- a. Interact with the patient to gain her/ his confidence an cooperation, and assure her/ him comfort and modesty
- b. Perform accurate examinations in a sensitive manner, including
 - a. Breast examination
 - b. Thyroid examination
 - c. Lymphoglandular examination
 - d. Abdominal examination
 - e. Genitourinary examination
 - f. Musckulo skeletal examination
- c. Use accurate medical terminology to describe
 - a. Normal anatomy
 - b. Appearance of common pathology of the skin, thyroid, genitalia, perianal region
 - c. Appearance of common breast changes and disorders
- d. Communicate the relevant results of the exam in a well-organized written and oral reports

3. Investigations and diagnosis

Rationale: the ability to order and interpret investigations that are relevant is one of the important essentials to reach to a diagnosis.

The student will be able to:

- a. Identify appropriate investigations for a list of differential diagnoses
- b. Advocate the importance of efficient resource utilization
- c. Describe the underlying mechanism of investigations
- d. Interpret investigation results

Unit two: Acute abdomen in Adults

1. Large bowel Obstruction

Rationale: Large bowel obstruction is one of the commonest causes of acute abdomen in adults.

Early, and accurate diagnosis and prompt intervention are known to prevent complications and significantly lower morbidity and mortality.

Students will be able to:

- a. Describe the signs and symptoms of Large bowel obstruction
- b. Diagnose the presence of bowel obstruction
- c. Interpret a plain abdominal X-ray with Large bowel obstruction
- d. Differentiate between large and small bowel obstruction
- e. Differentiate between complicated and simple obstruction
- f. Differentiate between malignant and benign causes of large bowel obstruction
- g. Provide early management of large bowel obstruction
- h. Comply to the urgent need of early recognition and management of large bowel obstruction
- i. Accept the importance of preventive surgery in the community by providing health education

2. Small Bowel Obstruction

Rationale: Small bowel obstruction is another common and important cause of acute abdomen in adults. Early recognition and prompt management is necessary to prevent complications that are associated with morbidity and mortality.

Students will be able to:

- a. Describe the signs and symptoms of Small bowel obstruction
- b. Identify the causes of small bowel obstruction
- c. Diagnose the presence of bowel obstruction
- d. Differentiate between large and small bowel obstruction
- e. Differentiate between complicated and simple obstruction
- f. Explain the volume and electrolyte derangement in small bowel obstruction
- g. Interpret a plain abdominal X-ray with small bowel obstruction
- h. Describe the principles of management of small bowel obstruction
- i. Manage conservatively patients with small bowel obstruction

3. Acute Appendicitis

Rationale: Acute appendicitis is another important emergency surgical illness commonly encountered especially in young adults. Delay in diagnosis and treatment results in serious complications like perforation and abscess collection which significantly increases the morbidity and can even lead to death of the patient.

Students will be able to;

- a. Describe the signs and symptoms of acute appendicitis
- b. List the differential diagnosis of RLQ pain in both sexes
- c. Diagnose acute appendicitis
- d. Differentiate between appendicular mass and abscess

- e. Differentiate between simple versus complicated appendicitis
- f. Apply Alvarado score in the diagnosis of acute appendicitis
- g. Comply to the urgent diagnosis and surgical management of acute appendicitis

4. Perforated PUD

Rationale: Perforated Peptic ulcer disease is a common emergency illness. Early diagnosis and treatment will prevent serious morbidity and mortality.

Students will be able to;

- Describe the signs and symptoms of perforated PUD
- List the differential diagnosis of epigastric pain
- Identify the risk factors and causes of PUD
- Describe the pathophysiology of peptic ulcer formation
- Diagnose Perforated PUD by doing Complete History and Physical exam
- Outline the investigation plan for a patient suspected of having perforated PUD
- Interpret an erect chest X-ray with air under diaphragm
- Outline the initial management of a patient diagnosed to have perforated PUD
- Describe the surgical treatment of Perforated PUD
- Facilitate the safe transfer of patients who are diagnosed to have perforated PUD
- Explain the prognostic factors for the treatment outcome of a patient with perforated PUD

5. Acute Cholecystitis

Rationale: Acute cholecystitis is a common surgical problem affecting mainly middle aged women. Prompt diagnosis and early management will decrease morbidity and mortality.

The students will be able to:

- Describe the signs and symptoms of acute cholecystitis
- List a differential diagnosis of a patient with RUQ abdominal pain
- Diagnose a patient with acute cholecystitis
- Describe the pathophysiology of gall stone formation
- Describe the possible complications Acute cholecystitis
- Describe the sonographic characteristics of acute cholecystitis
- Outline the investigation of patient suspected of acute cholecystitis
- Interpret the CBC result of a patient suspected to have acute cholecystitis
- Manage conservatively a patient diagnosed with acute cholecystitis
- Describe the surgical management of gall stone disease

6. Acute Pancreatitis

Rationale: Acute pancreatitis is a common surgical problem encountered in the emergency setting. It needs to be promptly diagnosed and managed to prevent the occurrence of serious complications.

The students will be able to:

- Describe the signs and symptoms of acute pancreatitis
- Diagnose a patient with acute Pancreatitis
- Outline the investigation modalities of a patient suspected of acute pancreatitis
- Provide early management of acute pancreatitis

7. Non-surgical conditions that mimic acute abdomen

Unit three: Benign and Malignant disorders of the gastrointestinal system

1. Gastric Ca

Rationale: Gastric ca is a common illness. High index of suspicion is mandatory to diagnose the illness at an early stage, as prognosis is stage dependent.

The student will be able to:

- Describe the signs and symptoms of gastric ca.
- Identify risk factors of Gastric Ca
- Describe the 'alarming symptoms' of gastric ca
- Diagnose gastric ca
- Differentiate between early and advanced gastric ca
- Describe the histologic and gross pathological classification of Gastric cas
- Outline the investigative modalities for diagnosing gastric ca
- Outline the investigative modalities for staging gastric ca
- Provide patient education on risk factors of gastric ca
- Interpret barium meal
- Describe the electrolyte disturbances that can be associated with vomiting
- Provide initial management of a patient with fluid and electrolyte disturbance related to vomiting
- Explain the prognosis of gastric ca
- Outline the principles of management of gastric ca by stage

2. Colorectal Ca

Rationale : Colorectal ca is one of the common GI related cancer that is affecting our people. High index of suspicion should be there

3. Pancreatic tumors
4. Tumors of the biliary tract and gall bladder
5. Esophagealca
8. Inflammatory Bowel Disease
9. Stomas

Unit four: Disorders of the hepatobiliary system

1. Gall stone disease

Rationale: Gall stone disease is a common surgical problem affecting middle aged women. Though majority are asymptomatic, when symptomatic they can lead to serious complications. Gall stone disease is an important differential diagnosis for patients presenting with supraumbilical abdominal pain

The student will be able to:

- Describe the sign and symptoms of gall stones
- Identify gall stone disease as a differential diagnosis for supraumbilical abdominal pain
- Diagnose a patient with symptomatic gall stone disease
- Identify the risk factors for gall stone disease
- Explain the pathophysiology of stone formation in bile
- Identify the complications of gall stone disease
- Outline the investigation of patients suspected of gall stone disease
- Describe the principles of management of gall stone disease

2. Obstructive Jaundice

Describe the signs and symptoms of OJ

Diagnose the presence of jaundice

Differentiate between medical and surgical jaundice

Differentiate between benign versus malignant obstructive jaundice

Identify the complications of obstructive jaundice

Describe the signs and symptoms of ascending cholangitis
Describe the pathophysiology of complications of obstructive jaundice
Diagnose ascending cholangitis

3. Liver abscess (pyogenic, amoebic)

4. Hydatid cyst of liver

Unit five: Acute abdomen in children

1. Intussusception

Student will be able to: Recognize that intussusception can be misdiagnosed as dysentery or other diarrheal diseases

2. Acute appendicitis

3. Bilious vomiting in neonates

4. Bowel obstruction(HSD, ARM)

Unit six: Anterior Neck swelling

1. Endemic Goiter

2. Solitary nodule

3. Toxic Goiters

4. Inflammatory and infectious thyroid disorders

5. Malignant tumors of the thyroid

Unit seven: Breast conditions

1. Benign Disorders of the breast

2. Malignant Disorders of the breast

Unit eight: Urology

Section A: Bladder outlet Obstruction

Section B: Stone diseases

Section C: Urologic malignancies (bladder ca, RCC)

Section D: Scrotal swelling

Unit nine: Neurosurgery

1. Head Injury

2. Spinal Cord injury

Unit ten: Chest surgery

1. Suppurative diseases (Empyema, lung abscess,...)

Unit Eleven: Trauma and accidents

1. Chest trauma

2. Abdominal trauma

3. Head injury

4. Burn

TEACHING /LEARNING METHODS:

The following teaching Methods will be used during the attachment:

- Interactive lecture
- Video show
- Demonstration
- Role play
- Case study
- Simulated (clinical skills lab) practice
- OPD teaching
- Teaching round
- Bedside teaching
- Community visit

- Portfolio
- Personal research and reflection exercise
- Inter-professional learning experience in the skills lab, clinic, community
- Morning report, conference
- Seminars series
- Grand rounds.
- Night Duty once per week
- Ambulatory clinical attachment
- Hospital ward attachment
- Skill labs and side labs practice
- Self-directed learning (SDL)
- E-Learning
- Journal club
- Community learning sessions
- Mentorship

Equally important is the unstructured teaching that happens all day every day on patient issues being managed by the health team (Medical Doctors Nurses etc.) every health team member should contribute to the education of the student .Teaching by example through good team spirit and leadership.

Assessment types

Formative

This will not contribute to the total grade of the student but provides continuous feed back to the student and it can be applied in the form of providing feed backs on assignments completed, students' professional behavior and etc.

- Direct observation of clinical skills
- Logbook
- Global rating of performance or 360 degree evaluation
- Portfolio
- Personal research and reflection exercise

Summative assessment

- **Progressive (continuous) assessment (Quizzes, Concept application exercise - 15%**
 - Professionalism mini-evaluation exercise (P-MEX)- Direct observation of student's professional behaviors -5%

- **end of attachment practical exam(long, short, viva ,OSCE)--50%**

a. Written exam -- 30%

Methods of assessment

1. DOCS (Direct Observation of Clinical Skills)
2. OSCE (Objective Structured Clinical Examination)
3. OSPE (Objective Structured Practical Exam)
4. PRRE (Personal Research and Reflection Exercise)
5. Project Work
6. Clinical examination (Long case and Short cases)
7. Oral exam (Viva)
8. Written exam (MCQ, Short Essay, Matching, True- False with reasoning)
9. Log books and portfolios
10. Global rating
11. 360° Assessment (Self, Peer, faculty, health facility staff and community member)

Module Schedule

Date	Learning Activity	Required Reading/ Assignment
Week 1 Basic Surgery	Interactive lecture (3 hrs) • Introduction to the module & History of surgery • Metabolic response to injury • Shock, electrolytes and fluid management of a surgical patient	Reflective portfolio
	Clinical practicum and teaching round (12hrs.)⁴² Student placement at surgical outpatient and inpatient departments	
	PBL/Case Study(3hrs)	
	Case Study	
	Bedside (4hrs) • Case related to principles and ethics and professionalism in surgery (Shifting the focus of the bed side discussion more towards patient social history, socioeconomic considerations, ethical dilemmas, in general patient as a human being) • Case related to the topics in the week	
	Clinical skills lab (4 hrs) • Simulated practice: Approaching a surgical patient (history taking and physical examination) • Simulated practice: Infection control and patient safety (hand washing, use of personal protective equipment, safe handling of sharps and other equipment, medical waste management, instrument processing) • Simulated practice: Wound care	
	Primary health care or community visit(1hr lecture/week and 4 hrs attachment/week) • Surgical OPD and ward • Home visits of patients with surgical conditions	
	Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities	
Week 2 Basic Surgery	Interactive lecture (3hrs) • Hemostasis, Bleeding and transfusion • Wound healing and principles of wound care • Surgical infections including soft tissue infection and use of antibiotics in surgery	Reflective portfolio

⁴² Logbook includes records of all clinical skills attended or performed and community visit activities and encounters

	Clinical practicum (12hrs) Student placement at surgical outpatient and inpatient departments	
	Round (4hrs)	
	PBL/Case Study(3hrs) • Perioperative management	
	Bedside (4hrs) • Case related to the topics in the week • Case related to surgical infections	
	Clinical skills lab (4hrs) • Simulated practice: Preoperative evaluation • Simulated practice: Repair of laceration (Suturing) • Simulated practice: Wound care	
	Primary health care or community visit(2hrs lecture/week and 4 hrs attachment/week) • Surgical OPD and ward • Home visits of patients with surgical conditions	
	Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities • Communicate within health care provider group, Leadership in effective team building & Team management skills ((2 hours of interactive lecture , Case study & role play)	
Week 3 Peripheral vascular and plastic surgery	Interactive Lecture (3hrs) • Common surgical conditions of the head and neck (cleft lip and palate, diseases of salivary glands) • Skin tumors • Musculoskeletal tumors • Peripheral arterial, venous and lymphatic diseases	Reflective portfolio
	Clinical practicum (12hrs) Student placement at surgical outpatient and inpatient departments	
	Round (6hrs)	
	PBL/Case Study(3hrs)	
	Case study • Burn	
	Bedside (6hrs) • Case related to the topics in the week	
	Clinical skills lab (4hrs) • Simulated practice: Assessing (history and physical examination) a burn patient including depth and area of burn; calculating rate and quantity of fluid to be	

	given; techniques for treating a burn Simulated practice: Subcutaneous abscess drainage	
	Primary health care or community visit(1hr lecture/week and 4 hrs attachment/week) Home visits of patients prioritizing patients with surgical conditions covered during the week	
	Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities	
Week 4 Endocrine and Breast	Interactive Lecture (3hrs) - Surgical diseases of the thyroid and parathyroid glands (goiter, disorders of thyroid metabolism) - Surgical diseases of the breast (benign and malignant tumors, breast anomalies)	Reflective portfolio
	Clinical practicum (12hrs) Student placement at surgical outpatient and inpatient departments	
	Round (4hrs)	
	PBL/Case Study(3hrs)	
	Case study 3hrs Adrenal gland tumors	
	Bedside (4hrs) - Case related to thyroid - Case related to breast lump	
	Clinical skills lab (4hrs) Simulated practice: Assessing a patient with neck mass (history and physical examination) Simulated practice: Assessing a patient with breast lump (history and physical examination)	
	Primary health care or community visit (1hrs lecture/week and 4 hrs attachment/week) - Surgical OPD and ward - Home visits of patients prioritizing patients with surgical conditions covered during the week	
	Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities	
Week 5 Cardio-thoracic	Interactive Lecture (3hrs) - Surgical diseases of chest, lungs and heart (injuries, hemoptysis, lung infections, diseases of the pleura and pericardium, mediastinitis) - Surgical diseases of the esophagus (injuries, esophagitis, achalasia, esophageal cancer, hiatal	Reflective portfolio

	hernia)	
	Clinical practicum (12hrs) Student placement at surgical outpatient and inpatient departments	
	Round (4hrs)	
	PBL/Case Study(3hrs) • A case of dysphagia	
	Case Study • Patient with chest trauma	
	Bedside (6hrs) • Case related to surgical diseases of the thorax • A case of dysphagia	
	Clinical skills lab (4hrs) Simulated practice: • History and chest examination • interpretation of radiologic imaging of chest trauma • Barium swallow	
	Primary health care or community visit (1hrs lecture/week and 4 hrs attachment/week) • Surgical OPD and ward • Home visits of patients prioritizing surgical conditions covered during the week	
	Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities	
Week 6	Interactive Lecture (3hrs) • Surgical conditions of Pancreas • Gall stone diseases • Surgical disorders of the liver	Reflective portfolio
GI	Clinical practicum (12hrs) Student placement at surgical outpatient and inpatient departments	
(Hepato biliary acute abdomen and pancreas)	Round (4hrs)	
	PBL/Case Study(3hrs) • Approach to patient with acute abdomen	
	Case Study • Abdominal wall hernia	
	Bedside (4hrs) • Case related to Hepato-biliary disease • Case related to Hepato-biliary disease	
	Clinical skills lab (4hrs) • Simulated practice: History taking and abdominal examination including interpretation of imaging studies	

	of different abnormalities Primary health care or community visit (1hrs lecture/week and 4 hrs attachment/week) • Surgical OPD and ward • Home visits of patients prioritizing surgical conditions covered during the week Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities	
Week 7	Interactive Lecture (3 hrs) • PUD and Gastric ca • Colorectal Ca • Perianal diseases Clinical practicum (12hrs) Student placement at surgical outpatient and inpatient departments Round (4hrs) PBL/Case Study(3hrs) • Intestinal obstruction Case study Perianal disease Bedside (4hrs) • Case related to GOO • Case related to rectal bleeding/colonic ca Clinical skills lab (4hrs) • Simulated practice: History taking and abdominal examination including interpretation of imaging studies of different abnormalities • Simulated practice: Primary health care or community visit (1hrs lecture/week and 4 hrs attachment/week) • Surgical OPD and ward • Home visits of patients prioritizing surgical conditions covered during the week Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities	
Week 8	Interactive Lecture (3hrs) ▪ BOO ▪ Urolithiasis ▪ Urologic tumors Clinical practicum (15hrs) Student placement at surgical outpatient and inpatient departments	Reflective portfolio

	Round (6 hrs)	
	PBL/Case Study(3hrs)	
	Bedside (6hrs) • Case of BOO • Case of flank pain	
	Clinical skills lab (4hrs) • Simulated practice: Assessing surgical patients with urinary symptoms • Simulated practice:DRE	
	Primary health care or community visit(1hr lecture/week and 4 hrs attachment/week) • Surgical OPD and ward • Home visits of patients prioritizing surgical conditions covered during the week	
	Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities	
Week 9	Interactive Lecture (4hrs) ▪ Head Injury & Spinal Injury ▪ Brain tumors ▪ Peripheral Nerve injury ▪ Common problems affecting hip, knee, foot and ankle ▪ Inflammation and infection of the bone and joints Clinical practicum (15hrs) Student placement at orthopedic outpatient and inpatient departments and follow up clinics	
	Round (6 hrs)	
	PBL/Case Study(3hrs) • Peripheral nerve injury ??? • Fracture of the long bones of the lower limb	
	Bedside (6hrs) • Case of Head injury • Case of Qaudri/Paraplegia • Case study osteomyelitis	
	Clinical skills lab (4hrs) • Simulated practice: Assessing surgical patients with Spinal and head injury • Simulated practice: POP application • Simulated practice: Interpreting radiologic findings of upper limb and lower limb fractures and disorders	
	Primary health care or community visit (1hrs lecture/week and 4 hrs attachment/week) • Surgical OPD and ward • Home visits of patients prioritizing surgical conditions	

	covered during the week	
	Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities	
Week 10	Monday-Wednesday: Study break for exam	Reflective portfolio
	Thursday: Written exam	
	Friday: OSCE/OSPE	

RESOURCES

Human resources

- Clinicians and specialists
- General Practitioner
- Public health professionals
- Other relevant Health Professionals (for Inter professional Pool)
- Education Experts

Training, Teaching and Learning Materials

- Trainer's Guide

Reference manuals and books

- Learners Guide
 - Handout
 - Books
 - Text books
 - E-books

Learning Facilities / Infrastructure

- Class room with the right class room environment.
- Skill laboratory
- Dissection room
- Equipped Multipurpose Integrated laboratory
- Library
- IT center with good internet connection
- Clinical set up (Primary and secondary where possible tertiary care health institutions)

Portable and stationary plants, machines and equipment

- Flip charts
- Video
- CD-ROM
- Computers
- DVD Player
- LCD projector Overhead Projector
- White board (2x1.5)

REFERENCE

- Norman J Williams, Christopher J.K. Bulstrode, P Ronnan O'Connell. Bailey and Love's Short Practice of Surgery. 26th ed. 2012
- Courtney M. Townsend Jr. [et al.]. Sabiston textbook of surgery: the biological basis of modern surgical practice.—19th ed. 2012
- Schwartz, Principles of Surgery. 10th edition. 2010
- Snell, Richard S. Clinical anatomy by regions. 9th edition. 2012
- Keith L. Moor, Arthur F. Dally, Anne M.R. Agur. Clinically oriented anatomy. 6th edition. 2010
- Eric Arthur Newsholme and Tony R. Leech. Functional biochemistry in health and disease. 2010.
- Stephen Goldber. Clinical biochemistry made ridiculously simple. 1999.
- Ashis Banerjee. Clinical physiology. An examination primer. 2005.
- Bertram G. Katzung. Basic and clinical pharmacology. 12th edition. 2011

- EFMOH, National compassionate and respectful care training participant manual for health work force, 2017
- EFMOH, National compassionate and respectful care facilitators guide, 2017

MODULE TITLE: PAEDIATRICS AND CHILD HEALTH I

MODULE CODE: PEDI 4011

DURATION: 10 weeks

MODULE PREREQUISITE

Completed Pre-clerkship courses successfully

MODULE DESCRIPTION:

This module is intended to equip students with profound knowledge and development of clinical skills relevant to understanding the science of pediatrics and child health and to enable the students to develop clinical knowledge, skills and attitude required for the understanding and diagnosis of pediatrics and childhood disorders with emphasis on those problems common in Ethiopia. Students will also develop relevant competencies with the Medical expertise, Scholar, Collaborator, Communicator, Manager, Health Advocate and Professional. This module is given during the third year of the clerkship period of the undergraduate medical education program.

MODULE DESIGN:

The teaching will focus on building the ability of the student in history taking, physical examination in the clinical setting, formulating differential diagnosis, reaching appropriate diagnosis, listing relevant investigation and constructing appropriate plan of management.

Education in Pediatrics and Child health clerkship will be delivered in various types of health care setting and in collaboration across disciplines. The students will have outpatient and inpatient experiences: Outpatient experience will be in the respective department of the hospital and also in the primary care settings (Health centers) which will be coordinated with the Pediatrics and Child health. In the outpatient setting, students will generally be supervised directly by staff physician/pediatricians. During their in-patient service attachments, students will be integrated into the ward team. All patients clerked by medical students will be reviewed by a pediatric resident or by a staff

physician. Students will be responsible for admitted patients and their follow-up. They will be expected to research each assigned patient's disease by the use of appropriate texts and journals.

MODULE OBJECTIVES

By the end of this module, students will be able to assess children and adolescents with medical disorder and formulate differential diagnoses and reach appropriate diagnosis, listing relevant investigation and constructing appropriate plan of management.

By the end of this module, students will be able to assess well child and adolescents.

Supporting Objectives

As a Medical Expert

- Apply knowledge of normal and abnormal structures and functions of the body in analyzing childhood medical problems
- Apply knowledge of growth and development in analyzing childhood medical problems
- Apply knowledge of growth and development and immunization in assessing well child and adolescents.
- Formulate plan of follow up of well child.
- Apply knowledge of physiological mechanisms in analyzing childhood medical problems
- Apply knowledge of the various causes (genetic, developmental, metabolic, toxic, infectious, autoimmune, neoplastic , traumatic etc) of disease in analyzing childhood medical problems
- Analyze important determinants and risk factors of common medical disorders in children
- Demonstrate appropriate pediatric history taking including from child patient with medical condition or care giver
- Demonstrate physical examination in a child
- Make an initial assessment of a child presenting with medical problem and suggest differential diagnosis
- Formulate a plan of investigation for a child with medical problem
- Attend/ Assist basic bed side tests and procedures being done
- Interpret the results of basic investigations of a child with medical disorder
- Formulate a likely diagnosis of a child with medical problem
- Formulate a plan for management for a child with medical problem according to established principles and best evidence.

As a communicator

- Demonstrate communication with a child , parents/care giver, peers, interns, residents, instructors and other hospital staff using a patient-centered approach that encourages patient trust and autonomy and is characterized with Compassionate Respectful and Caring(CRC) manner.
- Respond to patients' non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients
- Manage disagreements and emotionally charged conversations
- Use patient-centered interviewing skills to effectively gather relevant biomedical and psychosocial information
- Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information
- Provide a clear structure for and manage the flow of an entire patient encounter

- Seek and synthesize relevant information from other sources, including the patient's family, with the patients or parent/caretaker's consent.
- Effectively communication with ill patient and patient's family, and health care team

As a Collaborator

- Establish and maintain healthy relationships with physicians and other colleagues in the health care professions to support relationship-centered collaborative care
- Recognize one's own professional level and limits
- Share knowledge and information to other colleagues maintaining patients confidentiality rules.
- Show respect to ideas, order of command, colleagues and members of the health care team
- Respect and value diversity among other colleagues

As a Leader

- Develop their leadership skills
- Familiarize with the Ethiopian and international health delivery systems.
- Set priorities and manage time to balance practice and personal life
- Identify occupational hazards as a medical student for sustainable practice
- Demonstrate the ability to build and manage team

As a Health Advocate

- Work with pediatric patients and their families to address the determinants of health and their health seeking behavior Incorporate health promotion childhood illnesses prevention, , and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of childhood disease
- Participate in a process to improve childhood health and child rights in the community

As a Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling on learning
- Recognize disease out breaks and national reportable childhood diseases
- Promote a safe learning environment
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance
- Recognize knowledge gaps in biomedical, clinical and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care

- Pose questions amenable to scholarly investigation and select appropriate methods to address them

As a Professional

- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes (student code of conduct guideline), standards of practice, and laws governing practice
- Recognize unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need
- Apply preventive ethics in daily clinical practice
- Implement professional development plan that promote lifelong learning
- Demonstrate the ability to provide care in compassionate and respectful manner

TEACHING /LEARNING METHODS:

The following teaching Methods will be used during the attachment:

- Lectures (Problem Based Learning (PBL) is encouraged where feasible)
- Bedside teaching,
- Teaching round
- Seminars series
- Ambulatory clinical attachment
- Hospital ward attachment
- Skill labs and side labs practice
- Self-directed learning (SDL)
- E-Learning

Equally important is the unstructured teaching that happens all day every day on patient issues being managed by the health team (Medical Doctors Nurses etc.) every health team member should contribute to the education of the student. Teaching by example through good team spirit and leadership.

Assessment types

- **Summative assessment**
 - ❖ Progressive (continuous) assessment....20%
 - P-MEX-5%
 - ❖ practical exam(long ,short and OSCE).....50%
 - ❖ written exam.....30%
- **Formative assessment**
 - ❖ This will not contribute to the total grade of the student but provides continuous feed back to the student and it can be applied in the form of providing feed backs on assignments completed, students' professional behavior and etc.

Methods of assessment

1. OSCE (Objective Structured Clinical Examination)
2. OSPE (Objective Structured Practical Exam)
3. PRRE (Personal Research and Reflection Exercise)/ Log books and portfolios
4. Clinical examination (Long case and Short cases)
5. Oral exam (Viva)
6. Written exam (MCQ, Short Essay, Matching, True- False with reasoning)
7. 360 ° Assessment (Self, Peer, faculty, health facility staff and community member)
8. Professionalism mini-evaluation exercise (P-MEX)- Direct observation of student's professional behaviors

Week 1	Interactive Lecture (6 hrs)*, ➤ Introduction to pediatrics ➤ Extrauterine adaptation of the newborn and Perinatal Asphyxia ➤ Neonatal jaundice ➤ Neonatal sepsis	Reflective portfolio
	Clinical practicum (6hrs) • Assignment of students to pediatrics ward and NICU • History and physical examinations on all body systems. • Separate session on Neonatal history taking and physical examination.	
	Bed side teaching (4 hrs)	
	Teaching Round (4hrs)	
	Seminar (2hrs)-should be offered before the clinical practicum • History taking and physical examination of neonates and children	
	Simulation lab (2 hrs) • Simulation on anthropometry and interpretation, BP measurement and interpretation	
Week 2	Interactive Lecture (6 hrs) ➤ Prematurity ➤ Respiratory Distress in New borns ➤ Growth and Development ➤ PEM	Reflective portfolio
	Clinical practicum (6 hrs) ➤ Preceptorship (2 hrs) Friday afternoon after the Whole group discussion.	
	Seminar (2 hrs) Common childhood illnesses (Introducing IMNCI)	
	Bedside teaching (4 hrs)	
	Teaching Round (4hrs)	
	• Simulation (2hr) • on common neonatal problems (Respiratory distress, Jaundice, Perinatal asphyxia etc)	
	Primary health care or community visit Home visits of patients	
	Whole group session (3 hrs)	

	Faculty facilitated discussion and reflection on the week's learning activities	
Week 3	Interactive Lecture (6 hrs) ➤ Diarrheal diseases ➤ Upper respiratory tract infection ➤ Common lower respiratory tract disorders VPD (Measles, pertussis, Tetanus, polio, diphtheria) Case Study (2 hrs) Approach to a child with acute flaccid paralysis Clinical practicum (6 hrs) ➤ Preceptorship (2 hrs) Friday afternoon after the Whole group discussion.	Reflective portfolio
	Bedside teaching (4hrs)	
	Teaching Round (4hrs)	
	Simulation lab (2hr) Video on IMNCI (respiratory Distress)	
	Primary health care or community visit Home visits of patients	
	Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities	
Week 4	Interactive Lecture (6 hrs) ➤ HIV/AIDS in paediatrics ➤ Tuberculosis ➤ Extra pulmonary tuberculosis ➤ AFI (Malaria, TF, RF) Meningitis (Beyond Neonatal Age) Seminar(2 hrs) Immunization Clinical practicum (6 hrs) ➤ Preceptorship (2 hrs) Friday afternoon after the Whole group discussion. Bed side teaching (4 hrs) Teaching Round (4hrs) Simulation lab (2hr) - Video on Seizure and Coma (IMNCI)	
	Primary health care or community visit Home visits of patients	
	Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities	

	- Professional Lifelong learning & Continuous professional development and career planning (1 hour, Interactive lecture ,Video &Individual project) - Group Assignment: Provide a checklist for 4 groups of students. They will OBSERVE CRC on hospital Environment for TWO WEEKS. The students record their observation in the hospital environment from gate to speciality care. It will be presented and discussed	
Week 5	Interactive Lecture (6 hrs) - Congenital Heart Diseases - Acquired Heart Diseases - Anemia	Reflective portfolio
	Case Study (2 hrs)	
	nutritional requirements and feeding practices	
	Clinical practicum (6 hrs) ➤ Preceptorship (2 hrs) Friday afternoon after the Whole group discussion.	
	Bedside teaching (4 hrs)	
	Teaching Round (4hrs)	
	Simulation lab (2hr) Simulation on congestive heart failure and arrhythmias	
	Primary health care or community visit Home visits of patients	
Week 6	Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities	Reflective portfolio
	Interactive lecture (6 hrs) ➤ Leukemia and lymphoma ➤ Solid Tumors ➤ Endocrine disorders ○ Pituitary- adrenal axis DM, DI and Thyroid gland disorders,)	
	Case Study (2 hrs)	
	FTT	
	Clinical practicum (6 hrs) Clinical practicum Preceptorship (2 hrs) Friday afternoon after the Whole group discussion.	
	Bedside teaching (4 hrs)	
	Teaching Round (4hrs)	
	Simulation (2 hrs)	

	Simulation on Bone Marrow Aspiration and BM Smear preparation. Side lab to see slides of Complete blood count, anemia, leukemia,	
	Primary health care or community visit Home visits of patients	
	Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities	
Week 7	Interactive lecture (6 hrs) ➤ Common childhood Renal disease glomerulonephritis, nephrotic syndrome, UTI ➤ Seizure Disorder	Reflective portfolio
	Clinical practicum(6hrs) Preceptorship (2 hrs) Friday afternoon after the Whole group discussion.	
	Simulation lab-Child with renal compliant, Side lab to do urine dipstick (2 hrs)	
	Bed side teaching (4 hrs)	
	Teaching Round (4hrs)	
	Case Study (2hr)- bleeding disorders	
	Primary health care or community visit Home visits of patients	
	Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities	
Week 8	❖ Connective Tissue Disorders (JRA, SLE, Henock – Schonlein purpura) Chromosomal Disorders(DS, TS) Bone and joint infection (Osteomyelitis and septic arthritis) Clinical practicum (6hrs) Preceptorship (2 hrs) Friday afternoon after the Whole group discussion. Case Study (2 hrs) ➤ Burn and Drowning Simulation lab ➤ Videos on malnutrition and dehydration (IMCI) Bed side teaching (4 hrs) Teaching Round (4hrs) Primary health care or community visit Home visits of patients Whole group session (3 hrs) - Faculty facilitated discussion and reflection on the week's learning activities	

	• CRC behaviors of HCP (1 hours introductory lecture , let them make a 1 hour presentation of their findings in relation to CRC in four groups, 15 min for each.)	
Week 9	Interactive lectures (6 hrs) ➤ Common surgical emergencies (Appendicitis, Intussusception, IHPS etc) ➤ Common Congenital anomalies (cleft lip and palate, HSD, anorectal malformation)	
	Clinical practicum (6hrs) Preceptorship (2 hrs) Friday afternoon after the Whole group discussion.	
	Case Study ➤ The comatose child, Approach & management	
	Simulation lab ➤ Simulations on common pediatric procedures (NGT, Foley Catheter, LP, BF, Blood drawing etc)	
	Bed side teaching (4 hrs)	
	Teaching round (4hrs)	
	Primary health care or community visit Home visits of patients	
	Whole group session (3 hrs) • Faculty facilitated discussion and reflection on the week's learning activities •	
Week 10	Monday-Wednesday: Study break for exam	
	Tuesday: Written exam	
	Thursday: OSCE	

REFERENCES

- Nelson's Text book of Pediatrics, 20th edition
- Karen J. Marcante, Nelson Essentials of pediatrics, 6th edition
- Current Pediatrics, 6th edition
- Rudolph's Pediatrics, 20th edition
- Assaye Kasie Manual for Neonatology
- Feigin and Cherry, Pediatrics Infectious disease
- IMNCI Chart booklet
- Lynn S. Bickley, Bate's guide to physical examination and history taking, 10th edition
- Keith L. Moor, Arthur F. Dally, Anne M.R. Agur. Clinically oriented anatomy. 6th edition. 2010
- Eric Arthur Newsholme and Tony R. Leech. Functional biochemistry in health and disease. 2010.
- Stephen Goldber. Clinical biochemistry made ridiculously simple. 1999.
- Ashis Banerjee. Clinical physiology. An examination primer. 2005.
- Bertram G. Katzung. Basic and clinical pharmacology. 12th edition. 2011
- EFMOH, National compassionate and respectful care training participant manual for health work force, 2017

MODULE TITLE: OBSTETRICS AND GYNECOLOGY I

MODULE CODE: OBGY 4011

Duration: 10 weeks

Module Description: This first clerkship in obstetrics and gynecology is intended to enable medical students to develop clinical knowledge, skills and attitude required for the understanding, diagnosis and management of obstetric and gynecologic disorders with emphasis on problems common in Ethiopia. They will also develop essential knowledge and skills in the provision of obstetric care. During this clerkship, students will have opportunities to review and relate knowledge of biomedical sciences for the understanding of patient problems they encounter in clinical settings. Students will also develop relevant competencies in health promotion and disease prevention, communication, collaboration, leadership, women health advocacy, scholarship professionalism.

Prerequisites: Completed Pre-clerkship courses successfully

Module Objective

- Upon completion of this module, students will be able to assess patients with obstetrics and gynecologic disorders and formulate differential diagnoses. They also assess pregnant women in pregnancy and labor and make diagnosis.

Supporting Objectives

8. Medical expert

- Apply knowledge of normal and abnormal structures and functions of the human body system especially reproductive system in analyzing obstetrics and gynecologic conditions
- Apply knowledge of the human life cycle and effects of growth, development especially reproductive system in analyzing obstetrics and gynecologic conditions
- Apply knowledge of physiological mechanisms and processes of the human body system in analyzing obstetrics and gynecologic conditions
- Apply knowledge of the various causes and pathogenesis of abnormal obstetric and gynecologic disorders in analyzing patient problems
- Analyze important determinants and risk factors of obstetric and gynecologic conditions
- Apply knowledge of principles of drug pharmacokinetics and pharmacodynamics in the treatment of common obstetric and gynecologic conditions
- Demonstrate proper history taking skills from a patient with an obstetric and gynecologic disorder
- Demonstrate proper physical examination skills of a patient with an obstetric and gynecologic disorder
- Attempt to identify abnormal physical findings in a patient with obstetric and gynecologic disorders
- Attempt to make an initial assessment of a patient presenting with signs and symptoms of an obstetric and gynecologic disorder
- Propose a plan of investigation for a patient with an obstetric and gynecologic disorder in partnership with the patient, obtaining informed consent as an essential part of this process

- Show the ability to perform bedside tests and procedures useful for the workup of common obstetrics and gynecologic conditions
- Attempt interpreting the results of commonly used investigations
- Demonstrate the ability to formulate a likely diagnosis in a patient with an obstetrics and gynecologic disorder
- Demonstrate clinical judgments and decisions, based on the available evidence in analyzing problems of an obstetric and gynecologic patient
- Demonstrate provision of proper antenatal care
- Demonstrate the ability to attend normal labor and use partograph to monitor progress of labor
- Demonstrate the ability to perform neonatal resuscitation
- Attempt to outline a plan for management of a patient with obstetric and gynecologic disorder according to established principles and best evidence
- Demonstrate the ability to assess and counsel clients to help them reach a decision of their contraceptive choice and considering medical eligibility criteria
- Describe the characteristics, indication, unwanted effects and management of adverse effects of the various types of family planning methods
- Demonstrate the technique and skills to administer short and long acting contraceptive methods based on clients choice
- Analyze the indications and options in safe termination of pregnancy
- Demonstrate safe termination of pregnancy in a simulated setting
- Demonstrate a willingness to adhere to infection prevention principles in approaching patients (clients) with obstetrics and gynecology disorders

9. Communicators

- Demonstrate communication with clients seeking gynecologic and obstetric care using a patient-centered approach that encourages patient trust and autonomy and is characterized by empathy, respect, and compassion
- Optimize the physical environment for clients seeking gynecologic and obstetric care comfort, dignity, privacy, engagement, and safety
- Recognize when the values, biases, or perspectives of clients seeking gynecologic and obstetric care, physicians, or other health care professionals may have an impact on the quality of care, and modify the approach to the patient accordingly
- Respond to patients' non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients
- Use patient-centered interviewing skills to effectively gather relevant biomedical and psychosocial information
- Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information
- Seek and synthesize relevant information from other sources, including the patient's/client's family, with the patient's consent.
- Demonstrate sharing information that are clear, accurate, timely, and adapted to patient's/client's and his or her family's level of understanding and need in a simulated setting
- Effectively communicate with ill patient and patient's family, and health care team

10. Collaborators

- Show the willingness to work with students of other professions like midwives to maintain a climate of mutual respect and shared values

- Establish and maintain healthy relationships with physicians and other colleagues in the health care professions to support relationship-centered collaborative care
- Recognize one's own professional roles and limits
- Share knowledge and information to other colleagues
- Show respect toward other physicians and members of the health care team
- Respect and value diversity among other colleagues

11. Leaders

- Demonstrate leadership in professional practice
- Manage their practice and career of themselves, other health care providers and support staff
- Demonstrate the ability to effectively plan and efficiently manage one's own time and activities to cope with uncertainty, and the ability to adapt to change
- Promote occupational health as physicians for sustainable practice
- **Demonstrate the ability to build and manage team**

12. Women Health Advocate

- Demonstrate responding to the needs of women by advocating for policy and system-level change
- Demonstrate responding to the individual woman's reproductive health needs and rights by advocating with the woman within and beyond the clinical environment
- Demonstrate working with women and their families to address the determinants of health that affect women's reproductive health and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors
- Lead a healthy life style to be role models to promote women health through for example participating in breast and cervical screening

13. Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice
- Ensure patient safety is maintained while learning
- Provide effective feedback to enhance learning and performance
- Analyze global and national causes and trends in maternal morbidity and mortality
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them

- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry
- Contribute to the teaching of medical and other health sciences students regarding obstetrics and gynecology

14. Professional

- Integrate other intrinsic roles into their learning and practice of gynecology and obstetrics
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in gynecologic and obstetrics care
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients/clients seeking gynecologic and obstetric care, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice in gynecology and obstetrics
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the provision of maternal health services
- Participate in peer assessment and standard-setting
- Demonstrate a culture that recognizes, supports, and responds effectively to colleagues in need
- Apply preventive ethics in daily clinical practice
- Implement professional development plan that promote lifelong learning
- Demonstrate the ability to provide care in compassionate and respectful manner

Teaching and learning methods

- Interactive lecture
- Problem-based learning
- Case study
- Demonstration
- Role play
- Simulated (clinical skills lab) practice
- OPD teaching
- Teaching round
- Bedside teaching
- Personal research and reflection exercise/ Portfolio

Teaching and learning materials

- Videos of patient examinations and clinical procedures in obstetrics and gynecology
- Imaging studies of different obstetrics and gynecology conditions
- Mannequins for diagnostic and therapeutic procedures in obstetrics and gynecology

References

- Cunningham (et al). Williams Obstetrics (latest edition)
- Danforth, Gynecology and Obstetrics. (latest edition)
- Gabbe, Obstetrics. Normal and Problem pregnancies. (latest edition)

- Current Diagnosis & Treatment Obstetrics & Gynecology, (latest edition)
- *Berek, Jonathan S. Berek & Novak's Gynecology*, (latest edition)
- Keith L. Moor, Arthur F. Dally, Anne M. R. Agur. *Clinically oriented anatomy*. (latest edition)
- Eric Arthur Newsholme and Tony R. Leech. *Functional biochemistry in health and disease*. (latest edition).
- Stephen Goldber. *Clinical biochemistry made ridiculously simple* (latest edition)
- Ashis Banerjee. *Clinical physiology. An examination primer*. (latest edition)
- Bertram G. Katzung. *Basic and clinical pharmacology* (latest edition)
- EFMOH, National compassionate and respectful care training participant manual for health work force, 2017
- EFMOH, National compassionate and respectful care facilitators guide, 2017

Methods of assessment

Formative

- PBL progressive assessment
- Direct observation of clinical skills
- Logbook
- Global rating of performance or 360 degree evaluation
- Professionalism mini-evaluation exercise (P-MEX)- Direct observation of student's professional behaviors
- Portfolio/Personal research and reflection exercise

Summative

- Progressive assessment (10 %)
- 360 degree evaluation or global rating of performance (10 %)
- Written exam (50 %)
- OSCE (30%)

Every week, half of the students go to clinical skills lab and half of them go to hospital on a rotation basis

Depending on the number of students, it may be more efficient to distribute students between clinical skills learning labs and clinical practicum sites to ensure all of them are fully engaged with learning.

*** In week 7 and 9 while teaching abortion care and family planning, PBL session replaced with case studies to allow one more skills lab session****

***** Bedside teaching can be replaced by activities in clinical skills lab in week 7 and 9**

Module schedule

Date	Learning Activity	Required Reading/ Assignment
Week1	Interactivelecture - Fertilization, development of embryo and placenta and physiological changes during pregnancy - Antenatal care	74 Reflective portfolio
	Clinical practicum Student placement at outpatient and inpatient departments (Antenatal, FP clinic, maternity, gynecology OPD and inpatient)	
	Problem-based learning tutorial Case of pregnancy	
	Bedside teaching Case related to the topics in the week if available	
	Sidelabor clinical skills lab Simulated practice: history taking and physical examination from a pregnant woman Lab practice: Pregnancy test, hemoglobin/hematocrit, urinalysis, stool exam, VDRL	
	Primary Health Units/Community visit Home visit to take past obstetrics and gynecology history with focus on antenatal care	
	Whole group session Faculty facilitated discussion and reflection on the week's learning Activities	
Week2	Interactivelecture - Physiology of labor - Normal labor and delivery	Reflective Portfolio
	Clinical practicum Student placement at outpatient and inpatient departments (Antenatal, FP clinic, maternity, Gyn OPD and inpatient)	
	Problem-based learning tutorial Second session on the case of pregnancy	
	Bedside teaching Case related to the topics in the week	
	Sidelabor clinical skills lab Practice assisting labor	

	Primary Health care unit/community visit Home visit to take past obstetrics and gynecology history with focus on past deliveries and current pregnancy if any. Whole group session Faculty facilitated discussion and reflection on the week's learning Activities	
Week 3	Interactive lecture • Multiple pregnancy • Antepartum hemorrhage Clinical practicum Student placement at outpatient and inpatient departments (Antenatal, FP clinic, maternity, gynecology OPD and inpatient) Case study Case related to multiple pregnancy and APH Bedside teaching Case related to the topics in the week Sideline labor clinical skills lab Simulated practice: Assessing (history and physical examination) and monitoring laboring mothers (use of partograph and listening to fetal heartbeat) Lab practice: Blood group, Rh, HIV rapid test, Fern test Primary Health care unit/community visit Home visit to take past obstetrics and gynecology history with focus on signs of high risk in pregnancy and give health education for mothers, identify pregnant women with high risk pregnancies and advise them to go to the health facility Whole group session Faculty facilitated discussion and reflection on the week's learning Activities. Patient Counseling (2 hour Demonstration and guided practice)	
Week 4	Interactive Lecture • Premature rupture of membrane and preterm labor • Post-term pregnancy • IUGR and IUFD Clinical practicum Student placement at outpatient and inpatient departments (Antenatal, FP clinic, maternity, gynecology OPD and inpatient) Case study Case related to preterm/post-term labor	
	Bedside teaching Case related to the topics in the week Clinical skills lab Simulated practice: Neonatal resuscitation Primary Health care unit/community visit Home visit to take past obstetrics and gynecology history with focus on signs of high risk in pregnancy and give health education for mothers Whole group session	

	Faculty facilitated discussion and reflection on the week's learning Activities	
Week 5	InteractiveLecture • Dystocia (Labor abnormalities, induction, augmentation) • Postpartumcare	
	Clinical practicum Student placement at outpatient and inpatient departments (antenatal, labor, maternity, gynecology)	
	Case study Caserelated to postpartum care including family planning	
	Bedsideteaching Caserelated to thetopicsin theweek	
	Side lab or clinical skills lab Simulated practice: Counseling postpartum women; assisting women to establish breastfeeding; caring forthe breast,	
	Primary Health care unit/community visit Home visit to take obstetrics and gynecology history with focus on post-partal care and give health education for mothers	
	Whole groupsession Faculty facilitated discussion and reflection on the week's learning Activities	
Week 6	InteractiveLecture • Postpartumhemorrhage • Antepartum and intrapartum fetal surveillance	Reflective Portfolio
	Clinical practicum Student placement at outpatient and inpatient departments (Antenatal, FP clinic, maternity, gynecology OPD and inpatient)	
	Case study • Written clinical simulation on Caserelated to postpartum hemorrhage	
	Bedsideteaching Caserelated to thetopicsin theweek	
	Sidelabor clinical skills lab Simulated practice: Activemanagement of third stageof labor	
	Primary Health care unit/community visit Home visit to take obstetrics and gynecology history with focus on postpartal care and give health education for mothers	
	Whole groupsession Faculty facilitated discussion and reflection on the week's learning Activities	
Week 7	InteractiveLecture • Abortion • Ectopicpregnancyandgestational trophoblastic disease	
	Clinical practicum • Studentplacementatoutpatientandinpatientdepartments(Antenatal, FP clinic, maternity, gynecology OPD and inpatient)	
	Problem-basedlearning tutorial Case on first trimester bleeding -relatedtosafeabortion(legal and ethical	

	dilemmas, social determinants, patient presentation, management etc.) Bedsideteaching • Case related to abortion • Value clarification for attitudinal transformation (VCAT) Clinical skills lab Simulated practice: Assessing women with first trimester bleeding (history, physical examination); Students practice MVA in a simulated setting Students practice MVA in a simulated setting Primary Health care unit/community visit Home visit to take past obstetrics and gynecology history with focus on abortion and give health education on first trimester bleeding and abortion Whole group session Faculty facilitated discussion and reflection on the week's learning Activities	
Week 8	Interactive Lecture • Sexually transmitted infections • Pelvic inflammatory diseases Clinical practicum Student placement at outpatient and inpatient departments (antenatal, labor, maternity, gynecology) Problem-based learning tutorial Second meeting – first trimester bleeding/abortion Bedsideteaching Case related to the topic in the week Side labor clinical skills lab Standardized patient + written simulation: Assessing women with STIs (History, physical examination); condom demonstration Lab practice: Specimen collection; gram staining; wet mount (KOH) Primary Health care unit/community visit Home visit to take past obstetrics and gynecology history with focus on STI and give health education on STI Whole group session Faculty facilitated discussion and reflection on the week's learning Activities	
Week 9	Interactive Lecture • Family planning and contraception Clinical practicum Student placement at outpatient and inpatient departments (FP clinic, antenatal, labor, maternity, gynecology) Case study Case related medical eligibility in FP choice Bedsideteaching Case related to topics in the week if available Clinical skills lab Simulated practice: Students practice pre-insertion counseling, insertion and removal of IUCD and Implant, counseling clients for contraceptive use; applying medical eligibility criteria, inserting and	

	removing Implants and IUCD	
	Primary Health care unit/community visit Home visit to take past obstetrics and gynecology history with focus on family planning and counselling on contraceptive use	
	Whole group session Faculty facilitated discussion and reflection on the week's learning Activities . Discuss legal medicine	
Week 10	Monday-Wednesday: Studybreak for exam	
	Thursday: Written exam	
	Friday: OSCE	

⁷⁴Students are required to maintain in their portfolio learning experiences, learning outcomes achieved, evidence of learning and reflections longitudinally and their mentors would provide feedback occasionally. Clinical instructor should also ask students to submit reflective portfolio twice during the attachment for formative assessment and feedback.

⁷⁵As much as possible, PBL cases during clerkship should preferably be based on real cases observed during clinical Practicum. **In interactive presentation, faculty may use various methods that can engage students like case studies, clinical simulation allowing students to work in group**

CLERKSHIP II MODULES

Typical Clerkship II weekly Schedule

Time	Monday	Tuesday	Wednesd	Thursday	Friday
8:00-9:00am	Morning Meeting	Morning Meeting	Morning Meeting	Class room teaching (SPH)	Primary health care/Community visit
9:00am-12:00pm	Round (CP)	Outpatient/ Inpatient (CP)	Round	(CP)	
1:30-4:30 pm	PBL*/Case study	Case study	Side lab/ skills lab	Bedside	Whole group session
4:30-5:30pm -	Independent study/preceptor	Independent study/preceptorship		Independent study/perceptor	
5:30-11 pm	DUTY	DUTY	DUTY	DUTY	DUTY

CLERKSHIP II MODULES

MODULE TITLE: INTERNAL MEDICINE- II

MODULE CODE: INME 5011

CREDIT HOURS= 12 ECTS

PREREQUISITE-COMPLETION OF INTERNAL MEDICINE -I

PLACEMENT: YEAR IV

COURSE DURATION: 07 WEEKS

MODULE DESCRIPTION: This module is intended to endow students with intense knowledge of clinical medicine and development of clinical skills relevant to understanding the science of internal medicine with emphasis on those problems common in Africa.

It deals with the principles & practice of internal medicine as both a science and an art.

It also provides basic understanding about medical history taking and performing physical examination through patient evaluation and medical case report writing basic laboratory procedures.

And generally targeted to equip students with consolidated knowledge and skill of internal medicine and promote patient physician relation.

Objective:

On completion of this course, the student is expected to acquire:

Apply knowledge of normal and abnormal structures and functions of the body in analyzing adult medical problems.

- Ability of definition, etiology, epidemiology, pathology, pathogenesis, clinical presentation & complication of medical diseases (with emphasis to list DDx and management of common prevailing diseases in the country)
- Skill of retrieving appropriate history through effective communication
- Skill of performance, detection & interpretation of physical findings
- Ability to formulate relevant differential diagnoses, discuss the differential diagnoses and reach to the possible diagnosis or reasonable impression
- Ability to suggest, & interpret laboratory investigations
- Ability to manage different ailments
- Formulate and implement a management or treatment plan as approved and supervised by their instructors and qualified clinicians.
- Knowledge of proper character & ethical guidelines, healthy relationship with staffs & patients, cooperativeness & ability to work with others.

- Effectively communicate with diverse patient populations including seriously ill and difficult patients
- Apply legal aspects in health care practice
- Solve ethical dilemmas
- Provide care in a CRC manner in clinical setting
- Show commitment to autonomous maintenance and continuous improvement of competence
- Advocate and lead implementation of CRC in the health care system

Teaching and learning methods

- ◆ Interactive lecture
- ◆ Video show
- ◆ Demonstration
- ◆ Role play
- ◆ Simulated (clinical skills lab) practice
- ◆ Side lab
- ◆ OPD teaching

- ◆ Teaching round
- ◆ Bedside teaching
- ◆ Case studies/simulated case scenario
- ◆ Community visit
- ◆ Seminars
- ◆ Inter-professional learning experience in the skills lab, clinic, community.
- ◆ Case studies

Teaching and learning materials

- ◆ Videos of clinical presentation of patients/ examination
- ◆ Imaging studies of different health issues
- ◆ Mannequins for procedures
- ◆

References:

- EFMOH compassionate and respectful care training manual for health work force, 2017.
- World Medical Association Medical Ethics Manual, 3rd edition, 2015

Methods of assessment

An effort will be made to make a quality control of the course offered through periodic evaluation of activities by students & academic staff

- I. Progressive assessment = 30%(Based on Department Criteria)
 - Professionalism mini-evaluation exercise (P-MEX)- Direct observation of student's professional behaviors -5 %

- ✓ Monitoring log book(to be provided by the department to each students)
- ✓ Attendance, Clinical/academic activities/procedures/presentation
- ✓ Clinical activities and procedures
- ❖ II. Practical examination (LONG/SHORT) = 40%
- ❖ II. Written examination = 30%
- ❖ And finally the 100% of internal examination grade will be converted to 70%
- ❖ External examination will be 30%

❖ Lecture Topics:

- 1.Pulmonary neoplasms
- 2.Parathyroid, metabolic bone disease
- 3.Clinical approach to normal and abnormal hemostasis
- 4.Malabsorption
- 5.IBD
- 6.Peripheral neuropathy/NMJD/Myopathy
- 7.Introduction of Rheumatology, Rheumatoid arthritis/SLE

Seminar topics:

- 1.Clinical pharmacology and rationale use of antibiotics
- 2.Management of ACS
- 3.Mx DKA/HHS
- 4.Managment of sever malaria
- 5.Mananagme of acute stroke/Coma
- 6.Management of AKI
- 7.Management of UGI bleeding

Bedside Teaching topics:

- 1.MENINGITIS

- 2.SPINAL CORD DISORDERS
- 3.AFI
- 4.HIV/OI
- 5.TB
- 6.COPD
- 7.CHRONIC BRONCHIAL ASTHMA
- 8.HF/IHD
- 9.PERICARDITIS/MYOCARDITIS/CARDIOMYOPATHY
- 10.CLD
- 11.AKI
- 12.CKD
- 13.DM/METABOLIC SYNDROME
- 14.LYMPHOMA
15. CML(MPD)

N.B: Teaching round and major rounds done twice a week and final year students are expected to present cases and out line management and attach notes to the patients medical documents.

Typical CII(YR4) Schedule/Internal Medicine

Typical CH (PRV) Schedule/ Internal Medicine					
Date/Time	Monday	Tuesday	Wednesday	Thursday	Friday
8:00-9:30am	TR		Community visit/Side lab	TR	Interactive lecture
9:45-12:00		OPD/H C			OPD/HC
PM					
1:30-3:00	Independent study/Clerking	BS	Simulation	BS	Seminar
3:00-5:00					

MODULE- Schedule:

WK	Academic activities/topics	Methodology
1	Morning meeting Ward round/ Bedside teaching(inpatient/OPD) ✓ Clerk patient with HIV/OIs - TB(pulm.& extrapulm) Seminar: - None Whole Person Care (WPC) –1 hour of introductory ppt and case studies <u>Skill lab</u> Simulation: demonstration of thoracentesis & paracentesis Video: Clinical presentation of patient with Lung Ca Show pulmonary imaging(CXR, CT SCAN) Lab- - AFB	Interactive Lecture Bedside teaching Teaching/Major rounds Seminar Skill lab Simulation Video Side lab

	- HIV serology test	
2	Morning meeting Ward round/ Bedsides - Clerk patient with AFI and - Lymphoma Seminar - Clinical pharmacology and rationale use of antibiotics. - Skill lab Simulation: demonstration of physical findings lymphoma, AFI Video: show patient with bleeding disorder(petechiae, ecchymosis, etc) Lab. - Blood morphology/BF Whole group discussion : Communicating With diverse population & difficult patients/ clients (introductory ppt and Role play,check list)	Interactive Lecture Bedside/OPD Teaching/Major rounds Seminar Skill lab Simulation Video Side lab
3.	Morning meeting Ward round/Bedsides: Clerk patient with: - IHD(HTN/ DM/Dyslipidemia) - HF/pericarditis/myocarditis Seminar - Management of ACS Skill lab Simulation. - Demonstration of physical findings of hypocalcemia & different bone diseases - Demonstration of physical findings of pericarditis/tamponade/ACS. Video: - Demonstration of imaging findings of patients with pericarditis, ACS - Demonstration of patients with hypocalcemia - Show image of bone disease Lab: - ECG tracing	Interactive Lecture Bedsides Teaching/Major rounds Seminar Skill lab Simulation Video Side lab
4	Morning meeting Ward round/ Bed side: Clerk patient with: - COPD - Chronic bronchial asthma Seminar - Mx of DKA & HHS Skill lab Simulation. - Demonstration of patient with COPD/Bronchial	Interactive Lecture Bedsides Teaching/Major rounds Seminar Skill lab Simulation Video Side lab

	asthma Video. - Demonstration of Endoscopic & other imaging studies of Malabsorption, - Demonstration of imaging of COPD Lab .- - RBS, urine glucose, urine ketone Whole Group session: Medical error, Professional Negligence ,Medical Complications and Disclosing medical errors and complications (2 hrs of introductory ppt and cases studies)	
5	Morning meeting Ward round/Bedside: Clerk patient with: - CLD - MPD/CML Seminar - Management of UGIB <u>Skill lab</u> Simulation. - Demonstration of physical findings of CLD Video. - Show Endoscopic findings of IBD - Demonstration of Endoscopic & other imaging studies of IBD ,PUD & CLD Side lab. Paracentesis-SAAG Whole groups session : Maintaining fitness to work: Self compassion (2 hrs of introductory ppt with case studies)	Interactive Lecture Bedside Teaching/Major rounds Seminar Skill lab Simulation Video Side lab
6	Morning meeting Ward round/Bedside: Clerk patient with: - Meningitis - Spinal cord diseases Seminar - Management of acute stroke/ Coma <u>Skill lab</u> Simulation : - Demonstration of physical findings of peripheral neuropathy, myopathy - Demonstration of LP Video. - Demonstration of stroke/ coma examination - Show patients with peripheral neuropathy/signs,symptoms Side lab	Interactive Lecture Bedside Teaching/Major rounds Seminar Skill lab Simulation Video Side lab
7	END attachment EXAMINATION	

MODULE TITLE: SURGERY CLERKSHIP II

MODULE CODE: SURG 5011

DURATION: 7 weeks

Module Description:

This second clerkship in surgery is intended to enable medical students to advance their clinical knowledge, skills and attitude required for the management of surgical disorders with emphasis given to common life-saving procedures and interventions. During this clerkship, students will have opportunities to review and relate knowledge of biomedical sciences for the understanding of patient problems they encounter in clinical settings. Students will also develop relevant competencies in health promotion and disease prevention, communication, professionalism, management of information, critical thinking, practice-based improvement and research and inter-professional collaborative practice.

MODULE DESIGN:

This Module Is intended to prepare the student for increased measures of responsibility and prepares students for Internship. This provides the students to have the experience of being the primary caregiver for patients in a well-supervised setting.

Education in Surgery clerkship will be delivered in different setting of health care and in collaboration across disciplines. The students will have outpatient and inpatient experiences: Outpatient experience will be in the respective department of the hospital and also in the primary care (Health centers) which will be coordinated with their specific department of attachment.

MODULE PREREQUISITE:

- Students must have completed clerkship I

MODULE OBJECTIVES

Module Objective

1. MEDICAL EXPERT

- Demonstrate a commitment to high-quality surgical care of their patients
- Integrate the other six roles into their practice of medicine
- Apply knowledge of the clinical and biomedical sciences relevant to different disciplines they are exposed to
- Apply concepts and principles of public health sciences to protect and promote individual and population health
- Perform appropriately timed clinical assessments with recommendations that are well organized and properly documented in written and/or oral form
- Carry out professional duties in the face of multiple, competing demands
- Recognize and respond to the complexity, uncertainty, and ambiguity inherent in medical practice
- Identify and prioritize issues to be addressed in a patient encounter
- Elicit a history, perform a complete physical exam, make a differential diagnosis, select appropriate investigations, perform basic investigations and interpret their results for the purpose of diagnosis and management, disease prevention, and health promotion
- Diagnose and establish a patient-centered management plan based on available evidence
- Recognize and respond to adverse events and near misses
- Contribute to a culture that promotes patient safety
- Adopt strategies that promote patient safety and mitigate negative human and system factors

- **Demonstrate the ability to detect post op complications**
- **Demonstrate the ability to identify preoperative risk factors**

Communicator

- Communicate using a patient-centered approach that encourages patient trust and autonomy and is characterized by empathy, respect, and compassion
- Optimize the physical environment for patient comfort, dignity, privacy, engagement, and safety
- Recognize when the values, biases, or perspectives of patients, physicians, or other health care professionals may have an impact on the quality of care, and modify the approach to the patient accordingly
- Respond to patients' non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients
- Manage disagreements and emotionally charged conversations
- Adapt to the unique needs and preferences of each patient and to his or her clinical condition and circumstances
- Use patient-centered interviewing skills to effectively gather relevant biomedical and psychosocial information
- Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information
- Provide a clear structure for and manage the flow of an entire patient encounter
- Seek and synthesize relevant information from other sources, including the patient's family, with the patient's consent.
- Share health care information and plans with patients and their families
- Skillfully share information and explanations that are clear, accurate, timely, and adapted to patient's and his or her family's level of understanding and need
- Communicate appropriately in difficult circumstances, such as when breaking bad news, and when discussing sensitive issues, such as alcohol consumption, smoking or obesity
- Communicate appropriately with difficult/ violent patients and people with mental illness
- Facilitate discussions with patients and their families in a way that is respectful, non-judgmental, and culturally safe
- Assist patients and their families to identify, access and make use of information and communication technologies to support their care and manage their health
- Use appropriate communication skills and strategies to help patients and their families make informed decisions regarding their health
- Document clinical encounters in an accurate, complete, timely, and accessible manner, in compliance with legal and regulatory requirements
- Communicate effectively using a written health record, electronic medical record, or other digital technology
- **Effectively communicate with diverse patient populations including seriously ill and difficult patient**

Collaborators

- Establish and maintain healthy relationships with physicians and other colleagues in the health care professions to support relationship-centered collaborative care
- Negotiate overlapping and shared responsibilities with physicians and other colleagues in the health care professions
- Engage in respectful shared decision-making with other colleagues, patients and their families

- Recognize one's own professional roles and limits
- Share knowledge and information to other colleagues
- Show respect toward other physicians and members of the health care team
- Respect and value diversity among other colleagues
- Implement strategies to resolve conflicts in a manner that supports a collaborative culture
- Assess when care should be transferred to another physician or health care professional
- Demonstrate safe transfer of care, using both verbal and written communication, during a patient transition to a different health care professional, setting, or stage of care

Leader

- Apply the science of quality improvement to contribute to improving systems of patient care
- Analyze adverse events and near misses to enhance systems of care
- Use health informatics to improve the quality of patient care and optimize patient safety
- Allocate health care resources considering justice, efficiency, and effectiveness for optimal patient care
- Apply evidence and management processes to achieve cost-appropriate care
- Develop their leadership skills
- Design and organize elements of health care delivery
- Facilitate change in health care to enhance services and outcomes
- Set priorities and manage time to balance practice and personal life
- Manage career planning, finances, and health human resources in a practice
- Implement processes to ensure personal practice improvement
- Promote occupational health for sustainable practice
- **Show commitment to autonomous maintenance and continuous improvement of competence**
- **Advocate and lead implementation of CRC in the health care system**

Health Advocate

- Work with patients to address determinants of health that affect them and their access to needed health services or resources
- Work with patients and their families to increase their opportunities to adopt healthy behaviors
- Incorporate disease prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of health that affect them
- Improve clinical practice by applying a process of continuous quality improvement to disease prevention, health promotion, and health surveillance activities
- Participate in a process to improve health in the community or population they serve

SCHOLAR

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance using various internal and external data sources

- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Demonstrate effective teaching to facilitate learning
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize practice uncertainty and knowledge gaps in clinical and other professional encounters and generate focused questions that can address them
- Demonstrate proficiency in identifying, selecting, and navigating pre-appraised resources
- Integrate evidence into decision-making in their practice
- For a given practice scenario, formulate well-structured scholarly questions
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion and critique of their relevance to professional practice
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Evaluate the applicability (external validity or generalizability) of evidence from a wide range of information resources
- Discuss the barriers to and facilitators of applying study findings to professional practice
- Use quality-appraised evidence-alerting services that highlight new evidence appropriate to their scope of professional practice
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

7. **PROFESSIONAL**

- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of practice
- Recognize and respond to ethical issues encountered in practice
- Recognize and manage conflicts of interest
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the Profession
- Fulfill and adhere to the professional and ethical codes, standards of practice, and laws governing practice

- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Manage personal and professional demands for a sustainable practice throughout the physician life cycle
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need
- **Apply Legal aspects in health care practice**
- **Solve ethical dilemmas**
- **Provide care in CRC manner in clinical setting**

COURSE CONTENT

- Fluid & electrolyte management of the surgical patient
- Shock
- Acute abdomen
- Intestinal obstruction
- Poly trauma
- Thoraco abdominal injuries, Management of chest and abdominal injuries
- Burn
- Surgical diseases of the esophagus
- Tumors of the esophagus and stomach
- Peptic ulcer diseases and complications of PUD
- Diseases of the colon and rectum
- Perineal diseases
- Stone diseases of the gall bladder
- Surgical diseases of the liver and biliary tree , Obstructive Jaundice
- Benign and malignant diseases of the thyroid
- Breast diseases
- Stone disease of the GUS
- B.P.H. and Prostatic Ca
- Genitourinary tumors
- Supportive diseases of the lungs and pleura
- Lung and pleural tumors
- Head injury and spinal cord injury
- Adrenal gland tumors
- Venous, lymphatic diseases and arterial diseases
- Introduction to Pediatrics Surgery
- Introduction to Plastic surgery
- Brain tumors
- Thyrotoxicosis and diseases of the parathyroid
- Upper air way obstruction; Tracheostomy, indications, care
- Hernia: Types, Pathophysiology, Complications, Management
- DDX of Scrotal and testicular swellings and management
- Abdominal mass presentations, DDX, investigation and Management
- Mediastinal masses – clinical presentations, DDX and management
- Perioperative care

- Post-operative complications
- Surgical infections and the use of antibiotics in surgery
- Principles of anesthesia

TEACHING /LEARNING METHODS:

The following teaching Methods will be used during the attachment:

- Lectures
- Bedside teaching,
- Teaching round
- Morning report, conference
- Seminars series
- Grand rounds.
- Night Duty once per week
- Ambulatory clinical attachment
- Hospital ward attachment
- Skill labs and side labs practice
- Self-directed learning (SDL)
- E-Learning
- Journal club
- Community learning sessions
- Mentorship

Equally important is the unstructured teaching that happens all day every day on patient issues being managed by the health team (Medical Doctors Nurses etc.) every health team member should contribute to the education of the student .Teaching by example through good team spirit and leadership.

Assessment types

Summative assessment

- ✓ Progressive (continuous) assessment (Quizzes, Concept application exercise (20 %)
- ✓ Written exam (30 %)
- ✓ OSCE (50%)

Formative assessment

This will not contribute to the total grade of the student but provides continuous feed back to the student and it can be applied in the form of providing feed backs on assignments completed, students' professional behavior and etc.

- Direct observation of clinical skills
- Logbook
- Global rating of performance or 360 degree evaluation
- Portfolio
- Personal research and reflection exercise

Methods of assessment

1. DOCS (Direct Observation of Clinical Skills)
2. OSCE (Objective Structured Clinical Examination)
3. OSPE (Objective Structured Practical Exam)
4. PRRE (Personal Research and Reflection Exercise)
5. Project Work
6. Clinical examination (Long case and Short cases)

7. Oral exam (Viva)
8. Written exam (MCQ, Short Essay, Matching, True- False with reasoning)
9. Log books and portfolios
10. Global rating
11. 360 0 Assessment (Self, Peer, faculty, health facility staff and community member)
12. Professionalism mini-evaluation exercise (P-MEX)- Direct observation of student's professional behaviors

Date	Learning Activity	Required Reading/ Assignment
Week 1 Pediatric Surgery Case study topics - Overview of Pediatrics surgery - Intestinal obstruction in neonates and infants - Intestinal obstruction in children emphasizing on Intussusception - Acute appendicitis in children	Clinical practicum (12hrs) Student placement at surgical outpatient and inpatient departments	Reflective portfolio
	Round (4hrs)	
	PBL/Case Study(3hrs) Intestinal obstruction in neonates	
	Case study(3hrs) Acute abdomen in children	
	Bedside (3 hrs) Case related to the topics in the week	
	Clinical skills lab (4 hrs) - Simulated practice: fluid and electrolyte balance in children - Simulated practice: Pediatric Imaging for intestinal obstruction - Simulated practice: HX and PE in pediatric patient with surgical emergency	
	Primary health care or community visit(1hr lecture/week and 4 hrs. attachment/week) Home visits of patients with surgical conditions, preparing health education on early referral of infants and children	
Week 2 Endocrine and breast	Clinical practicum (12hrs) Student placement at surgical outpatient and inpatient departments	Reflective portfolio
	Rounds (4hrs)	
	PBL/Case Study (3hrs) Hyperparathyroidism	
	Case study(3hrs) Case related to breast lump(management approaches)	
	Bedside (3 hrs) Case related to thyroid (management approaches, indications for surgery)	
	Clinical skills lab (4hrs) Simulated practice: Assessing a patient with neck mass(with emphasis to thyroid diseases, detection of Metastasis and post op complications)	

	Simulated practice: Assessing a patient with breast lump including staging, detection of post complications	
	Primary health care or community visit (1hr lecture/week and 4 hrs attachment/week) - HE in early detection and referral of breast cancer - HE in prevention of iodine deficiency goiter	
	Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities	
Week 3 Cardio-thoracic	Clinical practicum (12hrs) Student placement at surgical outpatient and inpatient departments	Reflective portfolio
	Round (4hrs)	
	Bedside (3hrs) Case related to surgical diseases of the thorax	
	Case study(3hrs) A case of dysphagia	
	PBL/Case Study (3hrs) Mediastinal Mass	
	Clinical skills lab (4hrs) Simulated practice: insertion of chest tube :tracheostomy and crico-thyroidotomy	
	Primary health care or community visit (1hr lecture/week and 4 hrs attachment/week)	
	Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities skills on breaking bad news &Informed consent and Issue of confidentiality, anonymity, and privacy(2 hour of introductory ppt and Role plays)	
Week 4 GI (Hepato-biliary acute abdomen and pancreas)	Clinical practicum (12hrs) Student placement at surgical outpatient and inpatient departments	Reflective portfolio
	Round (4hrs)	

	Round on patients operated with emergency Abdominal conditions	
	PBL/ Case Study (3hrs) Surgical/obstructive jaundice	
	Case study(3hrs) Intestinal obstruction	
	Bedside (3 hrs) Case related to Hepato-biliary disease	
	Clinical skills lab (4hrs) - Simulated practice: - Simulated practice: NG tube insertion, Rectal Tube	
	Primary health care or community visit (1hr lecture/week and 4 hrs attachment/week) - Home visits of patients prioritizing surgical conditions covered during the week	
	Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities Relevant legal frameworks governing professional practice (1 hour of introductory of ppt and group discussion)	
Week 5	Clinical practicum (12hrs) Student placement at surgical outpatient and inpatient departments	
GI	Round (4hrs)	
	PBL/Case Study(3hrs) A case of rectal bleeding	
	Case Study (3hrs) Abdominal trauma	
	Bedside Case related to GOO	
	Clinical skills lab (4hrs) Simulated practice: - History taking and abdominal examination including - interpretation of imaging studies of different colonic ca and GOO	

	• Simulated practice:	
	• Primary health care or community visit (1hr lecture/week and 4 hrs attachment/week) • Delivering health education on personal hygiene to prevent H. Pylori infection • Early visit to specialized care in older and high risk patients and those with persistent rectal bleeding in all ages	
	Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities	
Week 6 ▪ Urology ▪ Neurosurgery	Clinical practicum (12hrs) Student placement at surgical outpatient and inpatient departments	Reflective portfolio
	Round (6 hrs)	
	Bedside (3hrs) • Case of BOO • Case of Head injury	
	Case study(3hrs) • Approach to patient with hematuria	
	PBL/Case discussion Case of flank pain Case of trauma	
	Clinical skills lab (2 hrs) • Simulated practice: Assessing surgical patients with urinary symptoms • Simulated practice: Catheterization	
	• Primary health care or community visit (1hr lecture/week and 4 hrs attachment/week) • Surgical OPD and ward Home visits of patients with surgical conditions • Surgical OPD and ward • Home visits of patients prioritizing surgical conditions covered during the week	
	Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities	
Week 7	Monday-Wednesday: Study break for exam	Reflective portfolio
	Thursday: Written exam	
	Friday: OSCE/OSPE	

RESOURCES

Human resources

- Clinicians and specialists
- General Practitioner
- Public health professionals
- Other relevant Health Professionals (for Inter professional Pool)
- Education Experts

Training, Teaching and Learning Materials

- Trainer's Guide

Reference manuals and books

- Learners Guide
 - Handout
 - Books
 - Text books
 - E-books

Learning Facilities / Infrastructure

- Class room with the right class room environment.
- Skill laboratory
- Dissection room
- Equipped Multipurpose Integrated laboratory
- Library
- IT center with good internet connection
- Clinical set up (Primary and secondary where possible tertiary care health institutions)

Portable and stationary plants, machines and equipment

- Flip charts
- Video
- CD-ROM
- Computers
- DVD Player
- LCD projector Overhead Projector
- White board (2x1.5)

REFERENCE

- Norman J Williams, Christopher J.K. Bulstrode, P Ronnan O'Connell. Bailey and Love's Short Practice of Surgery. 25th ed. 2008
- Courtney M. Townsend Jr. [et al.]. Sabiston textbook of surgery: the biological basis of modern surgical practice.—19th ed. 2012
- Schwartz, Principles of Surgery. 9th edition. 2010
- Snell, Richard S. Clinical anatomy by regions. 9th edition. 2012
- Keith L. Moor, Arthur F. Dally, Anne M.R. Agur. Clinically oriented anatomy. 6th edition. 2010
- Eric Arthur Newsholme and Tony R. Leech. Functional biochemistry in health and disease. 2010.
- Stephen Goldber. Clinical biochemistry made ridiculously simple. 1999.
- Ashis Banerjee. Clinical physiology. An examination primer. 2005.
- Bertram G. Katzung. Basic and clinical pharmacology. 12th edition. 2011
- EFMOH compassionate and respectful care training manual for health work force, 2017.

MODULE TITLE: PAEDIATRICS AND CHILD HEALTH II

MODULE CODE: PEDI 5011

DURATION: 7 weeks

COURSE PREREQUISITE:

For Pediatrics II –completed clerkship I courses successfully

COURSE DESCRIPTION:

Module Description: This module is given during the third year of the clerkship period of the undergraduate medical education program. It is intended to equip students with profound knowledge and development of clinical skills relevant to understanding the science of pediatrics and child health and to enable the students to develop clinical knowledge, skills and attitude required for the understanding and diagnosis of pediatrics and childhood disorders with emphasis on those problems common in Ethiopia. Students will also develop relevant competencies with the Medical expertise, Scholar, Collaborator, Communicator, Manager, Health Advocate and Professional.

COURSE DESIGN:

Clerkship II: Is intended to prepare the student for increased measures of responsibility and prepares students for Internship. Emphasis will be given to management of clinical presentations and specific diseases process. This provides the students with the experience of being the primary caregiver for patients in a well-supervised setting.

Education in Pediatrics and Child health clerkship will be delivered in various types of health care setting, and in collaboration across disciplines. The students will have outpatient and inpatient experiences: Outpatient experience will be in the respective department of the hospital and also in the primary care settings (Health centers) which will be coordinated with the Pediatrics and Child health . In the outpatient setting students will generally be supervised directly by staff physician/ pediatricians. During their in-patient service attachments, students will be integrated into the ward team .All patients clerked by medical students will be reviewed by a pediatric resident or by a staff the physician. Students will be responsible for admission of their patients and their follow-up. They will be expected to research each assigned patient's disease by the use of appropriate texts and journals.

Module Objective

- By the end of this module, students will be able to assess childhood disorders and assist with their management including in health promotion and disease prevention.

Supporting Objectives

As a Medical Expert

- Apply knowledge of normal and abnormal structures and functions of the body in analyzing childhood medical problems
- Apply knowledge of growth, development and ageing in analyzing childhood medical problem
- Apply knowledge of physiological mechanisms in analyzing childhood medical problems
- Apply knowledge of the various causes (genetic, developmental, metabolic, toxic, infectious, autoimmune, neoplastic and traumatic) of disease in analyzing childhood medical problems
- Analyze important determinants and risk factors of common medical disorders in children

- Demonstrate appropriate history taking including social issues from child patient with medical condition or care giver
- Demonstrate physical examination in a child
- Make an initial assessment of a child presenting with medical problem and suggest differential diagnosis
- Apply knowledge of principles of drug pharmacokinetics and pharmacodynamics in pharmacotherapy
- Formulate a plan of investigation for a child with medical problem
- Perform basic bedside tests and procedures
- Interpret the results of basic investigations of a child with medical disorder
- Formulate a likely diagnosis of a child with medical problem
- Make clinical judgments and decisions based on the available evidence in a child with medical problem
- Formulate a plan for management for a child with medical problem according to established principles and best evidence.
- Perform clinical procedures for the diagnosis of childhood disorders
- Manage certain pediatric emergencies
- Analyze global and national trends in morbidity and mortality due to childhood illnesses of public health significance in Ethiopia

As a communicator

- Demonstrate communication with a patient with a child, care giver, peers and preceptors using a patient-centered approach that encourages patient trust and autonomy and is characterized by empathy, respect, and compassion
- Respond to patients' non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients
- Manage disagreements and emotionally charged conversations
- Use patient-centered interviewing skills to effectively gather relevant biomedical and psychosocial information
- Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information
- Provide a clear structure for and manage the flow of an entire patient encounter
- Seek and synthesize relevant information from other sources, including the patient's family, with the patient's consent.
- Skillfully share information and explanations that are clear, accurate, timely, and adapted to patient's and his or her family's level of understanding and need in a simulated setting
- Demonstrate the ability to communicate appropriately in difficult circumstances in pediatrics practice, such as when breaking bad news, and when discussing sensitive issues
- Demonstrate the ability to communicate appropriately with difficult sick child or violent parents
- Demonstrate the ability to communicate appropriately with vulnerable children such as victims of sexual abuse
- **Effectively communicate with diverse patient populations including seriously ill and difficult patient**
-

As a Collaborator

- Establish and maintain healthy relationships with physicians and other colleagues in the health care professions to support relationship-centered collaborative care
- Recognize one's own professional roles and limits
- Share knowledge and information to other colleagues
- Show respect toward other colleagues and members of the health care team
- Respect and value diversity among other colleagues

As a Leader

- Develop their leadership skills
- Design and organize elements of health care delivery
- Set priorities and manage time to balance practice and personal life
- Promote occupational health as a medical student for sustainable practice
- **Show commitment to autonomous maintenance and continuous improvement of competence**
- **Advocate and lead implementation of CRC in the health care system**
-

As a Health Advocate

- Work with pediatric patients and their families to address the determinants of health that affect them and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors
- Incorporate childhood illnesses prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of childhood disease
- Participate in a process to improve childhood health in the community
- Contribute to their obligation to promote, protect and enhance the essential elements of the medical profession for benefit of medical patients, the medical profession and society at large

As a Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling on learning
- Promote a safe learning environment
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance
- Recognize knowledge gaps in biomedical, clinical and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care

- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry
- Demonstrate the ability to use knowledge of the physician's role and those of other professions to appropriately assess and address the healthcare needs of children with disorder
- Demonstrate the recognition of the roles of complexity, uncertainty and probability in management decisions regarding childhood disorders
- Demonstrate one's recognition of the power and limitations of the scientific thinking based on information obtained from different sources in establishing the causation, treatment and prevention of childhood disorders

As a Professional

- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in practice
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need
- Demonstrate a habit of self-reflection, responsiveness to feedback and an on-going development of new skills, knowledge and attitude in pediatrics
- Apply Legal aspects in health care practice
- Solve ethical dilemmas
- Provide care in CRC manner in clinical setting

TEACHING /LEARNING METHODS:

The following teaching Methods will be used during the attachment :

- Interactive lectures
- Bedside teaching
- Teaching round
- Morning report
- Night Duty
- Ambulatory clinical attachment
- Hospital ward attachment
- Skill labs and side labs practice
- Self-directed learning (SDL)
- E-Learning
- Community learning sessions

- Role play
- Case studies

Equally important is the unstructured teaching that happens all day every day on patient issues being managed by the health team (Medical Doctors Nurses etc.) every health team member should contribute to the education of the student .Teaching by example through good team spirit and leadership.

Assessment types

- **Summative assessment**
 - ✓ Progressive (continuous) assessment....15%
 - ✓ Professionalism mini-evaluation exercise (P-MEX)- Direct observation of student's professional behaviors -5%
 - ✓ practical exam(long ,short, viva and OSCE).....50%
 - ✓ written exam.....30%
- **Formative assessment**
 - ✓ This will not contribute to the total grade of the student but provides continuous feed back to the student and it can be applied in the form of providing feed backs on assignments completed, students' professional behavior and etc.

Methods of assessment

2. OSCE (Objective Structured Clinical Examination)
3. OSPE (Objective Structured Practical Exam)
4. PRRE (Personal Research and Reflection Exercise)/ Log books and portfolios
5. Clinical examination (Long case and Short cases)
6. Oral exam (Viva)
7. Written exam (MCQ, Short Essay, Matching, True- False with reasoning)
8. 360 0 Assessment (Self, Peer, faculty, health facility staff and community member)

Schedule	Learning Activity	Required Reading/ Assignment
Week 1	Clinical practicum (6 hrs) • Assignment of students to paediatrics ward ,NICU ,Emergency OPD, Speciality clinics and Health centre • Three morning session meetings	Reflective portfolio
	Seminar(2 hrs) Prescription writing and Rational use of antibiotics	
	Simulation lab Neonatal resuscitation • Simulation – umbilical vein catheterization, blood sample collection and lumbar puncture	
	Bedside Teaching (4 hrs) •	
	Primary health care or community visit Pediatric OPD in health centers	

	• Home visits	
	Whole group session (3 hrs) • Faculty facilitated discussion and reflection on the week's learning activities Whole Person Care (WPC) – 1 hour of introductory ppt and case studies	
Week 2	Clinical practicum (6 hrs) • Assignment of students to paediatrics ward, NICU, Emergency OPD, Speciality clinics and Health centre • Three morning session meetings	Reflective portfolio
	Case study Poisoning(2 hrs)	
	(6 hrs) Simulation lab Case Scenarios on different poisoning syndromes	
	Bed side teaching (4 hrs)	
	Primary health care or community visit Pediatric OPD in health centers • Home visits	
	Whole group session (3 hrs) • Faculty facilitated discussion and reflection on the week's learning activities	
Week 3	Clinical practicum (6 hrs) • Assignment of students to paediatrics ward, NICU, Emergency OPD, Speciality clinics and Health centre • Three morning session meetings	
	Case study (2 hrs) Ambulatory paediatrics management (skin infection, IP, OM, TP)	
	Simulation lab(2 hrs) Case scenario on a patient with congestive heart failure	
	Primary health care or community visit Pediatric OPD in health centers • Home visits	
	Whole group session (3 hrs)	

	Faculty facilitated discussion and reflection on the week's learning activities Ethical Dilemmas & Ethical Dilemmas analysis(2 hour of introductory ppt and case studies)	
Week 4	Clinical practicum (6 hrs) - Assignment of students to paediatrics ward ,NICU ,Emergency OPD, Speciality clinics and Health centre - Three morning session meetings Case study Management of acute bronchial asthma Simulation lab Case scenario on a wheezing child Bed side teaching (4 hrs) Primary health care or community visit Pediatric OPD in health centers - Home visits Whole group session (3 hrs) - Faculty facilitated discussion and reflection on the week's learning activities	Reflective portfolio
Week 5	Clinical practicum - Assignment of students to paediatrics ward ,NICU, Emergency OPD, Speciality clinics and Health centre Three morning session meetings (2 hrs) Seminar fluid & electrolyte disturbances and their management. Bed side teaching (4 hrs) Simulation lab (2 hrs) case scenario on a child with diarrheal and shock Primary health care or community visit Pediatric OPD in health centers - Home visits Whole group session (3 hrs) - Faculty facilitated discussion and reflection on the week's learning activities	Reflective portfolio
Week 6		
	Clinical practicum	

	• Assignment of students to paediatrics ward , NICU, Emergency OPD, Speciality clinics and Health centre Three morning session meetings	
	Case study ➤ Management of a convulsing child ➤ Management of DKA Simulation ➤ Case scenario of status epileptics Bed side teaching (4 hrs) Side lab/clinical skill lab (2 hrs) Video on intraosseus infusion Simulation – practice intra osseus infusion Primary health care or community visit Pediatric OPD in health centers • Home visits Whole group session (3 hrs) • Faculty facilitated discussion and reflection on the week's learning activities	
Week 7	Monday-Tuesday: Study break for exam Wednesday: Written exam Thursday and Friday: Standardized oral exam and structured long cases	

REFERENCES

- Karen J. Marcante, Nelson Essentials of pediatrics, 6th edition
- Current Pediatrics, 6th edition
- Nelson's Text book of Pediatrics, 19th edition
- Rudolph's Pediatrics, 20th edition
- AssayeKasie, Manual for Neonatology,
- Feigin and Cherry, Pediatrics Infectious disease
- IMNCI Chart booklet
- Lynn S. Bickley, Bate's guide to physical examination and history taking, 10th edition
- Keith L. Moor, Arthur F. Dally, Anne M.R. Agur. Clinically oriented anatomy. 6th edition. 2010
- Eric Arthur Newsholme and Tony R. Leech. Functional biochemistry in health and disease. 2010.
- Stephen Goldber. Clinical biochemistry made ridiculously simple. 1999.
- Ashis Banerjee. Clinical physiology. An examination primer. 2005.
- Bertram G. Katzung. Basic and clinical pharmacology.
- EFMOH compassionate and respectful care training manual for health work force, 2017.
- World Medical Association Medical Ethics Manual, 3rd edition, 2015

MODULE TITLE: OBSTETRICS AND GYNECOLOGY II

MODULE CODE: OBGY 5011

Duration: 7 weeks

MODULE PREREQUISITE:

Completed Pre-clerkship courses successfully

Module Description: This second clerkship in obstetrics and gynecology is intended to enable students to advance their clinical knowledge, skills and attitude on detection and management of obstetric and gynecologic disorders with emphasis on problems common in Ethiopia. During the rotation students will have opportunities to relate knowledge of biomedical sciences for management of patient problems. Students will also consolidate relevant competencies in health promotion and disease prevention, communication, collaboration, leadership, women health advocacy, scholarship professionalism.

Module Objective

- By the end of this module, students will be able to assess patients with obstetrics and gynecologic disorders and assist with management including health promotion and disease prevention.

Supporting Objectives

1. Medical expert

- Apply knowledge of normal and abnormal structures and functions of the human body system especially reproductive system in analyzing obstetrics and gynecologic conditions
- Apply knowledge of the human life cycle and effects of growth, development especially reproductive system in analyzing obstetrics and gynecologic conditions
- Apply knowledge of physiological mechanisms and processes of the human body system in analyzing obstetrics and gynecologic conditions
- Apply knowledge of the various causes and pathogenesis of abnormal obstetric and gynecologic disorders in analyzing patient problems
- Analyze important determinants and risk factors of obstetric and gynecologic conditions
- Apply knowledge of principles of drug pharmacokinetics and pharmacodynamics in the treatment of common obstetric and gynecologic conditions
- Demonstrate proper history taking skills from a patient with an obstetric and gynecologic disorder
- Demonstrate proper physical examination skills of a patient with an obstetric and gynecologic disorder
- Attempt to identify abnormal physical findings in a patient with obstetric and gynecologic disorders
- Attempt to make an initial assessment of a patient presenting with signs and symptoms of an obstetric and gynecologic disorder
- Propose a plan of investigation for a patient with an obstetric and gynecologic disorder in partnership with the patient, obtaining informed consent as an essential part of this process
- Show the ability to perform bedside tests and procedures useful for the workup of common obstetrics and gynecologic conditions
- Attempt interpreting the results of commonly used investigations including obstetrics

Ultrasound in obstetrics and gynecology

- Demonstrate the ability to formulate a likely diagnosis in a patient with an obstetric and gynecologic disorder
- Demonstrate clinical judgments and decisions, based on the available evidence in analyzing problems of an obstetric and gynecologic patient
- Demonstrate the ability to attend normal labor and use partograph to monitor progress of labor
- Demonstrate the ability to perform neonatal resuscitation
- Develop a management plan for a patient with obstetric and gynecologic disorder according to established principles and best evidence
- Assess, counsel and offer the required services for women and girls seeking safe abortion services according to the national guideline
- Assess and counsel clients to help them reach a decision of their contraceptive choice and, offer the contraceptive methods they have chosen or facilitate referral where the methods the clients have chosen are found
- Provide short and long acting contraceptive methods based on clients choice under supervision
- Adhere to infection prevention principles in approaching patients (clients) with obstetrics and gynecology disorders
- Formulate the prognosis of common obstetrics and gynecology disorders taking into account the patient, natural history of the disease and intervention alternatives
- Perform procedures like episiotomy, vacuum/forceps, attend breech delivery

2. Communicators

- Demonstrate communication with clients seeking gynecologic and obstetric care using a patient-centered approach that encourages patient trust and autonomy and is characterized by empathy, respect, and compassion
- Optimize the physical environment for clients seeking gynecologic and obstetric care comfort, dignity, privacy, engagement, and safety
- Recognize when the values, biases, or perspectives of clients seeking gynecologic and obstetric care, physicians, or other health care professionals may have an impact on the quality of care, and modify the approach to the patient accordingly
- Respond to patients' non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients
- Use patient-centered interviewing skills to effectively gather relevant biomedical and psychosocial information
- Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information
- Seek and synthesize relevant information from other sources, including the patient's/client's family, with the patient's consent.
- Demonstrate sharing information that are clear, accurate, timely, and adapted to patient's/client's and his or her family's level of understanding and need in a simulated setting
- Demonstrate breaking bad news to patients and families
- **Effectively communicate with diverse patient populations including seriously ill and difficult patient**
-

3. Collaborators

- Show the willingness to work with students of other professions like midwives

to maintain a climate of mutual respect and shared values

- Establish and maintain healthy relationships with physicians and other colleagues in the health care professions to support relationship-centered collaborative care
- Recognize one's own professional roles and limits
- Share knowledge and information to other colleagues
- Show respect toward other physicians and members of the health care team
- Respect and value diversity among other colleagues

4. Leaders

- Demonstrate leadership in professional practice
- Manage their practice and career of themselves, other health care providers and support staff
- Demonstrate the ability to effectively plan and efficiently manage one's own time and activities to cope with uncertainty, and the ability to adapt to change
- Promote occupational health as physicians for sustainable practice
- **Show commitment to autonomous maintenance and continuous improvement of competence**
- **Advocate and lead implementation of CRC in the health care system**

5. Women Health Advocate

- Demonstrate responding to the needs of women by advocating for policy and system-level change
- Demonstrate responding to the individual woman's reproductive health needs and rights by advocating with the woman within and beyond the clinical environment
- Demonstrate working with women and their families to address the determinants of health that affect women's reproductive health and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors
- Lead a healthy life style to be role models to promote women health through for example participating in breast and cervical screening

6. Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice
- Ensure patient safety is maintained while learning
- Provide effective feedback to enhance learning and performance
- Analyze global and national causes and trends in maternal morbidity and mortality
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program

- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry
- Contribute to the teaching of medical and other health sciences students regarding obstetrics and gynecology

7. Professional

- Integrate other intrinsic roles into their learning and practice of gynecology and obstetrics
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in gynecologic and obstetrics care
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients/clients seeking gynecologic and obstetric care, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice in gynecology and obstetrics
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the provision of maternal health services
- Participate in peer assessment and standard-setting
- Demonstrate a culture that recognizes, supports, and responds effectively to colleagues in need
- Apply Legal aspects in health care practice
- Solve ethical dilemmas
- Provide care in CRC manner in clinical setting

Prerequisites: Obstetrics and Gynecology I

Teaching and learning methods

- Interactive lecture
- PBL
- Case study
- Demonstration including video
- Roleplay
- Simulated (skills lab) practice
- Practice bedside tests at Sidelab
- OPD teaching
- Teaching ground
- Bedside teaching
- Morning session
- Duty rotation
- Portfolio
- Personal research and reflection exercise
- Inter-professional learning experience in the skill lab, clinic, and PBL tutorial

Teaching and learning materials

- Videos of patient examinations and clinical procedures in obstetrics and gynecology
- Imaging studies of different obstetrics and gynecology conditions
- Mannequins for diagnostic and therapeutic procedures in obstetrics and gynecology

References

- Cunningham (etal). Williams Obstetrics. (latest edition)
- Danforth, Gynecology and Obstetrics. (latest edition)
- Gabbe, Obstetrics. Normal and Problem pregnancies. (latest edition)
- Current Diagnosis & Treatment Obstetrics & Gynecology, (latest edition)
- *Berek, Jonathan S. Berek & Novak's Gynecology*, (latest edition)
- Keith L. Moor, Arthur F. Dally, Anne M. R. Agur. Clinically oriented anatomy. (latest edition)
- Eric Arthur Newsholme and Tony R. Leech. Functional biochemistry in health and disease. (latest edition).
- Stephen Goldber. Clinical biochemistry made ridiculously simple (latest edition)
- Ashis Banerjee. edition Clinical physiology. An examination primer. (latest edition)
- EFMOH compassionate and respectful care training manual for health work force, 2017.
- World Medical Association Medical Ethics Manual, 3rd, 2015

Methods of assessment

Formative

- Seminar
- PBL progressive assessment
- Direct observation of clinical skills
- Logbook
- Global rating of performance or 360 degree evaluation
- Portfolio

- Personal research and reflection exercise
- Professionalism mini-evaluation exercise (P-MEX)- Direct observation of student's professional behaviors
- **Summative**
 - PBL progressive assessment (10%)
 - 360 degree evaluation or global rating of performance (10%)
 - Written exam (20%)
 - Standardized oral exam (10%)
 - Structured long cases (20%)
 - External exam (long, short and oral exams) (30%)

Module schedule

Date	Learning Activity	Required Reading/ Assignment
Week 1	Morning meeting	Reflective portfolio
	Clinical practicum Student placement at outpatient and inpatient departments (family planning, antenatal, labor, maternity, gynecology, follow-up clinic)	
	Case study • Case related to prolonged labor (addressing CPD, the three delays, too few or too many cesarean section) Case study • Case study on obstructed labor	
	Night duty Student placement at outpatient and inpatient departments (Gyn OPD, gyn ward, labor ward, maternity ward, operation theatre)	
	Bedside teaching Case related to the topic in the week	
	Clinical skills lab Simulation on operative delivery: Instrumental vaginal delivery (Vacuum and forceps) and cesarean section	
	PHC unit/community visit: Students identify knowledge gaps and harmful practices in the community that are risk factors for obstructed labor	
	Whole group session Faculty facilitated discussion and reflection on the week's learning activities	
Week 2	Morning Meeting	Reflective portfolio

	Clinical practicum Student placement at outpatient and inpatient departments (family planning, antenatal, labor, maternity, gynecology, follow-up clinic)	
	Case study - Case related to diabetes in pregnancy Case study - Case related to Hypertension in pregnancy, preeclampsia, eclampsia	
	Bedside teaching Case related to the topics in the week	
	Clinical skills lab Simulated practice: IUCD insertion and removal Simulated practice: Implant insertion and removal	
	PHC unit/community visit: Identify pregnant women with Diabetes mellitus, hypertension, cardiac illness, health education on risk factors, screening, ANC follow up	
	Whole group session Faculty facilitated discussion and reflection on the week's learning activities	
Week 3	Morning meeting	
	Clinical practicum Student placement at outpatient and inpatient departments (family planning, antenatal/PMTCT, labor, maternity, gynecology, follow-up clinic)	
	Case study - Case with cardiac illness in pregnancy Case study - Case of Rh immunization	
	Bedside teaching Case related to the topic in the week	
	Clinical skills lab Simulated practice: Approach to a victim of sexual abuse (Communication skills)	
	PHC unit/community visit: Students identify women with history of fetal/neonatal death, Students identify women with HIV /AIDS, counsel pregnant women for HIV testing; infant feeding	
	Whole group session Faculty facilitated discussion and reflection on the week's learning activities skills on breaking bad news & Informed consent and Issue of confidentiality, anonymity, and privacy(2 hour of introductory ppt and Role plays)	

Week 4	Morning Meeting	
	Clinical practicum Student placement at outpatient and inpatient departments (family planning, antenatal, labor, maternity, gynecology follow-up clinic)	
	Case study - Gender-based violence Case study - Maternal health and HIV/AIDS (include, disclosure, dilemma to get pregnant or not, contraceptives, determinants of infection, testing and access to care)	
	Bedside teaching Case related to the topic in the week	
	Clinical skills lab Simulation practice on counselling a pregnant woman with HIV/AIDS Simulation practice on counselling a gender-abuse victim	
	PHC/community visit Health education on the need to prevent and report gender-based violence Simulated practice: Counseling pregnant women for HIV testing; infant feeding counseling for HIV+women, PMTCT	
	Whole group session Faculty facilitated discussion and reflection on the week's learning activities	
Week 5	Morning Meeting	
	Clinical practicum Student placement at outpatient and inpatient departments (family planning, antenatal, labor, maternity, gynecology follow-up clinic)	
	Case study - Case related to infertility and amenorrhea (including gender-based myths, misconceptions about cause and treatment, determinants) Case study - Relaxation of pelvic floor	
	Bedside teaching Case related to the topic in the week	
	Side lab or clinical skills lab Simulated practice: Taking history and counselling of a woman with complaint of infertility	
	PHC/community visit Identify women/couple with complaints of infertility, identify knowledge gaps, give health education to the couple on the need for visit to the health facility for investigations and treatment Identify women with complaints of prolapse of uterus, identify knowledge gaps, give health education to women on the risk factors, the need for visit to the health facility for investigations and treatment	

	Whole group session Faculty facilitated discussion and reflection on the week's learning activities Relevant legal frameworks governing professional practice (1 hour of introductory of ppt and group discussion)	
Week 6	Morning meeting	
	Clinical practicum Student placement at outpatient and inpatient departments (family planning, antenatal, labor, maternity, gynecology, follow-up clinic)	
	Case study • Cervical cancer, ovarian cancer Case study • Myoma, endometrial cancer, benign and malignant tumors of vulva and vagina • Acute abdomen in pregnancy	
	Bedside teaching Case related to the topics in the week	
	Side lab or clinical skills lab • Visual inspection with Acetic acid, cervical pap smear • Abdominal paracentesis • Culdocentesis	
	PHC/community visit Students promote cervical cancer screening and participate in screening programs if any	
	Whole group session Faculty facilitated discussion and reflection on the week's learning activities	
Week 7	Monday-Tuesday: Study break for exam	
	Wednesday: Written exam	
	Thursday and Friday: Standardized oral exam and structured long cases	

⁸⁴ Students are required to maintain in their portfolio learning experiences, learning outcomes achieved, evidence of learning and reflections longitudinally and their mentors would provide feedback occasionally. Clinical instructors should also ask students to submit reflective portfolio twice during the attachment for formative assessment and feedback.

⁸⁵ As much as possible, PBL cases during clerkship should preferably be based on real cases observed during clinical practicum.

Interactive presentation – use various methods including case studies, clinical simulation having students work in small group and summarize the session reinforcing key points using lectures

MODULE TITLE: EMERGENCY MEDICINE

MODULE CODE: EMME 5011

PRE-REQUISITE: COMPLETION OF CLERKSHIP-I

MODULE DURATION: 6 WEEKS

MODULE DESCRIPTION

This module deals with providing advanced medical services including rapid assessment and stabilization of patients with urgent or life threatening conditions, gives insight in to pre hospital emergency medical services(EMS),cost effective urgent care in specially designed “fast track area”, and detailed management of selected patients in emergency medicine based clinical decision unit pertinent to Ethiopia. The students will also be competent in areas of emergency patient care, medical knowledge, practice based learning, interpersonal and communication skills as well as system based practice for their appropriate level of training in emergency medicine.

MODULE OBJECTIVE

At the end of the course the students will be able do initial assessment and stabilization with sense of urgency, initiate emergency management and prepare the patient for appropriate disposition.

SUPPORTIVE OBJECTIVES

As a Medical Expert

- Triage and set appropriate priorities when dealing with single or multiple acutely ill patients.
- Perform accurate and complete clinical assessments of patients presenting with non-specific clinical complaints and syndromes
- Establish and maintain clinical knowledge, skills and attitudes necessary to rapidly assess and manage a full spectrum of patients, often concomitantly, with acute or undifferentiated illness and injury.
- Demonstrate the capacity to differentiate and treat common acute problems.
- Provide a comprehensive assessment of the undifferentiated patient.
- Demonstrate basic life support skills and cardiopulmonary resuscitation.
- Recognize and initiate first aid for airway obstruction.
- Recognize and initiate appropriate management for a patient with respiratory distress
- Recognize and be prepared to intervene for all causes of shock in any age group.
- Provide rapid stabilization with intravenous access and fluid/blood administration.
- Explain the principles of cerebral resuscitation in brain illness and injury.
- Demonstrate the use of an automatic external defibrillator (AED).
- Demonstrate basic wound care techniques.
- Demonstrate basic trauma management skills, such as initial assessment using the ABC approach and full spine immobilization.
- Demonstrate mastery of basic procedural skills, such as basic airway management and venous access.
- Recognize life-threatening illness or injury, and apply basic principles of stabilization to the early management of these entities.
- Demonstrate the capacity to prioritize attention to those patients with more urgent conditions.
- Explain the importance of the ED as a key link between the general population and the health care system.

- Contribute to the improvement of quality care and patient safety in Emergency Medicine practice, integrating the available best evidence and best practices
- Demonstrate effective clinical problem solving and judgment to address patient problems, including interpreting available data and integrating information to generate differential diagnoses and management plans

As a communicator

- Establish positive therapeutic relationships with patients and their defined family units that are characterized by understanding, trust, respect, honesty and empathy
- Respect patient confidentiality, privacy and autonomy
- Be able to use language and terminology that facilitates understanding and decision-making by patients and their defined family units
- Listen attentively and effectively
- Be aware and responsive to nonverbal cues
- Effectively facilitate a structured clinical encounter
- Be knowledgeable of and attentive to different ethnic, social and cultural backgrounds
- Gather information about a disease, but also about a patient's beliefs, concerns, expectations and illness experience.
- Act professionally and tactfully when screening for sensitive issues or information.
- Seek out and synthesize relevant information from other sources, such as a patient's defined families units, caregivers, family physician, non-hospital personnel (police, mass media, firefighters, EMS personnel) and other professionals
- Deliver information that is concise, relevant, useful and respectful to a patient, their defined family units, colleagues and other professionals.
- Deliver information to a patient and their defined family unit, colleagues and other professionals in a humane manner and in such a way that it is understandable, encourages discussion and participation in decision-making.
- Respect the patient's privacy in accordance with privacy and confidentiality legislation, regulations and policies.

As a collaborator

- Work with others in a multi-disciplinary approach to assess, plan, provide and integrate care for individual patients.
- Optimize and expedite patient care through involvement of other health care professionals and delegate appropriately.
- Apply the principles of team dynamics
- Respect team ethics, including confidentiality, resource allocation and professionalism
- Respond positively to requests for help or advice
- Accommodate requests from community or hospital physicians for assistance or advice in patient management
- Demonstrate a respectful attitude towards other colleagues and members of an inter-professional team
- Work with other professionals to prevent conflicts
- Employ collaborative negotiation to resolve conflicts
- Recognize that differences and limitations in scope of practice between professionals that may contribute to misunderstandings and inter-professional tensions.

As a Manager

- Recognize the importance of the fair allocation of health care resources, balancing effectiveness, efficiency and access with optimal patient care.
- Apply the structure and function of the health care system as it relates to Emergency Medicine, including the roles of physicians.
- Possess the fundamental knowledge and skills needed to provide medical leadership.
- Demonstrate the ability to address complaints from patients, family members and colleagues.
- Set priorities and manage time to balance patient care, practice requirements, outside activities and personal life.
- Identify priorities for change in emergency health care.

As a Health Advocate

- Identify the health needs of an individual patient.
- Identify opportunities for advocacy, health promotion and prevention with individuals to whom they provide care with regards to emergency medicine.
- Identify the determinants of health of the populations, including barriers to access to care and resources with regards to emergency medicine.
- Identify vulnerable or marginalized populations within those served and respond appropriately.
- Explain an approach to implementing a change in a determinant of health of the populations they serve
- Explain how public policy impacts on the health of the populations served.
- Identify points of influence in the health care system and its structure.

Scholar

- Describe principles of learning relevant to medical education.
- Collaboratively identify the learning needs and desired learning outcomes of others.
- Select effective teaching strategies and content to facilitate others' learning.
- Demonstrate an effective lecture or presentation.
- Assess and reflect on a teaching encounter.
- Provide effective feedback.
- Describe the principles of ethics with respect to teaching.
- Pose an appropriate learning question.
- Access and interpret the relevant evidence.
- Integrate new learning into practice.
- Evaluate the impact of any change in practice.
- Document the learning process.
- Describe the principles of critical appraisal.
- Critically appraise retrieved evidence in order to address a clinical question.
- Integrate critical appraisal conclusions into clinical care.

Professional

- Identify the principles of good medical practice
- Outline the elements of informed consent
- Demonstrate professional behaviour and attributes in the emergency floor
- Demonstrate dressing in acceptable standard manner in emergency room

- Recognize the scope of practice in emergency medicine
- Demonstrate insight into one's own limits of expertise
- Demonstrate knowledge of and attention to different ethnic, social, and cultural backgrounds
- Illustrate respect to team ethics, including confidentiality, resource allocation, and professionalism.
- Demonstrate compassionate and respectful patient centered care in emergency room
- Demonstrate knowledge and understanding of professional, legal, and ethical codes of practice

Teaching and learning methods

- Interactive lecture
- Bedside teaching
- Case study
- Simulation based training
- Clinical practicum

Teaching and learning materials

- Mannequins-including computerized ones-adult, paediatric, neonatal and obstetric
- Lecture –white, black and, smart board
- Small group Teaching rooms
- Practical emergency skill training center well equipped with emergency equipments.
- Overhead projector
- AV projectors
- Collection of AV slides
- Selection of DVDs
- Pre-hospital care services
- Emergency crash cart
- Defibrillator
- Portable ultrasound

Assessment methods

Formative

- PBL/case study
- Direct observation of clinical skills
- Logbook
- Global rating of performance or 360 degree evaluation
- Portfolio

Summative

Progressive

- PBL/Case study-5%
- Clinical Practicum-20%
- Skill lab-20%
- Logbook-5%

End Module

- Written-30%
- OSCE-20%

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Typical Week for Emergency Medicine in Clerkship II

Time	Monday	Tuesday	Wednesday	Thursday	Friday ⁴³
8.00am-9.30 am	Interactive Lecture	Morning presentation	Interactive lecture	Morning presentation	Interactive Lecture
9.30am-12.00p m	Clinical practicum in ER	Interactive lecture	Clinical practicum in ER	Interactive lecture	Clinical practicum in ER

⁴³ Clinical practicum in this document refers to all clinical learning activities students engage in, supervised or unsupervised (inpatient as well as outpatient) including clerking, observing, assisting and performing bedside procedures and Rounds. Bedside rounds are not included as they are considered planned and makeups may be provided.

2.00pm-3.30pm	Skill labpractice	Case study	Bed side teaching	Skill lab practice	Skill lab practice
3.30pm-5pm	Skill lab practice	Clinical practicum in ER	Clinical practicum in ER	Clinical practicum in ER	Skill lab practice
5-11PM	Duty	Duty	Duty	Duty	Duty

Module schedule

Date	Learning Activity	Required Reading/ Assignment
Week1	Interactivelecture • Introduction to emergency Medicine • Primary and secondary survey • Airway management in the ED • Emergency care of neuro-trauma • Approach to chest and abdominal trauma	Reflective portfolio
	Clinical practicum Student placement in the ER	
	Case study • Poly-trauma case	
	Bedsideteaching Caserelated to the topicsin theweek if available, focus on • Primary and secondary assessment • FAST scanning of trauma patient	
	Clinical skills lab Video: Basic and advanced airway management Video: Chest tube insertion and needle decompression Simulated practice: • Adult basic life support/CPR • Basic airway management • Oxygen therapy • Needle cricothyrotomy • Needle thoracostomy • Chest tube insertion	
	Duty	
Week2	Interactivelecture • Approach to patient with respiratory distress with cases on severe pneumonia, acute exacerbation of COPD, status asthmaticus, ARDS, PTE, cardiogenic pulmonary edema • Approach to patient with undifferentiated shock • Basic ECG interpretation	Reflective Portfolio

	• Management of common arrhythmias • Approach to Acute coronary syndrome, non-traumatic Chest pain and ED diagnostics	
	Clinical practicum Student placement in the ER	
	Case study Case of arrhythmia/shock	
	Bedsideteaching Cases related to the topics in the week	
	Clinical skills lab video: peripheral and central venous access and IO Simulated practice • foreign body airway obstruction management • peripheral and IO access • RUSH protocol • practical session on ECG • Cardio-version and defibrillation	
	Duty	
Week 3	Interactive lecture • EM approach to poisoning • Management of common electrolyte disturbances • EM approach to coma and status epilepticus • Endocrine emergency DKA, HHS, thyrotoxicosis and thyroid storm, myxedema coma and adrenal crisis • Hypertensive emergency and syncope	
	Clinical practicum Student placement in the ER	
	Case study Case of acute abdomen	
	Bedsideteaching Cases related to the topics in the week	
	clinical skills lab Simulation practice: Manage resuscitation cases in a team	
	Duty	
Week 4	Interactive Lecture • Procedural sedation and analgesia • Emergency radiology • ED triaging • Patient flow in the emergency department • The concept of Pre hospital care and disaster preparedness	
	Clinical practicum Student placement in the ER	
	Case study • Mass casualty case	

	• Small group discussion on patient flow in the emergency department	
	Bedsideteaching Caserelated to the topics in the week	
	Clinical skills lab Video: Awake and crush intubation Slides: head CTs, CXR ,c-spine x ray Simulated practice: • Pre-hospital care • Emergency department triaging in a team • Initial evaluation and management of orthopedic emergencies	
	Duty	
Week 5	Interactive Lecture • Pediatrics airway management • common Pediatric respiratory emergencies • Pediatric shock • Initial management of pediatric CHF • Neonatal resuscitation • Pediatric CNS emergencies	
	Clinical practicum Student placement in the ER	
	Case study • Case scenarios on shock, respiratory distress and comatose child • Child abuse case	
	Bedsideteaching Caserelated to the topics in the week	
	Clinical skills lab Video: pediatric BLS Simulated practice: • Pediatrics life support and CPR • basic airway management • peripheral iv access • Intraosseous access • foreign body airway obstruction relieving maneuvers	
	Primary Health care unit/community visit • Home visits trying to understand common causes of emergencies in the community and their social determinants and conduct health promotion and disease prevention activities	
	Whole group session • Faculty facilitated discussion and reflection on the Module's learning activities	
Week 6	Exam Written OSCE	

MODULE TITLE: OPHTHALMOLOGY

MODULE CODE: OPHT 5011

NOMINAL DURATION: Clerkship II: 3 weeks

COURSE DESCRIPTION:

Ophthalmology attachment during Clerkship is intended to build upon the student's knowledge, skill and attitude required to diagnose and manage common ocular problems, provide primary eye care including minor procedures for emergency cases and refer those who need further evaluation and management. The student will be able to explain the ocular manifestations of systemic illness and identify those patients who need ophthalmic screening/evaluation. It is also to make the student recognize the impact of impaired vision or blindness on patients' quality of life including that of their families and participate in activities of prevention of blindness in the community and in the world. To enhance skill development, this clerkship courses will be delivered integrated to Professional Competency development (PCD), Primary care attachment relevant to Ophthalmology and linked Social and population health (SPH).

COURSE DESIGN:

This will be delivered in Clerkship II:

The teaching will focus on building the ability of the student in history taking, physical examination in the clinical setting, formulating differential diagnosis, reaching to appropriate diagnosis, listing relevant investigation and is intended to enable the student for an increased measure of responsibility and prepares students for Internship. This provides the students to have the experience of being the primary caregiver for patients in a well-supervised setting.

Education in ophthalmology clerkship will be delivered in different setting of health care and in collaboration across disciplines. The students will have out-patient and inpatient experiences: Out-patient experience will be in the respective department of the hospital and also in the primary care (Health centers) which will be coordinated with their specific department of attachment.

COURSE PREREQUISITE:

Pre-clerkship modules

COMPETENCIES/MODULE OBJECTIVES

Module objectives

By the end of the module the students will be able to diagnosis and manage common ocular problems and provide initial treatment for emergency ocular conditions, communicate with patients with ocular problems and the healthcare team, recognize the impact impaired vision or blindness has on patients' life including that of their families and participate in activities of prevention of blindness in the community and in the world.

Supporting Objectives

As a Medical Expert

- Demonstrate a commitment to high-quality primary eye care of their patients
- Apply knowledge of the clinical and biomedical sciences relevant to the eye care
- Identify determinants and risk factors of causes of visual impairment and blindness to protect and promote individual and population eye health
- Perform timely ophthalmic clinical assessments for patients with common eye problems and recommend a well organized and properly documented plan of management in written and/or oral form
- Elicit ophthalmic history in patients with different eye complaints.
- Perform basic eye exam for patients
- Make a differential diagnosis for common eye problems
- Identify and prioritize emergency ophthalmic conditions,

- Provide primary eye care and refer timely for common eye problems and emergency ophthalmic conditions
- Diagnose and establish a patient-centered management plan based on available evidences
- Provide primary eye care within the practical context of patient physician relationship taking to account patient's culture and adapting to economic limitations, and encounter time constraints
- Establish goals of care in collaboration with patients and their families, which may include slowing disease progression, achieving cure, improving function, treating symptoms, and palliation
- Plan and perform minor ophthalmic procedures and interventions for the purpose of assessment and/or management of eye diseases
- Obtain and document informed consent, explaining the risks and benefits of, and the rationale for, the proposed options
- Prioritize minor ophthalmic procedures, taking into account clinical urgency, potential for visual impairment and blindness, and the available resources
- Perform procedures in a skillful and safe manner, adapting to unanticipated findings or changing clinical circumstances
- Establish their roles as physicians and other health care professionals in the plans for ongoing eye care and seek appropriate consultation or referral to the ophthalmologist recognizing the limits of their expertise.
- Actively participate, as an individual and as a member of a team providing eye care, in the continuous improvement of eye health care quality and patient safety
- Contribute to a culture that promotes patient safety

As a Communicator, the Student should be able to...

- Communicate using a patient-centered approach that encourages patient trust and autonomy and is characterized by empathy, respect, and compassion
- Optimize the physical environment for patient with visual impairment and blindness with comfort, dignity, privacy, engagement, and safety
- Recognize when the values, biases, or perspectives of patients, physicians, or other health care professionals may have an impact on the quality of eye care, and modify the approach to the patient accordingly
- Respond to patients' non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients
- Manage disagreements and emotionally charged conversations
- Adapt to the unique needs and preferences of patients and those related to visual impairment and blindness
- Develop rapport, trust and ethical therapeutic relationships with patients with ocular problems and their families ;
- Elicit and synthesize accurate and relevant information, incorporating the perspectives of patients with ocular problems, their families and communities
- Use patient-centered interviewing skills to effectively gather relevant biomedical and psychosocial information
- Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information
- Provide a clear structure for and manage the flow of an entire patient encounter
- Seek and synthesize relevant information from other sources, including the patient's family, with the patient's consent.

- Skillfully share information and explanations that are clear, accurate, timely, and adapted to patient's and his or her family's level of understanding and need
- Communicate appropriately with difficult/ violent patients and people with mental illness
- Facilitate discussions with patients and their families in a way that is respectful, non-judgmental, and culturally safe
- Assist patients with visual impairment and blindness and their families to identify, access and make use of information and communication technologies to support them .
- Use appropriate communication skills and strategies to help patients and their families make informed decisions regarding their eye health
- Document and share written and electronic information about the medical encounter to optimize clinical decision-making, patient safety, confidentiality, and privacy

As a collaborator

- Establish and maintain healthy relationships with physicians and other colleagues in the health care professions to support relationship-centered collaborative primary eye care
- Negotiate overlapping and shared responsibilities with physicians and other colleagues in the eye care professions
- Engage in respectful shared decision-making with other colleagues, patients and their families
- Recognize one's own professional roles and limits in eye care
- Share knowledge and information to other colleagues
- Show respect toward other physicians and members of the eye care team
- Respect and value diversity among other colleagues
- Implement strategies to resolve conflicts in a manner that supports a collaborative culture
- Assess when care should be transferred to another physician or health care professional
- Demonstrate safe transfer of care, using both verbal and written communication, during a patient transition to a different health care professional, setting, or stage of care

As a Manager/Leader

- Contribute to the improvement of eye health care delivery in teams, organizations, and systems
- Apply the science of quality improvement to contribute to improving systems of eye care
- Analyze adverse events and near misses to enhance systems of care
- Use health informatics to improve the quality of patient eye care and optimize patient safety
- Engage in the stewardship of health care resources
- Allocate eye care resources considering justice, efficiency, and effectiveness for optimal patient care
- Apply evidence and management processes to achieve cost-appropriate care
- Develop their leadership skills
- Design and organize elements of health care delivery
- Facilitate change in health care to enhance services and outcomes
- Set priorities and manage time to balance practice and personal life
- Manage career planning, finances, and health human resources in a practice
- Implement processes to ensure personal practice improvement
- Promote occupational health for sustainable practice

As a Health Advocate

- Work with patients with ocular problems to address determinants of eye health that affect them and their access to needed eye health services or resources
- Work with patients with ocular problems and their families to increase their opportunities to adopt healthy behaviors
- Incorporate prevention of visual impairment and blindness, eye health promotion, and eye health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of eye health that affect them
- Participate in community eye care services, and in prevention of visual impairment and blindness, eye health promotion, and eye health surveillance activities
- Participate in global and national initiatives, strategies, to improve eye health in the community or population they serve

As a Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance using various internal and external data sources
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Demonstrate effective teaching to facilitate learning
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize practice uncertainty and knowledge gaps in clinical and other professional encounters and generate focused questions that can address them
- Demonstrate proficiency in identifying, selecting, and navigating pre-appraised resources
- Integrate evidence into decision-making in their practice
- For a given practice scenario, formulate well-structured scholarly questions
- Identify scholarly sources that shed light on a given professional question
- Evaluate the applicability (external validity or generalizability) of evidence from a wide range of information resources
- Discuss the barriers to and facilitators of applying study findings to professional practice
- Use quality-appraised evidence-alerting services that highlight new evidence appropriate to their scope of professional practice
- Interpret study findings, including a discussion and critique of their relevance to professional practice
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms

- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspect of practice
- Recognize and respond to ethical issues encountered in practice
- Recognize and manage conflicts of interest
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the Profession
- Fulfill and adhere to the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the healthcare professions
- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Manage personal and professional demands for a sustainable practice throughout the physician life cycle
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Teaching and learning methods

- Interactive lecture
- Bedside teaching
- Case study
- Simulation based training
- Clinical practicum
- Mentorship

Equally important is the unstructured teaching that happens all day every day on patient issues being managed by the eye care team (general practitioners, clinical and ophthalmic nurses, residents and ophthalmologists) and every health team member should contribute to the education of the student, teaching by example through good team spirit and leadership.

Teaching and learning materials

- Mannequins and models of the eye, (Eye Examination Trainer)
- Lecture –white, black and, smart board
- Small group Teaching rooms
- Overhead projector

- AV projectors
- Collection of AV slides (images of eye pathologies)
- Selection of DVDs (videos for demonstration of clinical examination and procedural skills)

Assessment methods

Formative

- PBL/case study
- Direct observation of clinical skills
- Logbook
- Global rating of performance or 360 degree evaluation
- Portfolio

Summative

Progressive

- PBL/Case study-5%
- Clinical Practicum-20%
- Skill lab-20%
- Logbook-5%

End Module

- Written-30%
- OSCE-20%

Module Schedule

Date	Learning Activity	Required Reading/Assignment
Week One Day 1	Interactive lecture (2hrs) • Introduction to the module • Anatomy and physiology of the eye	⁴⁴ Reflective portfolio
	Clinical skills lab (3 hrs) Simulated practice: Anatomic models of the eye, videos	
	Interactive presentation (2hr) Ocular history taking and physical examination	
Day 2	Clinical skill lab (2 hrs) Simulated practice: Ocular history and physical examination	Reflective portfolio
	Clinical practicum⁴⁵ (3 hrs) Student placement at ophthalmic outpatient and inpatient departments and follow up clinics including refractive clinics⁴⁶ and Operation theatre (major and minor)	
	PBL (2 hr) first session The Red Eye	
Day 3	Interactive lecture (2 hrs) Common Eyelid and conjunctiva disorders including	Reflective portfolio

⁴⁴ Week 5 Friday will be dedicated to Primary Health care unit/community visit, Home visits trying to understand common causes of emergencies in the community and their social determinants and conduct health promotion and disease prevention activities and Whole group session Faculty facilitated discussion and reflection on the Module's learning activities

	TRACHOMA Clinical practicum (3 hrs) Student placement at ophthalmic outpatient and inpatient departments and follow up clinics Operation theatre (major and minor) Interactive presentation, Case based discussion (2hrs) Approach to Acute visual loss	
Day 4	Interactive lecture, case based discussion (2 hrs) Progressive loss of vision emphasis on cataract and glaucoma part I Clinical practicum (3 hrs)including refractive clinics and Operation theatre (major and minor) Student placement at ophthalmic outpatient and inpatient departments and follow up clinics Clinical skills lab (2 hr) Simulated practice: Clinical examination of an ophthalmic patient with focus on pupillary examination, intraocular pressure test, confrontation visual field testing and funduscopy examination	Reflective portfolio
Day 5	Primary health care or community visit (4 hrs) • Ophthalmic outreach clinics • Home visits of patients prioritizing ophthalmic conditions and skills covered during the week • Community-based eye programs • Blindness control programs Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities	Reflective portfolio
Week two Day 1	Interactive Lecture, case based discussion (2 hrs) Progressive loss of vision emphasis on cataract and glaucoma part II Clinical skills lab (3 hrs) Simulated practice: Clinical examination of an ophthalmic patient with focus on pupillary examination, intraocular pressure test, penlight examination of anterior segment, and funduscopy examination PBL (2 hrs) second session The Red eye	Reflective portfolio
Day 2	Interactive lecture (2 hrs) Principles of management of Ocular emergencies Clinical practicum (3 hrs) Student placement at ophthalmic outpatient and inpatient departments and follow up clinicsincluding refractive clinics and Operation theatre (major and minor) Bedside teaching (2 hrs)	Reflective portfolio

	On admitted patients e.g. Keratitis, Orbital Cellulitis, retinoblastoma, cataract, glaucoma	
Day 3	Interactive lecture (2 hrs) Global and national major causes of blindness and visual impairment Vision 2020	Reflective portfolio
	Clinical practicum (3 hrs) Student placement at ophthalmic outpatient and inpatient departments and follow up clinics including refractive clinics and Operation theatre (major and minor)	
	Case study (2 hrs) Ocular trauma	
Day 4	Interactive Lecture (2 hrs) Childhood blindness (including leucocoria, Xerophthalmia)	Reflective portfolio
	Clinical practicum (3 hrs) Student placement at ophthalmic outpatient and inpatient departments and follow up clinics including Operation theatre (Major and minor)	
	Clinical skills lab (2 hrs) Simulated practice: Perform minor procedures, eyelid eversion, subconjunctival injection, corneal/conjunctiva foreign body removal, chalazion excision, probing and irrigation of nasolacrimal duct, tarsorrhaphy and tarsotomy	
Day 5	Primary health care or community visit (4 hrs) • Ophthalmic outreach clinics • Home visits of patients prioritizing ophthalmic conditions and skills covered during the week • Community-based eye programs • Blindness control programs	Reflective portfolio
	Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities	
Week three Day 1	Interactive Lecture (2 hrs) Approach to Diplopia and squint/strabismus, Amblyopia	Reflective portfolio
	Clinical practicum (3 hrs) Student placement at ophthalmic outpatient and inpatient departments and follow up clinics including refractive clinics and Operation theatre (major and minor)	

⁴⁵ Students are required to maintain in their portfolio learning experiences, learning outcomes achieved, evidence of learning and reflections longitudinally and their mentors would provide feedback occasionally. Clinical instructors should also ask students to submit reflective portfolio twice during the attachment for formative assessment and feedback.

⁴⁶ Clinical practicum includes all clinical learning activities that students engage in inpatient, outpatient, refractive clinic, and/or Operation theatre settings supervised and/or unsupervised (clerking patients, observing, assisting, and/or performing procedures, patient chart review including investigation results)

	Case study (2 hrs) Approach to proptosis	
Day 2	Interactive lecture (2 hrs) Ocular manifestation of systemic diseases	Reflective portfolio
	Clinical practicum (3 hrs) Student placement at ophthalmic outpatient and inpatient departments and follow up clinics including refractive clinics and Operation theatre (major and minor)	
	Bedside teaching (2 hrs) Case related to the topics in the week	
Day 3	Interactive lecture (2 hrs) Ocular manifestation of HIV/AIDS	Reflective portfolio
	Clinical practicum (3 hrs) Student placement at ophthalmic outpatient and inpatient departments and follow up clinics including refractive clinics and Operation theatre (major and minor)	
	Clinical skills lab (2 hrs) Simulated practice: Perform minor procedures, eyelid eversion, subconjunctival injection, corneal/conjunctiva foreign body removal, chalazion excision, probing and irrigation of nasolacrimal duct, tarsorrhaphy and tarsotomy	
Day 4	Clinical practicum (5 hrs) Student placement at ophthalmic outpatient and inpatient departments and follow up clinics including refractive clinics and Operation theatre (major and minor)	
	Independent study	
Day 5	Written Exam	
	OSCE	

MODULE TITLE: DERMATOLOGY

MODULE CODE: DERM 5011

NOMINAL DURATION: Clerkship II: 3 weeks

MODULE DESCRIPTION: dermatology attachment during Clerkship is intended to build upon the student's knowledge, skill and attitude required to diagnose and manage common dermatologic and venereological diseases with an emphasis on common problems in Ethiopia. Students will also have additional knowledge and develop competencies on tropical dermatology, enable them manage common dermatological disorders in greater depth and to identify referral cases. The subject will be delivered in the form of lectures, ward rounds, bed side teaching, outpatient sessions and management session linked Social and population health (SPH).

MODULE DESIGN

This will be delivered in clerkship II.

The teaching will focus on building the ability of the student in history taking, physical examination in the clinical setting, formulating differential diagnosis, reaching to appropriate diagnosis, listing relevant investigation and is intended to enable the student for an increased measure of responsibility

and prepares students for Internship. This provides the students to have the experience of being the primary caregiver for patients in a well-supervised setting.

Education of dermatology in clerkship will be delivered in different setting of health care and in collaboration across disciplines. The students will have out-patient and inpatient experiences: Out-patient experience will be in the respective department of the hospital and also in the primary care (Health centers) which will be coordinated with their specific department of attachment.

MODULE PREREQUISITE:

Completed Year 1-3 MODULEs successfully.

COMPETENCIES/MODULE OBJECTIVES

Module objectives

- By the end of this module students will be able to assess patients with dermato- venereological disorders and assist with their management.

Supporting Objectives

As a Medical Expert

- Apply knowledge of the clinical and biomedical sciences relevant to the skin care and venereal disease.
- Identify determinants and risk factors of causes of skin disorder to promote individual and population care.
- Perform timely dermatologic assessments for patients with common skin and venereal disease and recommend a well-organized and properly documented plan of management in written and/or oral form
- Elicit dermatologic history in patients with different skin complaints.
- Perform basic skin exam for patients
- Make a differential diagnosis for common skin problems and sexual transmitted diseases
- Identify and prioritize emergency dermatologic condition.
- Diagnose and establish a patient-centered management plan based on available evidences
- Provide primary skin care within the practical context of patient physician relationship taking into account patient's culture and adapting to economic limitations, and encounter time constraints
- Establish goals of care in collaboration with patients and their families, which may include slowing disease progression, achieving cure, improving function, treating symptoms, and palliation
- Plan and perform minor dermatologic procedures and interventions for the purpose of assessment and/or management of skin diseases
- Obtain and document informed consent, explaining the risks and benefits of, and the rationale for, the proposed options
- Prioritize minor dermatologic procedures, taking into account clinical urgency and the available resources.
- Counsel individuals on how to maintain healthy skins

As a Communicator

- Communicate using a patient-centered approach that encourages patient trust and autonomy and is characterized by empathy, respect, and compassion.

- Recognize when the values, biases, or perspectives of patients, physicians, or other health care professionals may have an impact on the quality of care, and modify the approach to the patient accordingly
- Respond to patients' non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients
- Contribute to a culture that promotes patient safety
- Manage disagreements and emotionally charged conversations
- Adapt to the unique needs and preferences of patients and those related to skin disorder and sexually transmitted infection.
- Develop rapport, trust and ethical therapeutic relationships with patients with skin problems and their families ;
- Elicit and synthesize accurate and relevant information, incorporating the perspectives of patients with skin and sexually transmitted disorders.
- Use patient-centered interviewing skills to effectively gather relevant biomedical and psychosocial information
- Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information
- Provide a clear structure for and manage the flow of an entire patient encounter
- Seek and synthesize relevant information from other sources, including the patient's family, with the patient's consent.
- Skillfully share information and explanations that are clear, accurate, timely, and adapted to patient's and his or her family's level of understanding and need
- Communicate appropriately with difficult/ violent patients and people with mental illness
- Facilitate discussions with patients and their families in a way that is respectful, non-judgmental, and culturally safe.
- Use appropriate communication skills and strategies to help patients and their families make informed decisions regarding their skin problem.
- Document and share written and electronic information about the medical encounter to optimize clinical decision-making, patient safety, confidentiality, and privacy

As a collaborator

- Establish and maintain healthy relationships with physicians and other colleagues in the health care professions to support relationship-centered collaborative primary skin care
- Negotiate overlapping and shared responsibilities with physicians and other colleagues in the skin care professions
- Engage in respectful shared decision-making with other colleagues, patients and their families
- Recognize one's own professional roles and limits in skin care
- Share knowledge and information to other colleagues
- Respect and value diversity among other colleagues
- Implement strategies to resolve conflicts in a manner that supports a collaborative culture.
- Demonstrate safe transfer of care, using both verbal and written communication, during a patient transition to a different health care professional, setting, or stage of care

As a Manager/Leader

- Contribute to the improvement of skin care delivery in teams, organizations, and systems
- Apply the science of quality improvement to contribute to improving systems of skin care

- Analyze adverse events and near misses to enhance systems of care
- Use health informatics to improve the quality of patient skin care and optimize patient safety
- Engage in the stewardship of health care resources
- Allocate skin care resources considering justice, efficiency, and effectiveness for optimal patient care
- Apply evidence and management processes to achieve cost-appropriate care
- Develop their leadership skills
- Design and organize elements of health care delivery
- Facilitate change in health care to enhance services and outcomes
- Set priorities and manage time to balance practice and personal life
- Manage career planning, finances, and health human resources in a practice
- Implement processes to ensure personal practice improvement
- Promote occupational health for sustainable practice

As a Health Advocate

- Work with patients with skin problems to address determinants of skin health that affect them and their access to needed skin health services or resources
- Work with the community to increase their opportunities to adopt healthy behaviors towards skin and sexually transmitted diseases.
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of skin health that affect them
- Participate in community skin care services, and in prevention of sexually transmitted infections.
- Participate in global and national initiatives, strategies, to improve skin health in the community or population they serve.

As a Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance using various internal and external data sources
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Demonstrate effective teaching to facilitate learning
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize practice uncertainty and knowledge gaps in clinical and other professional encounters and generate focused questions that can address them
- Demonstrate proficiency in identifying, selecting, and navigating pre-appraised resources

- Integrate evidence into decision-making in their practice
- For a given practice scenario, formulate well-structured scholarly questions
- Identify scholarly sources that shed light on a given professional question
- Evaluate the applicability (external validity or generalizability) of evidence from a wide range of information resources
- Discuss the barriers to and facilitators of applying study findings to professional practice
- Use quality-appraised evidence-alerting services that highlight new evidence appropriate to their scope of professional practice
- Interpret study findings, including a discussion and critique of their relevance to professional practice
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Commitment to high-quality skin care patients.
- Assist patients with physical impairment due leprosy access and make use of available resource.
- Actively participate, as an individual and as a member of a team providing care, in the continuous improvement of skin care quality and patient safety.
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality.
- Assess when care should be transferred to another physician or health care professional
- Demonstrate a commitment to excellence in all aspect of practice
- Recognize and respond to ethical issues encountered in practice
- Recognize and manage conflicts of interest
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the Profession
- Fulfill and adhere to the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the healthcare professions
- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Manage personal and professional demands for a sustainable practice throughout the physician life cycle

- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Teaching and learning methods

- Interactive lecture
- Case study
- Demonstration including video
- OPD teaching

Teaching and learning materials

- Videos of patient examinations and clinical procedures

Assessment types

Summative assessment

- Written exam
- OSCE

Formative assessment

- This will not contribute to the total grade of the student but provides continuous feedback to the student and it can be applied in the form of providing feedbacks on assignments completed, students' professional behavior and etc.
- PBL progressive assessment - Case discussion

Methods of assessment

- DOPS (Direct Observation of Practical Skills)*
- OSCE (Objective Structured Clinical Examination)
- Written exam (MCQ, Short Essay, Matching, True- False with reasoning)
- Log books and portfolios

Module Schedule

Date	Learning Activity	Assignment
Week One Day 1	Interactive lecture (2hrs) • Introduction to the module ○ Anatomy and physiology of epidermis, dermis and subcutis. ○ Describe process of keratinization ○ Appendages of the skin ○ Functions of the skin	⁴⁷ Reflective portfolio
	Simulated practice(1hr): Anatomic models of the skin and genitalia, videos	
	Interactive presentation (2hr) history taking and physical examination ○ Skin lesion description:- morphology, shape, arrangement and distribution.	

⁴⁷ Students are required to read independently on REFRACTIVE ERRORS

Day 2	Interactive lecture(2 hrs)	Reflective portfolio
	Common bacterial infections of the skin and hair follicles ○ Superficial pyodermas Impetigo contagiosa, Ecthyma ○ Soft tissue infections Cellulitis, Erysipelas ○ Bacterial folliculitis, Furuncle, Carbuncle	
	Outpatient clinical practicum(2hrs)	
	Case discussion on Cellulitis and erysipelas (2hrs)	
Day 3	Interactive lecture (2 hrs) ○ <i>Dermatophytes</i> Ring worm (Tinea) of the skin, hair & nails ○ <i>Yeasts</i> Candidiasis (cutaneous, mucosal and nails) ○ Pityriasis versicolor	Reflective portfolio
	Clinical practicum(2hrs)	
	Student placement to dermatology clinic	
	Case discussion on tinea capitis and onychomycosis(2hrs)	
Day 4	Interactive lecture, case based discussion (2 hrs) ○ Herpes simplex ○ Chicken pox & Herpes zoster ○ Warts ○ Molluscum contagiosum	Reflective portfolio
	Clinical practicum (2 hrs)	
	Student placement at dermatologic clinic outpatient and inpatient departments and follow up clinics	
	case based discussion (2hrs)	
	○ Melasma ○ Vitiligo	
Day 5	Interactive lecture(2hrs) ○ Acne Vulgaris ○ Rosacea ○ POD	Reflective portfolio
	Clinical practicum (2 hrs)	
	Student placement at dermatologic clinic outpatient and inpatient departments and follow up clinics	
	PBL(2hrs) Acne vulgaris and rosacea	
Week two Day 1	Interactive Lecture, case based discussion (2 hrs) Eczema (<i>with emphasis to Atopic Dermatitis /AD/</i> ○ Atopic dermatitis ○ Seborrheic dermatitis ○ Lichen simplex chronicus	

	○ Contact Dermatitis (Allergic and Irritant)	Reflective portfolio
	Clinical practicum (2 hrs) Student placement at dermatologic clinic outpatient and inpatient departments and follow up clinics	
	School visit to screen students (3hrs)	
Day 2	Interactive lecture (2 hrs) Psoriasis and Lichen planus	Reflective portfolio
	Clinical practicum (2 hrs) Student placement at Dermatologic outpatient and inpatient departments and follow up clinics	
	Follow up clinic (2hrs)	
Day 3	Interactive lecture(2hr) Leprosy and leprosy reactions	Reflective portfolio
	Clinical practicum (2 hrs) Student placement at dermatologic clinic outpatient and inpatient departments and follow up clinics.	
	Clinical skill lab(1hr) Video on the different skin lesion discussed Study dermatology atlas on cases discussed Lab: slit skin smear examination Slides showing AFB	
	Case based discussion (2 hrs)	
Day 4	Interactive Lecture (2 hrs) Leshmaniasis and onchocerciasis	
	Clinical practicum (2 hrs) Student placement at Dermatologic outpatient and inpatient departments and follow up clinics	
	Clinical skills lab (2 hrs) Video on the different skin lesion discussed Study dermatology atlas on cases discussed Lab: slit skin smear examination Slides showing LD bodies	
Day 5	Interactive lecture(2hrs) ○ Scabies ○ Pediculosis ○ Myiasis and tungiasis	Reflective portfolio
	Clinical practicum (2 hrs) Student placement at Dermatologic outpatient and inpatient departments and follow up clinics	
	Seminar presentation on skin manifestation of HIV (2hrs)	
Week three Day 1	Interactive Lecture (2 hrs) ● Venereal diseases ○ Genital discharges	Reflective portfolio

	○ Genital ulcers ○ Inguinal bubo	
	Clinical practicum (2 hrs) Student placement at Dermatologic outpatient and inpatient departments and follow up clinics	
	Community attachment :- House hold visit.(3hrs)	
Day 2	Interactive lecture (2 hrs) ● Cutaneous Adverse drug reactions ○ Morbilliform eruption ○ Fixed drug reactions ○ Erythema multiforme ○ Steven-Jonson Syndrome (SJS) ○ Toxic Epidermal Necrolysis (TEN)	Reflective portfolio
	Clinical practicum (2 hrs) Student placement at dermatologic outpatient and inpatient departments and follow up clinics	
	Interactive lecture (2hrs) Introduction to Autoimmune skin diseases (immunobullous diseases, connective tissue diseases)	
Day 3	Therapeutic modalities in dermatology(2hrs) Perform minor procedures, Cryotherapy, Electrocautary	Reflective portfolio
	Clinical practicum(2hrs) Student placement to dermatology clinic. Case discussion	
Day 4	Self-study	
Day 5	Written Exam	
	OSCE	

ANNEX: MODULE CONTENT

Topic	Objectives	Methods	Resources
LECTURES			
Basic anatomy and physiology of the skin	Describe basic structure of the skin (epidermis, dermis and subcutis) Describe process of keratinization Appendages of the skin Functions of the skin	Interactive presentation, simulated practice	
History taking and physical exam	Pertinent history and Physical examination with emphasis to the examination of the skin, nail, hair and mucous membranes	Interactive presentation, simulated practice, and clinical practicum	
Common bacterial infections of the skin and hair follicles	Overview of Superficial pyoderms and Soft tissue infections.	Interactive presentation, and clinical practicum,	
Common skin	<i>Dermatophytes and Yeasts</i>	Interactive	

diseases resulting from fungi		presentation, ,clinical practicum	
Common skin diseases resulting from viruses	Knowledge on common viral infections.	Interactive presentation, clinical practicum	
Common parasitic infestations of the skin	Overview on common parasitic infestations of the skin	Interactive presentation, clinical practicum	
Eczema (<i>with emphasis to Atopic Dermatitis /AD/</i>)	General features, classification, diagnosis and treatment	Interactive presentation, clinical practicum	
Papulosquamous disorder	General features, triggering factors, clinical manifestations, common variants, diagnosis & treatment.	Interactive presentation, clinical practicum	
Hansen's disease (Leprosy)	Introduction, definition, mode of transmission, classification, clinical presentation, leprosy reactions, diagnosis and treatment	Interactive presentation, PBL, clinical practicum	
Adenexal disorders	Common Pilosebaceous unit disorder	Interactive presentation, clinical practicum	
Pigmentary disorders	Overview of common Pigmentary disorders	Interactive presentation,case discussion,clinical practicum	
HIV and the skin	Common skin manifestations of HIV	Interactive presentation, seminar	
Veneral diseases (STDs)	Etiology, epidemiology, transmission, clinical features, diagnosis and treatment.	Interactive presentation, PBL, clinical simulation	
Syndromic management of STDs	Describe common STD syndromes.	Interactive presentation, ,clinical practicum	
Drug reaction	Overview of Cutaneous Adverse drug reactions.	Interactive presentation, clinical practicum	
Minor OR	Perform minor procedures	Demonstration, coaching	

MODULE TITLE: PSYCHIATRY AND MENTAL HEALTH

Module duration: 4 weeks

Module code: PSYC 5011

Module description: This will be delivered in Clerkship II and is intended to equip students with Knowledge, skill and attitude required to diagnose common mental disorders. The teaching will focus on building the ability of the student in history taking, mental state examination in the clinical setting, formulating differential diagnosis, reaching to appropriate diagnosis, formulating the case in Biopsychosocial model, listing relevant investigation and is intended to enable the student for an increased measure of responsibility and prepares students for Internship. Education in psychiatry clerkship will be delivered in different setting of health care and in collaboration across disciplines. The students will have out-patient and inpatient experiences: Out-patient experience will be in the respective department of the hospital and also in the primary care (Health centers) which will be coordinated with their specific department of attachment Course

Prerequisites

Completed the following courses successfully. Pre medicine, clerk ship I, pre-clerkship II and clerk ship I

Module Objective

The student will be able to explain the psychiatric manifestations of systemic illness and identify those patients who need psychiatric evaluation. It is also to make the student recognize the impact of mental illnesses on patients' quality of life including that of their families and participate in activities of prevention of mental illness in the community and in the world. By the end of the course the students will be able do assessment and management of common psychiatric disorders including mental health promotion and disease prevention.

Supporting Objectives

As Medical expert

Take psychiatric history from the patient or from their attendants and perform mental state examination of a patient with a psychiatric disorder within the practical context of the individual patient-physician relationship

Perform physical examination and Identify abnormal physical findings of a patient with a psychiatric disorder within the practical context of the individual patient-physician relationship

Apply knowledge of the various causes (biological, psychological and social) of disease in assessing and managing psychiatric problems

Make an initial assessment of a patient presenting with signs and symptoms of a psychiatric disorder to triage and prioritize the patient

Perform psychiatric risk assessment for harm to self and others

Demonstrate capacity in dealing with psychiatric emergencies (including emergency substance use problems) such as providing rapid stabilization with acutely agitated and disturbed patient

Perform biopsychosocial formulation for a likely diagnosis in a patient with a psychiatric disorder
Formulate a plan of basic investigation for a patient with a psychiatric disorder in partnership with the patient, obtaining informed consent as an essential part of this process

Demonstrate clinical judgments and decisions, based on the available evidence in diagnosing and managing problems in patients with psychiatric disorders

Suggest a plan for biopsychosocial management for a patient with psychiatric disorders according to established principles and best evidence, in partnership with the patient, their care givers, and other health professionals as appropriate.

Recognize psychiatric comorbidities in patients with general medical conditions

Analyze important biopsychosocial factors and determinants of common psychiatric disorders such as life-style, genetic, demographic, environmental, health system, social, economic, psychological and cultural.

Outline rehabilitation strategies for patients with psychiatric disorders

As a communicator

Demonstrate good and effective communication skills in practice

Establish positive therapeutic relationships with patients and their families that are characterized by understanding, trust, respect, honesty, empathy and by not being judgmental

Respect patient confidentiality, privacy and autonomy

Use language and terminology that facilitates understanding and decision-making by patients and their families or care givers

Be aware and responsive to nonverbal cues of the patient

Effectively facilitate a structured clinical encounter

Responsive different ethnic, social and cultural backgrounds

Gather information about a disease, but also about a patient's beliefs, concerns, expectations and illness experience.

Act professionally and tactfully when screening for sensitive issues or information.

Seek out and synthesize relevant information from other sources, such as a patient's defined families' units, caregivers, family physician, non-hospital personnel (police, mass media, firefighters, take holders) and other professionals

Deliver information that is concise, relevant, useful and respectful to a patient, their defined family units, colleagues and other professionals.

Deliver information to a patient and their defined family unit, colleagues and other professionals in a humane manner and in such a way that it is understandable, encourages discussion and participation in decision-making.

Respect the patient's privacy

Respect patient right accordance with privacy and confidentiality mental health legislation, regulations and policies.

Concise exchange necessary information on expected, active and discharged patients.

Demonstrate ability to communicate other mental health professionals, clinical psychologists, social workers, psychiatric nurses

Demonstrate the ability to communicate clearly, sensitively and effectively with patients with psychiatric disorders, people with traumatic life experience, their relatives and peers from the psychiatric and other profession, by listening, sharing and responding (S3, A3)

Demonstrate the ability to communicate appropriately with difficult or violent patients with psychiatric disorder (S3).

Demonstrate the capacity to break bad news to patients and family

Contribute to their obligation to promote, protect and enhance the essential elements of the profession for benefit of psychiatric patients, the profession and society at large (A3)

Contribute to prioritizing patients' needs and safety in caring for patients with psychiatric disorder (A3)

Recognize the need to see whole person as a complete being

Demonstrate professional values in approaching patients with psychiatric disorder (A3)

Demonstrate mutual understanding and relationship with patient's family (A3)
Able to communicate in non-clinical settings such as security officers of the Hospital during assessment of high risk patients

As a collaborator

Work with others in a multi-disciplinary approach to assess, plan, provide and integrate care for individual patients.

Optimize and expedite patient care through involvement of other health care professionals and delegate appropriately.

Describe the principles of team dynamics

Respect team ethics, including confidentiality, resource allocation and professionalism

Respond on time positively to requests for help or advice

Accommodate requests from community or hospital physicians for assistance or advice in patient management

Demonstrate a respectful attitude towards other colleagues and members of an extraprofessional team

Work with other health professionals minimizing conflicts

Employ collaborative negotiation to resolve conflicts

Recognize that differences and limitations in scope of practice between professionals that may contribute to misunderstandings and interprofessional tensions.

As a Manager

Recognize the importance of the fair allocation of health care resources, balancing effectiveness, efficiency and access with optimal patient care.

Describe the structure and function of the health care system as it relates to psychiatry, including the roles of physicians

Possess the fundamental knowledge and skills needed to provide medical leadership.

Demonstrate the ability to address complaints from patients, family members and colleagues.

Set priorities and manage time to balance patient care, practice requirements, outside activities and personal life.

Identify priorities for change in psychiatric service.

As a Health Advocate

Identify the health needs of an individual patient with mental disorder

Identify opportunities for advocacy, health promotion and disease prevention with individuals to whom they provide care.

Identify the determinants of health of the populations, including barriers to access to care and resources.

Identify vulnerable or marginalized populations within those served and respond appropriately.

Describe an approach to implementing a change in a determinant of health of the populations they serve.

Describe how public policy impacts on the health of the populations served.

Identify points of influence in the health care system and its structure.

Scholar

Describe principles of learning relevant to medical education.

Collaboratively identify the learning needs and desired learning outcomes of others.
 Select effective teaching strategies and content to facilitate others' learning.
 Demonstrate an effective lecture or presentation.
 Assess and reflect on a teaching encounter.
 Provide effective feedback.
 Describe the principles of ethics with respect to teaching.
 Pose an appropriate learning question.
 Access and interpret the relevant evidence.
 Integrate new learning into practice.
 Evaluate the impact of any change in practice.
 Document the learning process.
 Describe the principles of critical appraisal.
 Critically appraise retrieved evidence in order to address a clinical question.
 Integrate critical appraisal conclusions into clinical care.

Teaching and learning methods

Interactive lecture

PBL

Demonstration/ Video show

Role play

Case study

Simulated (skills lab) practice

OPD/In patient teaching

Bed side teaching

Personal research and reflection exercise Portfolio

Inter-professional learning experience in the skill lab, clinic, community and PBL tutorial / social worker/psychologist

Teaching and learning materials

Small group Teaching rooms

practical skill training rooms /PBL

Selection of DVDs

References

Kaplan &Sadock's Synopsis of Psychiatry Behavioral Sciences/Clinical Psychiatry 11th Edition
 Copyright © 2015 Wolters Kluwer

Kaplan &Sadock's Comprehensive Textbook of Psychiatry, 9th Edition, Sadock, Benjamin J.;
 Sadock, Virginia A.; Ruiz, Pedro Copyright ©2009 Lippincott Williams & Wilkins

Diagnostic and Statistical Manual of Mental Disorders

Fifth Edition, DSM -5™, Copyright © 2013 American Psychiatric Association

Essential psychopharmacology, Neuroscientific Basis and practical Applications, Second Edition,
 Stephen M. Stahl © Cambridge University Press 2000.

Methods of assessment

Formative

PBL progressive assessment

Direct observation of clinical skills

Logbook

Global rating of performance or 360-degree evaluation

Portfolio /Personal research and reflection exercise

Summative
 Progressive assessment
 360-degree evaluation or global rating of performance
 Written exam
 OSCE
 Assessment
 Progressive
 PBL progressive assessment 5%
 Clinical skill Lab 15%
 Bed side/ case discussion 10%
 Log book -5%
 Global rating of performance or 360-degree evaluation 10%
 Portfolio /Personal research and reflection exercise 5%
 End Module
 Written-30%
 OSCE-20%
 Course content (Annex I)
 Course schedule

	Course schedule	
Schedule	Learning Activity	Required Reading/ Assignment
Week One Day 1	Interactive Lecture (2hrs) • Introduction to psychiatry • classification of Psychiatric disorders Patient Doctor relationship(2hrs) • Student placement to psychiatry clinic. Clinical Lab(3hrs) Case Discussion: • classification of psychiatric disorders • Community /cultural disease explanatory models	
Day 2	Interactive Lecture(2hrs) • Sign symptoms/ terminologies in psychiatry/ & Psychopathology Clinical practicum (2hrs) Student placement to the psychiatry clinic. • Case Discussion on sign symptoms of mental disorders Clinical skills lab(3hrs) • Skills in eliciting psychopathology- role play • Video show: Clerking a psychiatry patient	Reflective portfolio
Day 3	Interactive Lecture(2hrs) • Psychiatric evaluation (examination and diagnosis of psychiatric patient)	

	• Interview techniques	
	Clinical practicum (2hrs) • Student placement to psychiatry clinic	
	Problem based learning(3hrs) • Case discussion: Doing assessment in patient with Depression	
Day 4	Interactive Lecture (2hrs) Relevant psychosocial theories 1. Attachment theory 2. Learning theory	
	Clinical practicum(2hrs) • Student placement to psychiatry clinic	
	Clinical skill lab(3hrs) • Practice taking history and mental state examination from a patient with schizophrenia – role play • Application of theories to psychiatry -case study	
Day 5	Primary health care/community visit • Primary health care visit • Rehabilitation centers /organization working for people with mental illness	
	Whole group discussion Faculty facilitated discussion and reflection on the week's learning activities	
Week Two Day 1	Interactive Lecture (3hrs) • Schizophrenia • Other psychotic disorders	
	Clinical practicum(2hrs) • Student placement to psychiatry clinic. Case discussion on schizophrenia	
	Case study(3hrs) • Schizophrenic patient evaluation and management	
Day 2	Interactive Lecture(3hrs) • Mood disorders ○ Bipolar and related disorders ○ Depressive disorders	Reflective portfolio
	Clinical practicum(2hrs) • Student placement to psychiatry clinic. Case discussion on Schizoaffective/other psychotic disorders	
	Clinical skill lab(3hrs) • Practice taking history and mental state examination from a patient with Brief psychotic disorder (post-partum onset) psychosis • Giving psychoeducation to the patient and the family of patient with psychotic disorder	
Day 3	Interactive lecture(2hrs) •	

	Clinical practicum(2hrs) • Student placement to psychiatry clinic. • Case discussion on patient with bipolar disorder	
	Problem based learning(3hrs) • Case: bipolar disorder patient • Skill lab: video – showing patient with bipolar I disorder manic episode a physician interaction	
Day 4	Interactive presentation (2hrs) • Anxiety disorders • Post traumatic and stress related disorders	
	Clinical practicum (2hrs) • Student placement to psychiatry clinic: Case discussion	
	Clinical skill lab(3hrs) • Practice taking history and mental state examination from patient with depression with high suicide risk– role play • Video showing Suicide risk assessment	
Day 5	Primary health care/community visit (4hrs) • Health center/rehabilitation center	
	Whole group discussion • Faculty facilitated discussion and reflection on the week’s learning activities	
	Clinical practicum (2hrs) • Student placement to psychiatry clinic. Case discussion on stimulant use disorder	
	Patient with stimulant use disorder with medical complications (E.g. peptic ulcer disease)	
Week three Day 1	Interactive presentation (3hrs) • Substance use related disorders	
	Clinical practicum(2.5hrs) • Student placement to psychiatry clinic. Case discussion	
	Clinical skill lab(3hrs) • Video showing of Child with Autistic spectrum disorder improves with psychosocial intervention and parent education	
Day 2	Interactive presentation (2hrs) • Neurodevelopmental disorders (CAP) • Psychiatric Emergency (Suicide, para suicide, violence, Agitated and aggressive patient)	
	Clinical practicum(2hrs) • Student placement to psychiatry clinic/ Psychiatric OPD	
	Clinical skill lab(3hrs) Case study • Management of acutely agitated psychotic patient • Suicide and Violence Risk assessment	

Day 3	Interactive presentation (3 hrs) • Defense mechanisms • Behavioral and emotional problems in children and adolescents • Somatic Symptom and Related Disorders	
	Clinical practicum(2.5hrs) • Student placement to psychiatry clinic	
	Clinical lab (3hrs) • Case discussion immature defense mechanism	
Day 4	Interactive presentation (3hrs) • Obsessive compulsive related disorders • Psychotherapy, psychosocial interventions and psychopharmacology	
	Clinical practicum(2hrs) • Student placement to child psychiatry clinic	
	Clinical skill lab(3hrs) • Evaluation and management of an adolescent with ADHD • Parent education	
Day 5	Primary health care/community visit /schools who gave special need education	
	Whole group discussion • Faculty facilitated discussion and reflection on the week's learning activities	
Week four Day 1	Interactive presentation (2hrs) • Personality disorders	
	Clinical practicum(2.5hrs) • Student placement to psychiatry clinic	
	• Skill lab(3hrs) • Video Demonstration in skills of evaluation and management of patient with erectile dysfunction disorder • Case study	
Day 2	Interactive presentation (2hrs) • Sexual Disorders and paraphilia • Neurocognitive disorders(Delirium)	
	Clinical practicum(2.5hrs) • Student placement to psychiatry clinic	
	Case study(3hrs) Skill in • Management of the patient with delirium • Mini mental state examination, Bed side assessment of delirium	
Day 3	Interactive presentation (1.5hrs)	

	Sleep-Wake Disorders	
	Clinical practicum (2.5 hrs.) Student placement to psychiatry clinic	
	Clinical skill lab (3hrs) Case study Practice managing of patient with presented with insomnia	
Day 4	Self-study	
Day 5	Examination	

Typical week in year 5

Time	Monday	Tuesday	Wednesday	Thursday	Friday
8:00 -10:00 am	Interactive Lecture	Interactive lecture	Interactive lecture	Interactive lecture	⁴⁸ Primary Health Care/Community Visit
	Video/demonstrations		Video/demonstrations		
10:00 -12:00 am	Outpatient	Outpatient	Outpatient	Outpatient	
1:30 -3:30 pm	PBL ⁴⁹ /Case study	bedside	Role play	PBL/Case study	Whole Group Session
3:30 -5:30 pm	Self-study	Clinical skills lab	Self-study	Clinical skills lab	

Resources

Annex: Course content for psychiatry

Course content
Class room teaching and clinical attachment

⁴⁸ Students are required to maintain in their portfolio learning experiences, learning outcomes achieved, evidence of learning and reflections longitudinally and their mentors would provide feedback occasionally. Clinical instructors should also ask students to submit reflective portfolio twice during the attachment for formative assessment and feedback.

⁴⁹ Students go to either primary health care facilities or communities every week on rotation basis to see the psychosocial preventive

	Topic	Objectives	Methods	Resources
1.	Introduction to psychiatry	- Describe mental health mental illness, Barriers challenges and common myths in psychiatric diagnosis & treatment - Explain Burden of psychiatric disorders	Interactive presentation	
2.	classification of Psychiatric disorders	Describe standard classification in psychiatry	Interactive presentation	
3.	Patient Doctor relation ship	- Demonstrate the standard patient doctor relationship - Impact on treatment and outcome	- Simulated practice - Role play - clinical practicum	
4.	Community explanatory models of mental disorders	Describe common understanding in community	Case discussion	
5.	Sign symptoms/ terminologies in psychiatry/ & Psychopathology	- Recognize signs and symptoms of mental illness - Discuss possible causes symptoms - Discuss common psychopathologies	- Interactive lecture - Reading Assignments - Video demonstration	
6.	Sign symptoms of mental disorders Psychopathology	- Demonstrate ability to elicit signs and symptoms in psychiatry - Identify psychopathologies in psychiatry	- Case Discussion - Clinical practicum - Role play - Video show	
7.	- Psychiatric evaluation (examination and diagnosis of psychiatric patient) - Interview techniques	- Apply interview techniques - Determining the nature, the problem - Express important skills in interview	- Interactive Lecture - Clinical practicum - Role play	
8.	Assessment of patient with Depressive disorder	- Determine sign symptoms of depressive disorders - List sign and symptoms of depressive disorders - Plan the biopsychosocial	Observed interview	

		management for depressive disorders		
9.	Psychosocial theories Relevant to psychiatry Attachment theory Learning theory	Indicate application of psychosocial theories to psychiatry	- Interactive presentation - case study	
10	Taking history and mental state examination from a patient with schizophrenia	- Recognize the comprehensive psychiatric evaluation and relevant symptom review - Explain the components of psychiatric evaluation of a patient with schizophrenia	- Interactive presentation, simulated practice, and clinical practicum - Role play	
11	Schizophrenia	- List sign and symptoms of schizophrenia - Analyze the causes of schizophrenia - Plan the biopsychosocial management for schizophrenia	- Interactive lecture - PBL/Case study - Role play	
12	Other psychotic disorders	- Analyze the causes of other psychotic disorder - Plan the biopsychosocial management of other psychotic disorders - Describe the difference between psychotic disorders - Prioritize emergency psychotic conditions and provide management and timely referral (catatonia and post-partum psychosis)	Interactive presentation, - case study - Reflective portfolio	
13	Bipolar and related disorders	- Describe signs and symptoms of bipolar related disorders - Analyze the causes of bipolar and related disorder - Plan the biopsychosocial management of bipolar	video – Demonstration patient with bipolar I disorder manic episode a physician interaction	

		and related disorder	• Case discussion	
14	Anxiety disorders	• Describe signs and symptoms of different anxiety disorders • Analyze the causes of Anxiety disorders • Plan the biopsychosocial management of Anxiety disorders	• Interactive presentation, • case study • Role play patient with generalized anxiety disorder	
15	Post traumatic and stress related disorders	• Analyze the causes of Post traumatic and stress related disorders • Plan the management of Post traumatic and stress related disorders	• Interactive presentation • case study	
16	Obsessive compulsive and related disorders	• Analyze the causes of obsessive compulsive related disorders • Plan the biopsychosocial management of obsessive compulsive and related disorders	• Interactive presentation • Video demonstration of patient with obsessive compulsive disorder	
17	Substance use related disorders I (Over view of substance use disorders and alcohol use disorders)	• Analyze the causes of alcohol use disorder • Plan the biopsychosocial management of alcohol use disorder	• Interactive presentation • case study	
18	Substance related disorder _2(stimulant and other substance related disorder)	• Analyze the causes of stimulant use disorder • Plan the biopsychosocial management of stimulant use disorder	• Interactive presentation • case study	
19	Neurodevelopmental disorders(CAP)	• Analyze the causes of Neurodevelopmental disorders • Identify cases and refer to other disciplines accordingly • Plan the biopsychosocial management of Neurodevelopmental disorders • Recognize preventable cause of	Interactive presentation case study Video demonstration of child with autistic disorder	

		neurodevelopmental disorders		
20	Behavioral and emotional problems in children and adolescents	- Analyze the causes of Behavioral and emotional problems in children and adolescents - Plan the biopsychosocial management of Behavioral and emotional problems in children and adolescents	- Interactive presentation, - case study	
21	Psychiatric Emergency	- Analyze the causes of Psychiatric Emergency - Plan the biopsychosocial management of Psychiatric Emergency	- Interactive presentation - case study - Video demonstration of suicide risk assessment	
22	Defense mechanisms	List Defense mechanisms	- Interactive presentation,	
23	Somatic Symptom and Related Disorders	- Analyze the causes of Somatic Symptom and Related Disorders - Plan the biopsychosocial management of Somatic Symptom and Related Disorders - Demonstrate ability to manage medically unexplained symptoms	- Interactive presentation - case study	
24	Obsessive compulsive and related disorder	- Analyze the causes of obsessive compulsive disorder - Plan Biopsychosocial management from obsessive compulsive disorder	- Interactive presentation - case study	
25	Psycho pharmacology	- Review pharmacological treatment of mental illness	- Interactive presentation - Case study	
26	Psychotherapy and psychosocial interventions	- Analyze the types of psychosocial interventions in primary care - Demonstrate skills in giving psycho education	- Interactive presentation, - case study - Video demonstration of	

			psycho education for the patient and their families	
27	Personality disorders	- Describe the sign and symptoms of personality disorders - Recognize the potential influence of personality disorders on patient – physician interaction	- Interactive presentation, - case study - Video demonstration of patient with Border line personality disorder	
28	Sexual disorders and paraphilia	- Analyze the causes of Sexual disorder and paraphilia - Plan the biopsychosocial management of Sexual disorders and paraphilia	- Interactive presentation - case study	
29	Neurocognitive disorders(Delirium)	- Analyze the causes of Neurocognitive disorders(Delirium) - Recognize risk factors for delirium - Preform assessment of patient with delirium - Plan the management of Neurocognitive disorders(Delirium)	- Interactive presentation - case study	
30	Sleep-Wake Disorders	- Analyze the causes of Sleep-Wake Disorders - Plan the biopsychosocial management of Sleep-Wake Disorders	- Interactive presentation - case study	

MODULE TITLE: OTORHINOLARYNGOLOGY-HEAD and NECK SURGERY

MODULE CODE:

NOMINAL DURATION: Clerkship II: 2 weeks

Module Description

The module will entail common congenital, inflammatory, degenerative, metabolic, traumatic and neoplastic disease of ear, nose and throat organs. It facilitates the acquisition of skills to enable the student to examine patient, identify problems, understand routines and special tests, do simple procedures and manage patients.

To enhance skill development, this clerkship courses will be delivered integrated to Professional Competency development (PCD), Primary care attachment relevant to Otolaryngology and linked Social and population health (SPH).

Module DESIGN:

This will be delivered in Clerkship II:

The teaching will focus on building the ability of the student in history taking, physical examination in the clinical setting, formulating differential diagnosis, reaching to appropriate diagnosis, listing relevant investigation and is intended to enable the student for an increased measure of responsibility and prepares students for Internship. This provides the students to have the experience of being the primary caregiver for patients in a well-supervised setting.

Education in the Otolaryngology-head and neck surgery clerkship will be delivered in different setting of health care and in collaboration across disciplines. The students will have outpatient and inpatient experiences: Outpatient experience will be in the respective departments of the hospital and also in the primary care (Health centers) which will be coordinated with their specific department of attachment.

Module Prerequisite: - Completed Pre-clerkship courses successfully.

MODULE OBJECTIVES:

The objective of this module is to impart knowledge and skills to students in diseases related to the organs of the ear, nose and throat.

SPECIFIC OBJECTIVES: At the end of the course students will be able to :

- Demonstrate skills in management of common ENT problems

Supportive Objectives

As Medical expert

- Demonstrate a commitment to high-quality primary ENT care of their patients
- Apply knowledge of the clinical and biomedical sciences relevant to the ENT care
- Identify determinants and risk factors of causes of hearing impairment, Sino-nasal diseases, throat and laryngeal problems
- Perform relevant investigative and therapeutic procedures for the ENT patient
- Diagnose ENT problems based on the analysis of history, physical exam and investigative work up
- Interpret important image and laboratory results and common ENT specific investigations such as: Audiometry, Tympanometry and Otoacoustics emission
- Make a differential diagnosis for common ENT problems
- Identify and prioritize and manage emergency ENT conditions,
- Diagnose and establish a patient-centered management plan based on available evidences
- Provide primary ENT care within the practical context of patient physician relationship taking in to account patient's culture and adapting to economic limitations, and encounter time constraints
- Establish goals of care in collaboration with patients and their families, which may include slowing disease progression, achieving cure, improving function, treating symptoms, and palliation
- Obtain and document informed consent, explaining the risks and benefits of, and the rationale for, the proposed options
- Perform procedures in a skillful and safe manner, adapting to unanticipated findings or changing clinical circumstances
- Establish their roles as physicians and other health care professionals in the plans for ongoing ENT care and seek appropriate consultation or referral to the OTOLARYNGOLOGIST recognizing the limits of their expertise.
- Actively participate, as an individual and as a member of a team providing ENT care, in the continuous improvement of Otolaryngology health care quality and patient safety

- Contribute to a culture that promotes patient safety

As a communicator

- Demonstrate communication with a patient with a dental, periodontal and oral cavity diseases using a patient-centered approach that encourages patient trust and autonomy and is characterized by empathy, respect, and compassion
- Optimize the physical environment for patient comfort, dignity, privacy, engagement, and safety
- Recognize when the values, biases, or perspectives of patients, physicians, or other health care professionals may have an impact on the quality of care, and modify the approach to the patient accordingly
- Respond to patients' non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients
- Manage disagreements and emotionally charged conversations
- Use patient-centered interviewing skills to effectively gather relevant biomedical and psychosocial information
- Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information
- Provide a clear structure for and manage the flow of an entire patient encounter
- Seek and synthesize relevant information from other sources, including the patient's family, with the patient's consent.
- Skillfully share information and explanations that are clear, accurate, timely, and adapted to patient's and his or her family's level of understanding and need in a simulated setting

As a Collaborator

- Establish and maintain healthy relationships with physicians and other colleagues in the health care professions to support relationship-centered collaborative primary ENT care
- Negotiate overlapping and shared responsibilities with physicians and other colleagues in the Otolaryngology care professions
- Engage in respectful shared decision-making with other colleagues, patients and their families
- Recognize one's own professional roles and limits in otolaryngology care
- Share knowledge and information to other colleagues
- Show respect toward other physicians and members of the Otolaryngology care team
- Respect and value diversity among other colleagues
- Implement strategies to resolve conflicts in a manner that supports a collaborative culture
- Assess when care should be transferred to another physician or health care professional
- Demonstrate safe transfer of care, using both verbal and written communication, during a patient transition to a different health care professional, setting, or stage of care

As a Manager

- Contribute to the improvement of Otolaryngology health care delivery in teams, organizations, and systems
- Apply the science of quality improvement to contribute to improving systems of ENT care
- Analyze adverse events and near misses to enhance systems of care
- Use health informatics to improve the quality of patient ENT care and optimize patient safety
- Engage in the stewardship of health care resources
- Allocate Otolaryngology care resources considering justice, efficiency, and effectiveness for optimal patient care

- Apply evidence and management processes to achieve cost-appropriate care
- Develop their leadership skills
- Design and organize elements of health care delivery
- Facilitate change in health care to enhance services and outcomes
- Set priorities and manage time to balance practice and personal life
- Manage career planning, finances, and health human resources in a practice
- Implement processes to ensure personal practice improvement
- Promote occupational health for sustainable practice

As a Health Advocate

- Work with patients having Ear, nose and oral throat diseases and their families to address the determinants of health that affect them and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors
- Incorporate ear, nose and throat diseases prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of ear nose and throat conditions that affect them
- Participate in a process to improve ear, nose and throat health in the community or population they serve
- Incorporate prevention of hearing impairment and deafness, ear and hearing care promotion, and hearing impairment surveillance into interactions with individual patients

AS Scholar:

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize knowledge gaps in biomedical, clinical and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care

- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Integrate other intrinsic roles into their learning and practice of medicine
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in practice
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

COURSE CONTENTS:

Topic	Objectives	Methods	Resources
LECTURES: FIRST WEEK			
Clinical applied anatomy and physiology of the ear	Describe the basic anatomy Explain the physiology of the ear	Interactive presentation	
History taking and physical exam in ENT	Recognize the comprehensive ENT history taking and relevant systemic review Explain the components of the basic ENT examination	Interactive presentation, simulated practice, and clinical practicum	
Diagnosis and management of diseases of external ear	Explain the symptoms and signs of pathologies of the external ear(inflammatory and non-inflammatory) Plan management and referral of patients with pathologies of external ear	Interactive presentation, and clinical practicum	
Diagnosis and management of diseases of middle ear	Explain the symptoms and signs of pathologies of the middle ear(inflammatory and non-inflammatory)	Interactive presentation, case study	

	Plan management and referral of patients with pathologies of middle ear		
Diagnosis and management of diseases of inner ear	Explain the symptoms and signs of pathologies of the inner ear (inflammatory and non-inflammatory) Plan management and referral of patients with pathologies of inner ear	Interactive presentation, and clinical practicum	
Emergencies in Otolaryngology	Identify and prioritize common ENT emergencies Plan the management of common ENT emergencies	Seminar PBL	
Operating theater teaching	Pre operative assessment. Indication and contraindications of common ENT surgical procedures Anticipated complications of common ENT procedures	Interactive presentation, observation	
Skills lab demonstration	Explain indications of common ENT procedures Practice common ENT procedures on mannequins	Interactive presentation, demonstration	
SECOND WEEK			
Clinical anatomy and physiology of the nose and paranasal sinuses	Describe the basic anatomy Explain the physiology of the nose and paranasal sinuses	Interactive presentation, clinical simulation	
Diagnosis and management of diseases of nasal pathologies	Explain the symptoms and signs of pathologies of the nose (inflammatory and non-inflammatory) Plan management and referral of patients with pathologies of the nose	Interactive presentation, case study,	
Diagnosis and management of diseases of paranasal sinus pathologies	Explain the symptoms and signs of pathologies of the paranasal sinuses (inflammatory and non-inflammatory) Plan management and referral of	Interactive presentation, case study	

	patients with pathologies of the paranasal sinuses		
Operating theater teaching	Pre operative assessment. Indication and contraindications of common ENT surgical procedures Anticipated complications of common ENT procedures	Interactive presentation, observation	
Skills lab demonstration	Explain indications of common ENT procedures Practice common ENT procedures on mannequins	Interactive presentation, demonstration	
Clinical anatomy and physiology of pharynx, larynx ,salivary glands ,neck and facial nerve	Describe the basic anatomy Explain the physiology of the pharynx ,larynx, salivary glands, neck and facial nerve	Interactive presentation, clinical simulation	
THIRD WEEK			
Diagnosis and management of Common infectious and inflammatory pathologies of oral cavity, oropharynx and salivary glands	Explain the symptoms and signs of pathologies of the oral cavity, oropharynx and salivary glands (inflammatory and non-inflammatory) Plan management and referral of patients with pathologies of the oral cavity, oropharynx and salivary glands	Interactive presentation, PBL, case presentation	
Diagnosis and management of Common benign and malignant pathologies of oral cavity ,oropharynx and salivary glands	Explain the symptoms and signs of benign and malignant pathologies of the oral cavity, oropharynx and salivary glands Plan management and referral of patients with benign and malignant pathologies of the oral cavity, oropharynx and salivary glands	Interactive presentation, PBL, case presentation	
Common causes of obstructive sleep apnea syndrome	Identify the common causes of sleep disturbance Diagnose obstructive sleep breathing Identify surgically correctable causes of sleep apnea Indications and contra indications of sleep apnea surgery	Interactive presentation, PBL, case presentation	
Diagnosis and	Explain common causes of facial	Interactive	

management of facial nerve pathologies	palsy Explain clinical manifestations of facial palsy Diagnostic and management approaches of facial palsy	presentation, PBL, case presentation	
Skills lab demonstration	Explain indications of common ENT procedures Practice common ENT procedures on mannequins	Interactive presentation, demonstration	
Operating theater teaching	Pre operative assessment. Indication and contraindications of common ENT surgical procedures Anticipated complications of common ENT procedures	Interactive presentation, observation	

Teaching and learning methods

- Interactive lecture
- Seminar Presentation
- Video show
- Demonstration
- Role play
- Case study
- Simulated (skills lab) practice
- OPD teaching
- Personal Research and Reflection exercise /Portfolio

Methods of assessment

Formative

- Direct observation of skills
- Logbook
- Global rating of performance

Summative

- Continuous progressive assessment (Quiz, seminar presentation, Test, attendance) (40%)
- Written exam (40 %)
- OSCE (20%)

References

- ✓ Scott Brown's otolaryngology & head and surgery, 8TH Edition.
- ✓ Otolaryngology, Basic science & clinical review, 2006 Edition.
- ✓ Cummings Otolaryngology and head & neck surgery, 5th Edition.
- ✓ Rudolf Probst, A step-By-Step-Learning Guide, 2006 Edition
- ✓ Current Diagnosis and treatment Otolaryngology—Head and neck Surgery, 2012

MODULE SCHEDULE

Date	Learning Activity	Required Reading/ Assignment
Week One Day 1	Interactive lecture (2hrs) • Introduction to the module • Anatomy and physiology of the ear	⁵⁰ Reflective portfolio
	Clinical skills lab (2hrs) Simulated practice: Anatomic models of the EAR, videos, nose and sinuses	
	Operation theatre (major and minor) (1hr)	
	Interactive presentation (2hr) ENT history taking and physical examination	
Day 2	Interactive lecture (2hrs) Diagnosis and management of external and middle ear diseases	Reflective portfolio
	Clinical practicum (2 hrs) Student placement at ENT outpatient and inpatient departments	
	Operation theatre (major and minor) (1hr)	
	Clinical skills lab (2 hr) Simulated practice: Minor otology procedure (ear irrigation , foreign body removal, audiometry , caloric test	
Day 3	Interactive lecture (2 hrs) Diagnosis and management of inner ear diseases	Reflective portfolio
	Clinical practicum (2 hrs) Student placement at ENT outpatient and inpatient departments	
	Operation theatre (major and minor) (1hr)	
	Interactive presentation, Case based discussion (2hrs) Ear and hearing care	
Day 4	Interactive lecture, case based discussion (2 hrs) Diagnosis and management of nasal and paranasal sinus pathologies	Reflective portfolio
	Clinical practicum (2 hrs) Student placement at ENT outpatient and inpatient departments and follow up clinics	
	Operation theatre (Major and minor) (1hr)	
	Case based discussion I(2 hr) Otologic emergencies	
Day 5		Reflective portfolio
	Operation theatre (Major and minor) (1hr)	

	Whole group session (3 hrs) Faculty facilitated discussion and reflection on the week's learning activities	
	Clinical practicum (2 hrs) Student placement at ENT outpatient and inpatient departments	
	Operation theatre (Major and minor OR) (1 hr)	
	Clinical based discussion (2 hrs) Rhinology emergency (epistaxia , nasal bone fracture	
Week two Day 1	Interactive Lecture (2 hrs) Clinical anatomy and physiology of pharynx, larynx , salivary glands ,neck and facial nerve	Reflective portfolio
	Clinical practicum (2 hrs) Student placement at ENT outpatient and inpatient departments	
	Operation theatre (Major and minor) (1 hr)	
	Seminar(2 hrs) • Diagnosis and management of Common infectious and inflammatory pathologies of oral cavity, oropharynx and salivary glands • Diagnostic approach to common neck masses	
Day 2	Interactive lecture (2 hrs) Diagnosis and management of common benign and malignant pathologies of oral cavity , oropharynx and salivary gland	Reflective portfolio
	Clinical practicum (2 hrs) Student placement at ENT outpatient and inpatient departments	
	Operation theatre (Major and minor) (1 hr)	
	PBL (2 hrs) second session SYSTEMIC MANIFESTATION OF ENT DISEASES	
Day 3	Interactive lecture (2 hrs) Common causes of obstructive sleep apnea syndrome	Reflective portfolio
	Clinical practicum (2 hrs) Student placement at ENT outpatient and inpatient departments and follow up clinics	
	Operation theatre (Major and minor) (1 hr) CLINICAL BASED DISCUSSION Head and neck emergency	

Day 4	Interactive Lecture (2 hrs) Diagnosis and management of facial nerve pathologies ENT radiology	
	Clinical practicum (2 hrs) Student placement at ENT outpatient and inpatient departments	
	Operation theatre (Major and minor OR) (1 hr)	
	Clinical skills lab (2 hrs) Simulated practice: Perform minor procedures, FNA neck mass, cricothyroidectomy , tracheostomy ,peritonsillar abscess etc..	
Day 5	Written Exam	
	OSCE	

RESOURCES

Human resources

- Otolaryngologists and Head and Neck surgeon
- Residents
- General Practitioner
- General and ENT nurses
- Audiologist/Audiology technician
- Speech pathologist/therapist
- Education Experts

Training, Teaching and Learning Materials

- Trainer's Guide

Reference manuals and books

- Learners Guide
 - Handout
 - Books
 - Text books
 - E-books

Learning Facilities / Infrastructure

- Class room with the right classroom environment.
- Skill laboratory with mannequins, models
- Equipped Multipurpose Integrated laboratory
- Library
- IT center with good internet connection
- Clinical set up (Primary and secondary where possible tertiary care health institutions)

Portable and stationary plants, machines and equipment

- Flip charts
- Video
- CD-ROM
- Computers
- DVD Player
- LCD projector Overhead Projector
- White board (2x1.5)

MODULE TITLE: DENTISTRY AND ORAL HEALTH

MODULE CODE:

NOMINAL DURATION: Clerkship I : 2 weeks

Module Description

This module is intended to equip the student's with profound knowledge, skill and attitude required in the diagnosis and management of diverse dental and oral diseases. Students will also develop relevant competencies with the Medical expertise, Scholar, Collaborator, Communicator, Manager, Health Advocate, and Professional.

Module DESIGN:

This will be delivered in Clerkship I:

The teaching will focus on building the ability of the student in history taking, physical examination in the clinical setting, formulating differential diagnosis, reaching to appropriate diagnosis, listing relevant investigation and is intended to enable the student for an increased measure of responsibility and prepares students for Internship. This provides the students to have the experience of being the primary caregiver for patients in a well-supervised setting.

Education in the Dentistry and Oral Health clerkship will be delivered in different setting of health care and in collaboration across disciplines. The students will have outpatient and inpatient experiences: Outpatient experience will be in the respective department of the hospital and also in the primary care (Health centers) which will be coordinated with their specific department of attachment.

Module Prerequisite: - Completed Pre -clerkship courses successfully.

Module Objective

By the end of this module, students will be able to assess patients with dental disorders and assist with their management including health promotion and disease prevention.

Supportive Objectives

As Medical expert

- Apply knowledge of the normal structures and functions of oral cavity in assessing and managing dental and oral facial condition.
- Apply principles of physiologic mechanisms and processes in assessing and managing dental, periodontal and oral cavity problems.
- Identify the various causes (developmental, infectious, neoplastic, and traumatic) of disease in assessing and managing dental, periodontal and oral cavity problems.
- Analyze important determinants and risk factors of common dental, periodontal and oral cavity diseases in assessing and managing.
- Apply knowledge of principles of drug used on the management periodontal and oral cavity diseases.
- Take history from a patient with a dental, periodontal and oral cavity diseases within the practical context of the individual patient-physician relationship
- Perform physical examination of a patient with a dental, periodontal and oral cavity diseases within the practical context of the individual patient-physician relationship
- Identify abnormal physical findings in a patient with a dental, periodontal and oral cavity diseases

- Make an initial assessment of a patient presenting with signs and symptoms of a dental, periodontal and oral cavity diseases.
- Formulate a plan of investigation for a patient with dental, periodontal and oral cavity diseases in partnership with the patient, obtaining informed consent as an essential part of this process.
- Perform certain clinical procedures like tooth extraction and fixing temporo- mandibular joint dislocation.
- Interpret the results of commonly used investigations in dental, periodontal and oral cavity diseases.
- Formulate a likely diagnosis in a patient with a dental, periodontal and oral cavity disease.
- Demonstrate clinical judgments and decisions, based on the available evidence in diagnosing and managing dental, periodontal and oral cavity diseases.

As a communicator

- Demonstrate communication with a patient with a dental, periodontal and oral cavity diseases using a patient-centered approach that encourages patient trust and autonomy and is characterized by empathy, respect, and compassion
- Optimize the physical environment for patient comfort, dignity, privacy, engagement, and safety
- Recognize when the values, biases, or perspectives of patients, physicians, or other health care professionals may have an impact on the quality of care, and modify the approach to the patient accordingly
- Respond to patients' non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients
- Manage disagreements and emotionally charged conversations
- Use patient-centered interviewing skills to effectively gather relevant biomedical and psychosocial information
- Integrate and synthesize information about the patient's beliefs, values, preferences, context and expectations with biomedical and psychosocial information
- Provide a clear structure for and manage the flow of an entire patient encounter
- Seek and synthesize relevant information from other sources, including the patient's family, with the patient's consent.
- Skillfully share information and explanations that are clear, accurate, timely, and adapted to patient's and his or her family's level of understanding and need in a simulated setting

As a Collaborator

- Establish and maintain healthy relationships with physicians and other colleagues in the health care professions to support relationship-centered collaborative care
- Recognize one's own professional roles and limits
- Share knowledge and information to other colleagues
- Show respect toward other physicians and members of the health care team
- Respect and value diversity among other colleagues

As a Manager

- Develop their leadership skills
- Design and organize elements of health care delivery
- Set priorities and manage time to balance practice and personal life
- Promote occupational health as physicians for sustainable practice

As a Health Advocate

- Work with patients having dental, periodontal and oral cavity diseases and their families to address the determinants of health that affect them and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors
- Incorporate dental, periodontal and oral cavity diseases prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of oral health that affect them
- Participate in a process to improve oral health in the community or population they serve

Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize knowledge gaps in biomedical, clinical and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Integrate other intrinsic roles into their learning and practice of medicine
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in practice

- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Schedule	Learning Activity	Required Reading/Assignment
Week One Day 1	Interactive Lecture (2 hrs) • Introduction to the module • Review of the anatomy and physiology of the oral cavity, face and jaws	Reflective portfolio
	Clinical practicum/case discussion (3 hrs) • Student placement to dental clinic.	
	Seminar : local Anesthetic consideration – (advantages, limitations, methods) (2 hrs)	
Day 2	Interactive Lecture (2 hrs) • Patient examination and evaluation • Nomenclature of teeth	Reflective portfolio
	. Clinical skills lab (2 hrs) • Skill lab: video showing Dental patient examinations • Study atlas of oral cavity, face and jaw	
	Clinical practicum (3 hrs) • Student placement to the dental clinic	
Day 3	Interactive Lecture (2 hrs) • Disease of the hard tissues of teeth	Reflective portfolio
	Clinical practicum (3 hrs) Student placement to dental clinic.	
	Seminar : Common developmental problems of oral cavity(2 hrs)	
Day 4	Interactive Lecture (2hrs) • Disease of the dental pulp • Principles of extraction	Reflective portfolio
	Clinical skill lab (2hrs) • Practice evaluation of a patient with oral and dental problem • Skill lab: video – showing dental pulp diseases and extraction principle	
	Clinical practicum (3 hrs) • Student placement to dental clinic.	
Day 5	Clinical practicum (4hrs)	

	Student placement to dental clinic	
	Whole group discussion (3 hrs) Faculty facilitates discussion and reflection on the week's learning activities	
Week Two	Day 1	
	Interactive Lecture (2hrs) Gingivitis and periodontal disease	
	Clinical skill lab (2 hrs) Video show on Periodontal Diseases	
	Clinical practicum (3hrs) Student placement to dental clinic. Case discussion on topic of the week	
Day 2	Interactive Lecture (2 hrs) Trauma of orofacial region TMJ /mandible and maxilla, teeth and soft tissue	Reflective portfolio
	Clinical skill lab (2 hrs) Video on the trauma of orofacial region, teeth and soft tissue	
	Clinical practicum (3 hrs) Student placement to dental clinic/ emergency OPD.	
Day 3	Interactive lecture (2hrs) Oro-facial and neck infections	Reflective portfolio
	Clinical practicum (3 hrs) Student placement to the dental clinic. Case discussion on the topics discussed over the weeks since starting the course	
	Seminar : Common Neoplastic disorder of Dental and oral cavity (2hrs)	
Day 4	Interactive Lecture(2hrs) Oral health	Reflective portfolio
	Clinical skill lab (2hrs) • Skill lab: video – showing oral health care • Practice counseling individuals on keeping oral health	
	Clinical practicum (3 hrs) • Student placement to dental clinic	
Day 5	Written Exam	
	OSCE	

Teaching and learning methods

- Interactive lecture
- Seminar Presentation
- Video show
- Demonstration
- Role play
- Case study
- Simulated (skills lab) practice
- OPD teaching

- Community visit
- Personal Research and Reflection exercise /Portfolio

Methods of assessment

Formative

- Direct observation of skills
- Logbook
- Global rating of performance
- Personal research and reflection exercise/Portfolio

Summative

- Continuous progressive assessment (Quiz, seminar presentation, Test) (40%)
- Written exam (40 %)
- OSCE (20%)

References

- Brad W. Neville, Douglas D. Damm, Jerry E. Bouquot, and Carl M. Allen. Oral and maxillofacial Pathology, 3rd revised edition, 200
- Edward W. Odell, Clinical problem solving in Dentistry, 3rd revised edition 2010
- David Wray, David Stenhouse, David Lee, Andrew J. E. Clark. Textbook of General and Oral Surgery, 1st Edition 2003
- Barry K. Berkovitz, G. R. Holland, Bernard J. Moxham. Oral Anatomy, Histology and Embryology, fourth edition. 2009 Esther M. Wilkins. Clinical Practice of the Dental Hygiene

SOCIAL AND POPULATION HEALTH MODULES

MODULE TITLE: SPH I: DETERMINANTS OF HEALTH

Module Code: SPH-401/1

Duration: 16 weeks

Module Description: This module is designed to equip learners with the knowledge, skills and attitude needed to analyze determinants of health with full participation of the community and other stakeholders. Students will also develop relevant competencies with the Medical expertise, Scholar, Collaborator, Communicator, Manager, Health Advocate and Professional. This module is given on the first semester during the first year of the pre-clerkship period of the undergraduate medical education program.

Module Design

Prerequisite: None

Module Objective

At the end of this module, the medical student will be able to analyze broad determinants of health at individual, family and community levels and mobilize them for action.

Supporting Objectives

As a Medical Expert

- Apply important determinants of health and disease for protecting and promoting individual and community health
- Apply environmental and ecological determinants of health and disease at individual, family and community levels

- Identify appropriate actions to prevent and control determinants of health and disease among individuals and families
- Apply evidence based community health practice for the improvements of health and health related activities using a systematic methodology
- Demonstrate the application of information technology to manage and access online medical information, and support one's own education
- Demonstrate how to search, collect, organize and interpret health and health-related information from different sources
- Apply the use of information and communication technology to assist in health promotion and disease prevention measures at individual, family and community levels

As a communicator

- Demonstrate clear and effective communication skills for interacting with individuals, families, communities, PHCU staff, Woredas health department staff, Academic staff and peers
- Recognize when the values, biases, or perspectives of communities, patients, physicians, or other health care professionals may have an impact on the quality of care, and modify the approach to the patient and community accordingly
- Respond to community's non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients

As a Collaborator

- Demonstrate professional values and behavior in interaction with individuals, families and communities consistent with the future role of a physician
- Demonstrate key public health values, attitudes and behaviors such as commitment to equity and social justice, recognition of the importance of the health of the community as well as the individual, and respect for diversity, self-determination, empowerment, and community participation
- Show respect for colleagues and other healthcare professionals and the ability to foster a positive collaborative relationship with them

As a Manager

- Develop their leadership skills
- Design and organize resources of health care delivery
- Set priorities and manage time to balance practice and personal life
- Promote occupational health as physicians for sustainable practice

As a Health Advocate

- Work with patients and their families to address the determinants of health that affect them and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors
- Incorporate determinants of health disease prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of health that affect them
- Participate in a process to improve determinants of health in the community or population they serve

Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance

- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize knowledge gaps in public health and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Integrate other intrinsic roles into their learning and practice of public health
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in practice
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Module Schedule

Weeks	Topic	Time
Week 1-2	1. Introduction to public health (4hrs) ○ Health and disease: concepts, definitions and perspectives ○ Public health: definition, philosophy, history,	○ 4 hours of classroom activities

	development, core functions and services ○ Public health sciences, their scope and use in medicine	○ 4 hours of community-based learning
Week 3-4	2. Human health and socio-economic factors (Economic factors include, but are not limited to, unemployment, poverty, income inequality, neighborhood deprivation, assets, economic growth, globalization, healthcare cost) (4hrs)	○ 4 hours of classroom activities ○ 2 hours of community-based learning
Week 5-15	3. Ecology and Environmental health (22 hrs) ○ Introduction to Human ecology and environmental health (2hrs) ○ Human – environment interaction (1hr) ○ Ecology of health and disease (4hrs) ○ Biosphere and pollution (2hrs) ○ Components of environmental health (Housing and institutional sanitation, water sanitation, waste disposal, vector control, food sanitation) (6hrs) ○ Occupational health and safety (4hrs) ○ Infection prevention (3hrs)	○ 22 hours of classroom activities ○ 14 hours of community-based learning
Week 16	Final Module Exam (2hrs)	

Teaching-Learning Methods

- Interactive lecture and discussion
- Small group learning activities: assignment, exercise, case study etc
- Individual reading
- PHCU/Community-based learning and study trip: home visit, discussion with individuals and families to identify and solve problems, observation, PHCU visit, Zonal and District Health Department Visit, field visit, and targeted literature review based on community experience
- Student presentation
- Personal research and reflection exercise (PRRE)
- Reflective portfolio and mentoring

Teaching-Learning Materials

- Lecture power point presentations
- White board, black board, chalk and marker
- LCD projector and laptop computer
- Flip charts, TV and video sets
- Stationery for handouts
- Reference books
- Journals
- CDs
- Logbooks for entry of community experience

Assessment Methods

Formative assessment

- ❖ Exercise and assignment
- ❖ Logbook and portfolio

- ❖ 360 degree evaluation
- ❖ Student presentation
- ❖ Global rating of community experience midway during the module

Summative assessment

- Personal research and reflection exercise (PRRE)⁵¹ (15 %)
- Reflective portfolio (5%)
- Quiz/ Test (10%)
- Global rating of community experience(15 %)
- Assignment and student presentation (15 %)
- Written final module exam (40 %)

References

1. YemaneBerhane, DamenHailemariam and HelmuKloos. Epidemiology and ecology of Health and Disease in Ethiopia. 2006
2. EPHTI. Ecology. Lecture note series for health science students. 2007
3. White, P. **Biopsychosocial medicine: An integrated approach to understanding illness.** 2005 Oxford University Press.
4. [Bernice A. Pescosolido](#), [Jack K. Martin](#), [Jane D. McLeod](#), [Anne Rogers](#) (Editors). Handbook of the Sociology of Health, Illness, and Healing. A Blueprint for the 21st Century. 2011
5. Sobo, E. J. &Loustaunau, M. **Cultural context of health, illness, and medicine** (2nded.) 2010. Greenwood
6. WHO. Closing the gap in a generation: health equitythrough action on the social determinants of health: final report of the commission on social determinants of health. 2008.
7. Robert H Friis. Essentials of environmental health (2nd edition). The essential public health series. 2012.
8. Kathryn Hilgenkamp. Environmental Health: Ecological Perspectives. 2006
9. Herman Koren and Michael Bisesi.Handbook of environmental health. 2002

MODULE TITLE: SPH II: MEASURMENT OF HEALTH AND DISEASE

Module Code: SPH-401/2

Duration: 26 weeks

Module Description: This module is designed to equip learners with the knowledge, skills and attitude needed to measure disease and other health conditions in the community for public health action. Students will also develop relevant competencies with the Medical expertise, Scholar, Collaborator, Communicator, Manager, Health Advocate and Professional. This module is given in second semester during the first year of the pre-clerkship period of the undergraduate medical education program.

Module Design

Prerequisite: SPH I(Determinants of health and Diseases)

⁵¹ Students are required to maintain in their portfolio learning experiences, learning outcomes achieved, evidence of learning and reflections longitudinally and their mentors would provide feedback occasionally. Clinical instructors should also ask students to submit reflective portfolio twice during the attachment for formative assessment and feedback.

Module Objective

At the end of this module, learners will be able to apply public health methods for the measurement of health and disease at individual and community levels.

Supporting Objectives

As a Medical Expert

- Apply the notion of health from scientific and layman perspective
- Apply the history, evolution and functions of public health and its relevance to the practice of medicine
- Apply epidemiological approach to disease causation with emphasis on infectious diseases
- Apply levels of prevention regarding avoidance and control at different levels
- Discuss types of epidemiologic studies including their use and limitations
- Demonstrate measures of morbidity and mortality including from existing data sources
- Apply different methods of data collection in the community
- Apply basic biostatistics concepts, tools and methods
- Discuss criteria for establishing and evaluating screening programs and factors that affect validity and reliability of screening tests
- Apply the processes, uses, and evaluation of public health surveillance
- Apply the steps of an outbreak investigation and management
- Discuss epidemiology of diseases of public health significance in Ethiopia and Globally
- Contribute health promotion and disease prevention methods for major public health problems
- Analyze community practice experience and perform practice-based improvement activities using a systematic methodology
- Demonstrate how to search, collect, organize and interpret health and health-related information from different sources
- Demonstrate the use information and communication technology to assist in health promotion and disease prevention measures for individuals and families
- Demonstrate the use of information technology to manage and access online medical information, and support one's own education

As a communicator

- Demonstrate clear and effective communication skills in interactions with individuals, families, communities, PHCU staff, local health department staff, peers and faculty
- Recognize when the values, biases, or perspectives of communities, patients, physicians, or other health care professionals may have an impact on the quality of care, and modify the approach to the patient and community accordingly
- Respond to community's non-verbal communication and use appropriate non-verbal behaviors to enhance communication with patients
- Manage disagreements and emotionally charged conversations

As a Collaborator

- Demonstrate professional values and behavior in interaction with individuals, families and communities consistent with the future role of a physician
- Demonstrate key public health values, attitudes and behaviors such as commitment to equity and social justice, recognition of the importance of the health of the community as well as the individual, and respect for diversity, self-determination, empowerment, and community participation
- Show respect for colleagues and other healthcare professionals and the ability to foster a positive collaborative relationship with them

- Demonstrate a habit of self-reflection, responsiveness to feedback and an on-going development of new skills, knowledge and attitude

As a Manager

- Develop their leadership skills
- Design and organize elements of health care delivery
- Set priorities and manage time to balance practice and personal life
- Promote occupational health as physicians for sustainable practice

As a Health Advocate

- Work with patients and their families to address the determinants of health that affect them and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors
- Incorporate determinants of health disease prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of determinants of health that affect them
- Participate in a process to improve determinants of health in the community or population they serve

Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize knowledge gaps in public health and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Integrate other intrinsic roles into their learning and practice of public health
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in practice
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Module Schedule

Week	Contents	Time
Week 1-12	Biostatistics (24hrs) 1. Introduction to Biostatics (1 hr) • Basic definitions and concept of biostatistics • Purpose of statistics and its use in medical and public health sciences 2. Descriptive statistics (8hrs) • Data and types of data • Variable and types of variables (Qualitative and quantitative) • Discrete and continuous variables • Measurement and measurement scales • Methods of data organization and presentation • Frequency distribution and tables • Tables for categorical and numerical frequency distributions • Diagrammatic presentation of data • Bar chart, pie chart, histogram, frequency polygon, line graph • Cumulative frequency polygon, box plot and scatter plot • Measures of central tendency (arithmetic mean, median, mode) • Measures of variability (range, variance, standard deviation, coefficient of variation) • Percentiles, quartiles and inter-quartile range • Characteristics of a distribution (skew ness)	▪ 24 hours of classroom activities ▪ 16 hours of community-based learning

	3. Demography and vital statistics (4hrs) • Definition and uses of demography • Sources of demographic data • Census and survey • Use of vital statistics • Population growth rate • Life expectancy at birth • Population pyramid • Some demographic and health indicators for Ethiopia 4. Probability and probability distributions (3hrs) • Concept of probability • Objective and subjective probabilities • Classical definition of probability • Independent and mutually exclusive events • Properties of probability and probability rules • Conditional probability • Probability distributions • Binomial and normal distributions • Standard normal distribution 5. Sampling techniques (2 hrs) • Sampling definition and purpose of sampling • Types of sampling techniques • Non-probability sampling methods • Probability sampling methods • Errors in sampling 6. Statistical inferences (3hrs) • Point and interval estimation • Hypothesis testing 7. Measures of Association(1hr) 8. Sample size determination (1hr) 9. Interpreting and communicating statistical findings (1hr)	
Week 13-15	2. Introduction to principle of epidemiology (6 hours) ➤ The subject matter of epidemiology (Definition, history, type, scope and purpose) (1hr) ➤ Epidemiological concepts of disease causation (5 hours) ○ Concepts of disease causation ○ Epidemiological models in disease causation (epidemiological triangle, web of causation, wheel model) ○ Factors in causation ○ Time, Place and Person concept in disease causation	▪ 6 hours of classroom activities ▪ 6 hours of community-based learning including student presentations

	○ Establishing causation ○ Natural history of diseases (communicable and non-communicable) ○ Levels of disease prevention	
Week 16-19	3. Types of epidemiologic study designs, their use and limitations (8hrs) ➤ Types of epidemiologic study designs (3 hrs) • Observational • Experimental ➤ Measuring disease frequency (incidence, prevalence)(1hr) ➤ Using available information to measure health and disease (health information system) (1hr) ➤ Death rates, morbidity and disability measures (1hr) ➤ Comparing disease occurrence (absolute and relative comparisons, standardization) (1hr) ➤ Epidemiology of diseases of public health significance in Ethiopia (1hr)	▪ 8 hours of classroom activities ▪ 10 hours of community-based learning including student presentations
Week 20-21	4. Public health surveillance (4hrs) ➤ Principles of public health surveillance ➤ Integrated disease surveillance and response ➤ Timely warning and intervention	▪ 4 hours of classroom activities ▪ 4 hours of community-based learning
Week 22-23	5. Outbreak investigation and management (4hrs) • Patterns of occurrence of diseases • Disease outbreaks • Steps of investigation of an outbreak • Management and control of an outbreak or epidemic	▪ 4 hours of classroom activities ▪ 4 hours of community based learning
Week 24-25	6. Screening Program and Evaluation (4hrs) ➤ Definition ➤ Diseases appropriate for screening ➤ Criteria for establishing screening programs ➤ Screening tests • Concepts of validity and reliability • Sensitivity and Specificity • Factors affecting validity and reliability of screening tests ➤ Evaluation of screening	▪ 4 hours of classroom activities ▪ 4 hours of community-based learning including student presentations
Week 26	Final Module Exam (2hrs)	

Teaching-Learning Methods

- Interactive lecture and discussion
- Small group learning activities: assignment, exercise, case study etc
- Individual reading

- PHCU/Community-based learning and study trip: home visit, discussion with individuals and families to identify and solve problems, observation, PHCU visit, Zonal and District Health Department Visit, field visit, and targeted literature review based on community experience
- Student presentation
- Personal research and reflection exercise (PRRE)
- Reflective portfolio and mentoring

Teaching-Learning Materials

- Lecture power point presentations
- White board, black board, chalk and marker
- LCD projector and laptop computer
- Flip charts, TV and video sets
- Stationery for handouts
- Reference books
- Journals
- CDs
- Logbooks for entry of community experience

Assessment Methods

Formative assessment

- ❖ Exercise and assignment
- ❖ Logbook and portfolio
- ❖ 360 degree evaluation
- ❖ Student presentation
- ❖ Global rating of community experience midway during the module

Summative assessment

- Personal research and reflection exercise (PRRE)⁵² (15 %)
- Reflective portfolio (5 %)
- Quiz/ Test (10%)
- Global rating of community experience (15 %)
- Assignment and student presentation (15 %)
- Written final module exam (40 %)

References

1. Fletcher. Principles of Epidemiology
2. Charles H Hennekens and Julie E Buring. Epidemiology in Medicine
3. Rothman, Kenneth J.; Greenland, Sander; Lash, Timothy L. Modern epidemiology. 3rd edition. 2008
4. David G. Kleinbaum, Kevin M. Sullivan. A pocket guide to epidemiology. 2007
5. YemaneBerhane, DamenHailemariam and Helmut Kloos. Epidemiology and ecology of health and disease in Ethiopia. 2006
6. Daniel. Biostatistics: a foundation for analysis in health sciences.
7. Pagano. Principles of Biostatistics
8. Colton. Statistics in Medicine
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MODULE TITLE: COMMUNITY BASED TRAINING PROGRAM (CBTP)

Module Code: **Varies according to the phases /4 phases /**

Module Code: SPH- 402/3-SPH-402/6

Prerequisites:

For Phase I or CBTP 1: SPH I –II

Phase II or CBTP 2: SPH I –III

Phase III or CBTP 3: SPH I –IV

Phase V or CBTP 4: SPH I –V

Attachment Duration: 4 weeks in each phase

Introduction to Community Based Training Program Attachment: As higher learning institutions host various intellectual disciplines and undertake many researches in different aspects of social life, they pose all the capacity to bring a significant impact on the development of the community.

Community based Education (CBE) is a means of achieving educational relevance to community needs and consequently of implementing a community-oriented education program .It consists of learning activity which uses the community extensively as a learning environment and involves students, teachers, community and all other stakeholders in the community throughout the educational experience. It follows a problem solving approach that starts with the identification of community problem, setting of objectives and plan of action to implementation and monitoring and evaluating progress. The program is of obvious benefit to both the students and the community. The activity of the students in the community is neither peripheral nor casual experience. It is integral part of the whole educational process which is evidenced by the total weight given to the program in the form of credit hours (20% of the curriculum) and as requirement for graduation.

The community-based education (CBE) deploys three basic approaches, as strategies; to address the community needs by producing socially accountable citizens. These are: Community Based Training Program (CBTP), Team Training Program (TTP) and Students Research Program (SRP).

Community Based Training Program (CBTP) is an integrated program which runs in phases from first year to graduation at the end of each academic year along with in built regular follow up program. The student learning activities in this program is to be phased, spiral in nature and problem solving in approach. Each phase has a well-planned and defined educational objectives and goals. The students undergoes through well-defined performance phases consisting of data collection, community diagnosis, planning, implementation and evaluation. These phases should, however, be viewed as continuum of activities in which the student is supposed to spend some time in each phase depending on his competence and the needs of the community concerned.

Attachment Description: Through the process of this training, students acquire the ability to plan, implement and evaluate health program. In addition students develop analytical thinking in interpretation of data, report writing and presentation and also the aptitude to work as a team. Thus this program benefits both the students and the community and contributes to the community development which is the final goal of education.

The CBTP attachment is intended to enable medical students to develop competencies in assessing and analyzing community health problems. Students will also develop relevant competencies with the Medical expertise, Scholar, Collaborator, Communicator, Manager, Health Advocate and Professional. The CBTP module is basically divided in to three phases. Accordingly the students are expected to go for CBTP attachment at the end of each academic year for about four weeks' period during the undergraduate medical education program.

General Purpose of CBTP attachments:

- Define the demographic, socio-economic, political and environmental aspects of a given community (determinants of health status).
- Make community diagnosis and draw health action plan which would enable students to suggest appropriate intervention measures according to the different levels of prevention (primary, secondary and tertiary)
- Organize health intervention programs utilizing the concept, components and strategies of primary health care (PHC).
- Plan and conduct problem oriented and community based research.

Module Design

Attachment Objectives

At the end of this module, the medical student will be able to conduct community diagnosis, identify priority community health problems and their social determinants to initiate action for change.

Supporting Objectives

As a Medical Expert

- Conduct demographic and health survey to identify common causes of mortality and morbidity and their determinants in the target community
- Analyze determinants of major health problems
- Prioritize community health problems for intervention planning
- Analyze community practice experience and perform practice-based improvement activities using a systematic methodology
- Demonstrate a habit of self-reflection, responsiveness to feedback and an on-going development of new skills, knowledge and attitude
- Apply search, collect, organize and interpret health and health-related information from different sources

As a communicator

- Communicate effectively results of the assessment to initiate action by communities and local stakeholders in the spirit of self-reliance
- Demonstrate clear, sensitive and effective communication skills in interacting with individuals, families, communities, PHCU staff, local health office, peers and faculty

As a Collaborator

- Demonstrate professional values and behavior in interaction with individuals, families and communities consistent with the future role of a physician
- Demonstrate key public health values, attitudes and behaviors such as commitment to equity and social justice, recognition of the importance of the health of the community as well as the individual, and respect for diversity, self-determination, empowerment, and community participation
- Show respect for peers and other healthcare professionals and the ability to foster a positive collaborative relationship with them

As a Manager

- Develop their leadership skills
- Design and organize elements of health care delivery
- Set priorities and manage time to balance practice and personal life
- Promote occupational health as physicians for sustainable practice

As a Health Advocate

- Work with patients and their families to address the determinants of health that affect them and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors

- Incorporate determinants of health disease prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of health that affect them
- Participate in a process to improve determinants of health in the community or population they serve

Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize knowledge gaps in public health and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Integrate other intrinsic roles into their learning and practice of public health
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in practice
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice

- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Strategies of CBTP

During each phase, students as a group are assigned to urban, semi-urban or rural community. The same group of students will be assigned in a given community every year to understand the fundamentals of that community life at the grass-root level. Students during their study and work in the communities use the primary health care strategies: community involvement, intersectorial collaboration and appropriate health technologies to meet their objectives.

The approaches employed for conducting CBTP include mapping, numbering and zoning of the catchments area, data collection by house-to-house visits and from relevant bodies as well as center-based data collection. With the growing number of students, the data collection may not cover every house and a suitable sampling technique shall be used when appropriate. The students then analyze the collected data, write reports and present the activities for each phase. When appropriate, interventions for health-related problems detected are made. Subsequently, regular follow-up work is done by the students in their respective communities.

Module Codes and duration of CBTP by phase of training

Phase of training	Module Code	Duration in weeks	Remark
Year II- Phase I	SPH- 402/3	4	First phase
Year III- Phase II	SPH- 402/4	4	Second phase
Year IV, Phase III	SPH- 402/5	4	Third phase
Year V, Phase IV	SPH- 402/6	4	Fourth phase

Prerequisites of each phase of CBTP Module:

In order to strengthen the link between theoretical knowledge and practical field experience the theoretical class lectures and the practical training in the CBTP are complementary. Thus students are expected to take prerequisite modules before each phase. The modules offered at every stage of the training are shown below.

Phases	Prerequisites for each phase of the Module
Phase I	Medical psychology, sociology, and anthropology, Introduction to Public Health , Environmental health, Biostatistics and Epidemiology
Phase II	Health Education, Health policy and management, Nutrition , Reproductive Health and Research Methodology
Phase III	Pediatrics & child health Internal medicine Gynecology & obstetric Surgery
Phase IV	All the required courses of preclinical and clinical including SPH I- V

Attachment Schedule

Phases of CBTP	Community attachment Activities
Phase 1	Community diagnosis (20 days) • Orientation ----- 1 day • Developing a structured format and Instrument for interview-----2 days • Data collection ---- 7 days • Data cleaning and analysis and report writing -----6 days • Rehearsal- -----2 days • Presentation/Symposium- -----1 day • Contingency---- 1 day
Phase 2	Development of Action Plan and Implement health intervention (20 days) • Orientation ----- 1 day • Developing a structured format and Instrument for interview -----2 days • Data collection for up-dating ----- 3 days • Data cleaning and analysis and report writing-----3 days • Develop action plan-----2 days • Present, discuss and review findings and action plan-----1 day • Implement health intervention-----8 days
Phase 3	Continuation of health interventions (20 days) • Orientation ----- 1 day • Developing a structured format and Instrument for interview -----2 days • Data collection for up-dating ----- 3 days • Data cleaning and analysis and report writing-----3 days • Modify action plan-----2 days • Present, discuss and review findings and action plan-----1 day • Continue implementing health intervention-----8 days
Phase 4	Monitoring and Evaluation of the community diagnosis and intervention (20 days) • Evaluate the learning experience and service provided to the community-----19 days • Presentation and Exam -----1 day

Organization of Student Groups

A group composed of 20-25 students is assigned in a Kebele. Although there is close support to students from supervisors, health service institutions and the communities, students are expected to explore situations by themselves and to share their experiences with each other. In order to foster this, student grouping is done with the following considerations:

- Level of student motivation: - those students who are better motivated to the CBTP are identified and assigned to each group.
- Academic performance: - In the grouping of students, due consideration is paid to the academic achievements of each student.
- Work experience: - If there are students with some sort of work experience in health or related field they are distributed equally among the groups.
- Language skills: - Those students who are capable of speaking the local language will be identified and distributed uniformly in each group to facilitate communication.

Each team will have a leader, reporter and logistic personnel as selected by the team members.

Job Description of CBE Coordinators

1. He/she will be appointed by the faculty academic commission
2. He/she is accountable to the faculty dean on the following issues:

The academic activities of CBE:

- The annual operational plan and activity report
 - Compiling and submitting the students grades
 - Assignment and supervision of supervisors
 - Stationary materials
 - Accept students' grade claim, clear and decide with the supervisors
3. He/she is accountable to the central CBE office on the following issues:

Submission and utilization of annual budget:

- Monitoring quality of training
- As member of standing committee
- Intervention and per diem budget
- Transport and scarce resources

Project development and proposal

4. He/she is a member of the academic commission
5. He/she is a member of the CBE standing committee
6. Plan the annual CBTP and SRP budget and activities of the faculty
7. The budget will be presented to the university by CBE and the allocated money will be controlled by central CBE office
8. The faculty CBE the allocated budget for the faculty academic activities
9. Provide the list of supervisors and days on supervision to the central CBE office for per diem payment
10. Organize the overall faculty CBE activities (CBTP & SRP)
 - Provide orientation to staffs and students at each phase of CBTP
 - Assign students to selected communities
 - Support and monitor the intervention activities and the students
11. Monitor and evaluate the faculty CBE activities
12. He/she requests the necessary materials from the faculty
13. At the end of each rotation he/she receives the activity and evaluation reports from students & supervisors
14. Evaluate and compile the students' CBTP/SRP grade and notify to the central **CBE office:**
15. Co-ordinate resource and logistic support for implementation of activities
16. Requests transport and scarce equipment from the central CBE office
17. With the approval of the faculty dean assign supervisors
18. In collaboration with the CBE office select kebeles for the CBTP
19. Revise and update the curriculum, which should be approved by the central

CBE and faculty:

20. Compiles, and submits the annual operational and activity report to the central CBE office.

Job Description of Supervisory Team Members

Students' supervision is a process where by supervisors assist students during their attachments and help them learn from the process as well. The composition of supervisory team is multidisciplinary and is based on phase objectives including speciality from behavioural and social sciences, public health, laboratory technology, nurses, environmental health, basic medical sciences and clinical medicine. It is believed that the team could contribute to the success of the programme when acting together in union than as an individual expert. Here is shown the specific tasks of supervisory team members throughout CBTP attachment.

1. Based on specific objective of the training programme, team members provide guidance and support to students.
2. Organize and coordinate students group in the practical implementation of community diagnosis (Selection of appropriate study design, population/target, measurement etc).
3. Generate a conducive educational atmosphere through discussion on application of concepts and principles of community-based education.
4. Guide students in developing survey instruments
5. Guide students in application of statistical methods in data collection, analysis, interpretation & presentation.
6. Assist students to measure factors that affect community health status through data collected from field, laboratory or clinical investigations.
7. Guide students how to perform zoning, numbering and mapping of a specific community.
8. Assist students in understanding the status of the environment as related to health in the specific community.
9. Guide and support students in application of parasitological survey, specimen handling, identification of parasite and recording.
10. Assist students to deeply understand the culture, believes, taboos of the community related to health.
11. Assist students on scientific report writing.
12. Evaluate students' activity and performance and submit the report at the end of the training programme to the supervisory team leader.

Job Description of Supervisory Team Leader

The supervisory team leader is directly accountable to the office of faculty CBE. He/She has the following responsibilities:

- A. Plans, directs, coordinates and monitors overall activities of his/her team.
- B. Confirms that the fieldwork is started on time according to the schedule and facilitates the work of his/her team.
- C. Schedules appropriate time for supervisory team members.
- D. Organizes regular supervisory team member's meeting to discuss performance and problems.
- E. Receives and compiles report from each supervisory team members, students group leader and submit to the office of faculty CBE.
- F. Makes sure that students receive proper supervision, guidance and consultation from supervisory team members.

Job description of student team leader, reporter and logistic personnel

Team leader- Is responsible for organizing the team, taking student attendance and represent students when there is a need.

Logistics- is responsible for requesting and obtaining necessary material like stationeries, instruments, tools etc.

Reporter - is responsible for compiling and presenting the activities at the end of the phase

Supportive Staff and Materials

Supportive Staff

The program has non – technical staff that takes care of the administrative affairs of the CBTP under the direct responsibility of the co-coordinator.

- Secretary typist
- Office boy/girl

Materials

The requirements are:

1. Expendable items
 - Stationary
 - Laboratory reagents
 - Fuel for transport
 - Demonstration materials
2. Non – expendable items
 - Typing machine
 - Xerox Machine
 - Duplicating machine
 - Audio – Visuals
 - Over – head projector
 - Slide projector
 - Film projector
 - Camera with photo processing equipment
 - Video camera with play back screen (TV)
 - Calculators – ordinary and scientific
 - Weighting scale – adult and baby scales
 - Office furniture for co-coordinator and supportive staff
 - Statistical current publications on Ethiopia
 - Vehicle

Evaluation System

Evaluation of the students' performance

The evaluation scheme has the following major categories.

- i. Written examination: - is conducted for each phase of the CBTP courses. The examination accounts for 20% of the students' final grade.
- ii. Symposium evaluation: - this refers to the presentations and defense for each CBTP course and accounts for 30%.
- iii. Activity evaluation by the supervisors

This includes evaluations of the students by their respective supervisors during both the fieldwork as well as report writing. The evaluation is made as per the format provided and it accounts for 40% of the students' final grade (see Annex- B1 for the format).

- iv. Peer evaluations among the students themselves

This scheme provides opportunity for the students to evaluate each other for their participation in the CBTP courses. The evaluation is made as per the format provided and it accounts for 10% of the students' final grade (see Annex-B2 for the format).

Evaluation of the Program

The need for an inbuilt evaluating mechanism assessing the effectiveness and efficiency of the CBTP is obvious. Students, supervisors, health institutions and community leaders evaluate the program at the end of each phase of CBTP. This is done by filling the evaluation format provided by the Faculty's CBE office. The evaluation made is further processed and the whole feedback is then used for future improvement of the programs. The evaluation of this program has two components.

A.Short term evaluation

This is done continuously every year. Its main objectives are:

- A.1. To determine the degree to which specific objectives laid down for each year are attended.
- A.2. To assess the practicability of general principles and strategies such as problem oriented teaching, integration of service and training, community involvement and intersectoral collaboration

- A.3. To identify major problems encountered during operation of the program regarding planning and organization in order to take appropriate remedial actions in the subsequent years.
- A.4. To assess the level of student motivation in the program.

B. Long term evaluation

This evaluation is done at the end of attachment of group of students in a specific community. Its objectives are:

- B.1. To assess the behavioral and attitudinal changes the program was able to bring about on the students.
- B.2. To determine the impact of the program on the health status of the community included under this program.

Academic Rules and Regulations

The rules and regulations pertaining to the community Based Training program (CBTP) are summarized as follows. Violations of the rules and any misconduct by a student will result in disciplinary actions being taken by the faculty's CBE office as per the regulations stated. Moreover, any implicit or explicit implications of the rules and regulations are also enforceable by the office.

1. The course of community Based Training program one (CBTP phase I) is a prerequisite course to the CBTP phase II. Accordingly a student is not allowed to register and take the second phase before he/she completes the first one.
2. A student who joins medical school by transfer cases from other Universities should take the courses in the sequence stated above and any inconveniences arising from the transfer is born by the student. E.g. a student may be delayed for one year to complete the CBTP modules.
3. Any misconduct by the student during data collection or report writing will result in the student earning a grade of "F" in the course and repeating it. Examples include insulting and quarrelling with the community members, kebele or Woreda officials, supervisors and any kind of unnecessary behaviors
4. A student who scores grades "D" or "F" in any one of the CBTP courses is subject to repeating the course.

The supervisor assigned to closely follow and give guidelines to the students during the CBTP activities should avail himself or herself for the whole duration and should always be ready to help the students. Failure to supervise will be communicated to the respective office and appropriate measures will be taken.

Main Activities in each phase of the CBTP

A period of four weeks is exclusively allotted to the program at the end of each academic year. This four weeks period is further scheduled in respective phases as follows:

Phase I Year II

At the end of this phase the student should be able to:

1. Do house numbering zoning and mapping
2. Determine the socio-demographic characteristics and means of communication of the community
3. Determine basic vital statistics indicators
4. Determine community resources and their distribution
5. Collect, analyze and interpret health and health related data
6. Identify, prioritize, plan, organize and implement intervention measures for some common problem of the community
7. Use communication skills to effect necessary health related behavioral changes to solve community health problems

Phase II year III

1. Collect and update data, analyze, interpret and draw inferences.
2. Use appropriate techniques to assess health status of a community: health interview survey, health service record review, nutritional survey such as anthropometric measurements and parasitological survey with simple laboratory tests using appropriate sampling techniques: simple random sampling,
3. Determine the sanitary level of the given community: waste disposal, water supply system, housing condition, and food sanitation and vector control
4. Identify, prioritize, plan, organize and implement intervention measures for some of the common health problem in the community
5. Organize and guide the use of appropriate technology for
 - 5.1 Safe and adequate water supply
 - 5.2 Proper liquid and solid waste disposal
 - 5.3 Proper food preparation and preservation
 - 5.4 Insect and rodent control
6. Teach & demonstrate balanced diet for the family.

Phase III year IV

1. Further apply his/her knowledge and skill through the implementation of the various programs.
2. Identify and manage the common preventable antenatal and postnatal problems.
3. Organize and carry out immunization programs.
4. Conduct schools health services including health appraisal and detection of defects.
5. Perform surveillance and containment activities (case-finding, early treatment, quarantine and isolation chemoprophylaxis)
6. Make appropriate preventive measures in population groups exposed to occupational hazards through accident prevention, protection from possible pollutants and toxicants.

Phase IV year V

1. Measure the quantitative effect of changes due to health procedures introduced into the community.
2. Evaluate the learning experience and service provided to the community
3. Write a comprehensive final report
4. Evaluate Strategies for CRC practice improvement, Compassionate leader, Stakeholder invt & organization culture in the health facilities

Teaching-Learning Methods

- Community survey
- Literature review
- Student presentation and discussion
- Portfolio
- PPRE
- Checklists of CRC evaluation

Teaching-Learning Materials

- Logbooks for entry of community experience
- Stationeries for community survey
- Computer, statistical software and internet
- Equipment and tool for conducting health assessment

Assessment Methods

Formative assessment

- ✓ Logbook and portfolio

- ✓ Global rating midway during the attachment

Summative assessment

- Community survey, action plan and presentation (40 %)
- Written Exam (20%)
- Global rating or 360 degree evaluation (15 %)
- Reflective portfolio (10 %)
- Personal research and reflection exercise (15 %)

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MODULE TITLE: SPH III: HEALTH PROMOTION AND DISEASE PREVENTION

Module Code: SPH- 402/2

Duration: 28weeks

Module Description: The module is designed to equip learners with the knowledge, skills and attitude needed to promote health and prevent disease in individuals, families and population. Students will also develop relevant competencies with the Medical expertise, Scholar, Collaborator, Communicator, Manager, Health Advocate and Professional. This module is given on the first semester during the second year of the pre-clerkship period of the undergraduate medical education program.

Module Design

Prerequisite

- Determinants of Health and Disease
- Measurement of Health and Disease

Supporting Objectives

As a Medical Expert

- Discuss the history and evolution of health promotion, including the relationships between health education, health promotion and public health
- Discuss the concepts and models of disease prevention and health promotion
- Demonstrate the contribution of the social sciences to health promotion theory and practice
- Identify priority action areas for health promotion in Ethiopia
- Discuss the epidemiology of nutritional, reproductive/maternal, newborn and child health problems globally and nationally
- Apply health problems in their social context with focus on reproductive health and maternal, neonatal and child health and malnutrition
- Apply methods of nutritional assessment and interpret results
- Discuss evidence-based strategies to improve nutrition of individuals and population
- Discuss evidence-based strategies to improve reproductive/maternal, newborn and child health
- Discuss national reproductive health and nutrition strategies
- Discuss health promotion programs in Ethiopia
- Discuss application of different health education related theories in designing and assessing behavior change
- Discuss the planning of health education in the context of the Precede-Proceed Model
- Discuss the concepts of empowerment, participation, social capital, and capacity building
- Identify barriers for the implementation of health education in individuals and population groups, based on theories of diffusion and social change
- Identify appropriate health promotion measures effective for health problems of public health significance in Ethiopia
- Demonstrate the ability to promote the health of populations by influencing lifestyle, nutrition and socio-economic, physical and cultural environment through methods of health promotion, including health education, directed towards populations, communities and individuals
- Demonstrate the ability to plan, implement and evaluate health promotion activities
- Analyze community practice experience and perform practice-based improvement activities using a systematic methodology
- Demonstrate how to search, collect, organize and interpret health and health-related information from different sources

- Demonstrate the use information and communication technology to assist in health promotion and disease prevention measures for individuals and families
- Demonstrate a habit of self-reflection, responsiveness to feedback and an on-going development of new skills, knowledge and attitude (AS3)

As a communicator

- Demonstrate the ability to communicate effectively in writing and orally with linguistic and cultural proficiency
- Apply communication and group dynamic strategies in interactions with individuals and groups
- Demonstrate the ability to use effective communication for healthcare advocacy
- Demonstrate clear, sensitive and effective communication skills in interacting with individuals, families, PHCU staff, peers and faculty

As a Collaborator

- Establish and provide advice for individuals and families to promote health and prevent illness
- Demonstrate professional values and behavior in interaction with individuals, families and communities consistent with the future role of a physician
- Demonstrate key public health values, attitudes and behaviors such as commitment to equity and social justice, recognition of the importance of the health of the community as well as the individual, and respect for diversity, self-determination, empowerment, and community participation
- Show respect for peers and other healthcare professionals and the ability to foster a positive collaborative relationship with them

As a Manager

- Develop their leadership skills
- Design and organize resources of health care delivery
- Set priorities and manage time to balance practice and personal life
- Promote occupational health as physicians for sustainable practice

As a Health Advocate

- Work with patients and their families to address the determinants of health that affect them and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors
- Incorporate determinants of health disease prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of health that affect them
- Participate in a process to improve determinants of health in the community or population they serve

Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved

- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize knowledge gaps in public health and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Integrate other intrinsic roles into their learning and practice of public health
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in practice
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Module Schedule

Weeks	Topics	Time
Week 1-10	1. Health Education and Promotion (20 hrs)	20 hours of classroom activities
Week 1-2	➤ History, concepts, aims and principles of health promotion and health education (4 hrs) • History and evolution of health promotion and health education • Concepts of health promotion and health education • Health education in PHC • Health education in Ethiopia • Basic principles of health education • Aims of health education • Contribution of social sciences to health promotion	4 hours of classroom activities 4 hours of community-based learning
Week 3	➤ Application of health education theories and models in behavior change (2hrs) • Human behavior and health • Health education theories and models • Health Belief Model • Social Learning Theory • Stages of Change • Diffusion of Innovation Theory • Theory of Planned Behavior	2 hours of classroom activities 4 hours of community-based learning including 2 hours of students' presentations on group assignments mainly on health education theories and models
Week 4-5	➤ Health communication (4hrs) • Definitions and concepts of communication • Principles of health communication • Communication model and process • Individual and group communication strategies • Effective communication skills • Barriers of communication	4 hours of classroom activities 4 hours of community-based learning

Week 6-7	➤ Planning, implementing and evaluating health education (4hrs) • Methods and materials for health education • Adult learning theories • Peer education • Conducting health education • Evaluating health education	4 hours of classroom activities 4 hours of community-based learning
Week 8	➤ Health education in different settings (2hrs) • Patient education • School health education • Prison health education	2 hours of classroom activities 2 hours of community-based learning
Week 9-10	➤ Health promotion principles (4hrs) • Health perspectives and choice of strategies to address health issues • Models and theories of health promotion (PRECEDE-PROCEED Model) • Principles of advocacy • Principles of social marketing • Principles of social/community mobilization • Community diagnosis	4 hours of classroom activities 4 hours of community-based learning
Week 11-16	2. Reproductive Health (RH) (12hrs) ○ History, concepts, definitions, components of RH (1 hour) ○ History and development of MCH/RH in Ethiopia (1 hour) ○ Family planning (2 hours) ○ Maternal health (2 hours) ○ Child health (2 hours) ○ Adolescent and youth reproductive health (3 hours)	12 hours of classroom activities 16 hours of community-based learning
Week 17-27	Nutrition in Health and Illness (22hrs) 1. Types of Nutrients and their source: carbohydrates, fats, proteins, vitamins minerals and water (2hrs) 2. Nutritional requirement (principles of determination, what RDA is about)(2hrs) • Energy requirement • Requirement for other nutrients 3. Nutritional deficiency states(4hrs) • Micro-nutrient deficiencies • Protein energy malnutrition • Nutritional anemia 4. Assessment of nutritional status (5hrs) • Anthropometric methods • Food consumption (dietary) surveys • Biochemical methods	○ 22 hours of classroom activities ○ Student presentation ○ 32hours of community-based learning

	• Clinical methods • Quality control measure in collecting data about nutritional status • Mortality and morbidity in malnutrition 5. Nutritional surveillance(2hrs) • Survey VERSUS surveillance • Policies and programs affecting nutrition • Global Nature of malnutrition 6. Cultural factors affecting nutritional status (nutritional anthropology)(2hrs) • Food faddisms, food aversions, prejudices • Factors affecting food preferences • Omissions certain foods from the diets certain groups of the population (children, pregnant women, lactating women) 7. Nutritional interventions(3hrs) • Food based approaches • Dietary diversification • Dietary Modification • Supplementation (Supplementary feeding programs, Maternal and child supplementation) • Economic approaches • Essential nutrition actions approach (Seven doable actions at six or more contract points) 8. Food handling and food born diseases(2hrs) • Food born diseases • Food poisoning • Food spoilage and deterioration • Food preservation methods • Food additive and preservatives	
Week 28	Written Final Module Exam (2hrs)	

Teaching-Learning Methods

- Interactive lecture and discussion
- Small group learning activities: assignment, exercise, case study etc
- Individual reading
- PHCU/Community-based learning and study trip: home visit, discussion with individuals and families to identify and solve problems, observation, PHCU visit, Zonal and District Health Department Visit, field visit, and targeted literature review based on community experience
- Student presentation
- Personal research and reflection exercise (PRRE)
- Reflective portfolio and mentoring

Teaching-Learning Materials

- Lecture power point presentations
- White board, black board, chalk and marker

- LCD projector and laptop computer
- Flip charts, TV and video sets
- Stationery for handouts
- Reference books
- Journals
- CDs
- Logbooks for entry of community experience

Assessment Methods

Formative assessment

- ❖ Exercise and assignment
- ❖ Logbook and portfolio
- ❖ 360 degree evaluation
- ❖ Student presentation
- ❖ Global rating of community experience midway during the module

Summative assessment

- Personal research and reflection exercise (PRRE)⁵³ (15 %)
- Reflective portfolio (5 %)
- Quiz/ Test (10%)
- Global rating of community experience (15 %)
- Assignment and student presentation (15 %)
- Written final module exam (40 %)

References

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 20. Lancet series on maternal health
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 23. Lancet series on reproductive health

MODULE TITLE: SPH IV: HEALTH POLICY AND MANAGEMENT

Module Code: SPH- 403/9

Duration: 24 weeks

Module Description

This SPH module is designed to equip medical students with the knowledge, skills and attitude needed to lead and manage health services and organizations. Students will also develop relevant competencies with the Medical expertise, Scholar, Collaborator, Communicator, Manager, Health Advocate and Professional. This module is given on the second semester during the second year of the Clerkship I period of the undergraduate medical education program.

Module Design

Prerequisite: _____

Module Objective

At the end of this module, medical students will be able to apply principles and methods of management and leadership for effective and efficient management of the Ethiopian healthcare system.

Supporting objectives

As a Medical Expert

- Describe the Ethiopian health system including historical development, organization, structure, approaches, policy, strategy, regulations and programs
- Apply principles and functions of management as well as leadership in the health sector
- Apply the principles of economics in the health sector including issues of equity and financing
- Describe the principles and components of the national health management information system
- Analyze the concepts of health care coverage, utilization and quality with focus in the Ethiopian health system
- Analyze community practice experience and perform practice-based improvement activities using a systematic methodology
- Search, collect, organize and interpret health and health-related information from different sources

- Use information and communication technology to assist in health promotion and disease prevention measures for individuals and families

As a communicator

- Demonstrate skills for effective communication with individuals, families, communities, health sector staff, local leadership and development partners with sensitivity to personal and cultural factors for the promotion of health and prevention of diseases
- Advise individuals and families to promote health and prevent illness

As a Collaborator

- Demonstrate professional values and behavior in interaction with individuals, families and communities consistent with the future role of a physician
- Demonstrate key public health values, attitudes and behaviors such as commitment to equity and social justice, recognition of the importance of the health of the community as well as the individual, and respect for diversity, self-determination, empowerment, and community participation
- Show respect for peers and other healthcare professionals and the ability to foster a positive collaborative relationship with them
- Demonstrate a habit of self-reflection, responsiveness to feedback and an on-going development of new skills, knowledge and attitude

As a Manager

- Develop their leadership skills
- Design and organize elements of health care delivery
- Set priorities and manage time to balance practice and personal life
- Promote occupational health as physicians for sustainable practice

As a Health Advocate

- Work with patients and their families to address the determinants of health that affect them and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors
- Incorporate determinants of health disease prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of health that affect them
- Participate in a process to improve determinants of health in the community or population they serve

Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance

- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize knowledge gaps in public health and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Integrate other intrinsic roles into their learning and practice of public health
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in practice
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Module Schedule

Week	Contents	Teaching/Learning Activities
Week 1 – 6	Health Policy and Management (48 hrs) ➤ Introduction (3hrs) • Definitions of Management, health and public health • Meaning and scope of health service management ➤ Management Theories (3hrs) • Introduction • Classical management theory • The behavioral school	6 hours of classroom activities 6 hours of student seminar presentation 12 hours of community-based learning

	• The quantitative school • Other approaches to management theory ➤ Development and organization of the health system (6hrs) • Global typologies for health systems • Development and organization of the health system in Ethiopia (indigenous/modern, public/private) • Approaches in organization and delivery during various periods (modern health services, basic health services, PHC, SWAP); Initiatives in health development (including MDGs, PASDEP) • Policies, strategies and programs during various periods	
Week 7-11	➤ Management and leadership in the health sector (10hrs) • Health management with focus to the Ethiopian health system (planning, organizing, implementation, monitoring and evaluation) (6hrs) • Concepts and applications of leadership in the health sector (4hrs)	10 hours of classroom activities 10 hours of community-based learning
Week 12-15	Resource Management (human, finance, time and material (8hrs) ➤ Human Resource Management • Human Resource Planning • Recruitment and selection • Orientation and Socialization • Supervision • Performance appraisal • Promotion, transfer and supervision ➤ Theories of motivation • Content theories of motivation • EGG theories • Two factor theories • Process of theories of motivation • Implication for health managers ➤ Managing other resources • Managing materials/ Equipment • Managing drugs • Managing finance/money • Managing time • Material space	8 hours of classroom activities 16 hours of community-based learning
Week 16-17	➤ Health services coverage, utilization and quality(4hrs) • The concept of coverage and patterns of health	4 hours of classroom activities 4 hours of community-

	care utilization (including organization and use of the referral system) (3hrs) • Quality of health care (1hr)	based learning
Week 18-20	➤ Health Economics and Health Care financing (6hrs) • Application of economics to the health sector • Demand and supply in the health sector • Issues of equity in the health sector • Methods of economic evaluation and costing of health care programs • Principles and types of healthcare financing • Healthcare financing in Ethiopia	6 hours of classroom activities 6 hours of community-based learning
Week 21	➤ Health management information system (HMIS) (2hrs) • Principles and components of HIS/HMIS • Ethical and legal issues in HMIS • Application of IT in the analysis and management of health systems • Ethiopian HMIS	2 hours of classroom activities 2 hours of community-based learning
Week 22-23	➤ Managing a health team • Structure and functions • Staffing patterns • Controlling and assessing the work • Health information in the management process ➤ Public Health Laws/ National health legislations	4 hours of classroom activities 4 hours of community-based learning
Week 24	Final Written Exam (2hrs)	

Teaching-Learning Methods

- Interactive lecture and discussion
- Small group learning activities: assignment, exercise, case study etc
- Individual reading
- PHCU/Community-based learning and study trip: home visit, discussion with individuals and families to identify and solve problems, observation, PHCU visit, Zonal and District Health Department Visit, field visit, and targeted literature review based on community experience
- Student presentation
- Personal research and reflection exercise (PRRE)
- Reflective portfolio and mentoring

Teaching-Learning Materials

- Lecture power point presentations
- White board, black board, chalk and marker
- LCD projector and laptop computer
- Flip charts, TV and video sets
- Stationery for handouts
- Reference books
- Journals
- CDs

- Logbooks for entry of community experience

Assessment Methods

Formative assessment

- ❖ Exercise and assignment
- ❖ Logbook and portfolio
- ❖ 360 degree evaluation
- ❖ Student presentation
- ❖ Global rating of community experience midway during the module

Summative assessment

- Personal research and reflection exercise (PRRE)⁵⁴ (15 %)
- Reflective portfolio (5 %)
- Quiz/ Test (10%)
- Global rating of community experience (15 %)
- Assignment and student presentation (15 %)
- Written final module exam (40 %)

References (textbooks and articles)

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2. Wagstaff A, Van Doorslaer E. Equity in the finance and delivery of health care. 1995.
3. Shaw RP, Griffin CC. Financing health care in Sub-Saharan Africa through user fees and insurance. The World Bank, 1995.
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MODULE TITLE: SPH-V: RESEARCH METHODOLOGY

Module Code: SPH- 504/9

Duration: 19 weeks

Module Description: The module is intended to prepare learners to critically evaluate medical literature and design operational health research. Students will also develop relevant competencies with the Medical expertise, Scholar, Collaborator, Communicator, Manager, Health Advocate and

Professional. This module is given on the first semester during the third year of the Clerkship II period of the undergraduate medical education program.

Module Design

Prerequisite: SPH II (measurement of health and disease: Biostatistics & Epidemiology)

Module Objective

At the end of this module, the medical student will be able to critically evaluate medical literature and develop a research proposal.

Supporting Objectives

As a Medical Expert

- Apply the principle of health research including its rationale
- Identify and prioritize health research topic
- Applycritically review basic medical literature
- Develop research proposal
- Apply a software program to analyze data
- Interpret research findings appropriately

As a communicator

- Demonstrate effective research writing and presentation skills
- Demonstrate clear and effective communication skills in interactions with individuals, families, communities, PHCU staff, local health department staff, peers and faculty
- Recognize when the values, biases, or perspectives of communities, patients, physicians, or other health care professionals may have an impact on the quality of care, and modify the approach to the patient and community accordingly

As a Collaborator

- Demonstrate the principles of research ethics
- Demonstrate key public health values, attitudes and behaviors such as commitment to equity and social justice, recognition of the importance of the health of the community as well as the individual, and respect for diversity, self-determination, empowerment, and community participation
- Show respect for colleagues and other healthcare professionals and the ability to foster a positive collaborative relationship with them
- Demonstrate a habit of self-reflection, responsiveness to feedback and an on-going development of new skills, knowledge and attitude

As a Manager

- Develop their leadership skills
- Design and organize elements of health care delivery
- Set priorities and manage time to balance practice and personal life
- Promote occupational health as physicians for sustainable practice

As a Health Advocate

- Work with patients and their families to address the determinants of health that affect them and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors
- Incorporate determinants of health disease prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of determinants of health that affect them

- Participate in a process to improve determinants of health in the community or population they serve

As a Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize knowledge gaps in public health and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Integrate other intrinsic roles into their learning and practice of public health
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in practice
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting

- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Module Schedule

Week	Content	Time
Week 1	1. Introduction to research methods (2hrs) ➤ Definition of research ➤ Rationale for research ➤ Types of research ➤ Features of health systems research ➤ The research process	2 hours of classroom activities 2 hours of community based learning
Week 2-4	2. Planning a research (6hrs) ➤ Identifying and prioritizing topics for research (2hrs) ➤ Analysis and statement of the problem (2hrs) ➤ Review of available literature and information (1hr) ➤ Formulation of research objectives (1hr)	6 hours of classroom activities 6 hours of community based learning
Week 5-11	3. Research Methodology (14hrs) ➤ Types of study designs ➤ Study variables ➤ Data collection techniques ➤ Sample size and sampling methods ➤ Plan for data collection and organization ➤ Plan for data processing and analysis ➤ Ethical considerations ➤ Pre-testing the methodology ➤ Steps of student research proposal development	14 hours of classroom activities 14 hours of community based learning
Week 12	4. Work Plan (1hr) 5. Budget Plan (1hr)	2 hours of classroom activities 2 hours of community based learning
Week 13	6. Plan for Project Administration, Monitoring and Utilization of results (1hr) 7. Finalizing and reviewing the research proposal (1hr)	2 hours of classroom activities 2 hours of community based learning
Week 14-16	8. How to carry out a research (6hrs) • Principles of fieldwork activities • Methods of data processing and analysis (Epi Info, SPSS, qualitative data analysis) • Scientific writing skills • Skills of scientific presentation	6 hours of classroom activities 6 hours of computer lab
Week 17	9. Writing manuscript for publication (2hrs)	2 hours of classroom activities 2 hours of community based learning
Week 18	10. Critical evaluation of research (4hrs)	2 hours of classroom activities 2 hours of community based learning

Week 19	Exam(2hrs)	
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Teaching-Learning Methods

- Interactive lecture
- Exercises and assignments
- Computer lab practice(Statistical software programs such as Epi Info and SPSS and qualitative data analysis programs like Open Code)
- Practice, coaching and feedback: developing research proposal, conducting research, data analysis, report writing and oral presentation
- Portfolio
- PRRE

Teaching-Learning Materials

- Lecture power point presentations
- White board, black board, chalk and marker
- LCD projector and laptop computer
- Flip charts, TV and video sets
- Stationery for handouts
- Reference books
- Journals
- CDs
- Logbooks for entry of community experience
- Computers with appropriate statistical and word processing software

Assessment Methods

Formative assessment

- ❖ Exercise and assignment
- ❖ Logbook and portfolio
- ❖ 360 degree evaluation
- ❖ Student presentation
- ❖ Global rating of community experience midway during the module

Summative assessment

- Personal research and reflection exercise (PRRE)⁵⁵ (15 %)
- Reflective portfolio (5 %)
- Quiz/ Test (10%)
- Global rating of community experience (15 %)
- Assignment and student presentation (15 %)
- Written final module exam (40 %)

References

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2. Corlien M. Varkevisser, IndraPathmanathan, and Ann Brownlee. Designing and Conducting Health Systems Research Projects: Volume 1 (Proposal Development and Fieldwork). KIT/IDRC. 2003

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4. HenrykDancygier: Clinical epidemiology. How to do clinical practice research
5. Margaret L. Brandeau: Operations research and health care. Handbook of methods and applications
6. Nigel Bruce, Danniell Pope and Debbi Stanistreet: Quantitative methods for health research. A practical interactive guide to epidemiology and statistics
7. Ann Bowling: Research methods in health. Investigating health and health service.

INTERNSHIP MODULES

MODULE TITLE: TEAM TRAINING PROGRAM (TTP)

Module Code: IN-601/5

Duration: 4weeks

Place in the curriculum: at the end of Internship

Attachment Description: The TTP attachment is intended to provide medical interns experiential learning opportunities while providing primary health care services by teaming up with other health professionals. Students will also develop relevant competencies with the Medical expertise, Scholar, Collaborator, Communicator, Manager, Health Advocate and Professional. This module is given at the end of Internship period of the undergraduate medical education program. This supervised practice covers both clinical and public health tasks.

Module Design

Pre-requisite: Clerkship II and III

Attachment Objective

At the end of this attachment the medical intern will be able to provide essential clinical and public health services in primary health care setting in the context of effective teamwork with other healthcare students.

Supporting Objectives

As a Medical Expert

- Diagnose and manage patients at the primary health care unit level in an ethical and efficient manner
- Diagnose and manage common infectious diseases and nutritional deficiencies
- Recognize immediate life-threatening conditions, institute appropriate first-line treatment and arrange referral for those who require higher level of care
- Provide basic and comprehensive emergency obstetric care available at PHCU level
- Provide continued care for patients with common chronic health conditions)
- Identify and refer patients who require higher level of care
- Provide comprehensive reproductive health/maternal and child health
- Perform minor surgical and obstetric procedures
- Perform and interpret basic laboratory tests
- Identify priority community health problems and hazards and their determinants
- Design and implement effective and feasible health promotion and disease prevention interventions
- Design and implement health education sessions on priority health issues
- Advise individuals and families to promote health and prevent illness

- Evaluate effectiveness, efficiency, accessibility, equitability, and quality of health services
- Analyze community practice experience and perform practice-based improvement activities using a systematic methodology
- Use information technology to manage information, access online medical information, and support one's own education
- Search, collect, organize and interpret health and health-related information from different sources
- Use information and communication technology to assist in health promotion and disease prevention measures for individuals and families

As a communicator

- Communicate effectively with individuals, families, communities, PHCU staff, local health department staff, peers and faculty
- Demonstrate a habit of self-reflection, responsiveness to feedback and an on-going development of new skills, knowledge and attitude

As a Collaborator

- Interact with other healthcare professionals through effective teamwork
- Lead and manage healthcare team and health services at PHCU level
- Mobilize community partnerships and action to identify and solve community health problems
- Interact with individuals and families with sensitivity to personal and cultural factors
- Demonstrate professional values and behavior in interaction with individuals, families and communities consistent with the future role of a physician
- Demonstrate key public health values, attitudes and behaviors such as commitment to equity and social justice, recognition of the importance of the health of the community as well as the individual, and respect for diversity, self-determination, empowerment, and community participation
- Show respect for peers and other healthcare students and professionals and the ability to foster a positive collaborative relationship with them

As a Manager

- Develop their leadership skills
- Design and organize elements of health care delivery
- Set priorities and manage time to balance practice and personal life
- Promote occupational health as physicians for sustainable practice

As a Health Advocate

- Work with patients and their families to address the determinants of health that affect them and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors
- Incorporate determinants of health disease prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of health that affect them
- Participate in a process to improve determinants of health in the community or population they serve

Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice

- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize knowledge gaps in public health and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations
- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Integrate other intrinsic roles into their learning and practice of public health
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in practice
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Attachment Schedule

Attachment Weeks	Main Activities of TTP attachment
Week 1	Community diagnosis and Community entry

	❖ Orientation ❖ Developing a structured format and Instrument for interview ❖ Discussion with local administration, health office, PHCU staff and community representatives about attachment objectives and roles and responsibilities of all parties ❖ Data collection ❖ Data cleaning and analysis and report writing ❖ Develop action plan ❖ Presentation and Discussion
Week 2	Public health interventions (outreach & clinical services) ❖ Plan and implement PHCU and local health office activities in coordination with them ❖ Clinical service at OPD, clinic, wards and outreach sites throughout the week including duty ❖ Public health interventions: • Health education • School health • Prison health • EPI services • Epidemic investigation and management • Primary health care evaluation including clinical services, environmental health activities (inspection of water sources, food hygiene in public restaurants, public sanitation facilities, waste disposal, health facilities supervision, workplace safety) • Mini-project to solve priority community health problems ❖ Home visit on Fridays for half day ❖ Weekly activity report and seminar on Friday afternoons
Week 3	❖ Continue Week 2 activities
Week 4	❖ Same as Week 2 for the first three days Evaluation and Monitoring ❖ Evaluate effectiveness and efficiency of the service rendered and the community learning experience in the next two days ❖ Overall reporting and discussion

Typical Weekly Schedule of Week 2-3

	Monday	Tuesday	Wednesday	Thursday	Friday
AM	Group I: Clinic service at PHCUs including duty Group II: Public health interventions	Group I: Clinical service at PHCUs including duty Group II: Public health interventions	Group I: Public health interventions Group II: Clinical service at PHCUs including duty	Group I: Public health interventions Group II: Clinical service at PHCUs including duty	Home visit
PM	Group I: Clinic service at PHCUs including	Group I: Clinic service at PHCUs including duty	Group I: Public health interventions Group II:	Group I: Public health interventions Group II:	Weekly activity report Seminar

	duty	Group II: Public health interventions	Clinical service at PHCUs including duty	Clinical service at PHCUs including duty	
	Group II: Public health interventions				

NB.

- Depending on the number of interns and size of facilities, 2-4 interns will be assigned for duty at OPD and wards during evening hours and weekends.
- Seminar topics will be selected through discussion between interns and faculty based on national and local relevance
- Outreach and public health interventions will be coordinated with plans of the district and PHCUs.

Teaching-Learning Methods

- Community survey
- Mini-project
- Supervised clinical practice
- Supervised community practice
- Portfolio
- PRRE⁵

Teaching-Learning Materials

Can vary according to the targeted need of each activity

- Logbooks for entry of community experience
- Stationeries for community survey
- Drugs, equipment, tools and materials for clinical and public health interventions
- White board, black board, chalk and marker
- LCD projector and laptop computer
- Flip charts, TV
- Stationery for handouts
- Reference books
- Journals

Assessment Methods

Formative assessment

- Logbook and portfolio
- Global rating midway during TTP
- Weekly activity report
- Seminar

Summative assessment

- 360 degree evaluation (20 %)
- Global rating (20 %)
- Reflective portfolio (15 %)
- PRRE⁵⁶ (15 %)
- Community diagnosis (15 %)

- Mini-project (15 %)

References

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MODULE TITLE: STUDENT RESEARCH PROJECT

Module Code: 602

Duration: 44 weeks

Module Description

All medical students are required to do an independent research work. The research project is intended to enable medical students to design and conduct operational health research. While doing their research they will gain knowledge, attitude and skill on problem identification, data collection, data analyses, proposing solutions and implementation. The research project will also be a springboard for future elaborated research. The project should not be an expensive to finance by the college and research advisors will be assigned for each student researcher.

This module is intended to prepare learners to design and conduct operational health research. Students will also develop relevant competencies with the Medical expertise, Scholar, Collaborator, Communicator, Manager, Health Advocate and Professional. This module is expected to be summarized and presented at the end of the internship attachments of the undergraduate medical education program.

Module Design

Prerequisite: SPH I-V

Module Objective

At the end of this module, the medical student will be able to design and conduct an operational health research. Medical interns are expected to:

Supporting Objectives

As a Medical Expert

- Select research topic of public health importance
- Conduct literature review
- Develop research proposal
- Apply research principles
- Analyze data and make appropriate interpretation of findings
- Develop a scientific research report
- Disseminate research findings
- Effectively analyze community based health and health related problems and perform evidence -based interventions against pertinent health problems of the country using a systematic research methodology
- Demonstrate the ability to plan, implement and evaluate health research activities
- Identify valid and appropriate health research measures for health problems of public health significance in Ethiopia

As a communicator

- Demonstrate the ability to communicate effectively in writing and orally with linguistic and cultural proficiency
- Apply communication and group dynamic strategies in interactions with individuals and groups
- Demonstrate the ability to use effective communication for healthcare advocacy
- Demonstrate clear, sensitive and effective communication skills in interacting with individuals, families, PHCU staff, peers and faculty

As a Collaborator

- Establish and maintain advise individuals and families to promote health and prevent illness

- Demonstrate professional values and behavior in interaction with individuals, families and communities consistent with the future role of a physician
- Demonstrate key public health values, attitudes and behaviors such as commitment to equity and social justice, recognition of the importance of the health of the community as well as the individual, and respect for diversity, self-determination, empowerment, and community participation
- Show respect for peers and other healthcare professionals and the ability to foster a positive collaborative relationship with them

As a Manager

- Develop their leadership skills
- Design and organize elements of health care delivery
- Set priorities and manage time to balance practice and personal life
- Promote occupational health as physicians for sustainable practice

As a Health Advocate

- Work with patients and their families to address the determinants of health that affect them and their access to needed health services or resources and to increase their opportunities to adopt healthy behaviors
- Incorporate determinants of health disease prevention, health promotion, and health surveillance into interactions with individual patients
- Lead a healthy life style to be role models to promote population health
- Work with a community or population to identify the determinants of health that affect them
- Participate in a process to improve determinants of health in the community or population they serve

Scholar

- Develop, implement, monitor, and revise a personal learning plan to enhance professional practice
- Identify opportunities for learning and improvement by regularly assessing their performance
- Engage in collaborative learning to continuously improve personal practice and contribute to collective improvements in practice
- Recognize the power of role-modeling and the impact of the hidden curriculum on learners
- Promote a safe learning environment
- Ensure patient safety is maintained when learners are involved
- Collaboratively identify the learning needs of others and prioritize learning outcomes
- Provide effective feedback to enhance learning and performance
- Use assessment and evaluation tools and practices that are relevant to the teaching and learning context
- Recognize knowledge gaps in public health and other professional encounters and generate focused questions that can address them
- Identify scholarly sources that shed light on a given professional question
- Interpret study findings, including a discussion
- Determine the validity and risk of bias in a wide range of scholarly sources
- Describe study results in both quantitative and qualitative terms
- Demonstrate an understanding of the scientific principles of research and scholarly inquiry and the role of research evidence in contemporary health care
- Identify ethical principles for research and incorporate them into obtaining informed consent, considering potential harms and benefits, and considering vulnerable populations

- Contribute to the work of a research program
- Pose questions amenable to scholarly investigation and select appropriate methods to address them
- Summarize and communicate to professional and lay audiences, including patients and their families, the findings of relevant research and scholarly inquiry

As a Professional

- Integrate other intrinsic roles into their learning and practice of public health
- Exhibit appropriate professional behaviors and relationships in all aspects of practice, reflecting honesty, integrity, commitment, compassion, respect, altruism, respect for diversity, and maintenance of confidentiality
- Demonstrate a commitment to excellence in all aspects of learning
- Recognize and respond to ethical issues in practice
- Exhibit professional behaviors in the use of technology enabled communication
- Demonstrate accountability to patients, society, and the profession
- Recognize the professional and ethical codes, standards of practice, and laws governing practice
- Recognize and respond to unprofessional and unethical behaviors in physicians and other colleagues in the health care professions
- Participate in peer assessment and standard-setting
- Exhibit self-awareness and effectively manage influences on personal well-being and professional performance
- Promote a culture that recognizes, supports, and responds effectively to colleagues in need

Module Schedule

Week	Content	Time
Week 1-22	Research Title selection and Proposal Development	Year V Sem II
Week 23 -24	Proposal presentation	Year V Sem II
Week 25-28	Exam study time	Year V Sem II
Week 29-31	Revision of proposal	Internship Sem I
Week 32-41	Final submission of proposal	Internship
Week 42	Data collection	Research leave
Week 43-46	Analysis and Write-up	Internship
Week 47	Draft research report submission	Internship
Week 48-51	Final research report submission	Internship
Week 52-60	Preparation for oral presentation	Internship
Week 62	Research defense presentation	At the end of Internship attachment

Teaching - Learning Methods

- Periodical advice, supervised research
- Portfolio

Teaching-Learning Materials

- LCD and computer

- Writing board and marker or chalk
- Flip charts
- Handouts of lecture materials and other pertinent reference materials
- Computers with data analysis software and internet access

Assessment Methods

Formative assessment

- Portfolio
- Progressive assessments of research activities

Summative assessment

- Developing a research protocol or proposal (40 %)
- Final research report (30 %)
- Oral presentation of research finding (30 %)

References

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INTERNSHIP

DESCRIPTION AND DESIGN:

Internship is intended to prepare and enable the graduate to function independently in a supervised setting. Internship spans for 62 weeks. The intern is expected to consult and refer patient when needed.

CLINICAL:

It included: Internal Medicine, General Surgery, Pediatrics and Obstetrics & Gynecology attachments for 10 weeks each, emergency medicine for 4 weeks Orthopedics, Psychiatry, Ophthalmology, dermatology, ENT and dentistry for 2 weeks each.

PROFESSIONAL COMPETENCY DEVELOPMENT (PCD):

PCD will be continued during Internship will be offered through WGS, SGS, Seminars, this includes: Quality Improvement, Infection prevention, Teaching methodology introduction, Leadership and hospital management and other selected areas of relevance.

SOCIAL AND POPULATION HEALTH (SPH)

The interns will be attached to primary health care units for 4 weeks additional, 2 weeks for exclusive Research thesis work has been allotted and One week for research presentation. This

attachment helps the Interns to further consolidate and apply the various Biomedical and Clinical sciences, Professional Competency Development (PCD) and Social and Population Health leanings (SPH) and practice in the primary health care setting more independently.

OBJECTIVE OF INTERNSHIP

Internship will provide an opportunity for the students to: further develop the different competencies attained in the four years education

